POST-TRANSLATIONAL PROCESSING OF MICROTUBULE PROTEIN DURING PERIPHERAL NERVE REGENERATION

A thesis submitted in accordance with the requirements of

the University of Liverpool for the degree of

Doctor of Philosophy

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by

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September 1993

Acknowledgements

I wish to express my thanks to my supervisor, Dr.Graham McLean, for his help, encouragement, guidance shown to me throughout my studentship. Indeed his patience and good humour throughout this work, in particular, during the preparation of this thesis.

Furthermore, I would like to express my thanks to Dr. Alan Hargreaves my friend and colleage, for his constant support, unerring optimism, sympathetic ear throughout my studentship, oh! and the hours of fun we had arguing over politics.

I thank Professor Alasdair Breckenridge for the use of facilities in the Department of Pharmacology and Therapeutics and for his interest in this project.

The excellent technical assistance of Dave Trafford, Anne Ward, Lynn McLaughlin and Keith Blackburn is gratefully acknowledged. I would also like to thank Yvonne Janvier for typing this manuscript.

I am grateful to the Wellcome Trust for their financial support throughout this studentship.

Finally, I would like to thank my parents, Brian and Andrea Mullins, my sister Samantha Green and my fiancee Joy Paviour for their continual support and interest and to whom I will be forever grateful.

Publications

Mullins, F.H., Hargreaves, A.J. and McLean, W.G. (1991). Tubulin:tyrosine ligase activity in regenerating peripheral nerve. *Biochem. Soc. Trans.* **19** 1143-1146

Mullins, F.H., Li, J-Y., Hargreaves, A.J., Dahlstrom, A. and McLean, W.G. (1992). Tyrosination state of alpha-tubulin in peripheral nerve regeneration. *Neurochem. Int.* **21** (Suppl) A4 16.

Mullins, F.H., Hargreaves, A.J., Li, J-Y., Dahlstrom, A. and McLean, W.G. (1993). Tyrosination state of alpha-tubulin in regenerating peripheral nerve. *J. Neurochem.* **61** (In Press)

Post-translational processing of microtubule protein during peripheral nerve regeneration F.H.Mullins

In response to neuronal injury, a number of changes occur in the organization of the neuronal cytoskeleton. There is evidence from cell culture studies that post-translational modifications of tubulin and microtubule-associated proteins (MAPs) may have a role in these events. The aim of the work was therefore to examine two such modifications, the tyrosination of α-tubulin and the phosphorylation of β -tubulin and MAP-2 in regenerating peripheral nerve.

Adult rats were anaesthetized and the left or right sciatic nerve randomly selected and crushed to induce regeneration. Five days later the rats were killed and the regenerating and the contralateral (control) nerves were removed. Quantitative immuno-blotting of nerve homogenates revealed a significant increase ($P<0.01$) in the proportion of α-tubulin that was tyrosinated in nerve pieces distal (peripheral) to a nerve crush compared to nerve pieces proximal (central) to a nerve crush and to uncrushed nerve. No such difference occurred in ligatured (crushed but non-regenerating) nerve, implying that the increase was related to the presence of regenerating fibres. This effect was confirmed by cytofluorimetric scanning and fluorescence confocal laser scanning microscopy of fixed sections of control and regenerating nerve, stained with antibodies directed against tyrosinated α-tubulin. Furthermore, a significant ($P<0.01$) increase was found in the proportion of tyrosinated α-tubulin in non-assembled tubulin fractions in nerve pieces containing regenerating fibres. This occurred in the absence of a change in the proportion of assembled and non-assembled tubulin. Measurements of tubulin:tyrosine ligase (TTL) activity *in vitro*, indicated a decrease in tyrosine incorporation into tubulin from nerve pieces distal, compared to those proximal to a nerve crush. There was no such difference in ligatured nerves. It is proposed that the increased amount of tyrosinated α-tubulin is related to an alteration in assembly rate of microtubules required for neurite outgrowth and that the apparent decrease in the TTL activity *in vitro* reflects the increased tyrosination *in vivo*.

The study of post-translational phosphorylation of microtubule protein concentrated on the role of protein kinases in the phosphorylation of β-tubulin and MAP-2. From the same sciatic nerve regeneration model outlined above, homogenates of appropriate nerve pieces and of dorsal root ganglia were sub-fractionated into soluble and membrane-bound fractions. The protein kinase activity associated with each fraction was assayed independently under various conditions designed to enhance particular kinase activities.

Measurement of cyclic AMP-dependent protein kinase activity associated with the supernatant fractions, leading to incorporation of ^{32}P-ATP-derived phosphate into exogenous MAP-2 prepared from pig brain, showed a statistically significant increase in this activity in regenerating nerve ($P<0.05$) to uncrushed contralateral nerve. The presence of MAP-2 in sciatic nerve was confirmed by immunohistochemistry and of hyperphosphorylated MAP-2 was confirmed by Western blotting with specific antibodies. Under the same conditions there was no alteration in the phosphorylation of β-tubulin nor of Ca^{++}-calmodulin- or second messenger-independent protein kinase activity directed against MAP-2. But significant differences were noted in the association of total MAP-2 with the axonal cytoskeleton in regenerating nerve.

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INTRODUCTION

NERVE REGENERATION AND GROWTH

In the peripheral nervous system, when a nerve is injured by a crush or transection, the portion of the axon disconnected from the cell body (distal stump) undergoes Wallerian degeneration. The degeneration is characterized by the rapid recruitment of myelomonocytic cells (macrophages) to the injury site and into the distal stump (Perry *et al.*, 1987; Fawcett and Keynes, 1990). Macrophages accumulate in the distal stump, prior to Schwann cell proliferation and together with the Schwann cells, they are responsible for the removal and the recycling of the degenerated material (Hall, 1989; Fawcett and Keynes, 1990). Macrophages have been implicated in the recycling of lipids (Stoll *et al.*, 1989; Snipes *et al.*, 1986), indeed, the accumulation of apo-lipoprotein E synthesized by macrophages, is involved in receptor mediated uptake of cholesterol by growth cones for membrane biosynthesis (Mahley *et al.*, 1988). Furthermore, macrophages may modify the mRNA levels of nerve growth factor (NGF; Lindholm *et al.*, 1987), a retrogradely transported trophic factor, via the synthesis of interleukin-1.

The degenerated fibres in the distal stump are replaced by regenerating neurites (sprouts) that arise proximal to the injury site and regrow through the endoneurial tubes in the distal stump. The regenerating axons are usually in contact with the Schwann cell basal lamina and the Schwann cell membrane, that present surface adhesion molecules e.g. L1, N-CAM and N-cadherin which may help to promote neurite outgrowth (Bixby and Harris, 1991), until they re-establish their connections with the target tissue. The process known as regeneration is characterized by changes in the biochemical, physiological and structural properties

of the nerve.

Characteristic of the cell body response are morphological and metabolic changes. The main morphological response of the cell body is chromatolysis (Lieberman, 1971), while one of the earliest metabolic changes is an increase in the activity of ornithine decarboxylase (ODC; Gilad and Gilad, 1983) involved in the decarboxylation of ornithine to putrescine, important in the synthesis of polyamines necessary for cell growth and differentiation. Also characteristic of the cell body response to nerve injury, are changes in the expression and the synthesis of growth associated proteins and cytoskeletal proteins (Neumann *et al.*, 1983; Goldstein *et al.*, 1988; Hoffman and Cleveland, 1988; Miller *et al.*, 1989; Oblinger and Wong, 1990). In particular, the synthesis of the growth associated protein, GAP-43 is elevated during regeneration but decreases upon reinnervation (Bisby, 1988), and is thought to have a facilitatory effect on nerve regeneration.

Neurones possess an intracellular network of protein filaments collectively known as the neuronal cytoskeleton. The cytoskeletal components: microtubules, microfilaments and intermediate filaments are differentially and spatially organized within the axon (Mangeat, 1988; Cancalon, 1990; Baas and Ahmad, 1992).

Elongation of regenerating neurites is mediated by the growth cone, an enlarged terminus of the growing neurite containing large amounts of the cytoskeletal proteins tubulin and actin, but little if any neurofilament proteins (Bunge, 1973; Gordon-Weeks, 1987; Cheng and Sahyoun, 1988). It is the tubulin and actin cytoskeletal elements which provide the motile machinery of the growth cone, and are important for the generation and maintenance of the shape of growth cones (Mitchison and Kirschner, 1988; Smith, 1988).

The involvement of the cytoskeletal proteins actin and tubulin in the active process of axonal elongation can be sub-divided into three stages, orientation, consolidation and conversion. Principally, microtubules provide axonal extension while actin filaments direct the growth (Mitchison and Kirschner, 1988).

During the orientation stage actin microfilaments in the filopodia undergo retraction or extension dependent on specific guidance cues. Following interaction with a favourable guidance cue, the filopodia are stabilized and consolidation of the orientation of the growth cone is achieved by microtubule polymerization.

During consolidation, the unstable microtubules present in the growth cones are converted into stable microtubules (Mitchison and Kirschner, 1988). Consolidation can follow two mechanisms; instructive or selective. The instructive mechanism involves directing microtubule polymerization or already polymerized microtubules along paths designated by the orientation phase. The other selective mechanism involves the presence of a subset of dynamic microtubules, polymerizing and depolymerizing in random directions. These eventually become stabilized or stable microtubules according to the length of time the filopodia exist in a particular direction or as a result of receptor-dependent stabilization (Mitchison and Kirschner, 1988). The dynamic instability of microtubules has a greater importance in decision making, in the selective mechanism of consolidation.

After orientation and consolidation there is a partial stabilization of actin filaments and microtubules to direct axonal elongation. To maintain this growth or path the actin filaments and microtubules must be stabilized permanently; this is called conversion. The following sequence of events has been proposed for conversion (Mitchison and Kirschner, 1988). The consolidation of microtubules

inhibits the protrusive nature of the actin microfilament network. The inhibition of this activity leads to isotropic compression tension on microtubules resulting in bundling, probably further promoted by microtubule associated proteins (MAPs). The bundling of microtubules is correlated with the loss of adhesion of the regenerating fibre to the Schwann cell substratum, and this leads to the formation of a new section of axon.

Thus the formation of growth cones and subsequent axonal elongation following nerve injury requires a reorganization of the cytoskeleton for these events to proceed (Cancalon, 1990).

NEURONAL CYTOSKELETON

The neuronal cytoskeleton maintains neuronal integrity through the regulation of the assembly/disassembly, the interactions, and the stability of cytoskeletal components. In particular, cell morphology (Weber *et al.*, 1975; Kirschner and Mitchison, 1986; Hoffman *et al.*, 1984; 1985; 1987; Hirokawa, 1991), axonal transport (Schnapp *et al.*, 1985; Vale *et al.*, 1985a,b,c; Allen *et al.*, 1982; 1985 Stebbings, 1990; Kuznetsov *et al.*, 1992) and axonal growth and elongation (Yamada *et al.*, 1970; Mitchison and Kirschner, 1988; Gordon-Weeks, 1991) are influenced and governed by the neuronal cytoskeleton, and modifications to it.

Microtubules

Microtubules are the largest of the neuronal cytoskeletal filaments. They are arranged into helical cylinders consisting of 13 protofilaments, each protofilament being made up of linear arrangements of heterodimers comprising α-and β-tubulin subunits (tubulin molecules; Sloboda, 1980). They have a diameter of approximately 25-28nm, a hollow core of 15nm and a wall thickness of 5nm. (Sloboda, 1980).

Axonal microtubules are orientated longitudinally with the polarity of microtubules always in the same direction, + (or fast-growing) ends at the distal part of the axon and the - ends towards the cell body (Soifer, 1982; Baas and Ahmad, 1992).

Microtubules are involved in numerous cell functions, in particular, the maintenance of cell shape, and intracellular translocation of proteins; indeed, these functions are dependent on the ability of microtubules to polymerize and depolymerize (Avila, 1990; Kirschner and Mitchison, 1986).

Microtubules may undergo three forms of polymerization; self assembly, directional polymerization, and dynamic instability. During self-assembly tubulin monomers and polymers exist in equilibrium with one another, and when a critical concentration of monomer is reached and exceeded, microtubule polymerization/self assembly occurs (Oosawa and Kasai, 1962; Kirschner and Mitchison, 1986). In this equilibrium, tubulin monomers would associate and dissociate from both ends of the microtubule. Microtubules may also exhibit directional polymerization, whereby net microtubule depolymerization occurs at one end and polymerization at the other. This model of polymerization/depolymerization was termed 'treadmilling' (Margolis and Wilson, 1978). In this model, it was proposed tubulin assembly into microtubules was coupled with GTP hydrolysis based on previous work of Weisenberg and Deery, 1976.

Tubulin has two binding sites for guanine nucleotides, one on the β-tubulin subunit termed the E or exchangeable site and another on the α-tubulin subunit termed the N or non-exchangeable site (Luduena *et al.*, 1987; Linse and Mandelkow, 1988; Carlier, 1989; Osei *et al.*, 1990). Once steady state is reached, the tubulin incorporated at one end of the microtubule (+ end) is equivalent to that released at

the opposite end (- end), which suggests that the amount of tubulin that is polymerized, together with the average length and number of microtubules, remains effectively constant at any given time. This assembly mechanism is thought to be dependent on GTP hydrolysis because of evidence from experiments using non-hydrolysable GTP analogues (Sandoval and Weber, 1980; Terry and Punch, 1980) where treadmilling of microtubules does not occur.

A third model termed the 'dynamic instability' was proposed for microtubule assembly/disassembly (Mitchison and Kirschner, 1984). In this model, the amount of total polymerized tubulin at any given time remains constant, but the mean length and total number of microtubules may differ. Evidence in support of the dynamic instability model was obtained from kinetic studies of GTP hydrolysis and microtubule assembly (Carlier and Pantaloni, 1981; Carlier *et al.*, 1989; Osei *et al.*, 1990) which suggested GTP hydrolysis was not directly coupled with tubulin assembly, but was likely to occur after an alteration to the conformation of the tubulin molecule following assembly. Characteristic of the dynamic instability model is the presence of two states of microtubules: a growing state and shrinking state which exist in tandem with one another i.e. one microtubule may grow, while another shrinks.

The polymerization state of microtubules is dependent on the nature of the nucleotides present on the ends of the microtubule. The presence of nonhydrolyzed GTP-bound tubulin at the ends of microtubules (called caps), is characteristic of a growing state (Carlier *et al.*, 1989; Caplow *et al.*, 1989). The GTP-bound tubulin cap stabilizes the microtubule, providing favourable conditions for microtubule elongation. Conditions which increase the number of tubulin molecules with

unhydrolyzed GTP at the capped ends, result in an increase in the rate of tubulin assembly. Physical prevention of assembly leads to the hydrolysis of the GTP bound to tubulin molecules on the capped ends. The hydrolysis results in GDP-bound tubulin being present on the capped ends, which in turn leads to a rapid and complete depolymerization (this is characteristic of the shrinking state). However, when microtubule depolymerization is complete the GDP bound to tubulin can be exchanged for GTP, which makes tubulin assembly competent once again.

As previously mentioned, the regulation of microtubule stability (assembly/disassembly) is important in the maintenance of neuronal integrity, supported by studies that have shown that colchicine and vincristine, which promote tubulin disassembly (Gottschalk *et al.*, 1968; Moix *et al.*, 1991; Greeson *et al.*, 1992) impair axonal transport, affecting neuronal function. The nerve contains different populations of microtubules with different stabilities (Brady *et al.*, 1984; Donoso, 1980; Filliatreau *et al.*, 1988; Denoulet *et al.*, 1989; Baas *et al.*, 1991; 1992). The molecular basis for this stability is poorly understood, but a number of different mechanisms may influence the stability of microtubules. The heterogeneity of the tubulin subunits which constitute microtubules may be important in conferring different stabilities to assembled microtubules (Brady *et al.*, 1984; Detrich *et al.*, 1987; Binet and Meininger, 1988; Banerjee *et al.*, 1990). In particular, the heterogeneity of tubulin is increased by post-translationally modified forms of tubulin (Gozes and Sweadner, 1981). Furthermore, the interactions of microtubules with other cellular components, in particular, microtubule associated proteins (MAPs), may be important in conferring microtubule stability (Olmsted, 1986).

Microtubules in axonal transport

Microtubules support both anterograde and retrograde axonal transport, which transport newly synthesized proteins to and from the axonal terminal, respectively (Allen *et al.*, 1985; Allan *et al.*, 1991). The axonal transport system may be subdivided into the three methods of transport: fast anterograde (forward) transport, slow axonal transport and fast retrograde axonal transport (Allan *et al.*, 1991).

Fast anterograde transport conveys essentially newly synthesized membranous organelles, neuro-transmitters and growth-associated proteins. The microtubule based fast anterograde transport depends on ATP-dependent microtubule motors (Adams, 1982; Allan *et al.*, 1991; Vale, 1992) one of which is known as kinesin, an ATPase that consists of two large α-sub-units with an apparent molecular weight of 110-125 kDa and two smaller β-sub-units with an apparent molecular weight of 60-70 kDa (Vale *et al.*, 1985b, 1992; Kuznetsov and Gelfand, 1986, Kuznetsov *et al.*, 1988; Wagner *et al.*, 1989).

Fast retrograde transport (from nerve endings to the cell body) conveys membrane bound organelles back to the cell body for recycling or for their elimination. The microtubule motor molecule thought to be responsible for fast retrograde transport is MAP-1C a microtubule associated ATPase (Paschal and Vallee, 1987). Together with its scavenger role, retrograde transport provides a mechanism for sending signals from the nerve terminal to the cell body (Bisby *et al.*, 1987; Moix *et al*, 1991; Greeson *et al.*, 1992).

Slow axonal transport may be split into two kinetic components which may be differentiated by the rate of axonal transport, along the axon, of the proteins they carry. These components are termed slow components a and b (SCa and SCb;

Hoffman and Lasek, 1975; Black and Lasek, 1980). The slower component, SCa, travels at 0.1-15mm/day and carries principally tubulin, MAPs and neurofilaments, whereas the faster component, SCb travels at 2-30mm/day and carries mainly actin microfilaments. A role for microtubules in the slow axonal transport of the components is unclear. However, a microtubule-activated GTPase called dynamin (Shpetner and Vallee, 1989; 1992; Scaife and Margolis, 1990) has been implicated in microtubule translocation, and may be a putative motor for slow axonal transport mediated by microtubules. Recently, however, it has been questioned whether this protein possesses sufficient force generating capacity (Vale, 1992) and therefore requires further investigation.

The delivery of structural and growth-associated materials to the regenerating neurites is mediated by fast and slow axonal transport. Fast anterograde transport is responsible for the delivery of newly synthesized growth associated proteins, during nerve regeneration, to the regenerating fibres. One growth-associated protein in particular is GAP-43, (Skene, 1989, Skene and Verag, 1989) a protein implicated in growth cone-substrate interactions (Meiri and Gordon-Weeks, 1990), as a G-protein regulator (Strittmatter *et al.*, 1990), and as a calmodulin binding protein. The synthesis and expression of this protein is enhanced as a consequence of nerve injury and it is post-translationally modified to allow it to be fast axonally transported (Skene, 1989, Skene and Verag, 1989) to the growth cone. While the rate of fast transport is unaltered as a consequence of nerve injury (Chan *et al.*, 1989) the integrity of the microtubule-based delivery system must be maintained to provide fast axonal transport. Retrograde transport also has an important role in nerve regeneration, in particular signal-transduction mechanisms, in particular, the transport of signals from the injury site to the cell body (Moix *et al.*, 1991; Greeson *et al.*,

1992).

The rate of regeneration appears to be limited by the delivery of cytoskeletal proteins viz. tubulin and actin by slow axonal transport (Wujek and Lasek, 1983; McLean, 1985) and reflects their involvement in the locomotion of the axonal growth cone. As previously mentioned, tubulin, the major component of microtubules, is axonally transported in two distinct groups of proteins slow components a and b (SCa and SCb; Hoffman and Lasek, 1975; Black and Lasek, 1980; McLean *et al.*, 1983) which may be distinguished biochemically. One contains essentially stable forms (SCa), whereas the faster component of the two contains more dynamic forms (SCb; Tashiro and Komiya, 1989;1991). Following neuronal injury there is a shift in tubulin stability characterized by an increase in the proportion of tubulin transported in SCb (Hoffman & Lasek, 1980; McLean, 1985). The shift in tubulin transport during nerve regeneration also reflects the importance of the cytoskeletal protein tubulin in nerve growth.

Tubulin

Tubulin is a heterodimeric protein composed of α- and β-subunits (Luduena *et al.*, 1977) with apparent molecular weights of 57 kDa and 55 kDa, respectively. Resolution of tubulin by iso-electric focusing has revealed that tubulin subunits exhibit a wide heterogeneity (Wolff *et al.*, 1982; Field *et al.*, 1984; Field and Lee, 1985; Cleveland and Sullivan, 1985). The significance of the wide tubulin heterogeneity is unclear, but may be important in the formation of microtubule sub-populations with different dynamic properties specific for certain cellular functions. Part of tubulin heterogeneity may be attributed to the presence of multiple genes which encode α- and β-tubulin subunits.

At least six genes have been detected for mouse α-tubulins (Villasante *et al.*, 1986; Cleveland, 1987; Gu *et al.*, 1988) which display a high degree of homology (Little and Selhaus, 1988), with the sequence heterogeneity restricted mainly to the carboxyl-terminal region of α-tubulin (Villisante *et al.*, 1986). Only four of these genes appear to be expressed in neuronal tissue one of which encodes for a tubulin isoform which does not contain a C-terminal tyrosine residue unlike the other genes which encode for tubulin isoforms which contain C-terminal tyrosine residues (Villasante *et al.*, 1986). Six genes for β-tubulin have been characterized in mammalian tissue (Bond *et al.*, 1984; Wang *et al.*, 1986; Burgoyne *et al.*, 1988) of which only five are expressed in neuronal tissue (Wang *et al.*, 1986), and like α-tubulin gene products there is a large sequence homology among β-tubulin isoforms (Burgoyne *et al.*, 1988; Little and Selhaus, 1988).

Microtubules formed by polymerization of α- and β- subunits, may contain several different tubulin isotypes (Cleveland, 1987). Indeed, the heterogeneity of tubulin is further increased by post-translational processing of tubulin which can include: acetylation (L'Hernault and Rosenbaum, 1985; Edde *et al.*, 1991), tyrosination/detyrosination (Barra *et al.*, 1974; Raybin and Flavin, 1977) and phosphorylation (Gard and Kirschner, 1985; Matten *et al.*, 1990).

The synthesis and the expression of tubulin has been shown to be enhanced in nerve regeneration (Neumann *et al.*, 1983). The expression of particular tubulin genes is greater than others. The class II β-tubulin mRNA expression is selectively induced during nerve regeneration over the other β-tubulin mRNAs (Hoffman and Cleveland, 1988). The expression of Tα1 tubulin mRNA is also increased in nerve regeneration (Miller *et al.*, 1989). The increases of β(class II) and Tα1 tubulin

expression are also seen during development (Bond *et al.*, 1984; Miller *et al.*, 1987). The genetic control of the heterogeneity of tubulin may be responsible for the formation of populations of microtubules with differential stability, which may confer specificity of microtubules for particular cellular functions. Indeed, in support of this, it has been demonstrated that microtubules with distinct dynamic characteristics can exist within the same cells (Kreis, 1987; Schulze *et al.*, 1987). Therefore it is likely that the increases in Tα1 mRNA and β(class II) mRNA are important in active axonal elongation, and that these isoforms confer particular characteristics to microtubules, necessary for nerve regeneration.

Microtubule associated proteins

A group of polypeptides known as microtubule associated proteins (MAPs) have been shown to co-purify with tubulin when isolated from brain by cycles of *in vitro* assembly and disassembly. (Murphy and Borisy, 1975; Olmsted, 1986; Wiche, 1989). MAPs are present on microtubules in neurones and are thought to exert a functional effect on the assembly and the stability of microtubules (Olmsted, 1986; Wiche, 1989) and enable microtubules to interact with other cytoskeletal proteins e.g. actin microfilaments (Sattilaro, 1986), and mediate organelle translocation along microtubules (Vale *et al.*, 1985a,b,c; Paschal and Vallee, 1987; Vallee *et al*, 1988; Vale, 1992). The MAPs that have been studied in detail are the high molecular weight (HMW) MAPs: MAP-1, MAP-2, MAP-3, MAP-4 and MAP-5 (Vallee and Bloom, 1984; Olmsted,1986; Wiche, 1989) and the tau proteins (Weingarten *et al.*, 1975; Cleveland *et al.*, 1977a,b; Lichtenberg *et al.*, 1988; Couchie *et al.*, 1992). Other MAPs which have been studied in detail include, STOP proteins (Job *et al.*, 1982, 1983; Margolis *et al.*, 1986) and chartins (Magendantz and Solomon, 1985).

MAP-1 has been resolved into three separate isoforms termed MAP-1a, 1b

and 1c with apparent molecular weight of 320-350 kDa (Bloom *et al.*, 1984;1985). MAP-1a and MAP-1b have little common structural relationship with each other (Bloom *et al.*, 1985; Luca *et al.*, 1986) and differ in their co-assembly with microtubules; (Bloom *et al.*, 1985). MAP-1b co-assembles less with microtubules than MAP-1a, however, MAP-1b may be post-translationally phosphorylated (Diaz-Nido *et al.*, 1988) which enhances its ability to co-assemble with microtubules. MAP-1c is a microtubule activated ATPase (Paschal *et al.*, 1987) and is thought to play a role in organelle transport along microtubules. For example MAP-1c was shown to stimulate mobility, or gliding, of isolated microtubules on glass coverslips. This gliding occurs in one direction, suggesting that MAP-1c is an unidirectional motor (Paschal *et al.*, 1987; Vale, 1992;); in particular it is thought to be the retrograde motor equivalent to cytoplasmic dynein (Gibbons, 1988). Other workers have detected other MAP-1 isoforms e.g. MAP-1.2 (Asai *et al.*, 1985), MAP-1x (Calvert and Anderton, 1985); however, they appear to be closely related and similar to MAP-1b not only in terms of apparent molecular weight, but also in terms of their cellular distribution and control of expression during neuronal development (Safaei and Fischer, 1989; Diaz-Nido and Avila, 1989).

MAP-1 may play a role in nerve regeneration; the expression and phosphorylation of MAP-1b has been correlated with neurite extension (Alletta *et al.*, 1988; Diaz-Nido *et al.*, 1988; Safaei and Fischer, 1989). Indeed, the amount of MAP-1b mRNA increases in correlation with the amount of newly growing axons (Safaei and Fischer, 1989). During development MAP-1a and MAP-1b exhibit differential expression, as development progresses MAP-1a is associated with a more stable cytoskeletal structure while MAP-1b appears to be associated with a more dynamic cytoskeleton, (Bloom *et al.*, 1985; Nixon *et al.*, 1990) On the other hand,

the post-translationally phosphorylated form of MAP-1b (Diaz-Nido *et al.*, 1990) is associated with sub-populations of stable microtubules in axons. The change in the expression and the post-translational modification of MAP-1b may therefore reflect a change in cytoskeletal dynamics during regeneration.

MAP-2 consists of a group of related phosphoproteins, two main isoforms- MAP-2a and MAP-2b with molecular weight approximately 280 kDa and an isoform with a lower apparent molecular weight 70 kDa termed MAP-2c (Vallee, 1984; Couchie and Nunez, 1985; Garner *et al.*, 1988). Like MAP-1, MAP-2 can be post-translationally phosphorylated, and can modify microtubule stability (Vallee, 1980; Avila *et al.*, 1988). MAP-2 has the ability to bind to microfilaments (Satillaro, 1986) and neurofilaments (Heimann *et al.*, 1985) which is in some way regulated by phosphorylation of MAP-2 (Satillaro, 1986). MAP-2 isoforms exhibit differential distribution; MAP-2a and 2b appear to be localized preferentially in the cell bodies and dendrites as determined by immunohistochemistry (Bernhardt and Matus, 1984; Caceres *et al.*, 1984; DeCamilli *et al.*, 1984), although immunoreactivity has been seen in axons (Papasozomenos *et al.*, 1985). MAP-2c has been shown to be present in axons, particularly during development of brain (Tucker *et al.*, 1988). Furthermore, the expression of MAP-2 forms appears to be developmentally regulated (Binder *et al.*, 1984; Burgoyne and Cumming, 1984). MAP-2a, 2b and 2c contain microtubule binding domains, which are located at the carboxyl-terminus region of the MAP-2 molecule; this tubulin binding domain has been shown to exhibit homology with that on the tau microtubule associated proteins (Lewis *et al.*, 1988). It is thought that this binding domain is involved in promoting microtubule assembly and stabilization (Vallee, 1980, 1984; Vallee *et al.*, 1984).

MAP-2, in particular MAP-2c, expression and distribution have been correlated with axonal elongation (Tucker *et al.*, 1988). In neuronal cell cultures MAP-2 expression can be induced by ganglioside treatment (Ferreira *et al.*, 1990) and correlates with neurite extension. During development MAP-2 expression is correlated with neurite formation and the increase in neurite extension (Ferreira and Caceres, 1989). Indeed, MAP-2c (also known as immature MAP-2) has been detected in the growth cones and filopodia of developing neurons (Caceres *et al.*, 1986). It was suggested perhaps MAP-2 has a role in neuronal commitment by stabilizing microtubules (Ferreira and Caceres, 1989). However phosphorylation of MAP-2 inhibits microtubule assembly (Burns and Islam, 1985).

MAP-3 was identified in brain (Huber *et al.*, 1985) and is a high molecular weight MAP, although it has not been as extensively studied as MAP-1 and MAP-2 because of the difficulties in its purification. However, an antibody has been raised against MAP-3 (Huber *et al.*, 1985, 1986) which recognizes a doublet with an apparent molecular weight around 180 kDa. Although not directly tested, it is thought to stimulate assembly, since incubation with the MAP-3 antibody reduced microtubule assembly (Huber *et al.*, 1986).

MAP-4 like MAP-3 was identified using an antibody raised against microtubule proteins (Parysek *et al.*, 1984a,b; 1985) and has an apparent molecular weight of 215-240 kDa (Parysek *et al.*, 1984a,b). MAP-4 has both an N-terminal projection domain and a C-terminal tubulin (microtubule) binding domain (Aizawa *et al.*, 1990; West *et al.*, 1991) and has sequence homology in the microtubule binding domain with MAP-2 and tau (Lewis *et al.*, 1988; Himmler *et al.*, 1989; Aizawa *et al.*, 1990, 1991; Mori *et al.*, 1991).

MAP-5 has also been identified by use of an antibody raised against microtubule proteins. It has an apparent molecular weight of 320 kDa (Riederer *et al.*, 1986) and is thought to be equivalent to MAP-1b as well as MAP-1x (Calvert and Anderton, 1985; Garner *et al.*, 1989, Cleveland, 1990).

Tau proteins were the first microtubule-associated proteins to be identified (Weingerten *et al.*, 1975) with apparent molecular weights of 35-65 kDa. (Goedert *et al.*, 1989). They contain four homologous tubulin binding repeats as seen with MAP-2 and MAP-4 and the Tau heterogeneity is explained by alternative mRNA splicing of a single tau gene (Lee *et al.*, 1988; Himmler *et al.*, 1989; Goedert *et al.*, 1989, 1990; Couchie *et al.*, 1992), and also by post-translational phosphorylation of tau proteins (Hagestedt *et al.*, 1989). The phosphorylation of tau proteins can influence microtubule stability, resulting in microtubule disassembly. Tau protein mRNA has been shown to increase in the early stages of brain development with a similar expressional pattern to α-tubulin mRNA. The mRNA for tau proteins decreases as development continues (Couchie *et al.*, 1988). It is not known whether the expression of tau proteins is altered during nerve regeneration.

The functional role of the high molecular weight MAPs and Tau proteins, except of MAP-1c, are still unclear. It is known these MAPs promote assembly (Kuznetsov *et al.*, 1981; Olmsted, 1986; Aizawa *et al.*, 1987) and may cross link microtubules to other cytoskeletal components such as actin microfilaments (Satillaro *et al.*, 1986) and neurofilaments (Miyata, 1986). They are thought to be involved in the regulation of neuronal differentiation (Olmsted, 1986; Ferreira and Caceres, 1989) in particular, MAP-2 and Tau through their effects on microtubule assembly and dynamics. In summary, MAPs are beleived to promote stability by suppression

of dynamic instability (Horio and Hotani, 1986; Farrell *et al.*, 1987); however, this characteristic property of MAPs may be altered by post-translational phosphorylation, in particular the activities of MAP-2 and tau.

Actin microfilaments

Actin exists in either a monomeric globular form, termed G-actin, or as a polymeric filamentous form, termed F-actin (Oosawa and Asakura, 1975). Actin is highly conserved molecule (Pollard and Cooper, 1986). Up to six genes for actin are expressed in higher vertebrates (Vanderkerckhove and Weber, 1984), two of which express the β and γ cytoplasmic actin isoforms found in neurones. These isoforms differ from each other in the N-terminal region, a region postulated to contain the binding site for actin binding proteins (Sutoh and Mabuchi, 1986).

The assembly of actin from G-actin to F-actin is affected by the ionic concentration of the local environment. In conditions where there is a low ionic concentration, then actin exists in its depolymerized G-actin form. In conditions of higher ionic strength and the presence of divalent metal cations such as magnesium, G-actin polymerizes into the long, helical double stranded polymer F-actin form (Oosawa and Asakura, 1985; Cooper *et al.*, 1983). Furthermore, actin is able to bind ATP and the bound ATP may be hydrolysed into ADP and P_i. However, the hydrolysis of ATP is not necessary for actin assembly. The kinetics of actin disassembly and assembly at the barbed (+) end and pointed end (-) are different dependent on the state of actin; ADP-actin dissociates from the + end of the actin microfilament faster than ATP-actin. An analogy can be drawn between this dynamic instability model (Carlier *et al.*, 1984; Pollard and Weeds, 1984; Mitchison and Kirschner, 1984) and that of GTP-bound tubulin and microtubules.

Actin assembly may be regulated by actin-binding proteins. In growth cones of extending neurites, the assembly of actin is regulated by actin depolymerizing factor (ADF) (Bamburg and Bray, 1987) ADF is a major component of the growth cone and its action is inhibited by phosphorylation (Morgan and Bamburg, 1988). Other actin binding proteins which influence the assembly of G-actin into F-actin include profilin and gelsolin (Blikstad *et al.*, 1980; Yin *et al.*, 1981, Vandekerckhove, 1989). These seem to regulate assembly by binding monomeric G-actin, thus shifting the critical actin concentration required for assembly, or regulate assembly by severing assembled F-actin (profilin and gelsolin respectively).

The actin microfilament system has been implicated in exocytotic events in the nerve (Matter *et al.*, 1989; Trifaro *et al.*, 1992) and in rapid organelle movement, which is independent of microtubules and microtubule based motors, in nerve cells (Kuznetsov *et al.*, 1992). Indeed, actin is involved in the active process of axonal elongation, in particular, in the orientation of the growth cone (Mitchison and Kirschner, 1988), as outlined above.

Intermediate filaments

The intermediate filaments proteins consist of five types of proteins:

Type I	acidic keratins
Type II	basic keratins
Type III	vimentin, peripherin, glial fibrillary acidic protein and desmin
Type IV	neurofilaments
Type V	nuclear lamins

The neuronal intermediate filament family tends to consist of type III and type IV proteins only; in particular, neurofilaments (Type IV), vimentin and peripherin

(type III) are the most widely reported intermediate filaments to be found in neurones (Shaw and Weber, 1982; Lee *et al.*, 1982; Portier *et al.*, 1984; Cochard and Paulin, 1984; Traub, 1985; Greene, 1989).

The type IV intermediate filaments are composed of three neurofilament proteins. These triplet proteins (Hoffman and Lasek, 1975; Liem *et al.*, 1978) have different molecular weights, as determined by amino acid sequencing (Julien *et al.*, 1987; Lees *et al.*, 1988; Myers *et al.*, 1987). The lower molecular weight neurofilament protein termed NF-L has a molecular weight about 60 kDa, the second triplet protein termed NF-M has a molecular weight of 100 kDa and the high molecular weight neurofilament triplet protein has a molecular weight of about 115 kDa. The neurofilament triplet proteins assemble to form neurofilaments; NF-L can support assembly into neurofilaments without NF-M and NF-H, but NF-M or NF-H on their own are less able to form neurofilaments (Geisler and Weber, 1981; Liem and Hutchinson, 1982; Tokutake *et al.*, 1984).

The functional importance of neurofilaments would appear to be largely architectural, in that their main function is to maintain axon calibre (Hoffman *et al.*, 1987). Although a functional role in axonal transport or other cellular functions has not been elucidated, alterations in the post-translational phosphorylation of neurofilaments have been observed in a number of neuropathological diseases e.g. Alzheimers disease and diabetic neuropathy (Sternberger *et al.*, 1985; Pekiner and McLean, 1991; McLean *et al.*, 1992).

The mRNA levels for neurofilaments, in particular the 68 kDa neurofilament protein, decrease in the first few weeks of nerve regeneration (Hoffman and Cleveland, 1988; Oblinger and Wong, 1990). Furthermore, the translation/synthesis

of neurofilament proteins is reduced, in the initial stages of nerve regeneration (Greenberg and Lasek, 1988; Oblinger and Wong, 1990). The changes in transcription and translation are correlated with a decrease in the neurofilaments present in regenerating fibres, suggesting the role of neurofilaments in the axon is associated principally with maturation and radial growth of the axon rather than a major role in axonal elongation *per se* (Oblinger and Wong, 1990).

The down-regulation of neurofilament expression and synthesis may be important with respect to changes in the axonal transport of the other cytoskeletal proteins, in particular, tubulin. The model proposed for slow axonal transport of cytoskeletal elements (Lasek, 1986) would suggest the slowest moving elements regulate the speed of the axonal transport of the other cytoskeletal elements. Thus, the reduction in neurofilaments in the axon, reflected by a decreased expression and synthesis during nerve regeneration, would have the effect of increasing the speed of axonal transport of the other cytoskeletal elements such as tubulin. In fact, an increase in the transport velocity of tubulin has been seen during nerve regeneration (Hoffman and Lasek, 1980; McLean, 1985) and the reduction in neurofilaments may play a role in this phenomenon.

The type III intermediate filaments are less well studied. Peripherin is a neuronal intermediate filament protein which exists in three isoforms with apparent molecular weights of 56 kDa, 58 kDa, 61 kDa; the major form being the 61 kDa protein (Greene, 1989; Landon *et al.*, 1989). However, vimentin is not normally found in neurones except in cultured or immature cells (Cochard and Paulin, 1984).

The expression of type III and type IV intermediate filaments are differentially regulated. During development neuroblasts express the type III intermediate filament

protein vimentin, with the expression of type IV neurofilament proteins with the high molecular weight neurofilament protein NF-H appearing much later during development (Cochard and Paulin, 1984; Pachter and Liem, 1984; Bennett, 1987; Cochard and Paulin, 1984). The expression of peripherin has been shown to increase in PC12 cells upon NGF stimulation (Portier *et al.*, 1984; Troy *et al.*, 1992) and to be upregulated during nerve regeneration at a time when neurofilament mRNA expression is at its lowest (Wong and Oblinger, 1990; Oblinger and Wong, 1990). Furthermore, the expression of vimentin is increased in sciatic nerve following a crush injury, the increase observed in nerve segments distal to the crush injury (DeLeon *et al.*, 1991). The functional correlate of this upregulation of type III intermediate filaments has yet to be elucidated.

In summary, the formation of growth cones and subsequent axonal elongation following nerve injury requires a reorganization of the cytoskeleton for these events to proceed (Cancalon, 1990). The synthesis and expression of cytoskeletal proteins may be altered as a consequence of nerve injury and the rate of nerve regeneration is related to the rate of slow axonal transport of cytoskeletal components, in particular SCb. Furthermore, following neuronal injury there is a shift in the proportion of tubulin transported in SCb (Hoffman and Lasek, 1980; McLean, 1985) reflecting a change in the dynamics of tubulin transport. This too may be mediated by a change in the expression of specific tubulin isoforms, known to occur as a consequence of nerve regeneration (e.g. Tα1 and β (class II); Hoffman and Cleveland, 1988; Miller *et al.*, 1989). However, a shift in the stability of the pre-existing cytoskeleton is probably largely the result of post-translational processing of cytoskeletal components, particularly in instances where changes in cytoskeletal dynamics occur before the arrival of newly synthesized cytoskeletal components

(Cancalon, 1990). Post-translational processing is a mechanism that may dynamically alter the biochemical properties of proteins in areas that may be remote from the protein synthesis centre. Post-translational modification of microtubule proteins may play a role in the reorganization of the pre-existing cytoskeleton during nerve regeneration.

POST-TRANSLATIONAL MODIFICATION OF MICROTUBULE PROTEINS

Tyrosination of α-tubulin

The C-terminal of α-tubulin may be post-translationally tyrosinated or detyrosinated. This process is mediated by the enzyme tubulin:tyrosine ligase (Barra *et al.*, 1988).

Tubulin:tyrosine ligase (TTL) has a molecular weight of 35 kDa (Raybin and Flavin, 1975) and forms a stoichiometric 1:1 complex with the tubulin dimer; the stable complex has a ligase binding site on the β-tubulin subunit (Wehland and Weber, 1987). On digestion of the ligase with V8 protease a 14 kDa and a 30 kDa fragment are produced. The 30 kDa fragment contains the binding site for β-tubulin and for ATP, while the 14 kDa fragment contains the catalytic properties of the ligase (Wehland and Weber, 1987). TTL is an ubiquitous enzyme with the level of activity higher in brain than any other tissue (Raybin and Flavin, 1977; Barra *et al.*, 1988), and α-tubulin is the only substrate of the ligase (Arce *et al.*, 1975; Barra *et al.*, 1988). TTL prefers non-assembled tubulin to assembled tubulin as a substrate (Arce *et al.*, 1978, Gundersen *et al.*, 1987).

The detyrosination of C-terminal tyrosinated microtubules is not only performed via tubulin:tyrosine ligase but also by an enzyme termed tubulin:tyrosine

carboxypeptidase (TTCP) (Hallak *et al.*, 1977; Argarana *et al.*, 1978, 1980; Modesti and Barra, 1986). TTCP has an apparent molecular weight of 90 kDa (Argarana *et al.*, 1980), and prefers assembled tubulin as a substrate over non-assembled tubulin. The tyrosination state of microtubules should therefore depend on relative activities of TTL and TTCP, and also the assembly state of tubulin.

It has been suggested that not all the tubulin pool participates in the post-translational tyrosination/ detyrosination cycle (Rodriguez and Borisy, 1978, Barra *et al.*, 1988; Arregui and Barra, 1990) and that possibly only some of the translated tubulin isoforms may participate in this cycle i.e. certain β-tubulins may not bind the ligase, or α-tubulins may not be able to be tyrosinated. It has also been proposed that there is a difference in the non-tyrosinable tubulin present in embryonic and in adult chick brain, 23% and 55% respectively (Rodriguez and Borisy, 1978; Barra *et al.*, 1988). Furthermore, these workers also found that the tyrosination state of α-tubulin differed between developing and adult tissue. The level of tyrosinated tubulin during development was greater than that in adults (Rodriguez and Borisy, 1978; Cumming *et al.*, 1984). These observations can be correlated with changes in TTL activity (Deanin *et al.*, 1977).

Studies performed on neuronal cell cultures that undergo neurite extension have also revealed changes in both TTL activity and the tyrosination state of tubulin following differentiation induced with nerve growth factor (Deanin *et al.*, 1982; Cambray-Deakin and Burgoyne, 1987; Arregui *et al.*, 1991). Thus it is likely that, as a consequence of neuronal injury and subsequent nerve regeneration, the changes in TTL activity and/or the tyrosination state of tubulin observed in cell culture and developmental studies may be recapitulated during nerve regeneration in the sciatic

nerve.

A functional role for tyrosinated tubulin is still unclear, but it has been shown to be differentially localized into distinct sub-populations of microtubules (Gundersen *et al.*, 1987; Robson and Burgoyne, 1989) that can be distinguished biochemically. Tyrosinated microtubules are dynamic whereas detyrosinated microtubules are more stable, distinguished by their sensitivity to temperature and depolymerizing drugs (Kreis, 1987; Gundersen *et al.*, 1987; Watson *et al.*, 1990).

It has been shown that C-terminal tyrosination can modify the secondary structure of tubulin (Postingl *et al.*, 1979) with no apparent effect on its assembly characteristics *in vitro* (Barra *et al.*, 1988). *In vivo*, tyrosination of microtubules may have an indirect effect on the stability/assembly state of microtubules. The tyrosination state *in vivo* may affect the binding of MAPs, known to promote the assembly of microtubules. Indeed, work has shown a smaller proportion of MAPs were associated with microtubules formed from detyrosinated than tyrosinated tubulin (Kumar and Flavin, 1982). Phosphorylation of the C-terminal tyrosine of tubulin has been shown to inhibit assembly (Wandosell *et al.*, 1987) *in vitro*. This secondary modification may well regulate the assembly of tubulin *in vivo*.

The correlation of increased tyrosination with the dynamic events that occur in development and growth cone mobility and biochemical dynamic characteristics shown *in vitro*, does suggest that tyrosination of tubulin plays a role in the assembly of microtubules, albeit an indirect one.

Acetylation of α-tubulin

Acetylation of α-tubulin was first described in *Chlamydomonas*, and is mediated by an acetylase. This post-translational process was found to occur on the

Epsilon-amino group of a lysine residue located at position 40 in the primary structure of α-tubulin (L'Hernault and Rosenbaum, 1983; 1985).

The monoclonal antibody 6-11B-1 raised against α-tubulin (Piperno and Fuller, 1985) was used to demonstrate the presence of acetylated tubulin in neurones by immunocytochemical and immunoblotting analyses (Cambray-Deakin and Burgoyne, 1987; Black and Keyser, 1987; Cambray-Deakin *et al.*, 1988). The immunological analyses were supported by incorporation assays on cell cultures using [^{3}H] acetate and protein synthesis inhibition (Black and Keyser, 1987; Edde *et al.*, 1991).

Using the monoclonal antibody 6-11B-1, acetylated α-tubulin has been found to co-localize with detyrosinated tubulin (Cambray-Deakin and Burgoyne, 1987). Furthermore, acetylated tubulin has been shown to be enriched in stable microtubules (Piperno and Fuller, 1987; Cambray-Deakin *et al.*, 1988; Webster and Borisy, 1989; Falconer *et al.*, 1989; Baas *et al*, 1991).

Acetylation of α-tubulin appears not to be coupled to its assembly, but the tubulin acetylase does prefer assembled tubulin as a substrate (Maruta *et al.*, 1986; Black and Keyser, 1987). However, the possibility that acetylation of α-tubulin may enhance the stability of microtubules is supported by observations made on cytoplasmic microtubules treated with depolymerizing drugs: microtubules containing acetylated α-tubulin are more stable than assembled microtubules that are not acetylated (LeDizet and Piperno, 1986). This would suggest acetylation of α-tubulin may enhance the binding to tubulin of MAPs which enhance the stability of assembled microtubules.

The above observations correlate with the appearance of acetylated α-tubulin following NGF-induced neuronal differentiation in PC12 cells. The appearance of acetylated α-tubulin does not occur until later stages of neurite extension, probably reflecting the amount of tubulin assembled at the early and later stages. During the early stages of neurite extension in cell culture, the total tubulin that exists in an assembled state is lower than that in later stages. (Black and Keyser, 1987).

Tubulin acetylase has only been partially purified from *Chlamydomonas* (Maruta *et al.*, 1986) but as yet has not been purified or characterized from mammalian tissue. It has only recently been shown that an acetyltransferase activity exists in nerve brain which acetylates tubulin on a single residue (Lys_{40}) the same as that by the acetylase present in *Chlamydomonas* (Edde *et al.*, 1991). Doubt has also been shown over the acetate labelling of α-tubulin in cultured cells being due to $Lys_{(40)}$ residue acetylation (Edde *et al.*, 1991). It has been shown in cell culture that the majority of radioactive acetate is converted to glutamate. It is this radioactive product, glutamate which labels tubulin by post-translational glutamylation rather than acetylation (Edde *et al.*, 1991).

Polyglutamylation of α-tubulin

Polyglutamylation of α-tubulin as a post-translational process has only recently been identified (Edde *et al.*, 1990). This modification involves the post-translational addition of a varying (1-5) number of glutamyl units on the γ-C-terminal of the glutamate reside (Glu^{445}) of α-tubulin. Polyglutamylation may be present on both detyrosinated and tyrosinated α-tubulin and is, therefore, unlikely to alter the substrate specificity of TTL or TTCP (Edde *et al.*, 1992).

As yet no antibodies have been raised that are specific for this post-

translational modification, nor has the distribution or the characterization of the enzyme involved been determined. Therefore the possibility that polyglutamylation may form new distinct microtubule subsets is unproven, nor is it known whether there are any differences in the distribution or pattern of polyglutamylation during development.

Phosphorylation of tubulin

A number of protein kinases have been identified that can phosphorylate tubulin *in vitro* and *in vivo* (Goldenring *et al.*, 1983; Kadowaki *et al.*, 1985; Serrano *et al.*, 1987; Wandosell *et al.*, 1986; 1987).

Calcium/calmodulin-dependent kinase II can phosphorylate both α and β tubulin (Yamamoto *et al.*, 1985; Wandosell *et al.*, 1986). It is a soluble 640 kDa holo-enzyme, phosphorylating tubulin on serine and threonine residues, and is present in brain cytosol, presynaptic nerve terminals, synaptoplasm and synaptic vesicle preparations (Goldenring *et al.*, 1984, Yamagata *et al.*,1991). Phosphorylation of tubulin by the calcium/calmodulin-dependent kinase blocks the *in vitro* polymerization of tubulin (Wandosell *et al.*, 1986) and enhances its interaction with phospholipid vesicles (Hargreaves *et al.*, 1986).

A casein kinase II-like enzyme can also phosphorylate tubulin on serine and threonine residues in the C-terminal region (Serrano *et al.*, 1987; Avila, 1988), but in comparison with calcium/calmodulin-dependent protein kinase II, it phosphorylates different serine residues and does not block *in vitro* polymerization of tubulin. Furthermore, unlike calcium/calmodulin-dependent kinase II, casein kinase II preferentially phosphorylates the β-tubulin isotypes (Serrano *et al.*, 1987; Avila *et al.*, 1988).

V8 protease digestion of phosphorylated tubulin obtained either from differentiating neuroblastoma cells grown in the presence of $^{32}PO_4$, or obtained from *in vitro* phosphorylation by purified casein kinase II, reveals that they are phosphorylated on the same peptide on the β-tubulin isotype. Comparison of the *in vitro* phosphorylated tubulin by calcium/calmodulin dependent-protein kinase II with the *in vivo* phosphorylation of tubulin reveals a different peptide phosphorylation pattern (Serrano *et al.*, 1987). This suggests that casein kinase II is responsible for the *in vivo* phosphorylation. Casein kinase II preference for polymerized tubulin supports the view that phosphorylation of β-tubulin *in vivo* occurs on assembled microtubules (Avila *et al.*, 1988).

Tubulin may also be phosphorylated by tyrosine kinases such as insulin receptor kinase (Kadowaki *et al*, 1985). Tyrosine-specific kinases are also signal transducing enzymes and are thought to have a role in the regulation of cell proliferation and differentiation (Hunter and Cooper, 1984; Yarden and Ulrich, 1988). A non receptor tyrosine kinase $pp60^{c\text{-}src}$ is highly enriched in growth cone preparations (Matten *et al.*, 1990) in particular growth cone membranes. Tubulin is the main *in vivo* substrate for $pp60^{c\text{-}src}$ and is phosphorylated on both α- and β sub-units (Matten *et al.*, 1990).

The expression of $pp60^{c\text{-}src}$ increases during nerve regeneration and has been correlated with its kinase activity (LeBeau *et al.*, 1991). It has also been found that certain cell adhesion molecules viz N-CAM may modulate tyrosine phosphorylation by modulation of tyrosine kinases or phosphatases (Atashi *et al.*, 1992). This may suggest a mechanism for extracellular guidance where molecules such as N-CAM modulate the cytoskeleton through phosphorylation, and would be of particular

interest if phosphorylation of tubulin *in vivo* by $pp60^{c\text{-}src}$ tyrosine kinase modifies tubulin dynamics, similar to that of insulin receptor kinase which can inhibit the polymerization of tubulin. Indeed, growth cone tubulin that is post-translationally phosphorylated by tyrosine kinase is located mainly in the growth cone membranes (Matten *et al.*, 1990) indicating that the modification can influence the intracellular distribution of tubulin during development. Thus, by influencing post-translational processing of cytoskeletal elements, extracellular guidance cues may modify cytoskeletal dynamics and thereby possibly regulate neuronal pathfinding and commitment during axonal elongation.

Therefore, the phosphorylation state of tubulin is likely to be important in the assembly and stabilization of microtubules during neurite extension in differentiating cells in culture, and the association of tubulin with particular membrane structures (Hargreaves *et al.*, 1986; Serrano *et al.*, 1987).

Phosphorylation of MAPs

The existence of post-translationally phosphorylated forms of MAPs has been shown *in vitro* (Sloboda *et al.*, 1975; Herman *et al.*, 1985; Diaz-Nido *et al.*, 1990) and *in vivo* (Gard and Kirschner, 1985; Serrano *et al.*, 1987; Diaz-Nido *et al.*, 1988; and Avila *et al.*, 1988). This post-translational event can be mediated by a number of kinases, some of which appear to have specificity for particular MAPs.

MAP-1a, MAP-1b and MAP-1c can be phosphorylated by endogenous protein kinases associated with microtubules (Hermann *et al.*, 1985). MAP-1a and MAP-1b can be phosphorylated *in vitro* by casein kinase II, but only MAP-1b appears to be phosphorylated *in vivo* by a casein kinase II-like activity (Avila *et al.*, 1988). Furthermore, the phosphorylation state of MAP-1b is greater in embryonic compared

to adult brain, reflected by an increase in MAP-1b phosphorylation during differentiation and neurite extension in neuroblastoma cells (Alletta *et al.*, 1988; Diaz-Nido *et al.*, 1988). This increase in MAP-1b phosphorylation may induce microtubule assembly. In support of this, monoclonal antibodies specific for phosphorylated forms of MAP-1b react with the microtubule population in differentiated neuronal cells as well as microtubule organizing centres in non- neural cells. If MAP-1b phosphorylation did not induce assembly, it may be important in stabilising assembled microtubules during neuronal commitment (Diaz-Nido and Avila, 1989).

MAP-2 can be phosphorylated by type II cAMP-dependent protein kinase (Vallee, 1980; Theurkauf and Vallee, 1982) which has been found to be associated with MAP-2. It may also be phosphorylated by calcium/calmodulin-dependent protein kinase (Yamamoto *et al.*, 1983; Schulman *et al.*, 1984), protein kinase C (Tsuyami *et al.*, 1986; Hoshi *et al.*, 1988b) and growth factor sensitive MAP-2 kinase (Sano *et al.*, 1990; Miyasaka *et al.*, 1990).

Characteristic of MAP-2 phosphorylation by cAMP-dependent protein kinase, by calcium/calmodulin-dependent protein kinase and by protein kinase C, is a reduction in the ability of MAP-2 to promote tubulin polymerization i.e. a decrease in microtubule stability. (Jameson *et al.*, 1980; Murthy and Flavin, 1983; Tsuyami *et al.*, 1987; Hoshi *et al.*, 1988b).

MAP-2 can undergo phosphorylation at multiple sites; while some phosphorylation sites of protein kinases overlap there are also distinct MAP-2 protein phosphorylation sites characteristic of individual kinases (Walaas and Nairn, 1989). Indeed, phosphorylation by protein kinases may be restricted to certain residues e.g.

type II cAMP- dependent kinase only phosphorylates MAP-2 on serine residues, while calcium/calmodulin-dependent kinase phosphorylates MAP-2 on both serine and threonine residues (Yamamoto *et al.*, 1985). The different phosphorylation sites may differentiate the effects of the separate kinases on microtubule stability through phosphorylation.

MAP-2 phosphorylation also decreases the ability of actin to bind to it. Together with a decrease in microtubule stability mediated by MAP-2 phosphorylation, this may result in the production of a cytoskeleton with increased plasticity which may be important for changes in cell shape to occur (Seldon and Pollard, 1983). This may explain why high molecular weight MAP-2 is compartmentalized to the cell body in mature axons yet has been located in growth cones and filopodia of growing axons (Caceres *et al.*, 1986; Ferreira *et al.*, 1990) where the cytoskeleton undergoes dynamic changes associated with changes in cell shape. It has also been suggested that modulation of phosphorylation of MAP-2 may regulate microtubule length *in vivo* (Burns *et al.*, 1984).

In support of the role of MAP-2 phosphorylation in the dynamic process of neurite extension, are studies performed on neuronal cell cultures. In PC12 cells stimulated with nerve growth factor to induce differentiation and neurite extension, an enhanced phosphorylation of MAP-2 has been observed (Alletta *et al.*, 1988; Landreth and Reiser, 1985; Greene *et al.*, 1986; Sano *et al.*, 1990; Miyasaka *et al.*, 1990). The phosphorylation is mediated by two distinct kinases, one thought to be cAMP- dependent protein kinase and the other believed to be a growth factor (NGF)-dependent protein kinase (Sano *et al.*, 1990; Miyasaka *et al.*, 1990). The occurence of the former in the early stages of neurite extension (Greene *et al.*, 1986), supports

the idea that cAMP-dependent phosphorylation of MAP-2 by type II cAMP-dependent protein kinase is important in modifying the cytoskeleton post-translationally to allow neurite outgrowth to commence. Furthermore, cAMP-dependent phosphorylation of MAP-2 can be regulated by polyamines (Nickerson and Wells, 1986). Polyamine production has also been shown to be important in cell growth and differentiation (Maudsley, 1979) which could implicate a correlation between MAP-2 phosphorylation and polyamine levels during cell growth and differentiation.

Tau proteins, like MAP-2 can be phosphorylated post-translationally by cAMP-dependent and calcium/calmodulin dependent kinases. Tau undergoes multiple phosphorylation, on serine and threonine residues, but some of these sites are distinct from those of other MAPs (Yamamoto *et al.*, 1985).

The phosphorylation of tau proteins reduces their ability to promote tubulin assembly (Yamamoto *et al.*, 1985) and regulates their interaction with other cytoskeletal proteins. Phosphorylation of tau proteins, like MAP-2, reduces the interaction with actin and crosslinking (Seldon and Pollard, 1983). It is known that phosphatase action can reverse these effects (Yamamoto *et al.*, 1988) suggesting that the equilibrium between kinase activity and phosphatase activity may have a role in modulating cytoskeletal plasticity. Indeed, during neurofibrillary degeneration in Alzheimers disease, the phosphorylation state of tau proteins may be of great significance (Flament *et al.*, 1990).

In summary, there is an abundance of evidence to suggest that the neuronal cytoskeleton, in particular microtubule proteins, are important in the maintenance of neuronal function and integrity. The microtubule proteins are essential to the process

of axonal transport, cell growth and differentiation. Alterations to microtubule proteins which have an effect on microtubule stability are therefore likely to modify the axonal function and influence neuronal remodelling.

As previously mentioned, a mechanism that may alter the biochemical properties of microtubule proteins, is post-translational processing. A number of post-translational modifcations to microtubule proteins can occur, these include post-translational tyrosination, acetylation, glutamylation and phosphorylation of tubulin together with phosphorylation of MAPs.

There is considerable evidence that post-translational processing of microtubule proteins is modified during cell growth and differentiation. This evidence has been mainly obtained from studies on cell culture and development. Since many events during the development of the nervous system, and in neurite outgrowth in cell culture, recapitulate those that occur during mammalian peripheral nerve regeneration, it was the aim of this work to investigate the post-translational processing of microtubule proteins during peripheral nerve regeneration in the sciatic nerve of the rat.

The experiments outlined in this thesis have examined the hypothesis that post-translational processing of microtubule proteins may be altered during nerve regeneration, which could be, at least in part, responsible for the dynamic changes to the cytoskeleton that have to occur for nerve regeneration to commence. The work is presented in two parts: the first part concerns the examination and measurement of post-translational tyrosination and acetylation of α-tubulin *in vivo* and *in vitro* while the second part, examines the *in vivo* protein kinase activities directed against tubulin and MAPs.

MATERIALS

All materials unless otherwise indicated were purchased from Sigma Chemical Co., Poole, Dorset. U.K.

METHODS

1 SURGICAL PROCEDURES

All operative procedures were performed according to the guidelines and methods set out in the U.K. Home Office Licence PPL 40/00230.

1.1 Animals and anaesthesia

The experimental procedures were performed using female Wistar albino rats, weighing 200-250g (approx. 10 weeks old). Animals were sedated with an intramuscular (i.m.) injection of diazepam (Roche Ltd; 2.5mg/kg) and neuroleptanalgesia induced with Hypnorm® (Crown Chemical Co.; a combination of fentanyl citrate 60μg/kg and fluanisone 3.5mg/kg) by i.m. injection. Following each operative procedure the wounds were sutured with 4-0 silk thread and treated with a mixture of topical antibiotics (Tribiotic spray ®; Riker Laboratories). The animals were allowed to recover for a period of five days, maintained on a standard diet (R14; Labsure Ltd) of food and water *ad libitum*. After recovery the animals were killed by thoracic dislocation.

Animals undergoing perfusion-fixation (see section 1.3) prior to sacrifice were anaesthetized with Mebumal® (Sodium pentobarbitone; 300mg/kg i.m. Bayer, U.K.)

1.2 Sciatic nerve crush

The following model was used: under anaesthesia and aseptic conditions, one sciatic nerve was randomly exposed in the gluteal region and crushed approximately 40mm from L4,L5 dorsal root ganglia using a sterile 4-0 silk thread pulled tightly against a glass rod to give a perpendicular crush. After a 15 sec crush, the silk thread was released, and the site of crush marked by a small suture in the surrounding muscle tissue. In certain instances, following a crush, a ligature was applied about the crush with 4-0 silk thread, to prevent the outgrowth of regenerating fibres. The contralateral control nerve was left untouched and served as an internal control.

1.3 Perfusion-fixation of nerves for immunocytochemical analysis

Following a five day recovery period after nerve crush, animals were anaesthetized (see section 1.1) and subsequently perfused transcardially, under high pressure, initially with phosphate buffered saline (PBS; 100mM sodium phosphate buffer, pH 7.4 containing 0.9% w/v NaCl) with 10μg/ml xylocaine and 5000 I.U./ml heparin to optimize the circulation of the fixative. Perfusion-fixation itself was performed using 4% w/v para-formaldehyde in PBS. Nerve pieces approximately 2cm in length, and 1cm either side of a nerve crush, from regenerating nerves and their equivalent from contralateral control nerves were removed carefully following perfusion-fixation and allowed to fix for a further 4h at 4°C in fresh fixative. After this fixation period the nerves were carefully rinsed and then stored in PBS containing 0.1% w/v sodium azide and 10% w/v sucrose at 4°C.

2 PURIFICATION OF MICROTUBULE PROTEINS

2.1 Microtubule preparation

Fresh porcine brains were obtained from a local abattoir and transported back on ice. The superficial blood vessels were carefully removed and the brains rinsed in 100mM piperazine N,N'-bis-2-ethane sulphonic acid (PIPES; pH 6.8) containing 1mM ethylene glycol-bis-(β-aminoethyl ether) N,N' tetra acetic acid (EGTA) and 0.5mM $MgSO_4$. The brains were then partially homogenized in an assembly buffer consisting of 100mM PIPES; pH 6.8, 20% v/v glycerol (BDH Ltd, U.K.), 1mM EGTA, 0.5mM $MgSO_4$ and 1mM phenylmethane sulphonyl fluoride (PMSF) using a domestic food liquidiser (25,000 rpm for 20 seconds). They were then further homogenized by four passes with a motor-driven glass-teflon homogenizer at 600 rpm until fully homogeneous. To remove the membranes, whole cells and cell debris, the resultant homogenate was centrifuged at 100,000g for 30 min at 4°C. The pellet was discarded and the supernatant retained and incubated at 37°C in the presence of 1mM guanosine-5'-triphosphate (GTP) for 30 min to initiate microtubule polymerization. The polymerized microtubules were then centrifuged at 75,000g for 30 min at 37°C. The supernatant was discarded and the pellet resuspended in assembly buffer by a hand operated glass/glass homogenizer to $^1/_{10}{}^{th}$ of the previous supernatant volume. The resuspended microtubules were allowed to depolymerize on ice for 30 min. The supension was centrifuged at 100,000g for 30 min at 4°C and the resultant pellet, which contained cold-stable aggregates of microtubule proteins, was discarded. The supernatant was incubated at 37°C in the presence of 0.5 mM GTP for 30 min then centrifuged at 75,000g for 30 min at 37°C. The twice-cycled

microtubule pellet so obtained could be stored at -80°C for up to six weeks until use.

2.2 Purification of tubulin by ion-exchange phosphocellulose chromatography

2.2.1 *Phosphocellulose precycling*

P11 phosphocellulose resin (Whatman, Maidstone, U.K.) was mixed with 25 volumes of 0.5M NaOH and the mixture allowed to settle for a period of five minutes. The supernatant was decanted off and the remaining slurry mixed with distilled water and allowed to settle before decanting the water off. This procedure of washing/rinsing was repeated until the pH of the supernatant fell below 11.0. The slurry of phosphocellulose was then mixed with 25 volumes of 0.5M HCl and allowed to settle over a period of five minutes. The HCl was decanted off and the slurry washed with distilled water, as outlined above, until the pH of the supernatant was above 3.0. The phosphocellulose was mixed twice and then stored in 0.5M sodium phosphate buffer (pH 7.0) at 5°C for up to one week until use.

2.2.2 *Preparation of a mini-phosphocellulose column.*

Approximately 10ml of phosphocellulose was decanted from the sodium phosphate buffer stock, and allowed to settle. The phosphate buffer supernatant was discarded and the remaining slurry was dispersed in column buffer consisting of 100mM 2[N-morpholino] ethane sulphonic acid (MES; pH 6.8), 1mM EGTA and 1mM $MgSO_4$. The suspension was loaded into a 10ml syringe (with glass fibre packing to support the column) to a settled volume of 7ml. The column was equilibrated immediately with 30-40 volumes of column buffer at 4°C. The eluate was checked to ensure the pH was 6.8 before a sample was applied.

2.2.3 *Phosphocellulose chromatroghraphy.*

Twice-cycled microtubule pellets (see section 2.1) were resuspended in column buffer by gentle teflon/glass homogenization. An aliquot was taken and the protein content calculated according to the method of Lowry *et al*, (1951; see section 10). A protein concentration of 5-10 mg/ml was normal. The suspension was incubated on ice for 30 min and then clarified by centrifugation at 50,000g for 15 min at 4°C. The supernatant fraction containing microtubule protein was then loaded onto the ion-exchange (phosphocellulose) column to a maximum of 5mg/ml bed volume, to prevent saturation of the column. The chromatography was performed at 4°C and the elution under gravity. Tubulin was eluted in the void volume while the microtubule-associated proteins were retained on the columm; 0.5ml fractions of the column eluant were taken, stored on ice and the protein content of each fraction estimated by the absorbance at 280nm, measured using a Cecil CE 599 automatic scanning spectrophotometer. The tubulin fractions were pooled and aliquots taken to determine the purity, by sodium-dodecyl sulphate (SDS) - polyacrylamide gel electrophoresis (see section 4.1) and to determine the protein content according to Lowry *et al*, (1951; see section 10). The remainder of the sample was stored on ice and used within 2h.

2.2.4 *Carboxypeptidase A-treated purified tubulin.*

Microtubule protein was prepared as described in section 2.1. Following first cycle polymerization/depolymerization purification of microtubules, the cold supernatant, obtained after separation from cold-stable aggregates of microtubule protein, was incubated at 37°C with 1mM GTP. After a 20 min incubation period,

carboxypeptidase A (Type II-DFP, EC 3.4.17.1) was added to the incubation mixture (20 ng/mg microtubule protein). Following a further 10 min incubation phenyl acetate (to a final concentration of 19mM) was added to inhibit carboxypeptidase A activity. The polymerized microtubules obtained after incubation were centrifuged at 75,000g for 30 min at 37°C. The pellet obtained was rinsed with, then resuspended in assembly buffer (see section 2.1) and the microtubules allowed to depolymerize on ice for 30 min. The suspension was then clarified by centrifugation at 100,000g for 30 min at 4°C, and the cold stable aggregates of microtubule proteins were discarded. The resultant supernatant containing the carboxypeptidase A-treated tubulin was subjected to ion-exchange phosphocellulose chromatography as outlined in section 2.2.3.

2.3 Preparation of heat-stable microtubule-associated proteins.

Heat-stable microtubule associated proteins (MAPs) were prepared according to the method of Vallee *et al*, 1985 with slight modifications. The microtubule pellet obtained after two cycles of polymerization/depolymerization purification (see section 2.1) was resuspended at 4°C in 100 mM MES; pH 6.8 containing 1mM EGTA, 1mM $MgSO_4$, 10mM dithiothreitol and 0.35M NaCl. The suspension was incubated at 4°C for 30 min, then centrifuged at 200,000g at 4°C for 15 min. The supernatant was boiled for 5 min, then chilled on ice for 5 min and centrifuged at 100,000g for 30 min at 4°C to remove aggregates.

The supernatant, containing the heat-stable MAPs, was then dialysed against 500 volumes of 100mM MES, 1mM EGTA and 1mM $MgSO_4$ (pH 6.8) for 3h at 4°C, then repeated with fresh dialysis buffer for a further 3h. An aliquot of the

resultant sample was resolved by sodium dodecyl sulphate polyacrylamide gel electrophoresis (SDS-PAGE; section 4) to determine the purity of the sample, whilst the remainder was stored at -80°C until use, for a period of up to six weeks

3 ANALYSIS OF ASSEMBLED AND NON-ASSEMBLED TUBULIN IN RAT SCIATIC NERVE EXTRACTS

Pieces of nerve (1cm) either immediately distal (peripheral) or immediately proximal (central) to a nerve crush from regenerating rat sciatic nerve and the equivalent peices from the contralateral control nerve were homogenized (immediately) by hand at 37°C in 200μl/ nerve piece of microtubule stabilising buffer (100mM MES; pH 6.8, 1mM $MgCl_2$, 1mM EGTA and 0.5% v/v Nonidet P-40 (NP-40)) containing 1mM GTP and protease inhibitors (0.1mg/ml N-para-tosyl-L-arginine methyl ether, 1mM PMSF, 1μM pepstatin, 1μM leupeptin, 0.05 U/ml aprotonin, 10mM soybean trypsin inhibitor and 0.1mM chloroquinin).

The homogenate was incubated at 37°C for 30 min to allow the extraction of assembled microtubules. Following incubation the suspension was centrifuged at 13,000g for 5 min at 37°C. The supernatant was separated and subjected to high speed centrifugation at 100,000g at 37°C for 30 min.

The supernatant containing non-assembled tubulin was separated from the pellet, and the volume recorded. The pellet containing the assembled tubulin was resuspended in approximately 50 - 150μl of microtubule stabilising buffer. Aliquots from the supernatant and the resuspended pellet were taken and subjected to 1-dimensional SDS-PAGE (see section 4). The remaining supernatant and resuspended pellet samples were stored at -80°C until use.

4 1-DIMENSIONAL SODIUM DODECYL SULPHATE POLYACRYLAMIDE GEL ELECTROPHORESIS

4.1 Gel preparation

One-dimensional sodium dodecyl sulphate polyacrylamide gel electrophoresis (1D SDS-PAGE) was performed using the Hoeffer SE250 'Mighty Small II' gel electrophoresis unit. Polyacrylamide gels usually consisted of 6% or 7.5% polyacrylamide resolving gels overlaid with a 4% polyacrylamide stacking gel.

The 6% and 7.5% resolving gels contained mixtures of 6% acrylamide (5.85% w/v acrylamide and 0.15% w/v N',N- methylene bisacrylamide) and 7.5% acrylamide (7.30% w/v acrylamide and 0.20% w/v N',N- methylene bisacrylamide) respectively with 0.375M Tris-HCl, 0.1% w/v SDS, 0.1% w/v ammonium persulphate and 0.07% v/v N,N,N',N',- tetramethyl ethylene diamine (TEMED); pH 8.8. The 4% stacking gel comprised 3.9% w/v acrylamide and 0.1% w/v N',N-methylene bisacrylamide with 0.125M Tris-HCl, 0.1% w/v SDS, 0.05% w/v ammonium persulphate and 0.05% v/v TEMED; pH 6.8.

4.2 Preparation and SDS-PAGE of samples

Protein samples to be electrophoretically resolved were first boiled in the presence of 2% w/v SDS, 10% v/v glycerol, 5% v/v β-mercaptoethanol and 1% w/v bromophenol blue, for 5 min. After boiling the samples were clarified by centrifugation at 13,000g for 5 min.

The SDS-polyacrylamide gels were pre-electrophoresed at 10mA for 20 min. in order to remove any traces of impurity (max. voltage, current limiting). Fixed

volumes from the supernatants of the clarified samples were then applied to separate wells of the stacking gel and electrophoresed in parallel with prestained molecular weight markers which comprised myosin (205 kDa), β-galactosidase (116 kDa), bovine serum albumin (77 kDa) and ovalbumin (46 kDa), or alternatively, comprised α-macroglobulin (180 kDa), β-galactosidase (116 kDa), fructose-6-phosphate kinase (84 kDa), pyruvate kinase (58 kDa), fumarase (48.5 kDa), lactic acid dehydrogenase (36.5 kDa) and triosephosphate isomerase (26.6 kDa).

The gels were electrophoresed at 30 mA/gel (maximum voltage/current limiting) at a constant temperature of 10°C maintained by a cooling unit. This was performed in a running buffer containing 25 mM Tris, 0.192M glycine and 0.1% w/v SDS; pH 8.3. Electrophoretic resolution of the protein samples was terminated when the marker dye front was within approximately 0.2cm from the end of the resolving gel. The length of the gel, the distance moved by the dye front relative to the stacking gel were measured prior to the removal of the gel from the apparatus. The gels were then subjected either to further electrophoretic transfer onto nitrocellulose membrane filters (see section 4.4) or stained for protein (see section 4.3 below).

4.3 Gel fixation and Coomassie blue staining

Gels of electrophoretically resolved proteins were placed in fixing solution containing 15.4% w/v trichloroacetic acid (TCA), 3.4% w/v sulphosalycylic acid and 30% v/v methanol, for 1h at room temperature. The fixed gels were transferred to a staining solution containing 45.4% v/v methanol, 7.2% v/v glacial acetic acid

and 0.25% w/v Coomassie blue R250, at room temperature overnight. The resultant stained gels were washed repeatedly in a destaining solution to reduce the non-specific background staining. The destaining solution consisted of 30% v/v methanol and 10% v/v glacial acetic acid. The gels were then rinsed with distilled water and sealed in moist airtight bags for storage at 10°C until photographed.

The distance moved by each polypeptide band in the electrophoretic pattern was measured perpendicularly from the stacking gel/ resolving gel interface to the centre of the appropriate band. The exact length of the gel was also measured and the relative mobilities of the polypeptide bands were then calculated using the following equation.

$$\text{Relative mobility (Rf)} = \frac{\text{distance moved by protein x original length of gel}}{\text{distance moved by dye front x gel length after fixation/staining}}$$

A standard curve was plotted, between Rf and $\log_{10}$ molecular weight of the protein standards. Linear regression analysis from the resultant semi-log plots allowed the apparent molecular weight of individual proteins to be estimated.

4.4 Electrophoretic transfer of proteins from polyacrylamide gels to nitrocellulose membrane

The electrophoretically resolved proteins were transferred from the resolving gel to nitrocellulose membrane (Schliecher and Schuell, mean pore size 0.45μm) by the application of an electric field. This procedure was performed using the Biorad Transblot system (Biorad, U.K.) with a transfer buffer which contained 16mM Tris,

120 mM glycine and 20% v/v methanol; pH 8.3. The resolving gel was overlaid with nitrocellulose membrane, which had been presoaked with transfer buffer. Air bubbles at the interface between the nitrocellulose membrane and the resolving gel were carefully removed.

The proteins in the resolving gel were then transferred to nitrocellulose in the transfer apparatus at 100mA (voltage maximum, current limiting) overnight at room temperature. The technique was a modification to that used by Towbin *et al.*, (1979). After transfer, the nitrocellulose membrane termed 'Western' blot was cut to the size of the resolving gel, and transferred to a 'blocking' solution for preincubation prior to the immunological incubations. This procedure, described in section 5.1 was performed to occupy protein binding sites on the nitrocellulose membrane not bound by the proteins transferred from the resolving gel, with immunologically inert proteins e.g. bovine serum albumin and casein.

5 IMMUNOLOGICAL PROCEDURES

5.1 Specificity of antibodies against tyrosinated tubulin

Pieces of nerve 1cm immediately distal (peripheral) and 1cm immediately proximal to a nerve crush were homogenized in 100mM MES, pH 6.8 containing 1mM EGTA, 1mM $MgCl_2$ and 1mM PMSF. The homogenates were clarified at 13,000g for 5 min, and the supernatant was incubated at 37°C for 10 min in the presence or absence of carboxypeptidase A (20ng/mg protein). The action of carboxypeptidase A was terminated with 19mM (final concentration) phenyl acetate and the samples were prepared for 1-D SDS-PAGE (see Section 4.2) and electrophoresed.

Resolved proteins were transferred to nitrocellulose sheets (see section 4.4) then incubated as outlined below in section 5.2 with monoclonal antibodies YL1/2, TUB 1-A2 and YOL 1/34. The resultant immunoblots were processed using the protocols outlined in section 5.2.[i].

5.2 Western blot immunobinding analysis

Sequential 5mm nerve pieces proximal and distal to a nerve crush from regenerating rat sciatic nerves and equivalent pieces from the contralateral control nerves were homogenized with a motor-driven glass/glass homogenizer in 2% w/v SDS and 10% v/v glycerol containing 1mM PMSF. Following homogenization, 5% v/v β-mercaptoethanol and 1% v/v bromophenol blue was added and the sample boiled then clarified by centrifugation, as described in section 4.2. The samples were subjected to 1-D SDS-PAGE, then the resolved proteins were transferred to nitrocellulose membrane as described in section 4.4. The resultant Western blots were preincubated for 40 min in Tris-buffered saline (TBS; 10mM Tris base, 0.9% w/v NaCl; pH 7.4) containing 3% w/v bovine serum albumin, termed TBS-BSA. This was followed by a 1h incubation in TBS containing 0.05% v/v Tween-20, termed TBS-T; both incubations were performed at room temperature.

Each Western blot was then incubated overnight at 4°C with one of the following antibodies: rat monoclonal antibodies YL1/2 and YOL 1/34 (Dako Ltd., Slough, U.K.) directed against tyrosinated and total α-tubulin, respectively (diluted 1:50 in TBS-BSA); and mouse monoclonal antibodies 6-11B-1 (a gift from Dr. R.B. Burgoyne, University of Liverpool; diluted 1:20 in TBS-BSA) HM1, AA6 and HM2

(diluted 1:500 in TBS-BSA) against acetylated α-tubulin, MAP-1a, MAP-1b and MAP-2, respectively. Western blots preincubated with TBS-BSA and TBS-T but not incubated with monoclonal antibodies (primary antibodies) served as controls for the immunological analyses.

The unbound monoclonal antibodies, following the primary antibody incubation, were removed by 3 x 10 min washes in TBS-T, after which the blots were incubated with either horseradish peroxidase (HRP)- conjugated rabbit anti-rat IgG (Dako Ltd., Slough, U.K., diluted 1:500 in TBS-BSA) or with HRP-conjugated rabbit anti-mouse IgG (Dako Ltd., Slough, U.K., diluted 1:500 in TBS-BSA) for 3h at room temperature in the dark. After 3 x 10 min washes with TBS-T, the resulting immunoblots were visualized using either of the following protocols:

[i] The immunoblots were incubated with ECL reagent (enhanced chemiluminescence kit, Amsersham, Aylesbury, U.K.) for 1 min at room temperature, before being blotted dry with filter paper (Whatman, Merseyside, U.K.). The enhanced chemiluminescence was then visualized following exposure to Kodak X-0mat RP photographic film over a period of 30 sec to 2 min, and development of the film using standard photographic techniques.

[ii] The immunoblots were incubated in 100mM TBS containing 3.4mM 4-chloronaphthol, 17% v/v methanol and 0.05% v/v hydrogen peroxide for 5-10 min. The reaction was terminated by washing the immunoblot in distilled water for 5 min.

The visualized immunobinding was analyzed by laser densitometry as described in section 5.5.

5.3 Dot immunobinding analysis

5.3.1 *Nerve homogenates*

One cm nerve pieces immediately proximal and distal to rat sciatic nerve crushes and equivalent pieces from contralateral control nerve were homogenized in a motor-driven glass/glass homogenizer in 100mM MES buffer; pH 6.8 containing 1mM EGTA, 1mM $MgCl_2$ and 1mM PMSF. The homogenates were centrifuged at 200,000g for 20 min at 4°C, after which the supernatant fraction was assayed for protein content according to the method of Lowry *et al.*, (1951; described in section 10). All samples were then diluted to the same protein concentration using the original homogenization buffer.

Aliquots from serial dilutions of the samples (5μl) were applied as spots to strips of nitrocellulose membrane. The nitrocellulose membranes were dried and then fixed with 10% v/v acetic acid and 25% v/v isopropyl alcohol for 15 min at room temperature, followed by several quick rinses with distilled water. The strips were preincubated for 40 min at room temperature in TBS-BSA, followed by a 1h incubation at room temperature in TBS-T. The strips were then incubated with a range of monoclonal antibodies: rat monoclonal antibodies YL 1/2 and YOL 1/34 (see section 5.2), mouse monoclonal antibodies 6-11B-1, HM1, AA6, HM2 (see section 5.2). Nitrocellulose strips which were not incubated with the primary antibodies served as controls.

The nitrocellulose strips were processed as outlined in section 5.2 for Western blots with the immunobinding visualized using ECL reagent and fluorography (see section 5.2.[i]), then quantified by laser densitometry (see section 5.5).

5.3.2 *Assembled and non-assembled sciatic nerve tubulin*

Samples of assembled and non-assembled tubulin fractions of nerve were prepared as described in section 3. Aliquots from serial dilutions of the assembled and non-assembled nerve fractions (5μl) were spotted on to strips of nitrocellulose membrane. The strips of nitrocellulose were fixed with 10% v/v acetic acid and 25% v/v isopropyl alcohol then preincubated in TBS-BSA as described in section 5.3.1.

The nitrocellulose strips were then probed with either [i] rat monoclonal antibodies YL1/2 or YOL 1/34 (1:50 in TBS-BSA), or [ii] mouse monoclonal antibodies HM2 and AP-20 (1:250 in TBS-BSA) directed against tyrosinated α-tubulin, total α-tubulin, MAP 2a, 2b, 2c and MAP 2a, 2b, respectively. Nitrocellulose strips not incubated with the monoclonal (primary) antibodies served as controls.

The strips were washed and incubated with secondary antibodies as outlined for Western blots in section 5.2. The immunobinding was visualized using ECL reagent and fluorography (see section 5.2.[i]), then quantified by laser densitometry (see section 5.5).

5.4 Immunocytochemical analyses

Pieces of regenerating and control nerve were prepared for immunocytochemical analyses as outlined in section 1.3. Following fixation the

nerve pieces were removed from the fixative and as much epineurium as possible was carefully removed, to reduce the possibility of interference of fluorescent cells (e.g. mast cells, macrophages) in the subsequent immunofluorescent measurements.

The nerve peices were mounted onto blocks of similarly fixed liver tissue using a cryogenic embedding medium (Tissue-Tek, Miles A.B., Sweden), to ensure the nerves were absolutely straight when embedded. The liver tissue was then mounted onto a cryostat chuck and then frozen in powdered CO_2 before cryostat sectioning. The cryostat sectioning was performed using a Jung-Reichardt (Germany) cryostat, from which 10μm longitudinal sections were obtained from the central region of the nerve pieces as described by Larsson *et al.*, (1984) a region which represents the typical distribution of axons along the nerve trunk. Straight and unwrinkled cryostat sections were collected on gelatin-coated glass slides as described by Dahlström *et al.*, (1982). The slides were stored in 0.1M PBS; pH 7.4 containing 0.1% w/v sodium azide at 4-8°C.

Sections were preincubated for 1h with normal horse serum (1:50) and 5% w/v non-fat milk (Marvel, Premier Brands, U.K.), diluted in incubation buffer which consists of 0.1M PBS; pH 7.4, 0.1% v/v Triton X-100, 1% w/v BSA, 10% w/v sucrose and 0.1% w/v sodium azide. Following preincubation, the sections were incubated overnight, at 4°C, with a range of mouse monoclonal antibodies: TUB1-A2 directed against tyrosinated α-tubulin (1:1600), 10E8/E7 directed against growth associated polypeptide-43 (GAP-43; 1:10, a gift from Dr. K. Meiri, University of Syracuse, U.S.A.), HM-1, HM-2 and AA6 (see section 5.3.1). All dilutions were

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performed in incubation buffer. Sections incubated in incubation buffer alone (i.e. no monoclonals) served as controls.

The sections were then rinsed carefully (3 x 10 min washes) with incubation buffer, before being incubated with biotin-conjugated horse anti-mouse IgG (1:200; Dako Ltd., Slough, U.K.) for 1h in the dark. The sections were rinsed, then incubated for 1h with fluorescein isothiocyanate (FITC) - labelled streptavidin (1:200; Dako Ltd., Slough, U.K.) in the dark. All incubations were performed in a moist chamber. The slides were carefully rinsed with incubation buffer, then finally with distilled water before mounting the slides with 0.25% v/v DABCO antifluorophore (diazobiacyclo [2-2-2] - octane).

The immunofluorescence observed was quantified using cytofluorimetric scanning techniques as described by Dahlström *et al.*, (1982). In brief, longitudinal sectioned nerves prepared for immunocytochemical analysis, as described above, were placed on a motor driven cross table. The immunofluorescence was scanned using fluorophotometer equipment based on a Nikon Optiphot fluorescence microscope, by passing the immunofluorescent nerve sections at 40μm/s accross a diaphragm (1000 x 100 μm) through which the emitted fluorescence was recorded by a photomultiplier tube and integrated via a computer (MacIntosh II®) for graphical reproduction using the Lab View® program (Dahlström *et al.*, 1989).

After their immunofluorescent intensity had been recorded, the same sections were subjected to confocal laser microscopy (Biorad, Lasersharp MRC 600) and their immunofluorescence patterns were recorded photographically.

5.5 Densitometry

Densitometric analyses of the fluorographs and autoradiographs obtained from the experimental protocols outlined in secion 5.1, 5.2 and 9.3, and Western blots developed with chloronaphthol as outlined in secion 5.2 [ii] were performed using the LKB Ultrascan XL laser densitometer (Model 2222-020, Bromma, Sweden) and the Pharmacia LKB Gelscan XL• program.

A very important consideration when using laser densitometry as a quantative tool, is to ensure densitometric evaluations are performed within a linear range of sample concentrations. To determine this, samples were always applied at more than one concentration and the absorbance obtained by densitometry was plotted against the sample concentrations for each peak of interest. A line was drawn to connect these points, and the portion which produced a straight line represented the linear range within which the densitometric evaluations were accurate. The absorbance data presented in RESULTS were obtained within the linear range.

6 TUBULIN: TYROSINE LIGASE ACTIVITY

The determination of tubulin: tyrosine ligase (TTL) activity was based on the method of Barra *et al.*, (1974) which involves the measurement of the incorporation of radiolabelled tyrosine into α-tubulin.

6.1 Tissue preparation

Fresh rat brain or 1cm pieces of rat sciatic nerve either side of a crush (proximal or distal) from regenerating nerve and their equivalents from uncrushed control nerve were homogenized in a motor driven (600 rpm) glass/glass

homogenizer, in the presence of 10mM sodium phosphate buffer; pH 7.0 containing 0.24M sucrose and 1mM PMSF. Preliminary experiments examined the use of rat brain and rat sciatic nerve tissue previously frozen, as a source of TTL.

The homogenates were centrifuged at 200,000g for 20 min at 4°C. The resultant supernatants were passed through a cooled (4°C) gel filtration Sephadex G-25 column, pre-equilibrated with 100 mM MES; pH 6.8 containing 1mM EGTA and 1mM $MgCl_2$, to remove free tyrosine. Aliquots of the eluant were removed and their protein content evaluated according to the method of Lowry *et al.*, (1951; see section 10). The remaining eluant was used immediately in the enzyme assay.

6.2 Tubulin:tyrosine ligase assay

The eluants of the nerve or brain tissue (obtained as in section 6.1) were incubated at 37°C in the presence of 100μM KCl, 12.5mM $MgCl_2$, 2.5 mM ATP, and 1.6μM L-[3,-5^{3}H] tyrosine (0.25μCi [S.A 51 Ci/mmol] Amersham, Aylesbury, U.K.) in a total volume of 250μl. At the start of the incubation and at various time intervals, aliquots (25-100μl) were removed from the reaction media and added to 1ml of ice-cold 20% w/v trichloroacetic acid (TCA), to terminate the assay and precipitate the protein. Proteins were allowed to precipitate overnight at 4°C.

Preliminary experiments examined the use of exogenous purified tubulin as a substrate. In these experiments, 250μl of purified tubulin (obtained as in section 2.2.3) or purified carboxypeptidase A- treated tubulin (obtained as in section 2.2.4) were added to the incubation mixture. The incorporation of [^{3}H]-tyrosine into these exogenous substrates was compared to [^{3}H]-tyrosine incorporation in the absence of

added substrate. The controls in these experiments were incubations in the absence of eluants obtained from nerve or rat brain. In some instances the effect of a 15mM concentration of the N-carbobenzoxy dipeptide (Z-GLU-TYR) which inhibits endogenous carboxypeptidase activity associated with C-terminal tubulin residues, was also investigated.

6.3 Quantification of radioactivity

The TCA-insoluble material obtained as in section 6.2 was collected on filter discs (Whatman GF/B discs; Whatman, Maidstone, U.K.) by vacuum filtration (Millipore vacuum manifold; Millipore, Bedford, U.K.), and each disc was washed sequentially three times with 5ml 10% w/v TCA, 5ml 5% w/v TCA, 5ml ethanol (absolute) and finally rinsed with diethyl ether. The resultant discs were dried, and the radioactive tyrosine incorporated into the TCA-insoluble material was measured by scintillation counting, using Aqualuma$^{(R)}$ scintillant fluid (M&B Laboratory Products, U.K.). The disintegrations per minute (DPM) were calculated by the channels ratio method using the Beckman LS 1801 liquid scintillation system.

6.4 Electrophoretic analysis of radiolabelled protein

To determine if [^{3}H]-tyrosine was incorporated into tubulin, samples obtained as outlined in section 6.1 were incubated as described in section 6.2 except that the assay was terminated and the protein precipitated using 9 vols of ice-cold acetone instead of ice-cold 20% w/v TCA. Following overnight precipitation at -20°C, the sample was clarified by centrifugation at 13,000g for 5 min and the acetone was evaporated off (at 60°C, 1h) after which the precipitate was resuspended in 2% w/v

SDS, 10% v/v glycerol and 5% v/v β-mercaptoethanol and boiled for 5 min. Bromophenol blue (1% v/v) was then added as a marker dye and the radiolabelled sample subjected to electrophoretic resolution, as described in section 4.2, in parallel with prestained molecular weight markers and purified tubulin (obtained as in section 2.2.3).

The gels were fixed and stained as outlined in section 4.3. The tracks of resolved radiolabelled protein were sectioned into 2.5mm pieces from the top to the bottom of the gel. The proteins from the gel pieces were eluted into 0.5ml of 30% w/v hydrogen peroxide overnight and the radioactivity determined by scintillation counting (as described in section 6.4).

7 TUBULIN:TYROSINE CARBOXYPEPTIDASE ACTIVITY

The determination of tubulin:tyrosine carboxypeptidase (TTCP) activity was performed using two methods, Method A was based on the procedure of Argarana *et al.* 1980, and method B was a pulse-chase experiment performed in conjunction with a TTL assay.

7.1 Tissue preparation

Fresh 1cm pieces of rat sciatic nerve either side of a crush (proximal or distal) from regenerating nerve and their equivalents from uncrushed control nerve were homogenized in a motor driven (600 rpm) glass/glass homogenizer, in the presence of 10mM sodium phosphate buffer; pH 7.0 containing 0.24M sucrose and 1mM PMSF. The homogenates were centrifuged at 13,000g for 5 min at 4°C. The resultant supernatants were passed through a cooled (4°C) Sephadex G25 gel

filtration column, pre-equilibrated with 100 mM MES; pH 6.8, containing 1mM EGTA, to remove free tyrosine. Aliquots of the eluant were removed and the protein content of these were evaluated according to the method of Lowry *et al.*, (1951; see section 10). The remaining eluant was used immediately in the enzyme assay.

7.2 Preparation of [^{3}H]-tyrosinated tubulin

Porcine brain or rat brain tissue was prepared as outlined in 6.1. The eluants of porcine brain or rat brain were incubated in the presence of 100μM KCl, 12.5mM $MgCl_2$, 2.5 mM ATP, and 1.6μM L-[3,-5^{3}H] tyrosine (0.25μCi [S.A 51 Ci/mmol] Amersham, Aylesbury, U.K.) for 20 min. at 37°C. After incubation, the mixture was cooled to 4°C and then passed through a pre-cooled Sephadex G-25 column (4°C) pre-equilibrated with 100mM MES (pH 7.0) containing 1mM EGTA, to remove free tyrosine.

The eluants were incubated in the presence of 1mM GTP and 4mM glycerol and purified by 2 cycles of assembly and disassembly, as described in section 2.1, and then purified by ion-exchange chromatography as described in section 2.2.3. Prior to use in the assay, the purified ^{3}H-tyrosinated tubulin was incubated in the presence of 1mM GTP and 10μM taxol in order to induce microtubule assembly.

7.3 Tubulin:tyrosine carboxypeptidase activity

7.3.1 *Method A*

The eluants, prepared as outlined in section 7.1, were incubated at 37°C at pH 7.0 with 2mM $MgCl_2$ and [^{3}H]- tyrosinated tubulin (50,000 DPM; approx. 20μg of protein) in a total volume of 200μl. At the start of the incubation (zero time) and at

30 min, 80μl aliquots were removed from the reaction media and added to 1ml of ice-cold 20 % w/v TCA, to terminate the assay and to precipitate the protein. The protein was allowed to precipitate overnight at 4°C. A control blank containing [^{3}H]-tyrosinated tubulin alone with the incubation buffer, served as a control, together with the zero time controls.

7.3.2 *Method B*

The TTCP activity associated with the tissue eluants prepared using the protocol outlined in section 6.1, was also measured using a pulse-chase experiment. The incubation was performed under the the same conditions described for the assay of TTL activity in the same tissue eluants (see section 6.2) with the exception that after 20 min incubation with [^{3}H]-tyrosine, non-radioactive tyrosine was added to the incubation mixture to a final concentration of 1mM. Aliquots of the incubation mixture were taken at 10 and 20 min after addition of non-radioactive tyrosine and added to ice-cold 20 % w/v TCA to terminate the assay and to precipitate the protein. The TCA precipitates were processed as outlined in section 6.3.

8 ACETYLATION OF α-TUBULIN

8.1 Tissue preparation

Rat brain or rat sciatic nerve were homogenized in either 100mM PIPES; pH 6.8 containing 1mM $MgSO_4$, 1mM EGTA, 1mM GTP and 1mM PMSF or in 100mM MES; pH 6.8 containing 1mM $MgSO_4$, 1mM EGTA, 1mM GTP and 1mM PMSF by means of a motor driven glass/glass homogenizer at 4°C. The homogenates were centrifuged at 13,000g for 5 min at 4°C, and the supernatant stored and used in subsequent incubations. Aliquots of supernatant were taken and

the protein content of them determined according to the method of Lowry *et al.*, (1951; see section 10).

8.2 Acetylase assay

The rat brain or rat sciatic nerve homogenates were incubated in the presence or absence of exogenous tubulin protein (2mg/ml) in either 100mM PIPES (pH 6.8) or 100mM MES (pH 6.8), both containing 1mM $MgSO_4$, 1mM EGTA and 1mM GTP, together with either [^{14}C] 1-sodium acetate (0.25μCi [S.A. 55mCi/mmol]; Amersham, Aylesbury, U.K.) or [^{14}C] acetyl-coenzyme A (1μCi; [S.A. 2.13 Ci/mmol]; Amersham, Aylesbury, U.K.), in a total volume of 250μl.Incubations in the absence of added rat brain or rat sciatic nerve homogenates served as controls. Aliquots of the reaction media were removed at zero time and at various time intervals thereafter. These were added to ice-cold acetone and the proteins allowed to precipitate overnight at -20°C.

8.3 Quantification of radioactivity

The TCA-insoluble material obtained in section 8.2 was collected and the radioactivity measured using the methods outlined in section 6.3. The samples precipitated in acetone were clarified by centrifugation at 13,000g for 5 min and the acetone evaporated off (60°C, 1h). The precipitates remaining were resuspended in 2% w/v SDS and 10% v/v glycerol and subjected to electrophoresis as outlined in section 4.2 and the radioactivity measured by processing the resultant gels as outlined in section 6.4.

9 PHOSPHORYLATION OF MICROTUBULE PROTEINS

9.1 Tissue preparation

The L4 and L5 dorsal root ganglia, nerve pieces 1cm immediately proximal and 1cm immediately distal to a nerve crush from regenerating rat sciatic nerve, and equivalent pieces from the contralateral uncrushed nerves, were homogenized in 100mM MES; pH 6.4 containing 0.5 mM $MgCl_2$ and with or without 2.5 mM EGTA together with the following protease inhibitors: 1mM PMSF, 1μM pepstatin, 1μM leupeptin, 0.05 U/ml aprotinin, 10mM soybean trypsin inhibitor, 0.1mM chloroquinin and 0.1mg/ml N-para-tosyl-L-arginine methyl ether.

The homogenates were centrifuged at 200,000g for 20 min at 4°C to separate the supernatant and particulate (pellet) fractions. The supernatant contained mostly cytosolic protein kinases, while the particulate fraction contained mostly membrane-associated protein kinases. The particulate fraction was resuspended in the homogenization buffer, then centrifuged at 13,000g for 1 min and the supernatant recovered. Aliquots from both supernatants were taken and assayed for protein content according to the method of Lowry *et al* (1951; see section 10). Samples from regenerating and control nerve were then diluted to the same protein concentration, using homogenization buffer. The samples were then stored at -80°C for up to 1 week until use.

9.2 Protein kinase assay

9.2.1 *Mg^{2+}- dependent Ca^{2+}- independent activity*

Soluble and particulate fractions (containing cytosolic and membrane-associated kinase activities, respectively) were obtained from the ganglia or pieces

of regenerating or control nerve, in the presence of 2.5 mM EGTA.

Either the soluble or the particulate fractions were incubated with either exogenous heat-stable MAPs or purified tubulin (as prepared in sections 2.3 and 2.2.3, respectively) or on their own, in the presence of 10mM $MgCl_2$, 20mM NaF (to inhibit ATPases), 1μM okadaic acid (to inhibit phosphatases) and 10μM ATP containing ^{32}P-γ-ATP (1μCi, [S.A. 5000 Ci/mmole] Amersham, Aylesbury, U.K.) in a total volume of 100μl at 37°C.

At zero time and after 30 min of incubation, aliquots (20-40μl) from the reaction media were taken and placed into an equal volume of boiling 4% w/v SDS and 20% v/v glycerol for 5 min, to terminate protein kinase activity and to prepare the samples for electrophoretic resolution (final concentration 2% w/v SDS and 10% v/v glycerol), and subsequent visualization as outlined in section 9.3. The difference in the protein kinase activities were quantified by laser densitometry of the resultant autoradiographs using similar methods to those outlined in section 5.5.

In certain instances purified tubulin was pre-incubated in the presence of 10μM taxol (a kind gift from Dr. A. Hargreaves, Trent University, U.K.) and 1mM GTP prior to use in the protein kinase incubation. The purpose of the preincubation was to polymerize tubulin prior to the kinase assay. Incubations performed in the absence of added nerve sample served as controls.

9.2.2 *Ca^{2+}- and calmodulin-dependent kinase activity*

Nerve samples obtained as outlined in section 9.1 in the absence of added

EGTA were used in these experiments. Either the soluble or the particulate fractions were incubated either alone or with exogenous heat-stable MAPs or purified tubulin (as prepared in section 2.3 and 2.2.3, respectively) in the presence of 10mM $MgCl_2$, 20mM NaF, 1μM okadaic acid, 0.2mM dithiothreitol, 0.2mM $CaCl_2$, 50μg/ml calmodulin and 10μM ATP containing ^{32}P-γ-ATP (1μCi) in a total volume of 100μl at 37°C. At zero time and at 30 min post-incubation, aliquots (40μl) were taken and processed as outlined in section 9.2.1.

In certain instances purified tubulin was pre-incubated in the presence of 10μM taxol and 1mM GTP prior to use in the incubation.

9.2.3 *cAMP-stimulated kinase activity*

Nerve samples obtained as outlined in section 9.1 in the presence of 2.5mM EGTA were used in the following incubations. Either soluble or particulate fractions were incubated with exogenous heat-stable MAPs or purified tubulin (as prepared in section 2.3 and 2.2.3, respectively) in the presence of 10mM $MgCl_2$, 20mM NaF, 1μM okadaic acid, 10μM cAMP and 10μM ATP containing ^{32}P-γ-ATP (1μCi) in a total volume of 100μl. At zero time and at 30 min post-incubation, aliquots (40μl) were taken and processed as outlined in section 9.2.1.

In certain instances purified tubulin was pre-incubated in the presence of 10μM taxol and 1mM GTP prior to use in the incubation.

9.3 Electrophoretic resolution of radiolabelled proteins

To visualize the incorporation of ^{32}P-radiolabelled phosphate into endogenous

and exogenous protein, radiolabelled protein samples obtained from section 9.2 were subjected to electrophoresis, as outlined in section 4, and the resultant gels were either fixed in 15.4% w/v TCA, 3.4% w/v sulphosalycylic acid and 30% v/v methanol for 1 h at room temperature, or transferred to nitrocellulose and processed as outlined in section 9.4. The fixed gels were rinsed several times in distilled water and then dried onto standard Whatman filter paper (Whatman 4MM, Merseyside, U.K.) under a vacuum using a gel drier (Bio-Rad; Model 224). The dried gel was exposed to Kodak X-Omat RP autoradiographic film for up to 21 days, and the development of the film was performed using standard photographic techniques. The resultant autoradiographs were analyzed by laser densitometry where appropriate, as described in section 5.5.

9.4 Western blot anlysis

The electrophoretically resolved radiolabelled proteins obtained as shown in section 9.3 were transferred to nitrocellulose membrane as outlined in section 4.4. The resultant Western blots were preincubated for 40 min in TBS-BSA. This was followed by a 20 min incubation in TBS-T; both incubations were performed at room temperature. The blots were allowed to dry at room temperature, and were then exposed to Kodak X-Omat RP autoradiographic film for up to 21 days. The development of the film was performed using standard photographic techniques. The resultant autoradiographs were compared with an equivalent autoradiograph obtained using the technique obtained in 9.3. The blots were incubated again for 40 min in TBS-BSA followed by a 1 h incubation in TBS-T; both incubations were performed at room temperature. They were then subjected to immunobinding analysis as

described in section 5.2 using antibodies directed against MAP-2, MAP-1 and tau as described in section 5.2 and 5.3.

The fluorographs obtained after immunobinding analysis were compared to the autoradiographs obtained after exposure of the radioactive blots to X-ray film in order to identify the phosphorylated polypeptides.

10 PROTEIN ASSAY

Protein concentrations of rat sciatic nerve, rat brain and purified tubulin samples outlined in earlier sections were assayed, according to the method of Lowry *et al*, (1951) using bovine serum albumin as standards.

Reagents:

a) Reagent A - 2% w/v sodium carbonate in 0.1M sodium hydroxide.

b) Reagent B - 1% w/v copper sulphate in 2% w/v sodium-potasium tartrate.

c) Reagent C - 49ml of A + 1ml of B.

d) Reagent D - Folin-Ciocalteau reagent diluted 1:1 with distilled water.

Aliquots of protein extract were made up to 100μl with distilled water, and then mixed with 1ml of freshly prepared reagent C. After 10 min incubation at room temperature, 100μl of freshly prepared reagent D was added, and vortex-mixed.

After a further 30 min incubation at room temperature the absorbance of each sample was read at 750nm in a single-beam spectrophotometer (Cecil CE599) at room temperature. The protein concentration of the sample was estimated from a calibration curve, obtained from a series of protein standards containing bovine serum albumin (0-100μg/ml) which were assayed in parallel.

11 STATISTICAL ANALYSIS

The experimental data were subjected to statistical analysis. Where data were normally distributed (Shapiro test; Z-test) they were analyzed by the students t-test. Otherwise the results were tested for statistical significance by the non-parametric Mann-Whitney U test/Wilcoxon-Rank test. Multiple comparisons were also tested for significance by analysis of variance (ANOVA test). In all cases, $p < 0.05$ was considered significant.

RESULTS

1 TYROSINATION AND ACETYLATION OF α-TUBULIN

1.1 Tyrosination of α-tubulin

The experiments outlined in the following section were designed to investigate whether any changes in the post-translational tyrosination of α-tubulin occurred as a consequence of mammalian peripheral nerve regeneration.

1.1.1 *1-Dimensional SDS-PAGE and Western blotting of crushed and uncrushed sciatic nerve extracts*

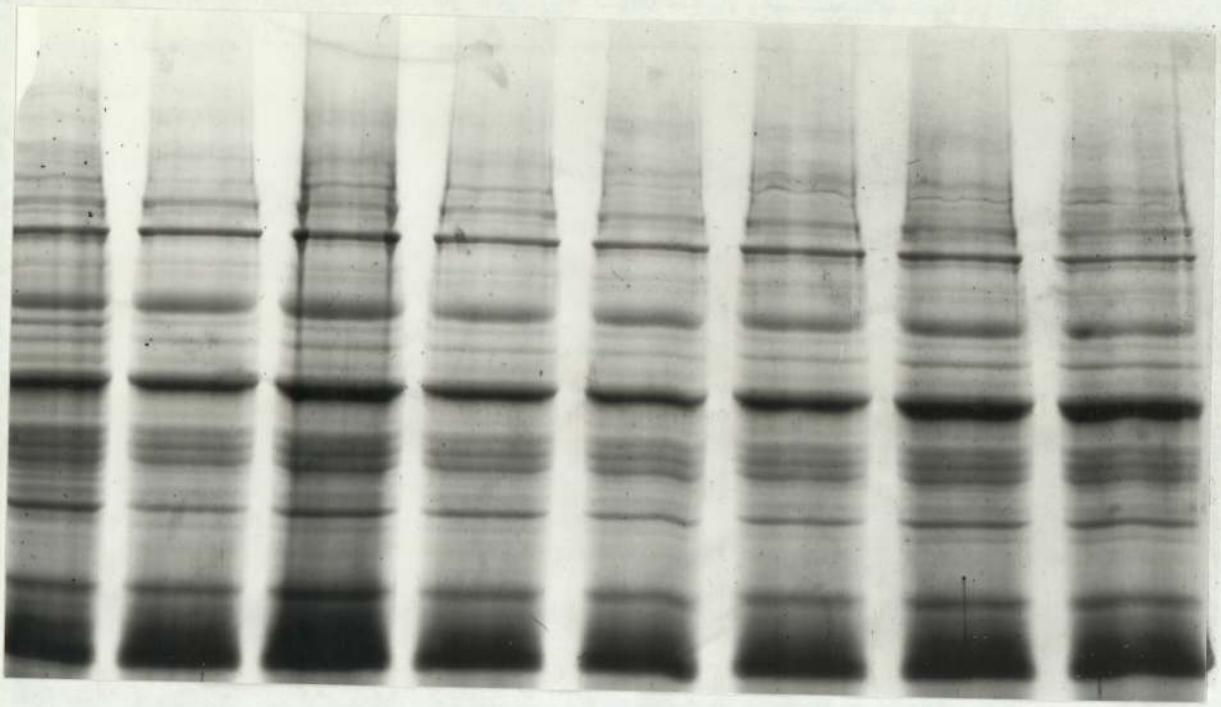
Homogenates of the L4,L5 dorsal root ganglia and sequential 5mm pieces of regenerating nerve proximal (central side) and distal (peripheral side) to a nerve crush and their equivalents from the contralateral uncrushed control nerve were prepared for electrophoresis (see METHODS section 4.2). The nerve proteins from these homogenates were resolved on 7.5% polyacrylamide gels, by electrophoresis. Gels of the resolved proteins were stained and visualized using Coomassie blue stain, and examined by eye to determine whether there were any gross differences in the protein content between pieces of regenerating and control nerve. As shown in figures 1 A-D, there appeared, qualitatively, to be very little difference in the protein concentrations between pieces of regenerating and control nerve, when examined by eye.

The specificity of the antibodies YL1/2 and YOL1/34, directed against tyrosinated and total α-tubulin, respectively, were determined using Western blotting techniques. Nerve extracts were resolved by SDS-PAGE and transferred to nitrocellulose, then probed with YL1/2 and YOL1/34 and visualized.

Figure 1 A-D

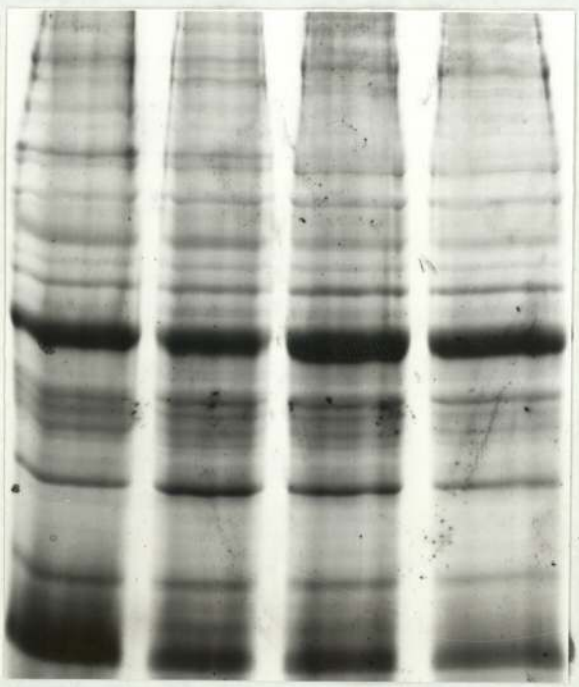
1-Dimensional SDS-PAGE of nerve proteins from regenerating and control nerve, visualized with Coomassie Blue stain. Tracks 1-8 in gel A present L4,L5 dorsal root ganglia (1) and sequential 5mm nerve pieces from the dorsal root ganglia to the site of the crush (2-8) from regenerating nerve. Tracks 1-4 in gel B represent sequential 5mm nerve pieces distal to a nerve crush. Gels C and D represent the equivalent pieces, of uncrushed contralateral uncrushed nerve, to those in Gels A and B respectively.

A



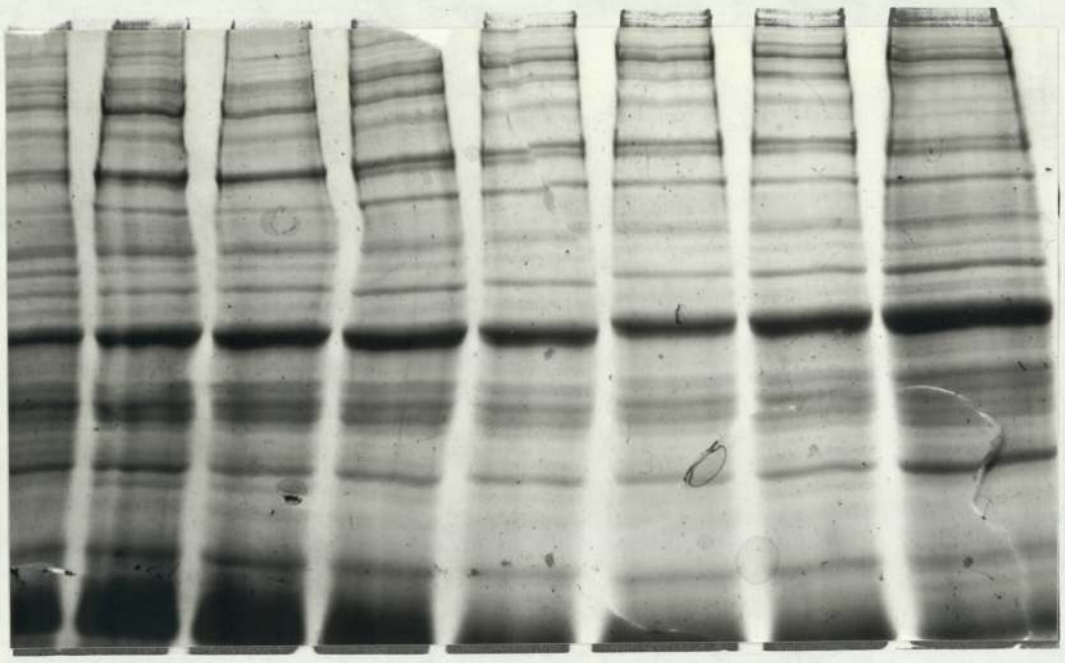
1 2 3 4 5 6 7 8

B



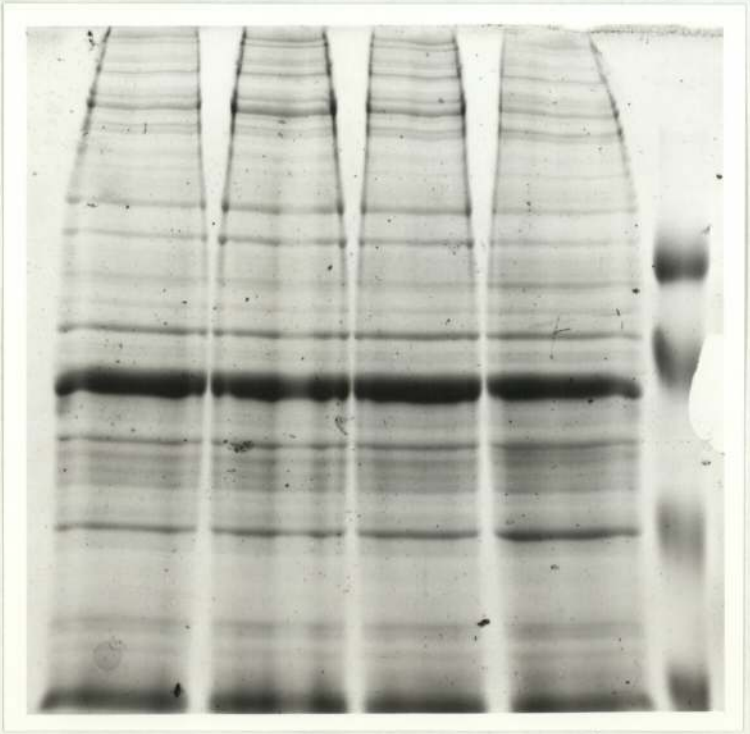
1 2 3 4

C



1 2 3 4 5 6 7 8

D



1 2 3 4

As shown in Figure 2 a single band corresponding to α-tubulin was recognized by each antibody. Limited Carboxypeptidase A treatment of purified tubulin, to remove the C-terminal tyrosine residues from the tubulin molecule, removed the immunoreactivity observed with YL1/2 on untreated purified tubulin, without affecting the immunoreactivity of YOL/134 as shown in figure 3. These preliminary experiments confirmed that both antibodies, YL1/2 and YOL1/34 were specific for C-terminal tyrosinated tubulin and total tubulin respectively, and that the epitopes for YL1/2 and YOL1/34 were conserved in the pieces of nerve examined.

Preliminary experiments were performed to validate the use of laser densitometry for the analysis of immunoreactivity patterns and quantification. A range of concentrations of nerve homogenate were resolved by SDS-PAGE, and then transferred to nitrocellulose. The resultant nitrocellulose blots were incubated with YL1/2 and YOL1/34, and the immunoreactivity analyzed by laser densitometry. An example of the densitometric absorbance data, obtained with separate dilutions of nerve, is shown in figure 4. The absorbance data displayed 'linearity' in relation to nerve concentration i.e. the absorbance was directly proportional to the concentration of nerve homogenate. The concentration of nerve homogenate used in subsequent experiments gave absorbance data that was approximately in the middle of the linearity range, shown in figure 4, for both antibodies.

Nerve proteins obtained from pieces of regenerating and control nerve were electrophoretically resolved, transferred to nitrocellulose, then probed with YL1/2 and YOL1/34. Examples of the immunoreactivity associated with tyrosinated α-tubulin (YL1/2) and total α-tubulin (YOL1/34) are shown in figures 5 and 6,

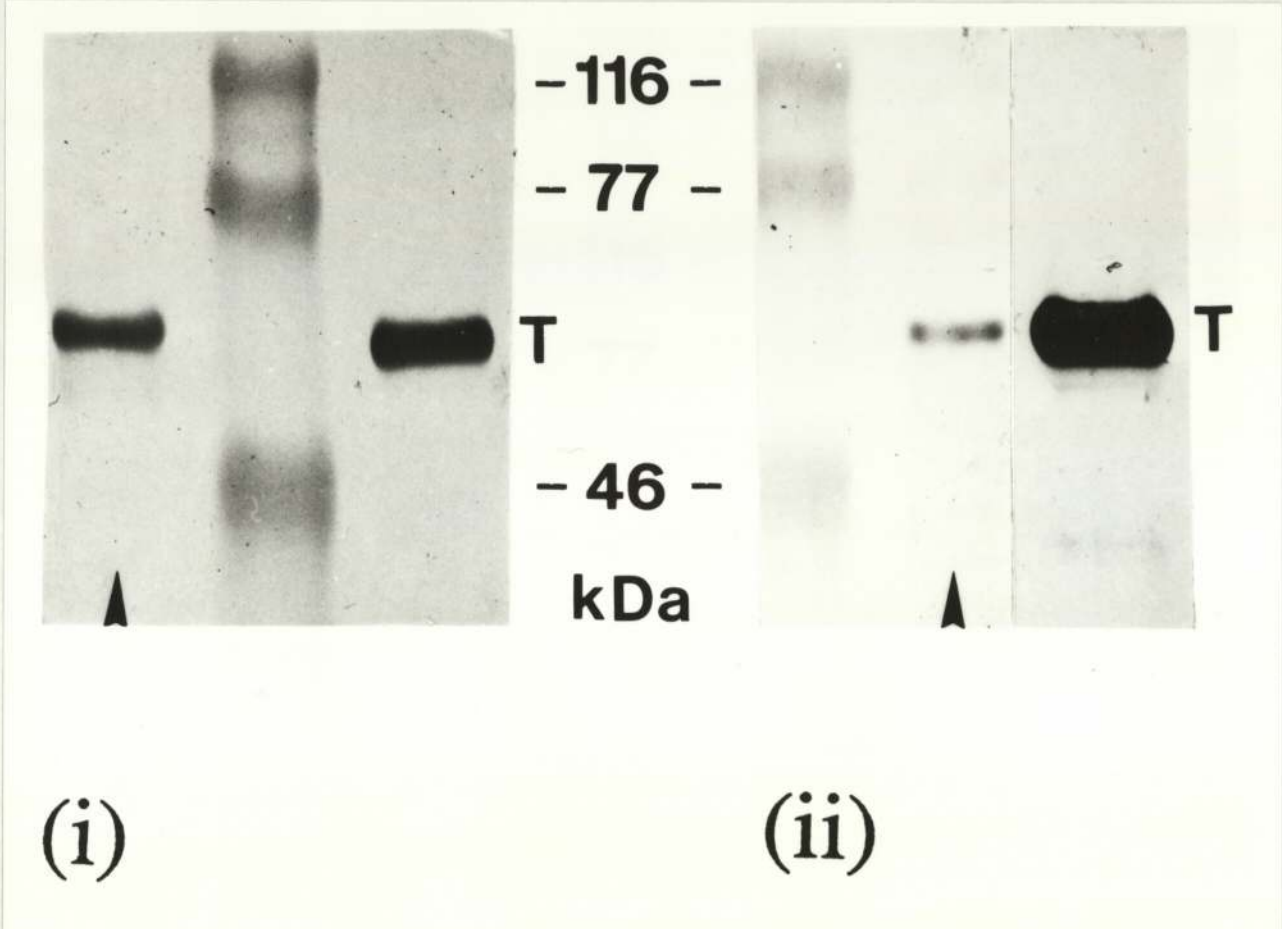

-116-
-77-
T
-46-
kDa
T
(i)
(ii)

Figure 3

Immunoblots of electrophoretically resolved purified tubulin, incubated either with (c) or without (u) Carboxypeptidase (20ng/mg purified tubulin) for 10 min at 37°C and probed with rat monoclonal antibodies YL1/2(i) against tyrosinated tubulin and YOL 1/34(ii) against total α-tubulin, visualized with HRP-conjugated rabbit anti-rat IgG and developed with chloronaphthol.

c u c u

(i) (ii)

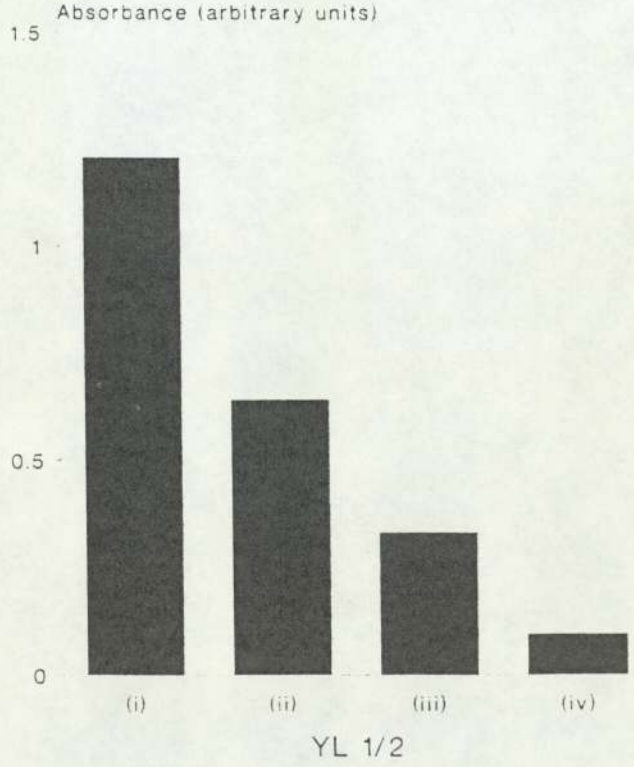
Figure 4

Absorbance for the intensity of fluorography obtained by densitometric analysis of an immunoblot of a range of concentrations (1:1 serial dilutions) of nerve homogenate.

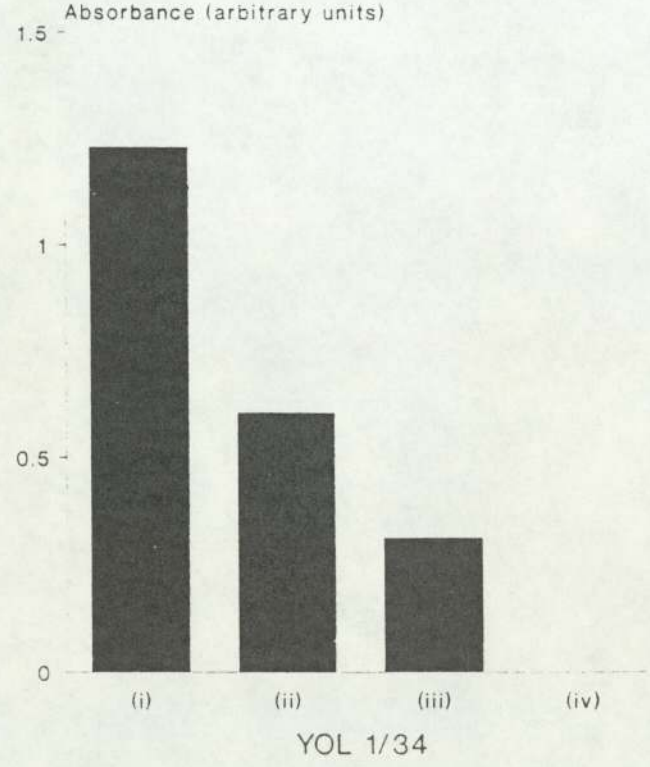
A range of decreasing concentrations (i) to (iv) of nerve homogenate were resolved by SDS-PAGE and then transferred to nitrocellulose. The resulting blots were probed with YL1/2 (A) and YOL 1/34 (B) and visualized with HRP-conjugated rabbit anti-rat IgG with development with ECL and fluorography. Shown below is the absorbance (arbitrary units) obtained by laser densitometry.

A

B



YL 1/2



YOL 1/34

Figure 5

Immunoblot detection of tyrosinated α-tubulin in rat sciatic nerve homogenates from one rat. Nerve homogenates were separated by 1-Dimensional SDS-PAGE. Immunoblots of electrophoretically resolved nerve proteins obtained from sequential 5mm pieces of nerve proximal to a crush (A) and distal to a crush (B) and equivalent pieces of contralateral uncrushed control nerve (C and D respectively) were probed with rat monoclonal antibody YL 1/2, visualized with HRP-conjugated rabbit anti-rat IgG and development with ECL and fluorography. Purified tubulin markers are included.

A

B

C

D

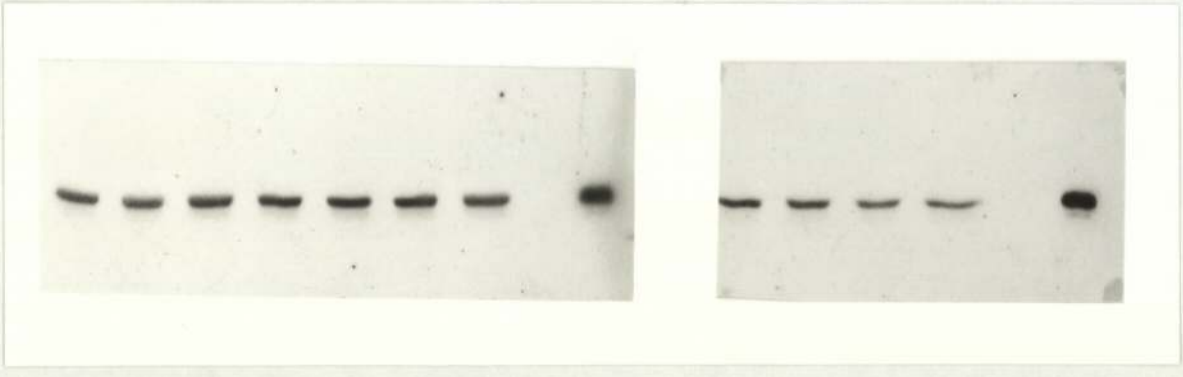
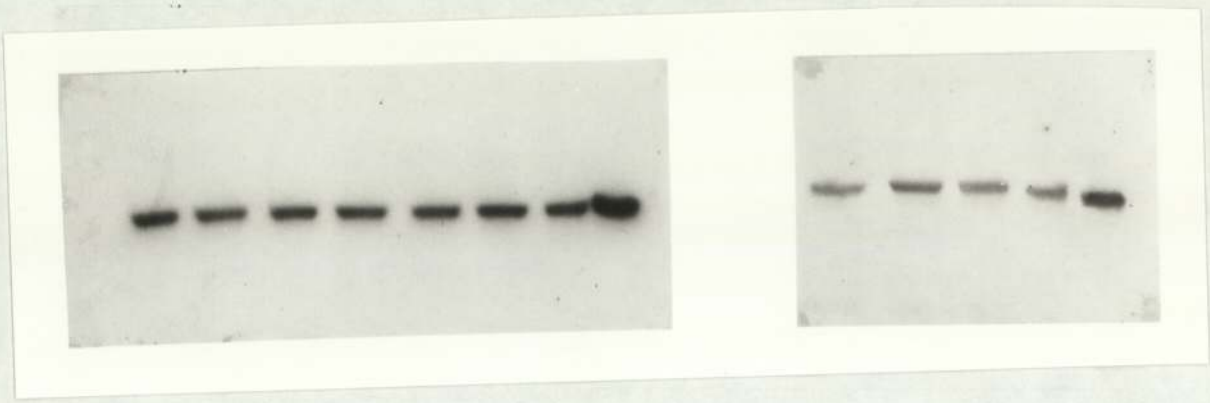


Figure 6

Immunoblot detection of total α-tubulin in rat sciatic nerve homogenates from one rat. Nerve homogenates were separated by 1-Dimensional SDS-PAGE. Immunoblots of electrophoretically resolved nerve proteins obtained from sequential 5mm pieces of nerve proximal to a crush (A) and distal to a crush (B) and equivalent pieces of contralateral uncrushed control nerve (C and D respectively) were probed with rat monoclonal antibody YOL 1/34, visualized with HRP-conjugated rabbit anti-rat IgG and development with ECL and fluorography. Purified tubulin markers are included.

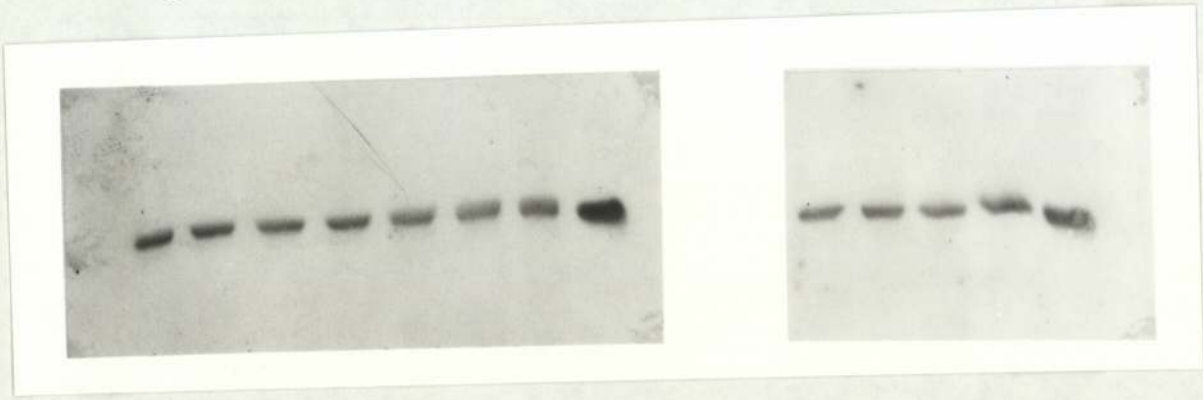
A

B



C

D



respectively.

In order to quantify this effect the immunoreactivity was analyzed by laser densitometry. The ratio of the immunoreactivity observed for YL1/2 relative to that for YOL1/34, was calculated in pieces of regenerating and control nerve. The measurements reflected the proportion of tubulin that was tyrosinated in each piece of nerve. The mean ratio of tyrosinated tubulin to total tubulin was calculated for pieces of nerve proximal to a crush and compared to the mean ratio in pieces of nerve distal to the crush, in regenerating and control nerves. As shown in figure 7, there was an elevated level of tyrosinated tubulin present in regenerating fibres, distal to the crush, compared to equivalent pieces of control nerve. The difference however, was not statistically significant ($P > 0.05$). The analysis itself was only semi-quantitative because of the intrinsic variance present when using nerve pieces that have not been assayed for protein concentration (see figure 8).

1.1.2 *Dot-immunobinding analysis of crushed, uncrushed and ligatured sciatic nerve extracts*

The semi-quantitative data obtained from Western blots of nerve homogenates, suggested that the levels of tyrosinated α-tubulin increased during nerve regeneration. This finding was investigated further using a quantitative dot-immunobinding technique with the same antibodies, YL1/2 and YOL1/34 (see METHODS section 5.3.1.).

This method enabled us to overcome the problems of variance in the protein content of pieces of nerve. The protein content of nerve pieces could be assayed, prior to immunochemical analysis, and adjusted to the same protein concentration.

Figure 7

The ratio of immunobinding of YL 1/2 (specific for tyrosinated α-tubulin) relative to that of YOL 1/34 (specific for total α-tubulin) in electrophoretically resolved homogenates of pieces of rat sciatic nerve proximal or distal to a nerve crush or from equivalent pieces of uncrushed (control nerve). The immunobinding was quantified by laser densitometry and expressed as the immunobinding of YL 1/2 relative to YOL 1/34 $\pm$ S.E.M. (n = 4).

respectively.

In order to quantify this effect the immunoreactivity was analyzed by laser densitometry. The ratio of the immunoreactivity observed for YL1/2 relative to that for YOL1/34, was calculated in pieces of regenerating and control nerve. The measurements reflected the proportion of tubulin that was tyrosinated in each piece of nerve. The mean ratio of tyrosinated tubulin to total tubulin was calculated for pieces of nerve proximal to a crush and compared to the mean ratio in pieces of nerve distal to the crush, in regenerating and control nerves. As shown in figure 7, there was an elevated level of tyrosinated tubulin present in regenerating fibres, distal to the crush, compared to equivalent pieces of control nerve. The difference however, was not statistically significant ($P > 0.05$). The analysis itself was only semi-quantitative because of the intrinsic variance present when using nerve pieces that have not been assayed for protein concentration (see figure 8).

1.1.2 *Dot-immunobinding analysis of crushed, uncrushed and ligatured sciatic nerve extracts*

The semi-quantitative data obtained from Western blots of nerve homogenates, suggested that the levels of tyrosinated α-tubulin increased during nerve regeneration. This finding was investigated further using a quantitative dot-immunobinding technique with the same antibodies, YL1/2 and YOL1/34 (see METHODS section 5.3.1.).

This method enabled us to overcome the problems of variance in the protein content of pieces of nerve. The protein content of nerve pieces could be assayed, prior to immunochemical analysis, and adjusted to the same protein concentration.

Figure 7

The ratio of immunobinding of YL 1/2 (specific for tyrosinated α-tubulin) relative to that of YOL 1/34 (specific for total α-tubulin) in electrophoretically resolved homogenates of pieces of rat sciatic nerve proximal or distal to a nerve crush or from equivalent pieces of uncrushed (control nerve). The immunobinding was quantified by laser densitometry and expressed as the immunobinding of YL 1/2 relative to YOL 1/34 $\pm$ S.E.M. (n = 4).

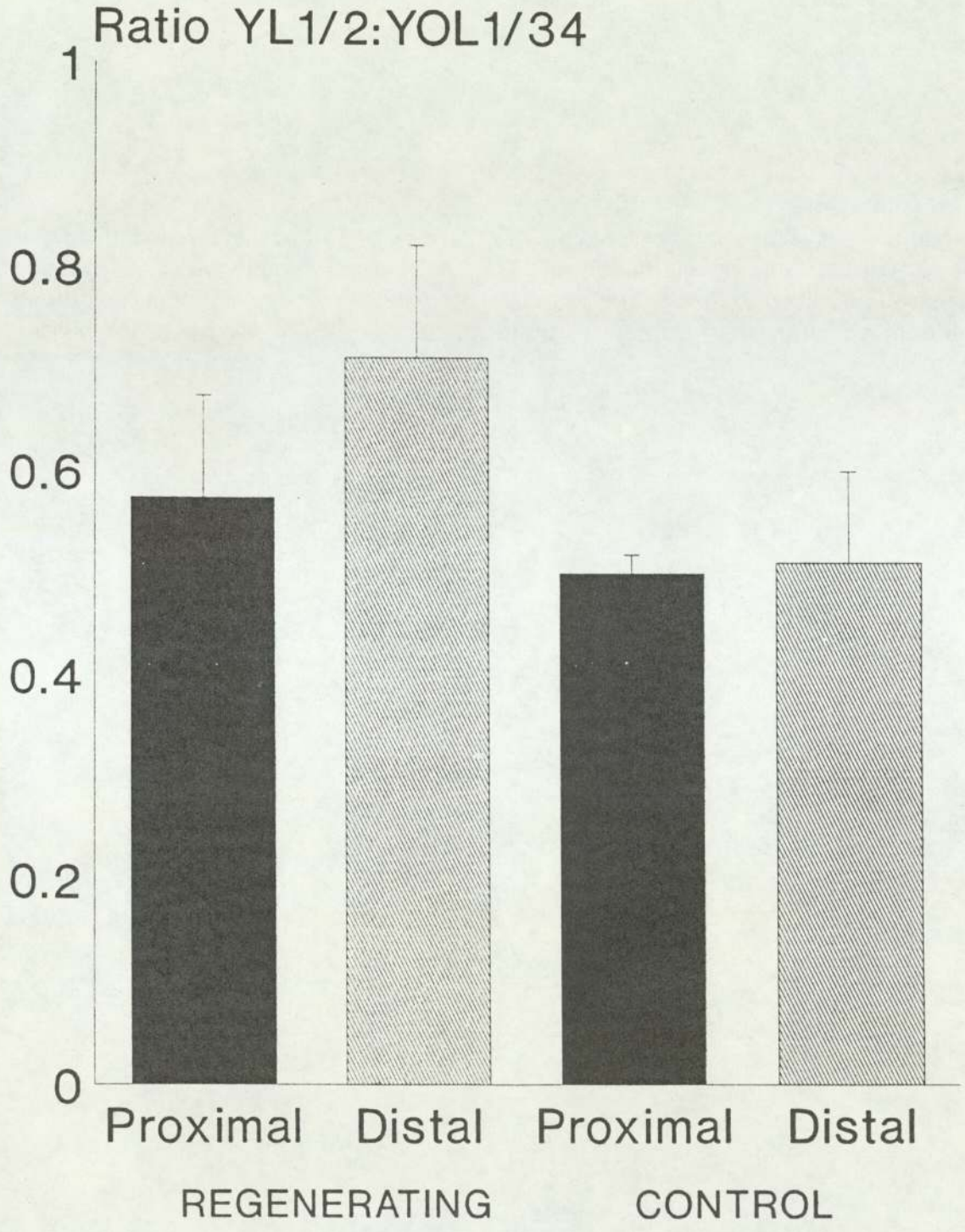
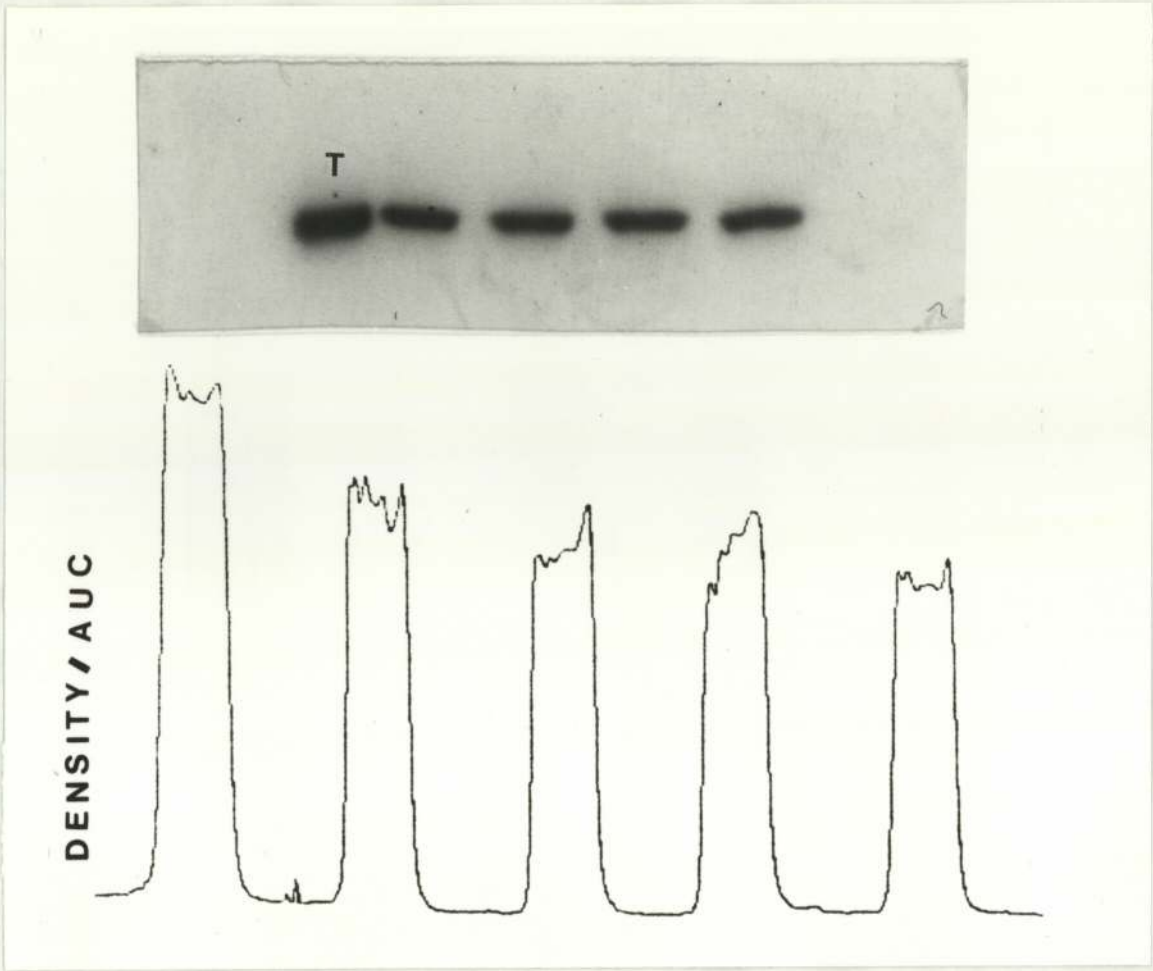

Ratio YL1/2:YOL1/34
1
0.8
0.6
0.4
0.2
0
Proximal
Distal
Proximal
Distal
REGENERATING
CONTROL

Figure 8

An immunoblot of rat sciatic nerve proteins separated by 1-dimensional SDS-PAGE probed with rat monoclonal antibody YOL 1/34 development with ECL and fluorography. Shown below is the graphical representation of absorbance of the immunoblot (represented as the area under the curve: AUC). Purified tubulin is denoted by T.


T
DENSITY / AUC

Furthermore, this method allowed several concentrations of the piece of nerve to be analyzed, to guarantee that the relationship between protein concentration and absorbance was linear, without prior validation (i.e. only data obtained from concentration of nerve pieces which displayed proportional linearity with the absorbance obtained were used). An example of data obtained from dot immunobinding analysis is shown in figure 9.

1.1.2.1 *Crushed nerves*

The proportion α-tubulin that was tyrosinated in pieces of crushed (regenerating) and uncrushed (control) nerve is shown in figure 10. There was an increase in the proportion of α-tubulin that was tyrosinated in pieces of nerve distal to a crush, on comparison with pieces of nerve proximal to a crush ($p < 0.05$) and equivalent pieces of control nerve ($p < 0.01$). The piece of nerve distal to a crush represents the region of nerve where regenerating fibres are present.

1.1.2.2 *Ligatured nerves*

After determining that there was an increase in the proportion of tyrosinated α-tubulin in pieces of nerve distal to the crush, we addressed the question of whether or not the increase in the proportion of tyrosinated α-tubulin would occur following a nerve crush, but in the absence of nerve regeneration (i.e. as a consequence of neuronal degeneration resulting from neuronal injury). To examine this, the proportion of α-tubulin that was tyrosinated was measured in pieces of nerve that had been ligatured about the crush, at the time of injury, to prevent nerve regeneration.

Figure 9

Dot immunobinding analysis of total α-tubulin serial dilutions ($^1/_2$; $^1/_4$; $^1/_8$; $^1/_{16}$; $^1/_{32}$) of rat sciatic nerve homogenate nerve spotted onto nitrocellulose strips. The strips were probed with YOL 1/34, visualized with HRP-conjugated rabbit anti-rat IgG and developed with ECL and fluorography. Shown below is the absorbance data obtained by densitometric analysis of the dot blot (represented as the area under the curve : AUC).

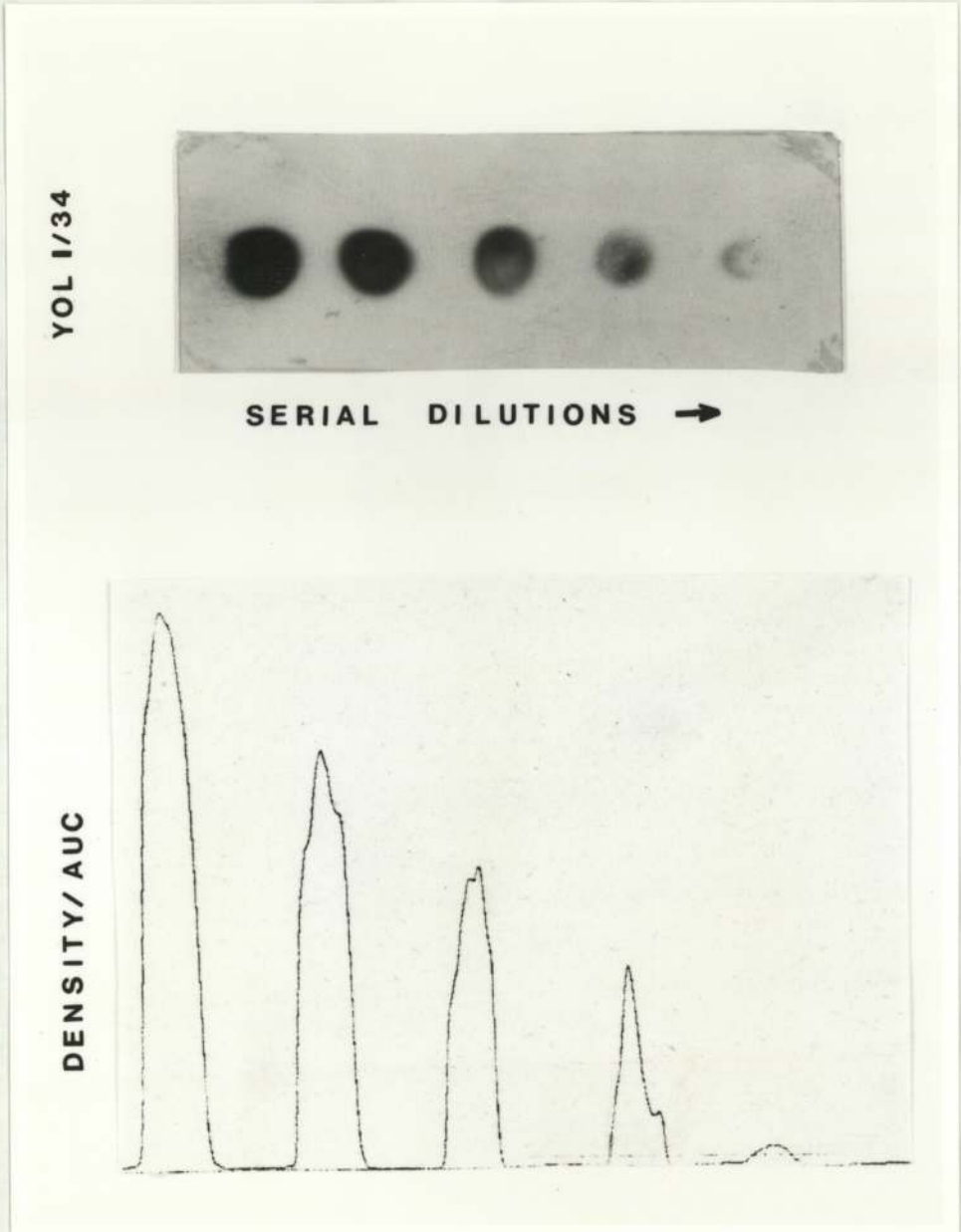
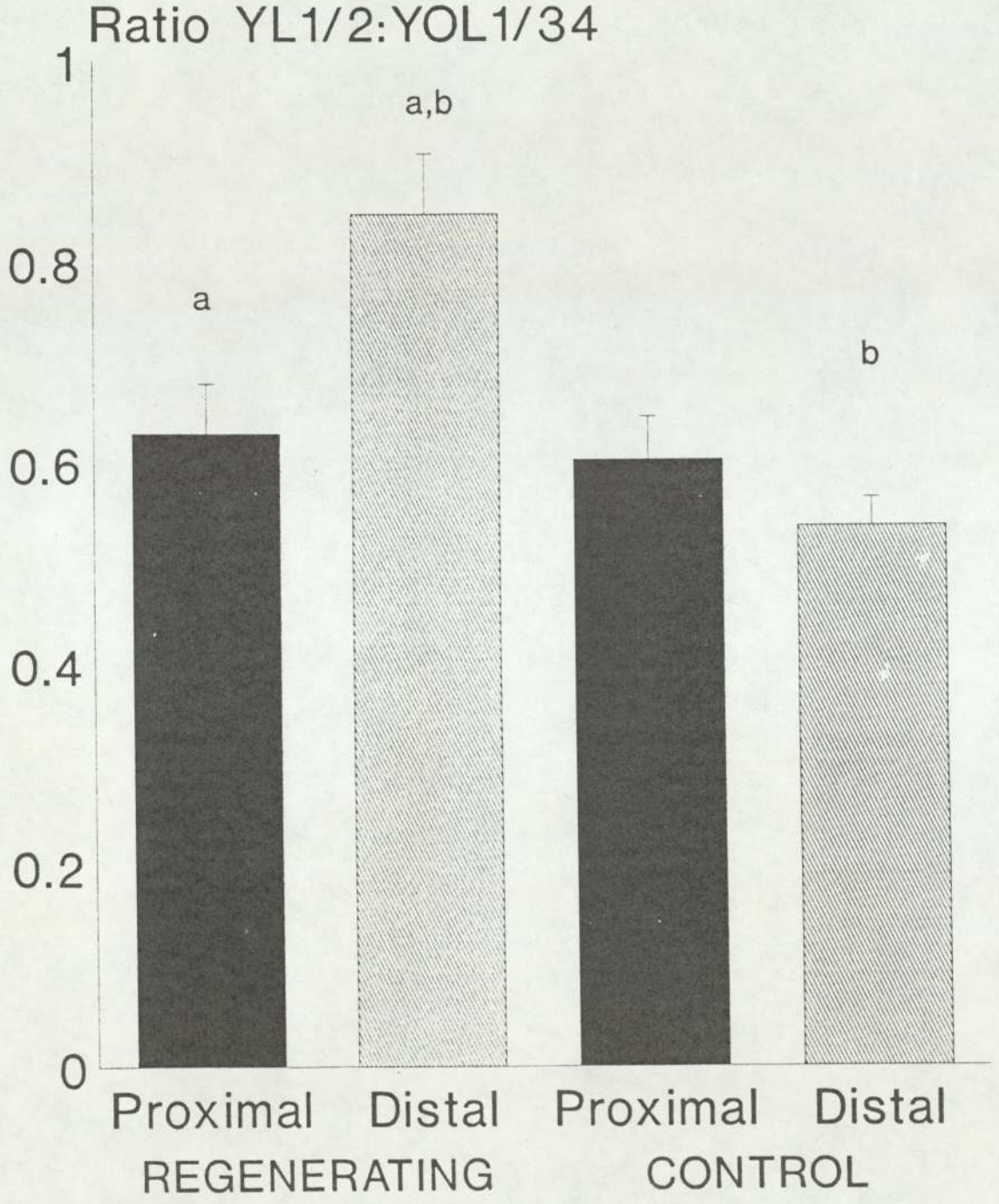

YOL I/34
SERIAL DILUTIONS →
DENSITY/ AUC

Figure 10

Comparison of dot immunobinding of YL 1/2 specific for tyrosinated α-tubulin, relative to that of YOL 1/34 specific for total α-tubulin, in homogenates of pieces of rat sciatic nerve proximal or distal to a nerve crush (regenerating), or from equivalent regions of uncrushed (control) nerves. The immunobinding was determined by laser densitometry and expressed as the immunobinding of YL 1/2 relative to YOL 1/34 (arbitrary units) $\pm$ S.D. Comparison of values marked with suffix 'a' had a statistically significant difference $P<0.05$; values with suffix 'b' had a statistically significant difference $P<0.01$ ($n = 8$ i.e from 8 separate animals).


Ratio YL1/2:YOL1/34
1
0.8
0.6
0.4
0.2
0
a
a,b
b
Proximal
Distal
Proximal
Distal
REGENERATING
CONTROL

Comparison of the proportion of α-tubulin that was tyrosinated between pieces of ligatured nerve, distal and proximal to a crush, revealed no increase in the proportion of tyrosinated α-tubulin such as that observed in regenerating nerves (figure 11). This result suggests that the increase in the proportion of tyrosinated α-tubulin observed in regenerating nerves, is not the direct consequence of neuronal degeneration and Schwann cell proliferation.

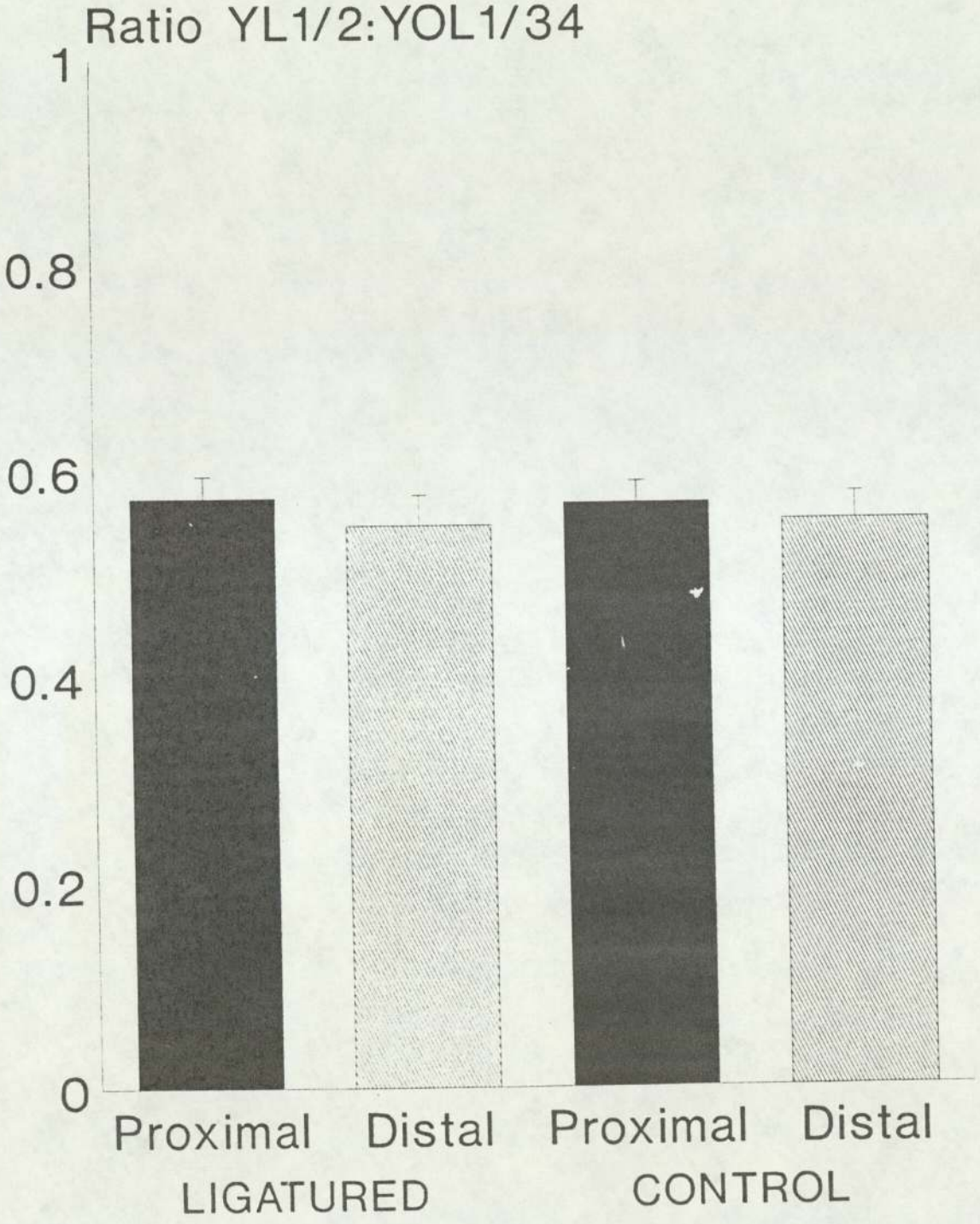
1.1.3 *Immunohistochemical analysis of crushed, uncrushed and ligatured sciatic nerve*

To support our previous biochemical observations on the proportion of α-tubulin that was tyrosinated in crushed (regenerating) and uncrushed (control) nerves an immunohistochemical study was performed. The data presented in this section were obtained in association with Professor A. Dahlström (University of Göteborg, Sweden), using a cytofluorimetric scanning technique pioneered by Dahlström *et al.*, (1989) together with confocal laser microscopy.

The distribution of tyrosinated α-tubulin in regenerating and control nerve was confirmed by cytofluorimetric scanning and confocal microscopy. Fixed longitudinal sections of regenerating and control nerve were probed with the mouse monoclonal antibody TUB 1-A2, directed against tyrosinated α-tubulin (this antibody was found to exhibit greater sensitivity than YL1/2 in this study).

Figure 11

Comparison of dot immunobinding of YL 1/2 specific for tyrosinated α-tubulin, relative to that of YOL 1/34 specific for total α-tubulin, in homogenates of pieces of rat sciatic nerve proximal or distal to a nerve crush (ligatured), or from equivalent regions of uncrushed (control) nerves. The immunobinding was determined by laser densitometry and expressed as the immunobinding of YL 1/2 relative to YOL 1/34 (arbitrary units) $\pm$ S.D. (n = 6).


Ratio YL1/2:YOL1/34
1
0.8
0.6
0.4
0.2
0
Proximal
Distal
Proximal
Distal
LIGATURED
CONTROL

1.1.3.1 *Specificity of TUB 1-A2*

The specificity of the mouse monoclonal antibody against tyrosinated α-tubulin was determined using the same method outlined for YL1/2 monoclonal antibody in section 1.1 of this chapter. The resulting immunoblot, of nerve extract incubated with TUB 1-A2, is shown in figure 12-A. A single band corresponding to α-tubulin was recognized by TUB 1-A2. The treatment of purified tubulin with carboxypeptidase A to remove C-terminal tyrosine molecules from α-tubulin and resulting immunoblots following incubation with TUB 1-A2 and YOL 1/34 are shown in Figure 12-B. Treatment with carboxypeptidase A of purified tubulin removed the immunoreactivity associated with TUB 1-A2 without affecting the immunoreactivity associated with YOL 1/34 (specific for total tubulin).

1.1.3.2 *Crushed nerves*

The immunofluorescence associated with TUB 1-A2 (tyrosinated α-tubulin) was examined in sections of regenerating and control nerves. The distribution of tyrosinated tubulin in those sections was observed by cytofluorimetric scanning. The immunofluorescent patterns were graphically represented as immunofluorescence profiles.

The immunofluorescence profiles obtained from sections of regenerating and control nerves are shown in figures 13 and 14, respectively. The immunofluorescence associated with tyrosinated α-tubulin was found to increase in the section of nerve distal to a crush, as shown in figure 13, in comparison to that proximal to a crush. In control nerves, however, the immunofluorescence profile was found not to differ between equivalent sections of nerve, as shown in figure 14.

Figure 12

Immunoblots of electrophoretically resolved purified tubulin, incubated either with (c) or without (u) carboxypeptidase (20ng/mg purified tubulin) for 10 min at 37°C and probed with rat monoclonal antibodies TUB-1-A2(i) against tyrosinated tubulin and YOL 1/34(ii) against total α-tubulin, visualized with HRP-conjugated rabbit anti-rat IgG and developed with chloronaphthol.

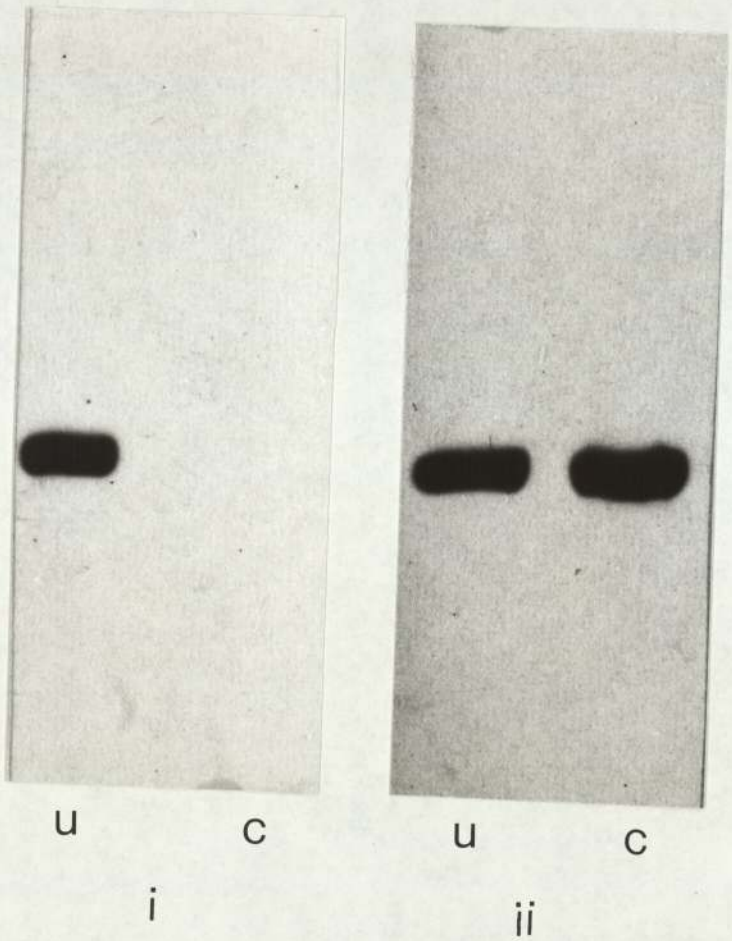

u
c
u
c
i
ii

Figure 13

Immunofluorescence profile of tyrosinated tubulin in sections of regenerating sciatic nerves. 2cm sections of regenerating nerve spanning areas proximal and distal to a nerve crush were probed with mouse monoclonal antibody TUB-1A2 (specific for tyrosinated α-tubulin), visualized with biotin-conjugated horse anti-mouse IgG with a final incubation with FITC-labelled streptavidin. The immunofluorescence was detected by cytofluorimetric scanning. The scanning profiles represent the immunofluorescence (arbitrary units) along regenerating nerve (proximal→distal: left to right) The crush is denoted by '^' (n = 8).

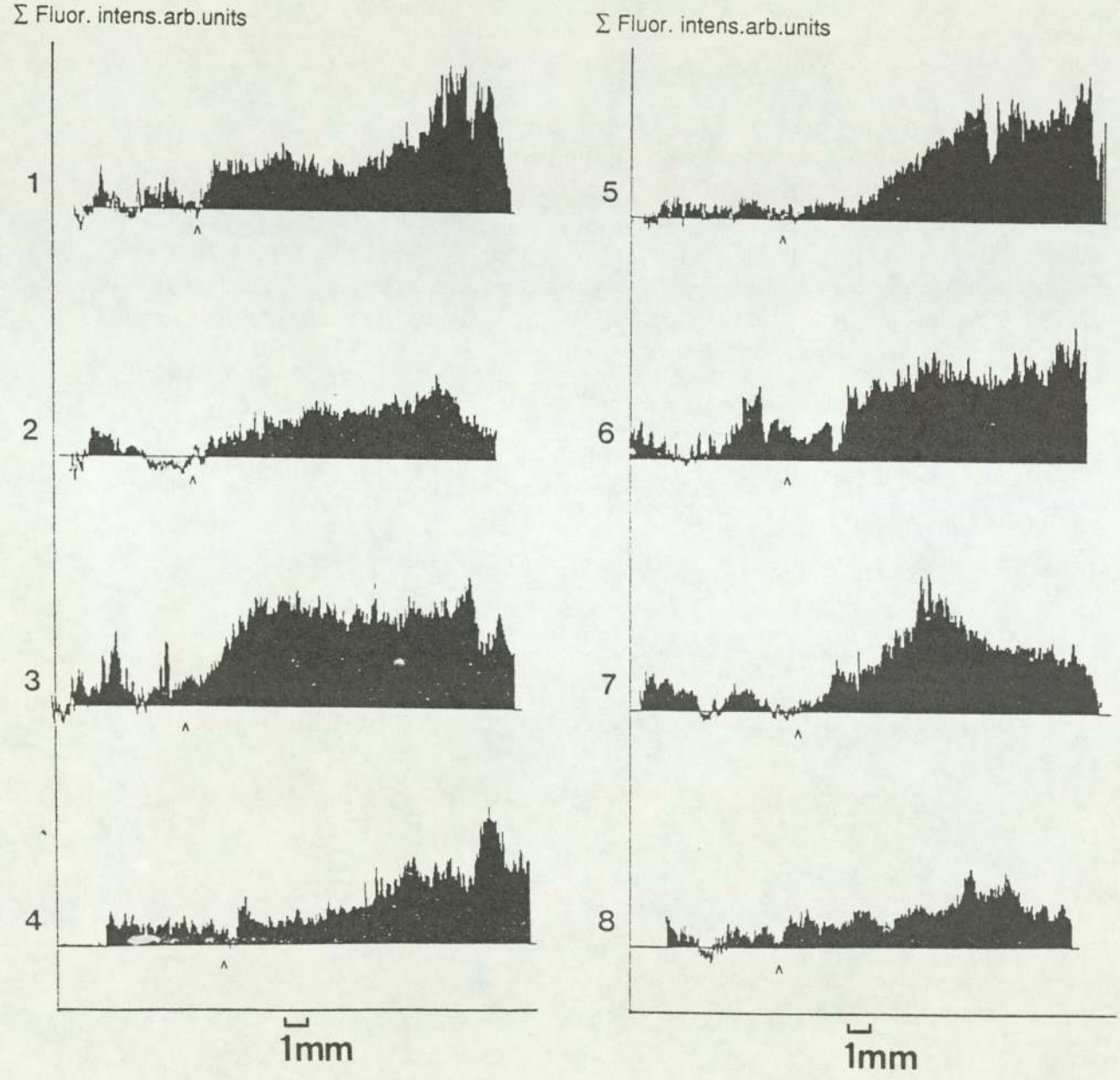
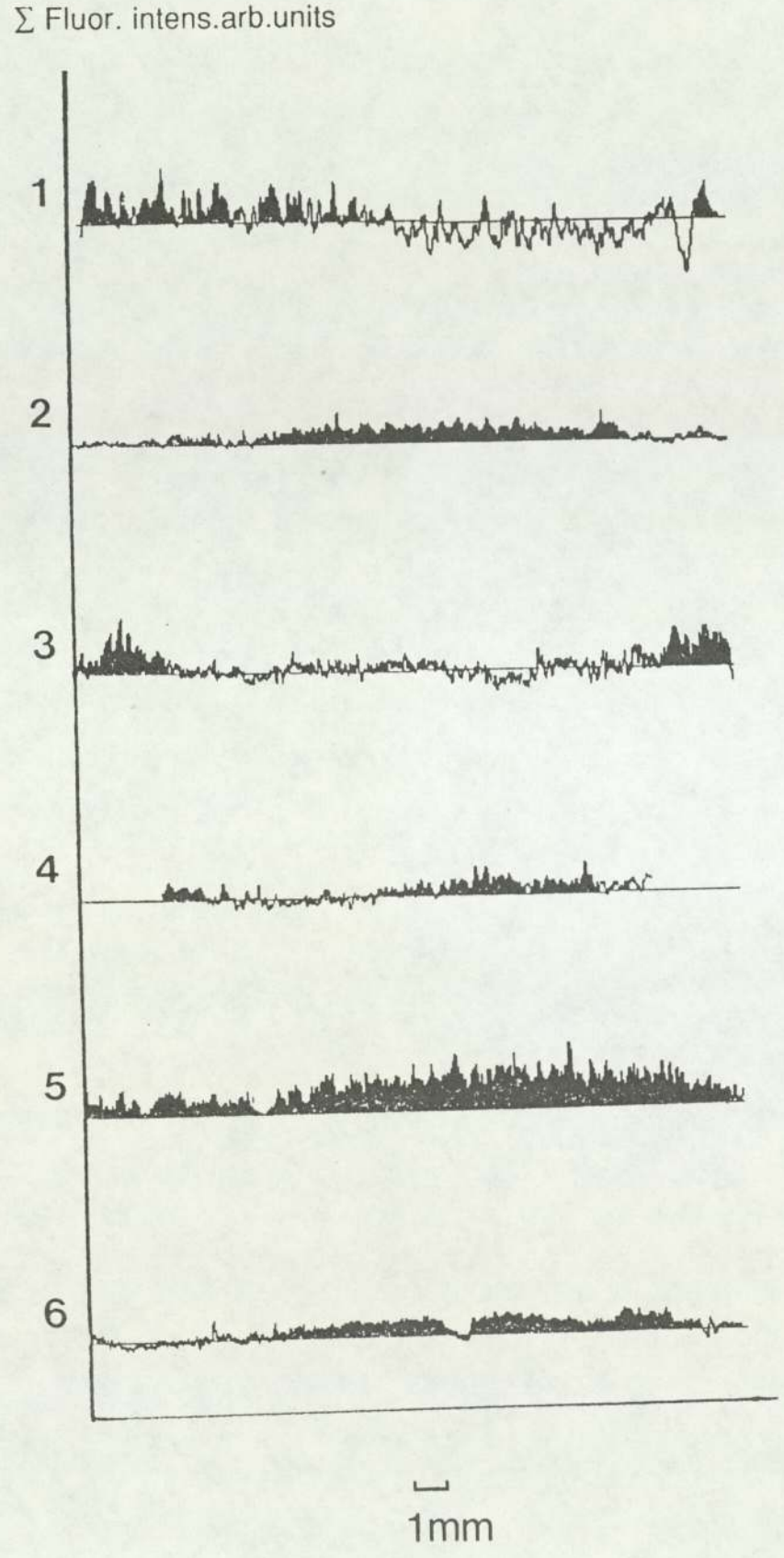

Σ Fluor. intens.arb.units
1
2
3
4
1mm
Σ Fluor. intens.arb.units
5
6
7
8
1mm

Figure 14

Immunofluorescence profile of tyrosinated tubulin in sections of control sciatic nerve. 2cm sections of control nerve spanning area equivalent to that proximal and distal to a nerve crush (see fig 13) were probed with mouse monoclonal antibody TUB-1A2 (specific for tyrosinated α-tubulin), visualized with biotin-conjugated horse anti-mouse IgG with a final incubation with FITC-labelled streptavidin. The immunofluorescence was detected by cytofluorimetric scanning. The scanning profiles represent the immunofluorescence (arbitrary units) along control nerve (proximal→distal: left to right; n = 6).


Σ Fluor. intens.arb.units
1
2
3
4
5
6
1mm

No attempt was made to obtain statistical data from cytofluorimetric scanning, but this phenomenon was observed in all eight nerves examined. Collectively, these observations correlated with our previous biochemical data.

The immunofluorescence observed by cytofluorimetric scanning was photographically recorded by confocal laser microscopy (see METHODS section 5.4). The immunofluorescence seen in sections of nerve proximal and distal to a crush, and their equivalents from uncrushed control nerve is shown in figure 15. A general immunofluorescence pattern of fibres, corresponding to tyrosinated α-tubulin, was observed in sections of both regenerating and control nerve. The pattern of immunofluorescence in nerve pieces distal to a crush was more dense and punctate, suggesting a shift in the tyrosinated tubulin that is associated with the assembled and disassembled fractions.

In some sections of regenerating nerve, distal to a crush, the immunofluorescence associated with tyrosinated α-tubulin was observed to increase initially and then decline. The decline in immunofluorescence associated with tyrosinated tubulin correlated with a decrease in the immunofluorescence associated with GAP-43, a growth-associated polypeptide, the expression of which is increased during nerve regeneration. This correlation, shown in figure 16, occurred 8 mm from the crush site, and the level of tyrosinated tubulin declined proximal to the decrease in GAP-43. These observations suggested the change in the tyrosination state of α-tubulin observed distal to a crush, was localized to the region where regenerating fibres are present.

Figure 15

Micrographs of cryostat sections of nerve proximal and distal to a crush and of equivalent regions of uncrushed nerve. Sections were incubated with TUB 1A2, against tyrosinated α-tubulin, and visualized with biotin-conjugated horse anti-mouse IgG and FITC-labelled streptavidin. The immunofluorescence-stained sections were analyzed by con-focal laser microscopy. Shown in **A** and **B** is the immunoreactivity in regions of control nerve equivalent to those shown in regenerating nerve **C**: proximal and **D**: distal to the crush site. The bar represents 250 microns.

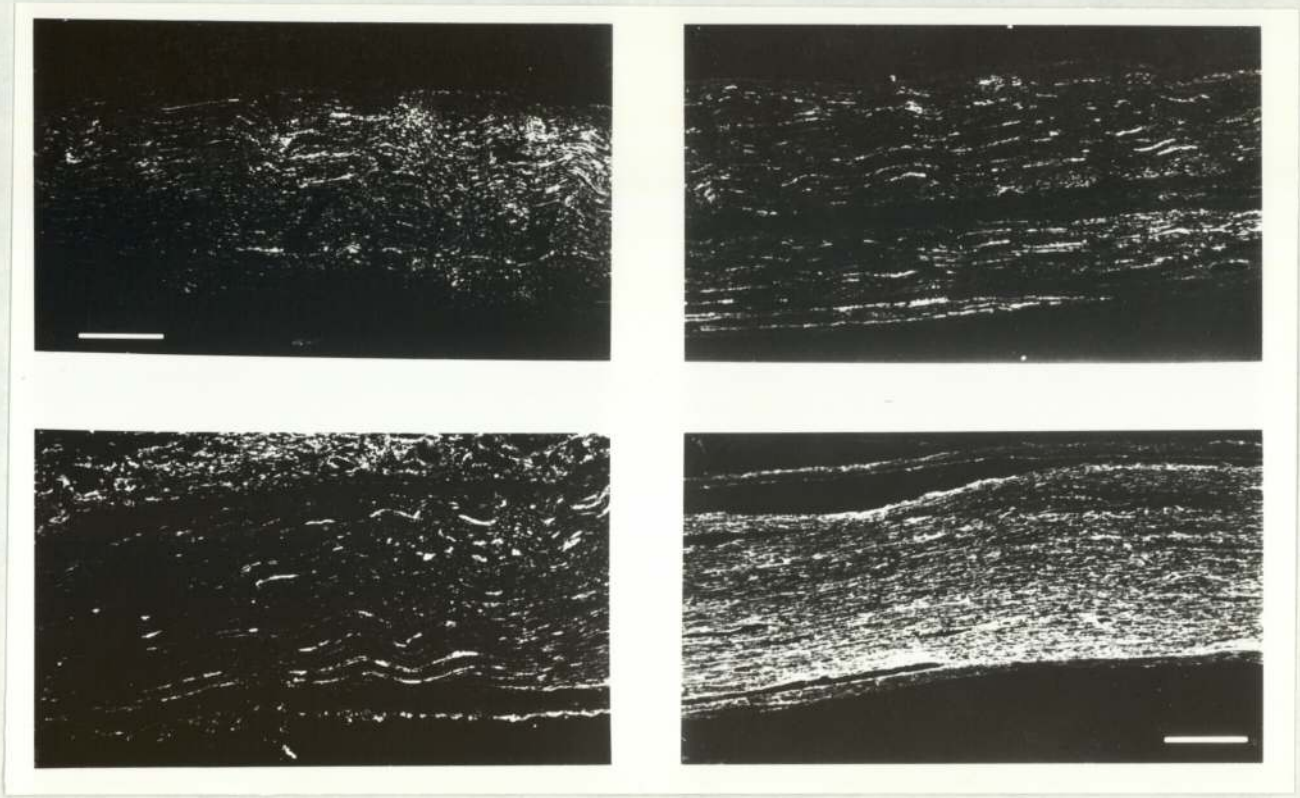
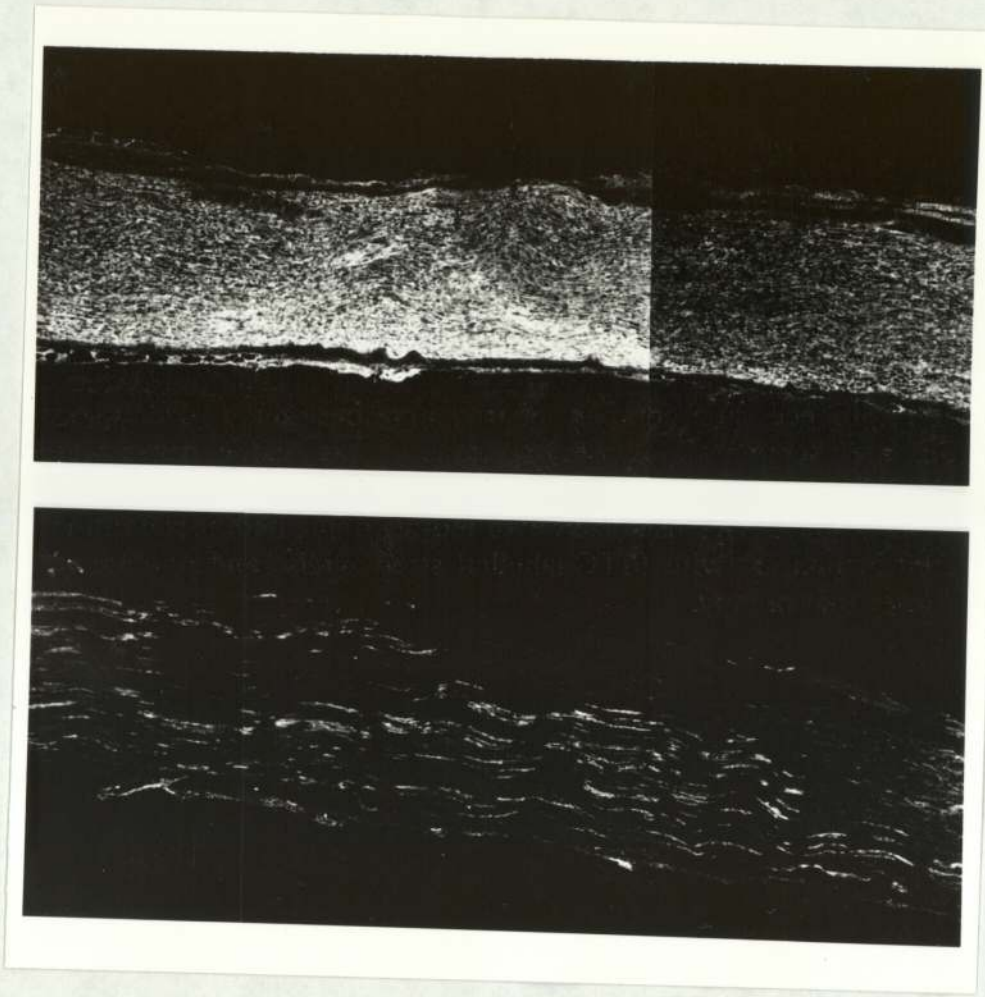

A
B
C
D

Figure 16

Immunofluorescence micrographs of regenerating nerve distal to a crush. Equivalent and sequential cryostat sections of rat sciatic nerve were probed separately with TUB-1A2 (A) and 10 E8/E7 (B) specific for tyrosinated α-tubulin and GAP-43 respectively, visualized with biotin-conjugated horse anti-mouse IgG with a final incubation with FITC-labelled streptavidin, and photographed using confocal laser microscopy.

A

B



1.1.3.3 *Ligatured nerve*

Examples of the immunofluorescence profile associated with tyrosinated α-tubulin in ligatured nerve sections, is shown in figure 17. The immuno-fluorescence profile suggests there is an accumulation of tyrosinated α-tubulin immediately proximal to the ligatured crush. The immunofluorescence did not increase distal to the ligature. This observation was in contrast to the immunofluorescence profile observed in regenerating nerve.

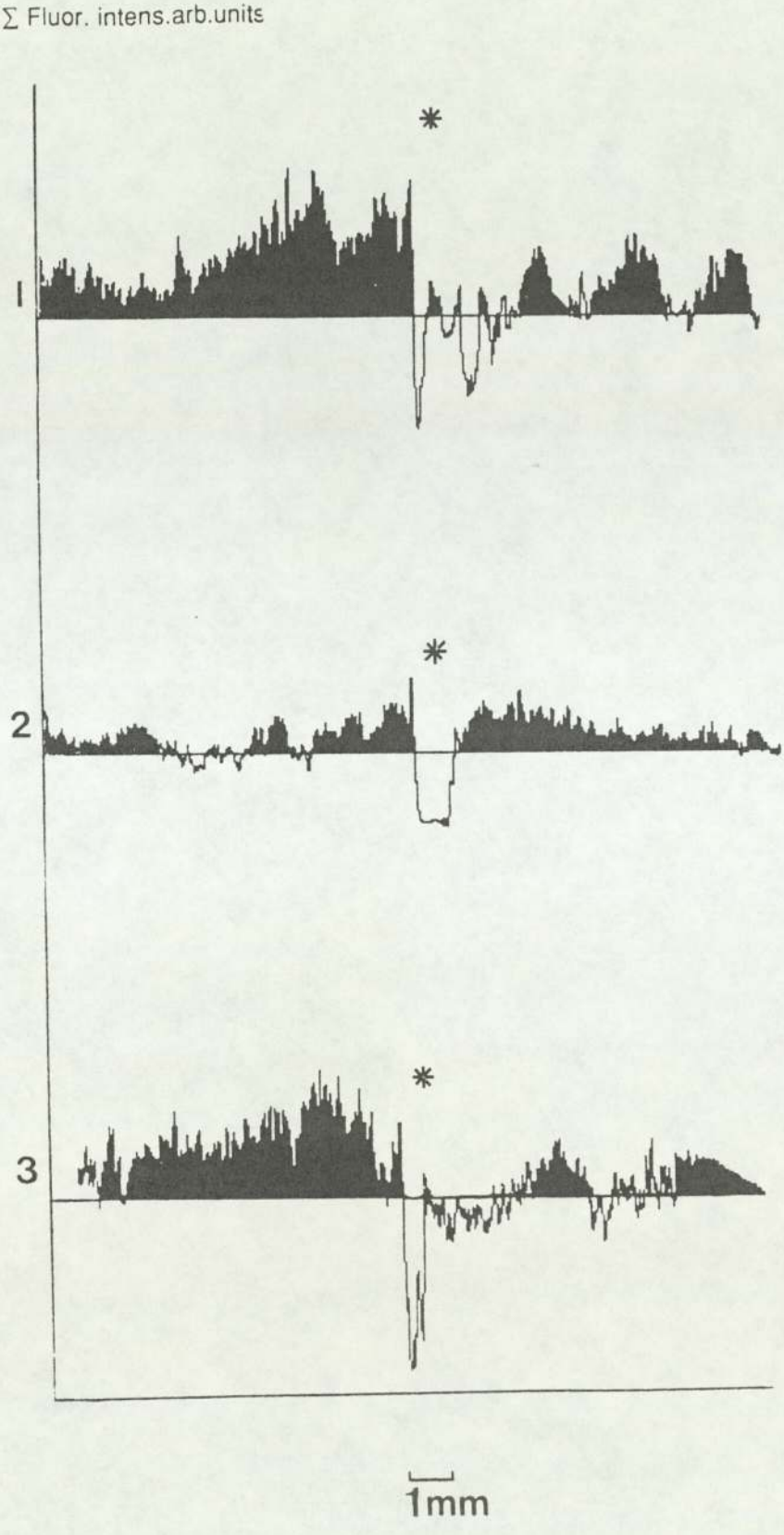
1.1.4 *Distribution of tyrosinated α-tubulin between assembled and non-assembled microtubule fractions*

The biochemical and immunohistochemical analysis of the proportion of α-tubulin that was tyrosinated in pieces of regenerating and control nerve, indicated there was an increase in the tyrosination state of α-tubulin in pieces of regenerating nerve, distal to a crush.

The possibility that the alteration in the biochemical composition of α-tubulin, which arises from the change in the tyrosination state of α-tubulin, may result in a shift in the proportion of tyrosinated tubulin associated with the assembled and non-assembled microtubule fractions was investigated, in pieces of regenerating and control nerve. In addition, the proportion of total α-tubulin associated with the assembled and non-assembled microtubule fractions was determined; a shift would indicate a shift in stability of the microtubule population.

Figure 17

Immunofluorescence profile of tyrosinated tubulin in sections of ligatured sciatic nerve 2cm. Sections of ligatured nerve spanning areas proximal and distal to a nerve crush were probed with mouse monoclonal anti-body TUB-1A2 (specific for tyrosinated α-tubulin), visualized with biotin-conjugated horse anti-mouse IgG with a final incubation with FITC-labelled streptavidin. The immunofluorescence was detected by cytofluorimetric scanning. The scanning profiles represent the immunofluorescence (arbitrary units) along ligatured nerve (proximal→distal: left to right) The crush is denoted by '*' (n = 3).


Σ Fluor. intens.arb.units
1
2
3
1mm

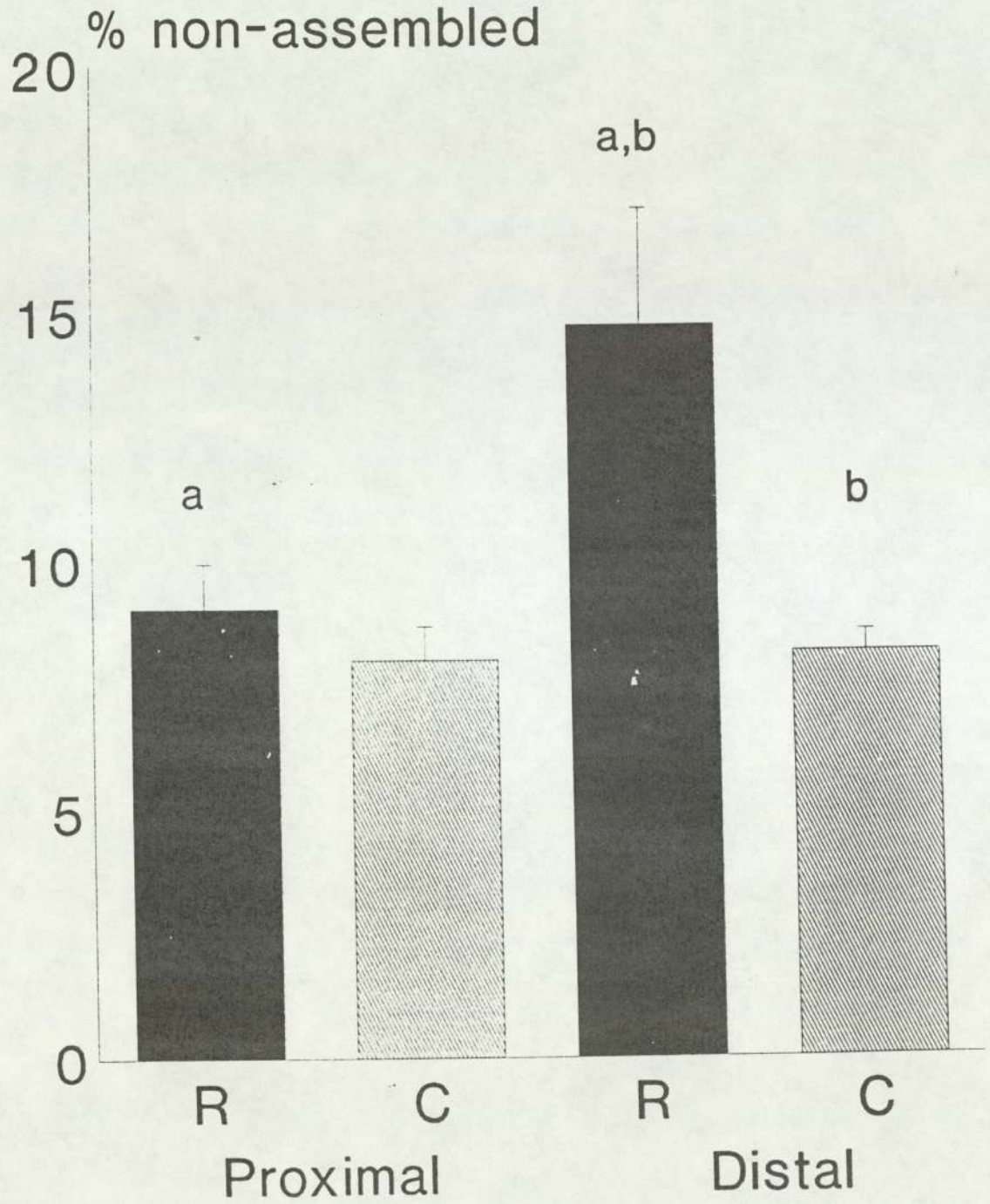
Pieces of regenerating nerve, proximal and distal to a crush and their equivalents from uncrushed control nerve were solubilized in NP-40 and separated into assembled (pellet) and non-assembled (supernatant) fractions following centrifugation (see METHODS section 3). The distributions of tyrosinated α-tubulin and total α-tubulin between assembled and non-assembled fractions, from regenerating and control nerve pieces were determined by dot-immunobinding assays (see METHODS section 5.3.2.), and compared with their electrophoretic profiles from 1-D SDS-PAGE (see METHODS section 4.).

1.1.4.1 *Immunobinding analysis*

The percentage of tyrosinated α-tubulin that was associated with non-assembled microtubule fractions from pieces of regenerating and control nerve is shown in figure 18. In pieces of regenerating nerve distal to a crush 14.8 ± 2.4% of tyrosinated α-tubulin was associated with the non-assembled microtubule fraction, whereas in pieces proximal to a crush, only 9.0 ± 0.9% of tyrosinated α-tubulin was associated with the non-assembled microtubule fraction. In control nerve equivalents, 8.0 ± 0.7% of tyrosinated α-tubulin was associated with the non-assembled microtubule fraction proximal to a crush, while 8.1 ± 0.4% of tyrosinated α-tubulin was associated with the non-assembled microtubule fraction in pieces of control nerve equivalent to distal to a crush. The percentage of tyrosinated α-tubulin associated with the non-assembled microtubule proteins was significantly greater in pieces of regenerating nerve, distal to a crush, compared to pieces of control nerve ($p<0.01$) and pieces of regenerating nerve proximal to a crush ($p<0.05$).

Figure 18

Immunobinding analysis of non assembled fractions from pieces of rat sciatic nerve proximal or distal to a nerve crush (regenerating: R) or from equivalent regions of uncrushed (control: C) nerves, using the monoclonal antibody YL 1/2, specific for tyrosinated α-tubulin. The immunobinding in the fractions was calculated by laser densitometry and is presented as the percentage of YL1/2 immunobinding of total tubulin in pieces for regenerating and control nerve that exist in a non-assembled state $\pm$ S.D. Comparison of values marked with suffix 'a' had a statistically significant difference $P<0.05$; values with suffix 'b' had a statistically significant difference $P<0.01$ ($n = 8$ i.e from 8 separate animals).


% non-assembled
20
15
10
5
0
a
a,b
b
R
C
R
C
Proximal
Distal

The percentage of total α-tubulin that was associated with the non-assembled microtubule fractions from pieces of regenerating and control nerve, is shown in figure 19. In regenerating nerve, 4.4 $\pm$ 0.4% and 3.4 $\pm$ 0.4% of total α-tubulin was associated with the non-assembled microtubule fractions from pieces of nerve proximal and distal to a crush, respectively. In control nerve, 4.5 $\pm$ 0.5% and 5.0 $\pm$ 0.6% of total α-tubulin was associated with the non-assembled microtubule fractions, from pieces of nerve equivalent to those proximal and distal to a crush, respectively. The distribution of total tubulin between the assembled and the non-assembled microtubule protein fractions was not statistically significant between regenerating and control nerves ($P > 0.05$).

1.1.4.2 *1-Dimensional SDS-PAGE*

The assembled microtubule fractions from pieces of regenerating nerve, proximal and distal to a crush, and equivalent pieces of uncrushed control nerve were prepared and subjected to 1-dimensional SDS-PAGE on 7.5% polyacrylamide gels (see METHODS section 4.1). The resolved proteins were stained with Coomassie blue, as shown in figure 20. Similar electrophoretic patterns were observed for equivalent pieces of regenerating and control nerve, both proximal and distal to a crush.

This observation supported our previous observation that there was no shift in the amount of total α-tubulin associated with the assembled fraction of microtubules, between regenerating and control nerve i.e. a shift in the assembly state.

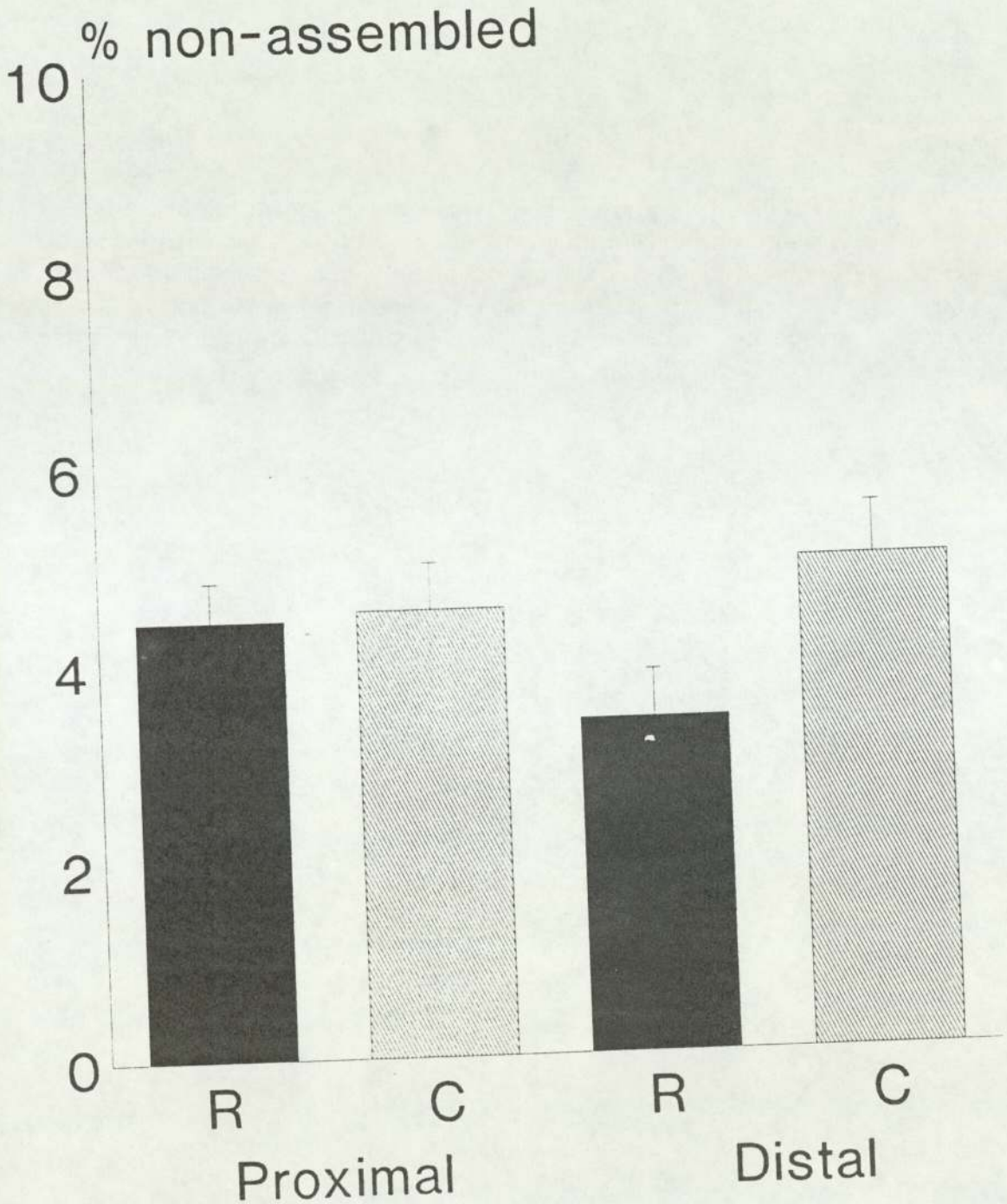
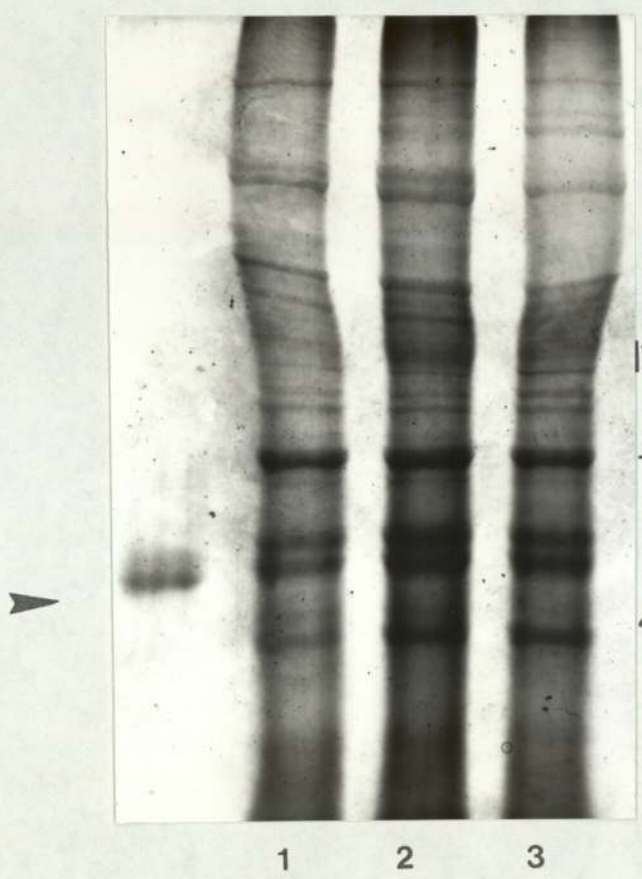

% non-assembled
10
8
6
4
2
0
R
C
R
C
Proximal
Distal

Figure 20

1-Dimensional SDS-PAGE of proteins associated with assembled tubulin obtained from pieces of regenerating and control nerve, visualized with Coomasie Blue stain. Tracks 1-3 represent L4,L5 dorsal root ganglia (1), nerve pieces proximal to a crush (2) and distal to a crush (3), respectively, from regenerating nerve. Tracks 4-6 are the equivalent respective pieces of control nerve. Molecular weight markers are included, tubulin is indicated by the arrow.


116 kDa
77 kDa
46 kDa
1
2
3

1
2
3

1.1.5 *Tubulin:tyrosine ligase activity*

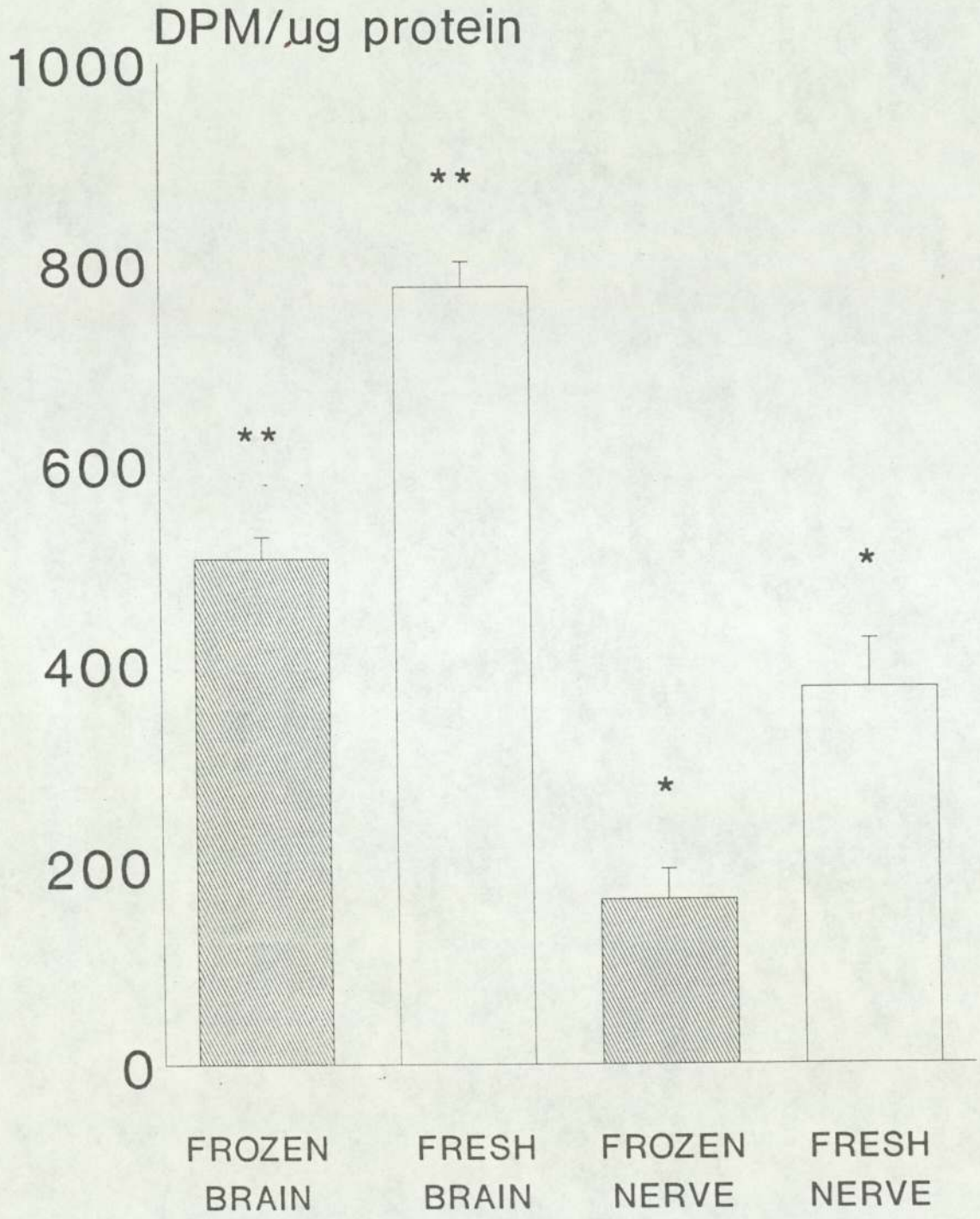
In view of the previous observations on the tyrosination of α-tubulin, further experiments were aimed to discover if the differences in the levels of tyrosinated α-tubulin, observed previously, were associated with a change in the activity of tubulin:tyrosine ligase (TTL), the enzyme responsible for the post-translational tyrosination of α-tubulin. This activity had not been previously detected in rat sciatic nerve. Preliminary experiments established the conditions necessary for this assay i.e. to measure the incorporation of [^{3}H]-tyrosine into tubulin.

1.1.5.1 *Assay development*

Homogenates obtained from fresh or frozen sciatic nerve and brain (as prepared in METHODS section 6.1) were incubated with [^{3}H]-tyrosine in the presence of phosphocellulose-purified porcine tubulin (see METHODS section 6.2) to determine if TTL activity decreased on storage at -80°C. Figure 21 shows the TTL activity in fresh and frozen rat brain and rat sciatic nerve, directed against exogenous phosphocellulose-purified porcine tubulin. The rat brain served as a positive control for TTL activity. The TTL activity obtained with homogenates from frozen rat brain and rat sciatic nerve was reduced in comparison to that obtained from homogenates of fresh rat brain and rat sciatic nerve ($P < 0.05$). In subsequent TTL assays fresh tissue was therefore used, rather than tissue stored at -80°C.

Figure 21

Tubulin-tyrosine ligase activity in homogenates of fresh and frozen rat sciatic nerve and rat brain. Homogenates were incubated with ^{3}H-tyrosine in the presence of exogenous purified porcine brain tubulin. The activity is represented as D.P.M/μg protein $\pm$ S.E.M. (n = 4). Comparison of values marked with * had a statistically significant difference $P<0.05$ between fresh and frozen; values with ** also had a statistically significant difference $P<0.05$ between fresh and frozen.


DPM/ug protein
1000
800
600
400
200
0
**
**
*
*
FROZEN BRAIN
FRESH BRAIN
FROZEN NERVE
FRESH NERVE

To confirm that [^{3}H]-tyrosine was in fact incorporated into tubulin in our assay system, we analyzed the distribution of radiolabelled proteins by 1-dimensional SDS-PAGE (see METHODS section 6.4) and subsequent scintillation counting. As can be seen in figure 22, the bulk of the radioactivity was associated with a protein of an apparent molecular weight of 55 kDa with a relative mobility identical to that of tubulin.

Although TTL may act reversibly, a net incorporation of [^{3}H]-tyrosine into tubulin was observed with time, under the conditions employed in this study, (figure 23). It was thought that carboxypeptidase A treatment of purified porcine tubulin would increase its acceptor capacity for [^{3}H]-tyrosine, by removing the C-terminal tyrosine residues that were already present on α-tubulin. Therefore an assay was designed to investigate whether purified carboxypeptidase A-treated porcine tubulin would be a better substrate than untreated purified porcine tubulin for TTL.

The incorporation of radiolabelled [^{3}H]-tyrosine into purified porcine tubulin and purified carboxypeptidase A-treated porcine tubulin (see METHODS 2.2.4.) mediated by rat brain TTL is shown in figure 24. The carboxypeptidase A-treated porcine tubulin did not incorporate any extra [^{3}H]-tyrosine than the untreated porcine tubulin. However, there was a net increase in the incorporation of [^{3}H]-tyrosine in the presence of exogenous porcine tubulin substrate ($P < 0.01$). The incorporation of [^{3}H]-tyrosine into both carboxypeptidase A-treated and untreated porcine tubulin mediated by TTL from rat sciatic nerve is shown in figure 25. Again, as with rat brain, carboxypeptidase A treatment had no effect on the net incorporation of [^{3}H]-

Figure 22

Extracts of rat sciatic nerve were incubated with ^{3}H-tyrosine and the resulting radiolabelled proteins were separated by polyacrylamide gel electrophoresis on 7.5% gels along with molecular weight markers and purified tubulin. Gels were cut into 2.5mm pieces and the radioactivity in each piece measured by scintillation counting and expressed as D.P.M. The apparent molecular weight of the major peak of radioactivity is compared with molecular weight standards and corresponds to that of tubulin.

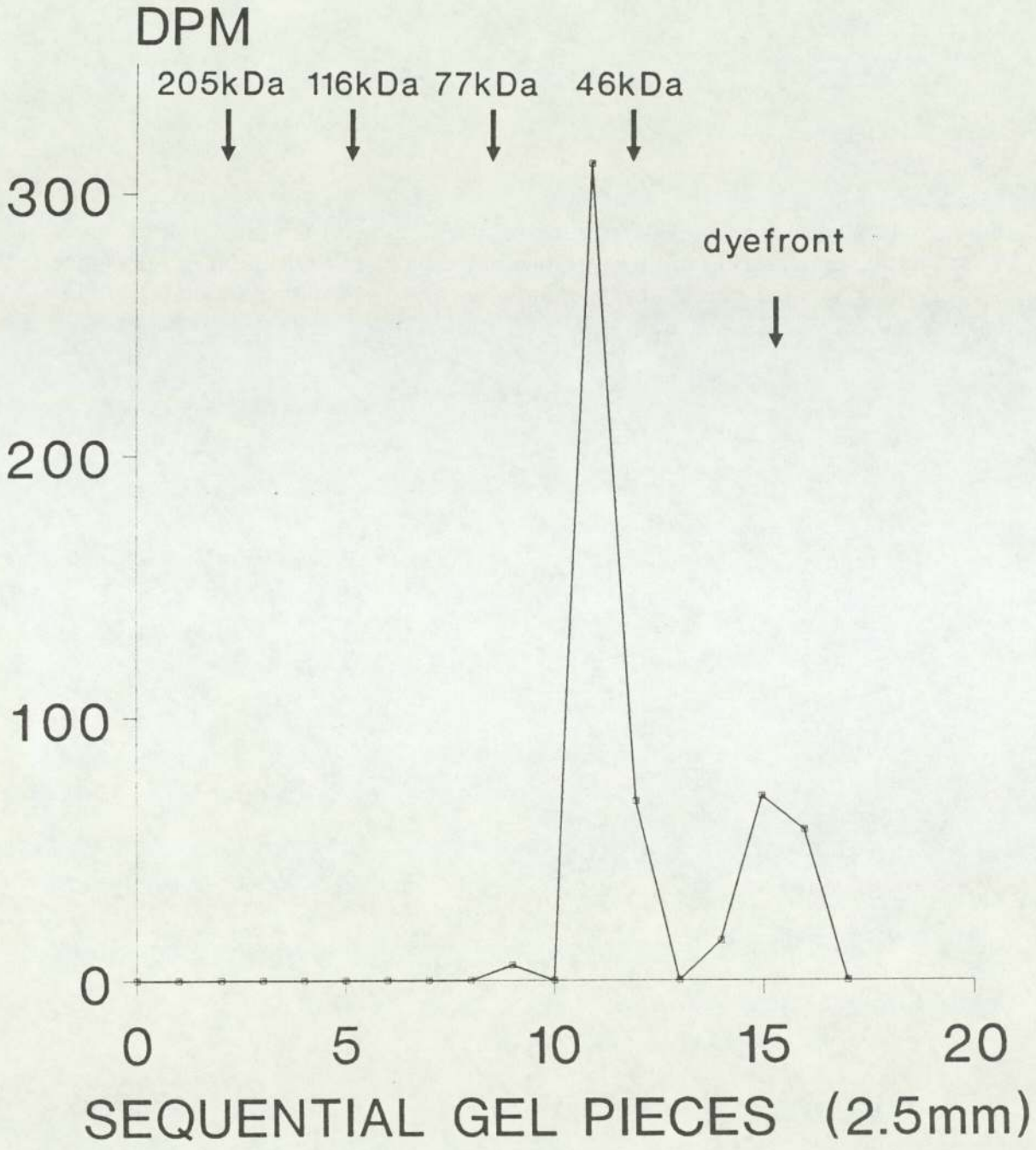

DPM
205kDa
116kDa
77kDa
46kDa
dyefront
300
200
100
0
0
5
10
15
20
SEQUENTIAL GEL PIECES (2.5mm)

Figure 23

Incorporation with time of ^{3}H-tyrosine into purified brain tubulin. Homogenates of rat sciatic nerve and rat brain, were incubated with ^{3}H-tyrosine in the presence of exogenous purified porcine brain tubulin for up to 50 minutes. The activity is represented as D.P.M/μg protein $\pm$ S.E.M. (n = 4).

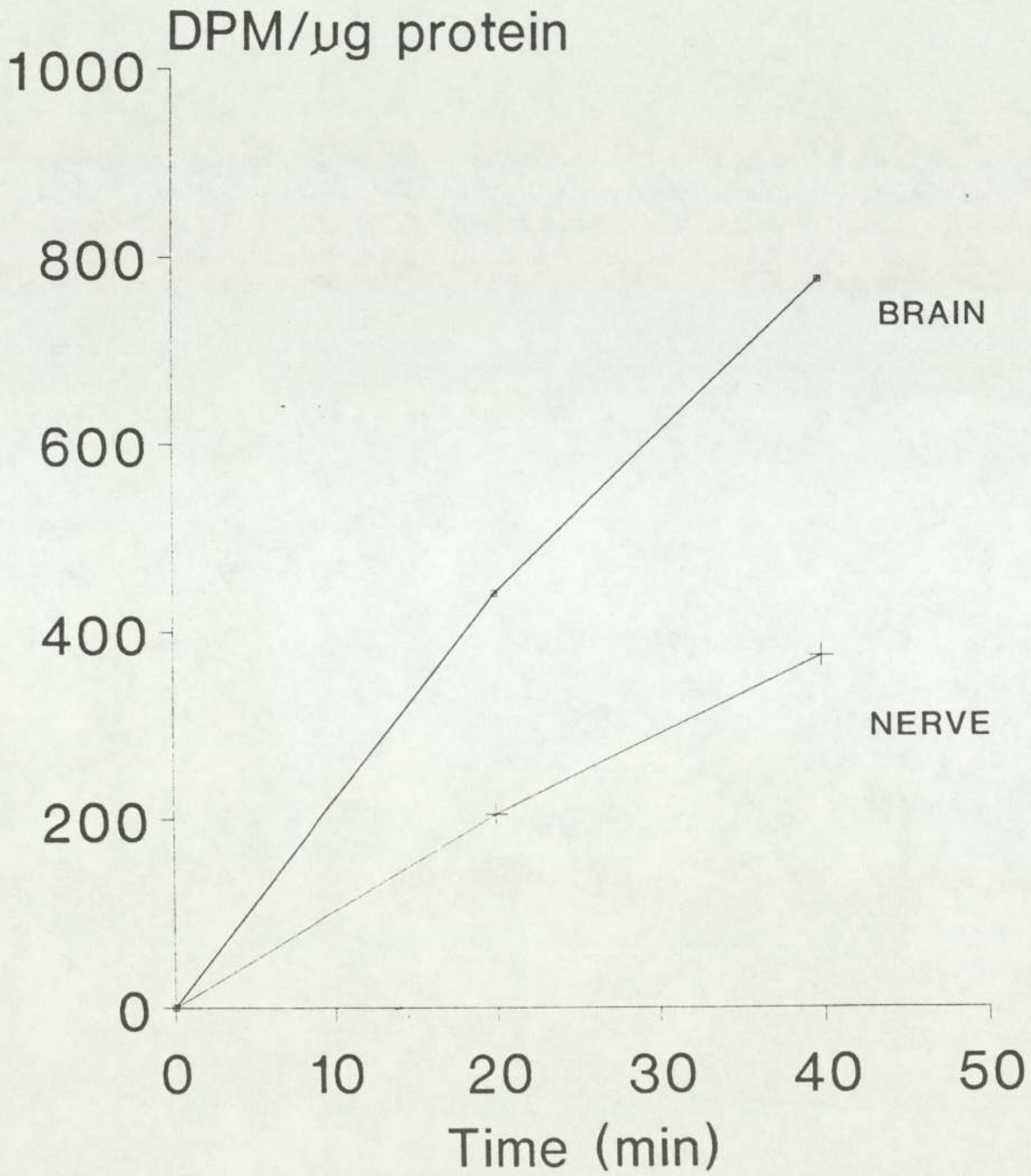

DPM/µg protein
1000
800
600
400
200
0
BRAIN
NERVE
0
10
20
30
40
50
Time (min)

Figure 24

Tubulin:tyrosine ligase activity in homogenates of rat brain. Homogenates were incubated with ^{3}H-tyrosine in the presence of porcine brain tubulin, exogenous purified carboxypeptidase A treated and untreated tubulin. The activity is represented as D.P.M/μg protein $\pm$ S.E.M. (n = 4). * $P<0.01$ between presence and absence of exogenous tubulin.

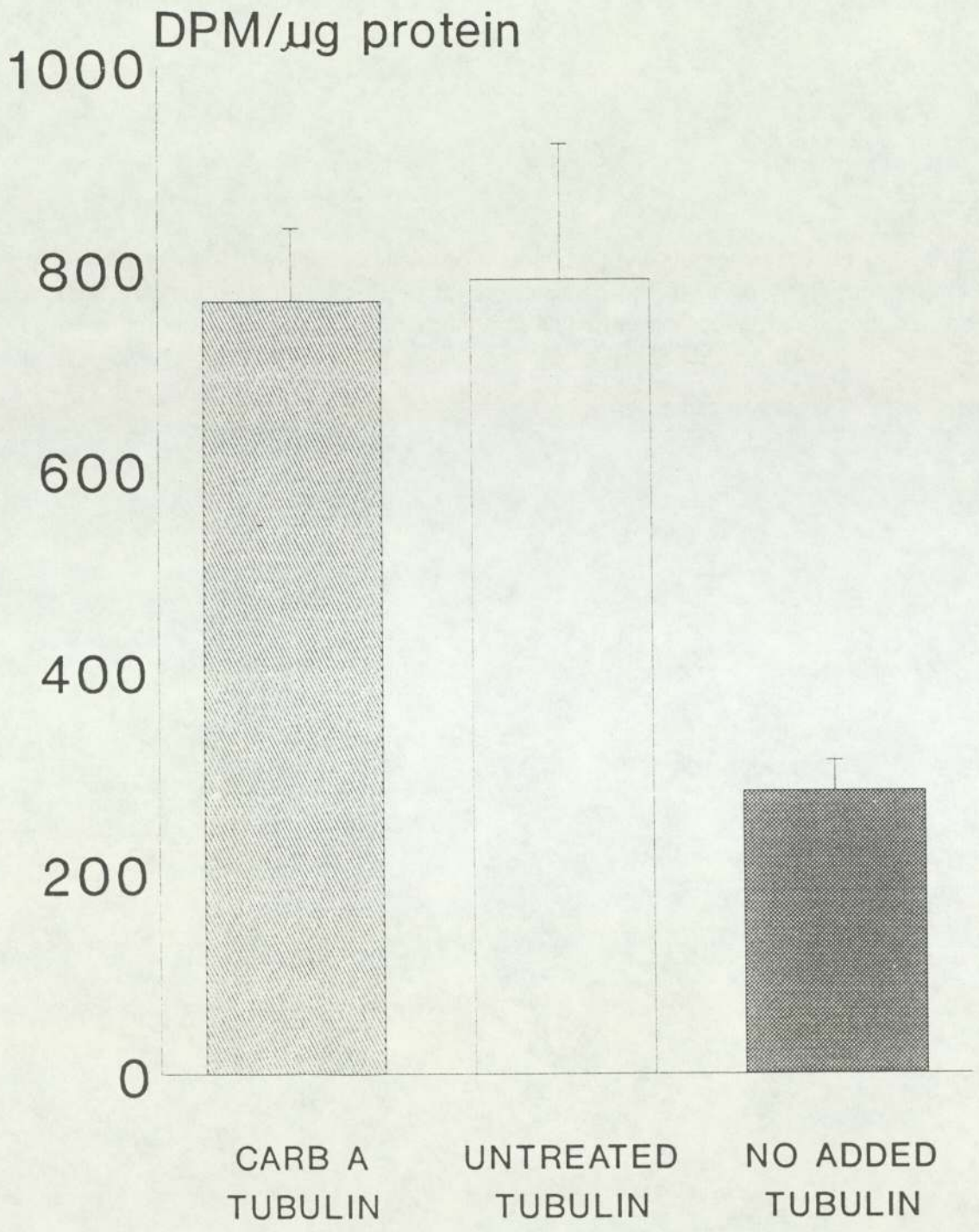
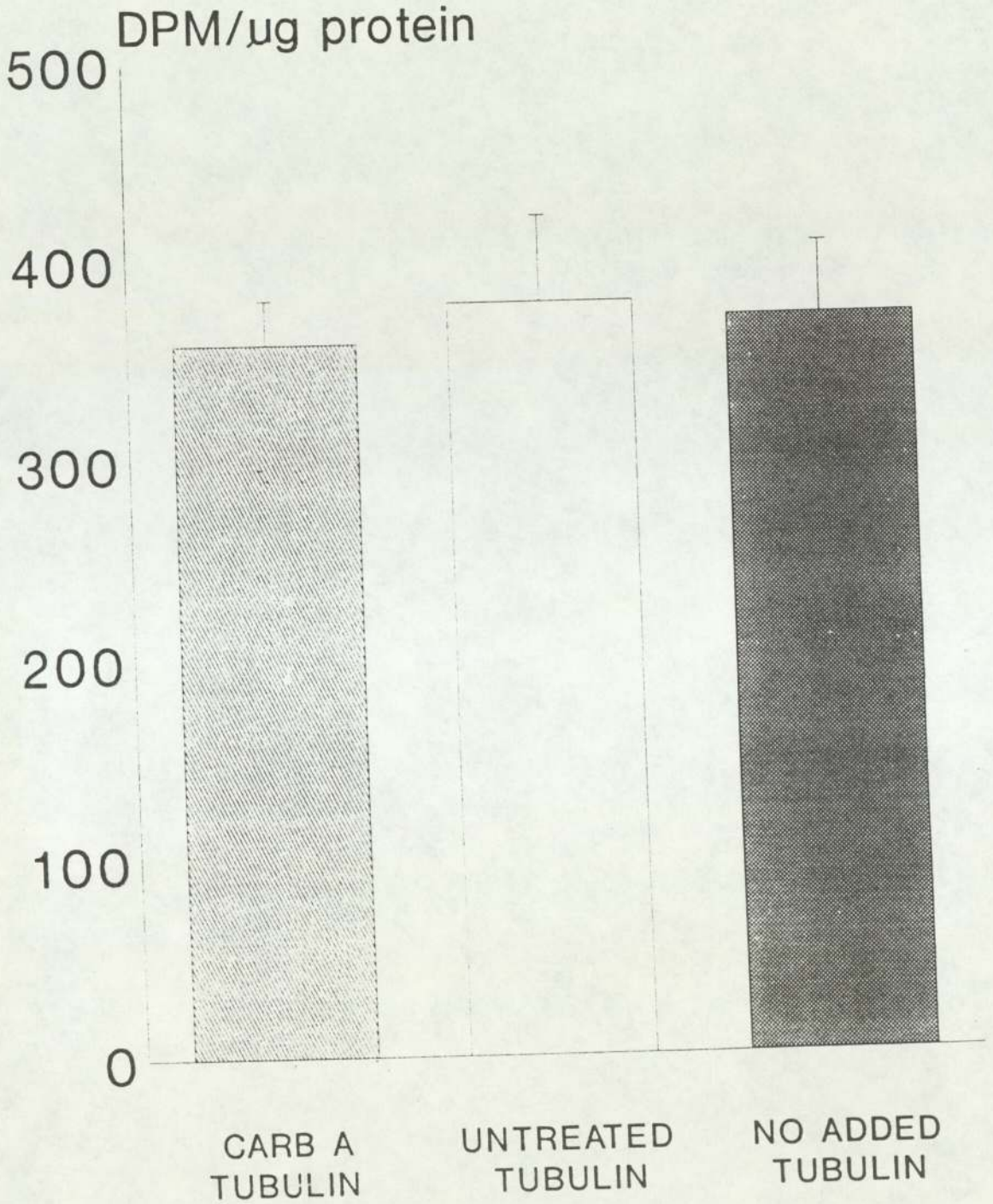

DPM/µg protein
1000
800
600
400
200
0
CARB A TUBULIN
UNTREATED TUBULIN
NO ADDED TUBULIN

Figure 25

Tubulin:tyrosine ligase activity in homogenates of rat sciatic nerve. Homogenates were incubated with ^{3}H-tyrosine in the presence of rat brain tubulin, exogenous purified carboxypeptidase A treated and untreated porcine tubulin. The activity is represented as D.P.M/μg protein $\pm$ S.E.M. (n = 4).


DPM/µg protein
500
400
300
200
100
0
CARB A TUBULIN
UNTREATED TUBULIN
NO ADDED TUBULIN

tyrosine into porcine tubulin. However, the presence of an exogenous tubulin substrate did not result in an increased incorporation of [^{3}H]-tyrosine compared to that seen in rat sciatic nerve homogenates alone.

In the light of the observations with rat sciatic nerve and an exogenous porcine tubulin substrate (figure 25), an assay was designed to determine whether the incorporation of [^{3}H]-tyrosine into an exogenous substrate obtained from the same species would be greater than that into endogenous tubulin alone. Purified rat brain tubulin and purified carboxypeptidase A-treated rat brain tubulin were both used as exogenous tubulin substrates. The incorporation of [^{3}H]-tyrosine observed in this assay is shown in figure 26. The incorporation of [^{3}H]-tyrosine was not increased in the presence of either of the exogenous tubulin substrates, from the same species, in comparison to that seen in rat sciatic nerve alone.

It was hoped that we could increase the incorporation of [^{3}H]-tyrosine into endogenous tubulin by inhibiting any endogenous carboxypeptidase activity that could cleave the C-terminal tyrosine of tubulin, using a N-carbobenzoxydipeptide termed Z-GLU-TYR which had been previously shown to inhibit endogenous carboxypeptidase activity *in vitro*. An assay was designed to investigate the effects of Z-GLU-TYR on the specific activities of both rat brain and rat sciatic nerve TTL; the results are shown in figure 27. The presence of Z-GLU-TYR actually reduced

Figure 26

Tubulin:tyrosine ligase activity in homogenates of rat sciatic nerve. Homogenates were incubated with ^{3}H-tyrosine in the presence of exogenous purified carboxypeptidase A treated and untreated rat brain tubulin. The activity is represented as D.P.M/μg protein $\pm$ S.E.M. (n = 4).

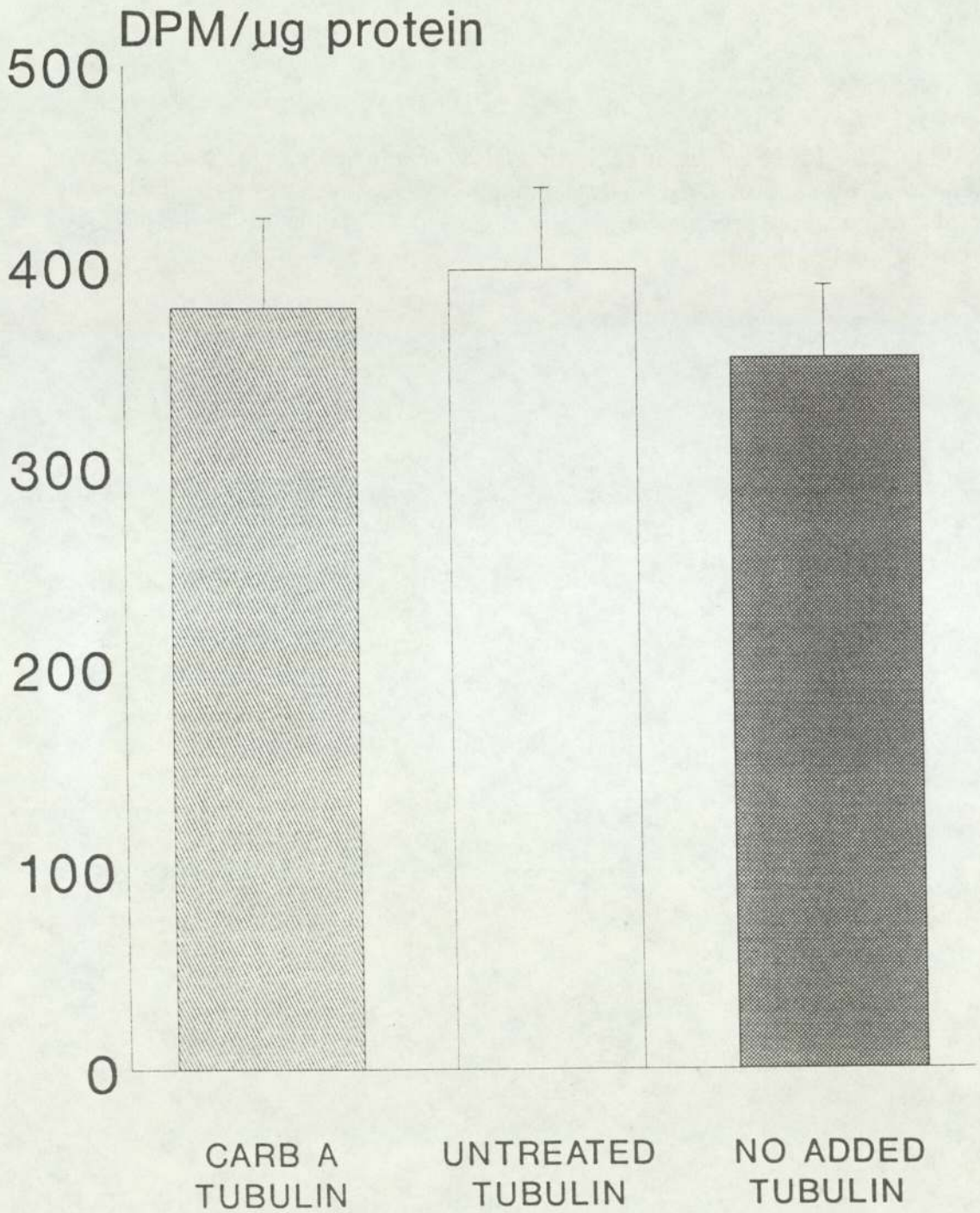
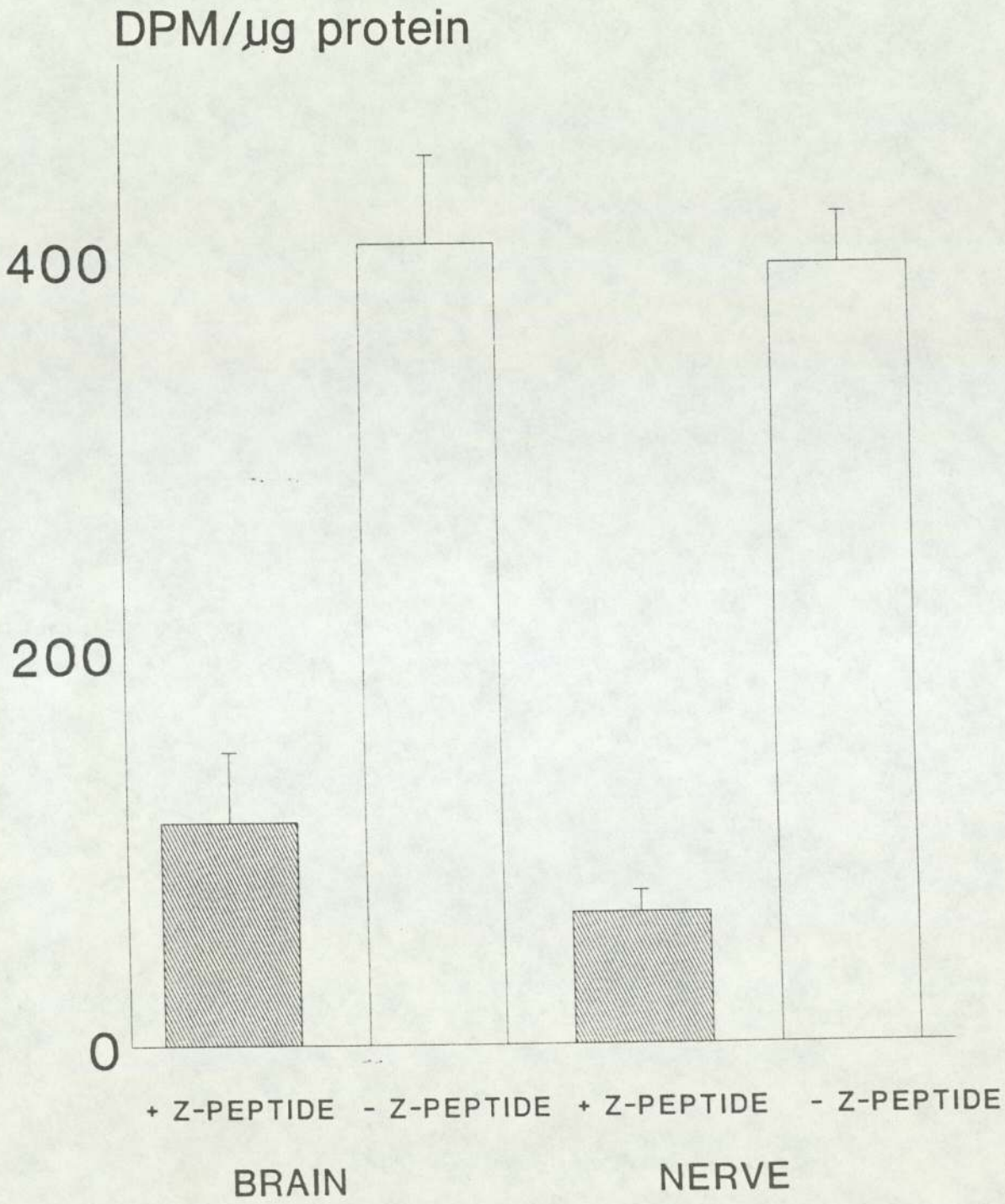

DPM/µg protein
500
400
300
200
100
0
CARB A TUBULIN
UNTREATED TUBULIN
NO ADDED TUBULIN

Figure 27

Tubulin:tyrosine ligase activity in rat brain and rat sciatic nerve homogenates in the presence and absence of Z-GLU-TYR dipeptide. Homogenates were incubated with ^{3}H tyrosine in the presence or absence of Z-GLU-TYR dipeptide and purified exogenous porcine tubulin.


DPM/µg protein
400
200
0
+ Z-PEPTIDE
- Z-PEPTIDE
+ Z-PEPTIDE
- Z-PEPTIDE
BRAIN
NERVE

the specific TTL activity of both rat brain and rat sciatic nerve, and was therefore not used in subsequent analyses.

Thus, in subsequent assays we measured the incorporation of [^{3}H]-tyrosine into endogenous nerve tubulin, expressing the incorporation as nmoles of tyrosine per mg of total nerve protein; this was then termed as the specific TTL activity.

1.1.5.2 *TTL activity in nerve pieces*

The specific activity of TTL (as defined above) was measured in pieces of regenerating nerve immediately proximal and distal to a crush injury, and in equivalent pieces of uncrushed control nerve (as defined previously). There was a decrease in the specific activity of TTL found in nerve pieces distal to a crush compared to nerve pieces proximal to a crush and equivalent pieces of control nerves ($P < 0.05$), as shown in figure 28.

To determine whether this phenomenon was linked with nerve regeneration or was a consequence of nerve degeneration, the TTL activity in pieces of ligatured nerve were measured. The specific activity of TTL in nerve pieces immediately proximal and distal to a crush from ligatured nerve, and their equivalent control nerve pieces are shown in figure 29. There was an apparent, slight decrease in the TTL activity proximal and distal to the crush in ligatured nerve, in comparison with equivalent pieces of control nerve, but this was not statistically significant ($P > 0.05$).

Figure 28

Specific tubulin: tyrosine ligase activity in homogenates of pieces of rat sciatic nerve proximal or distal to a nerve crush (regenerating) or from equivalent pieces of uncrushed (control) nerve. Homogenates were incubated in the present of ^{3}H-tyrosine; the activity is represented as nmoles of tyrosine bound per mg total protein $\pm$ S.E.M. (n = 8 in each case). * $P < 0.05$ compared with proximal pieces of regenerating nerve and equivalent pieces of control nerve.

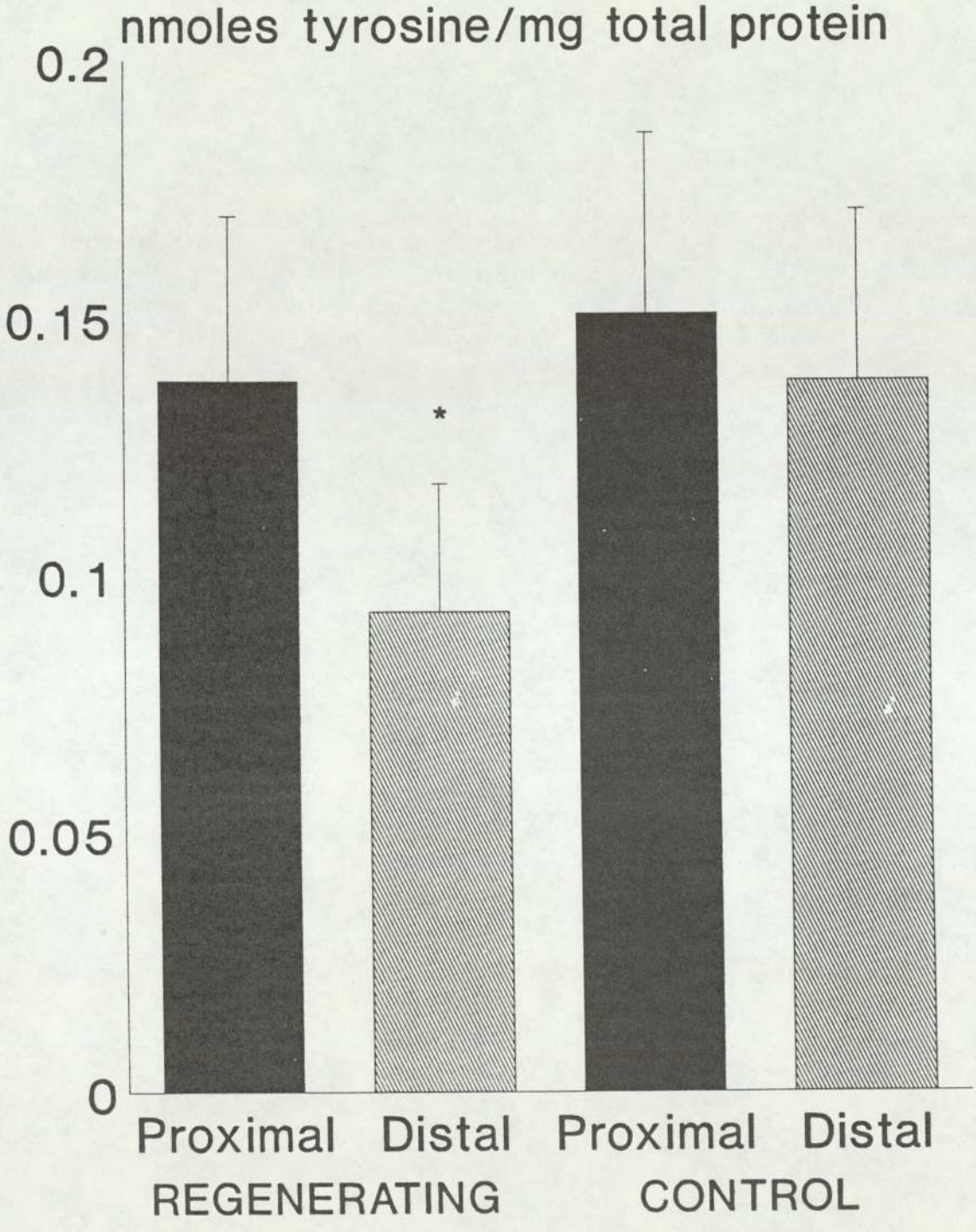
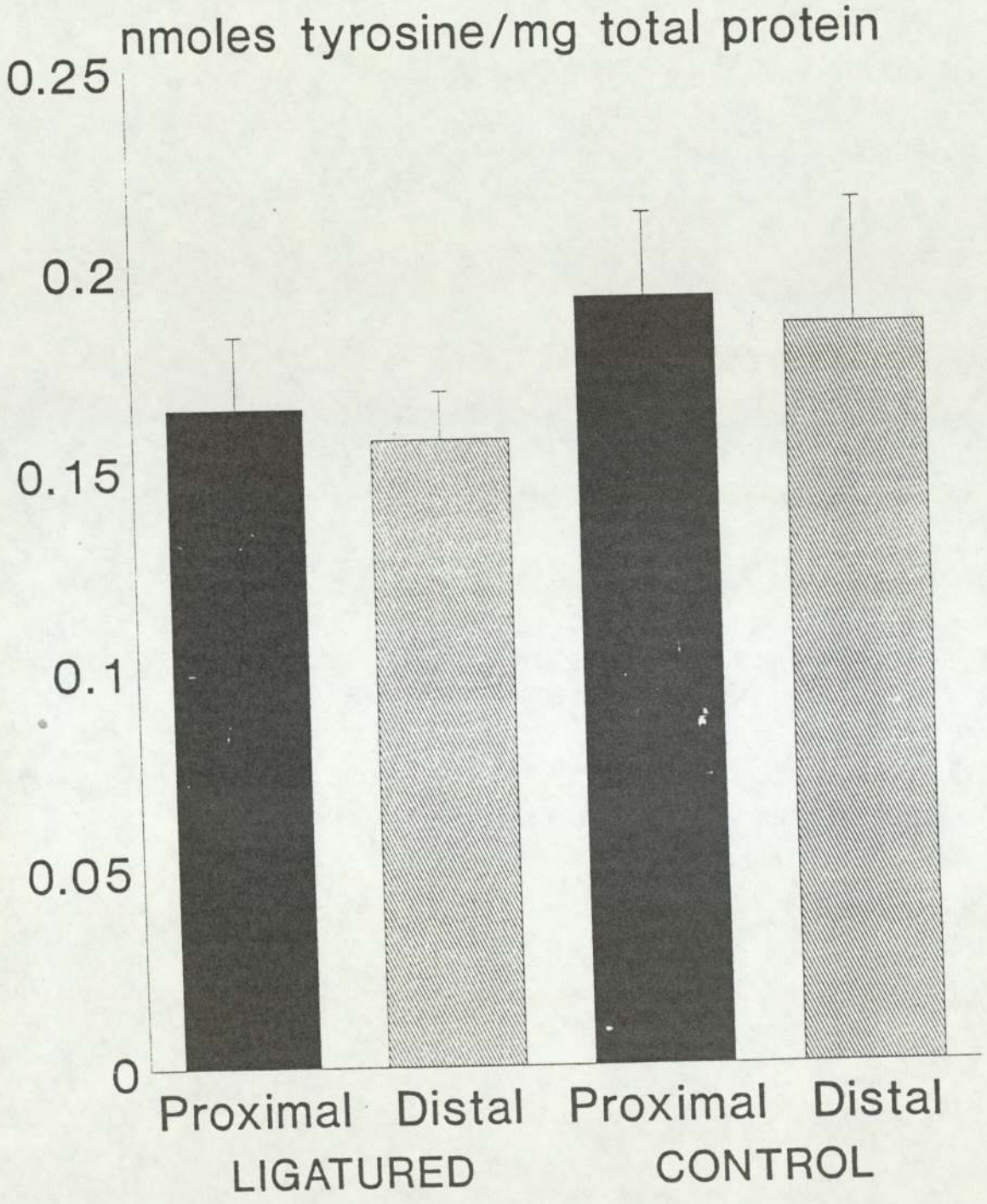

nmoles tyrosine/mg total protein
0.2
0.15
0.1
0.05
0
*
Proximal
Distal
Proximal
Distal
REGENERATING
CONTROL

Figure 29

Specific tubulin:tyrosine ligase activity in homogenates of pieces of rat sciatic nerve proximal or distal to a nerve crush (ligated), or from equivalent regions of uncrushed (control) nerve. Homogenates were incubated in the presence of ^{3}H tyrosine and the activity represented as nmoles of tyrosine bound per mg of total protein $\pm$ S.E.M. n = 4 in each case.


nmoles tyrosine/mg total protein
0.25
0.2
0.15
0.1
0.05
0
Proximal
Distal
Proximal
Distal
LIGATURED
CONTROL

1.1.6 *Tubulin:tyrosine carboxypeptidase activity*

We attempted to measure tubulin:tyrosine carboxypeptidase (TTCP) activity in rat sciatic nerve, and thus determine if a change in TTCP activity occurred as a consequence of neuronal injury.

For this, fresh homogenates of pieces of regenerating nerve immediately proximal and distal to a crush injury, and equivalent pieces of uncrushed control nerve (as described in METHODS section 7.1) were assayed for TTCP activity using two separate methods as outlined in METHODS 7.3.1 and 7.3.2., the data from which are shown in figures 30A-B and 31A-D, respectively. TTCP activity could not be detected in pieces of rat sciatic nerve from regenerating or control nerve using either of the two methods.

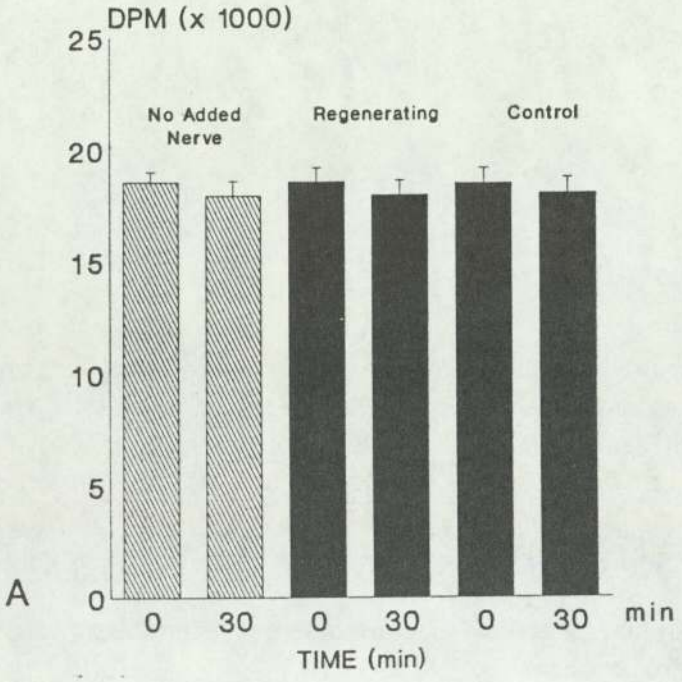
Figure 30

Tubulin:tyrosine carboxypeptidase (TTCP) activity in rat sciatic nerve. TTCP activity was assayed in pieces of nerve proximal and distal to a nerve crush and in equivalent pieces of control nerve. Homogenates were added to ^{3}H-radiolabelled tyrosinated rat brain tubulin and TTCP activity assayed at 0 and 30 min and compared with a control with no added homogenate.

Data is represented as D.P.M. $\pm$ S.E.M; n = 4 in each case.

A. TTCP activity in pieces of nerve proximal to a crush and their equivalents from control nerve.

B. TTCP activity in pieces of nerve distal to a crush and their equivalents from control nerve.


DPM (x 1000)
25
20
15
10
5
0
No Added Nerve
Regenerating
Control
A
0
30
0
30
0
30
min
TIME (min)

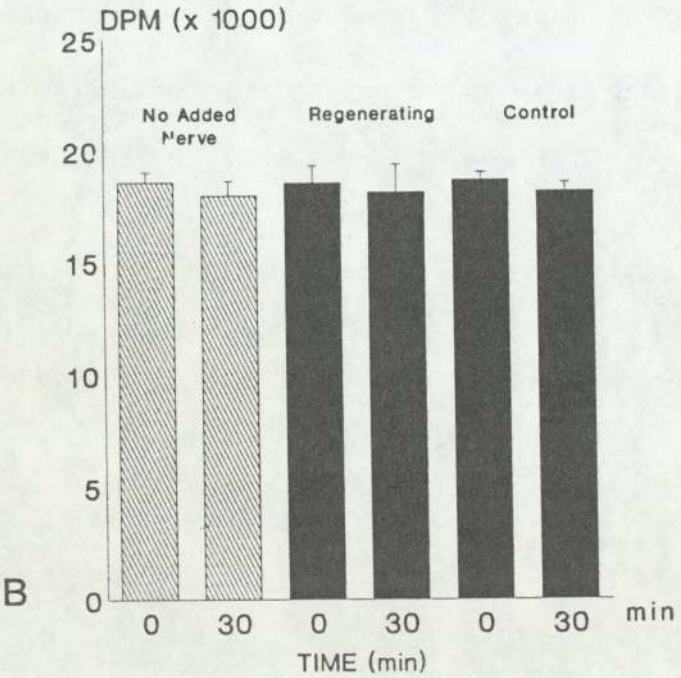

DPM (x 1000)
25
20
15
10
5
0
No Added Nerve
Regenerating
Control
B
0
30
0
30
0
30
min
TIME (min)

Figure 31

Tubulin:tyrosine carboxypeptidase activity in pieces of regenerating and control nerve by pulse-chase assay. Homogenates were incubated with ^{3}H-tyrosine, and after 20 min with the addition of 100X excess of unlabelled 'cold' tyrosine. TTCP activity was assayed at 0, 10 and 20 min following the addition of 'cold' tyrosine.

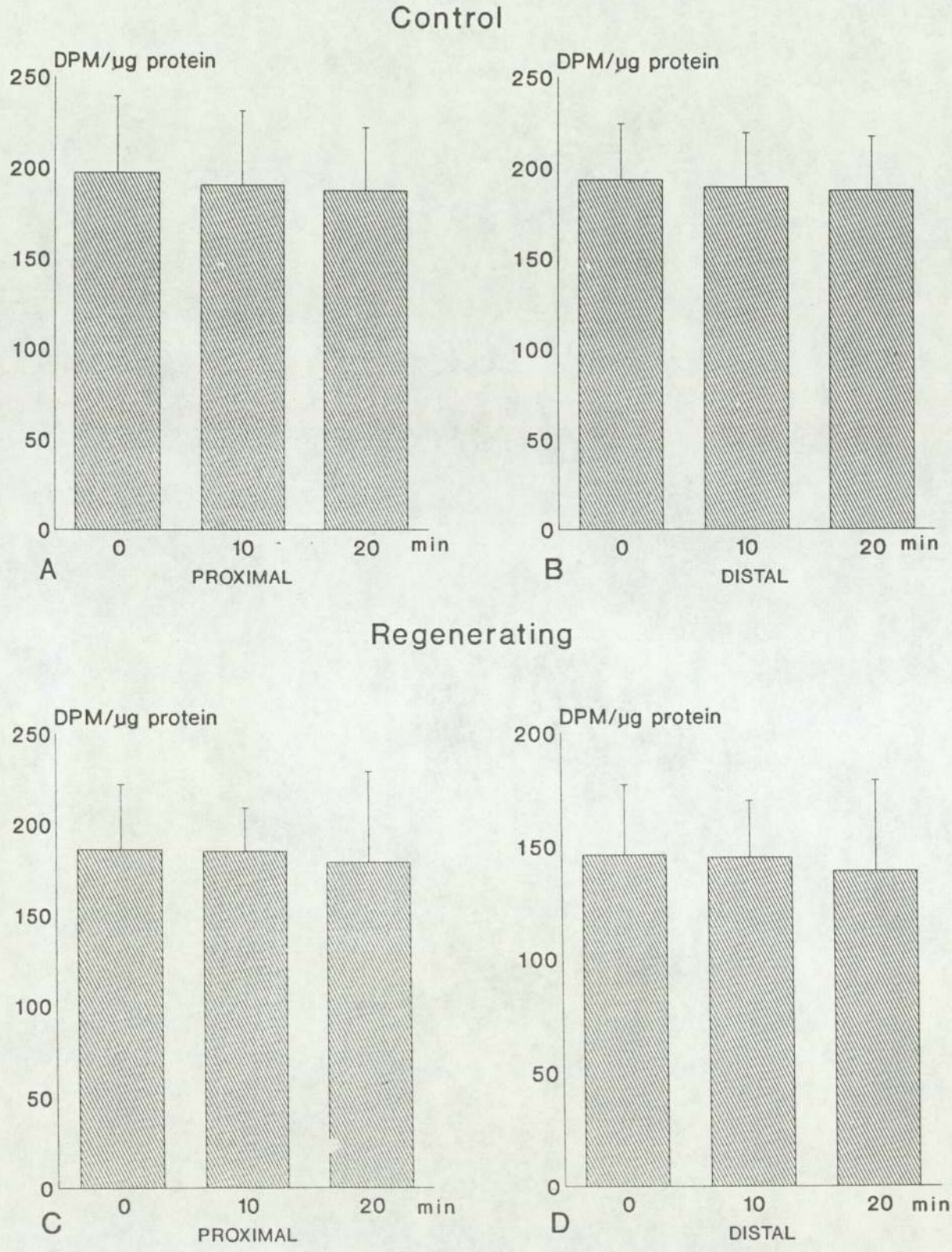
The data is presented as D.P.M/μg protein $\pm$ S.E.M; n = 6 in each case.

A. TTCP activity in pieces of control nerve equivalent to pieces of nerve proximal to a crush.

B. TTCP activity in pieces of control nerve equivalent to pieces of nerve distal to a crush.

C. TTCP activity in pieces of regenerating nerve proximal to a crush.

D. TTCP activity in pieces of regenerating nerve distal to a crush.


Control
DPM/µg protein
250
200
150
100
50
0
0
10
20
min
A
PROXIMAL
DPM/µg protein
250
200
150
100
50
0
0
10
20
min
B
DISTAL
Regenerating
DPM/µg protein
250
200
150
100
50
0
0
10
20
min
C
PROXIMAL
DPM/µg protein
200
150
100
50
0
0
10
20
min
D
DISTAL

1.2 Acetylation of α-tubulin

Post-translational acetylation of α-tubulin is another biochemical modification to tubulin. Unlike tyrosination, this modification has been associated with stable microtubule populations. There is also evidence that acetylation of α-tubulin is associated with axonal maturation in cell culture and development of the nervous system. The purpose of the following investigations was to determine whether a change in the post-translational acetylation of tubulin occurred as a consequence of the neuronal remodelling that takes place during nerve regeneration.

1.2.1 *1-dimensional SDS-PAGE and Western blotting*

Homogenates of L4,L5 dorsal root ganglia and sequential 5mm nerve pieces from the dorsal root ganglia to the crush (proximal) and sequential 5mm pieces past the crush (distal) were resolved by 1-dimensional SDS-PAGE on 7.5% polyacrylamide gels (see METHODS section 4.1 and 4.2). The resolved proteins were transferred to nitrocellulose (see METHODS section 4.4) and then subjected to immunochemical analysis using the mouse monoclonal antibody 6-11B-1, directed against acetylated α-tubulin. An example of an immunoblot obtained with this antibody is shown in figure 32. The epitope for 6-11B-1 was conserved in all L4,L5 dorsal root ganglia and nerve pieces that were examined. The pattern of immunoreactivity in pieces of nerve proximal and distal to a crush did not appear to differ between regenerating and control nerves, when examined by eye.

Figure 32

Fluorographs of acetylated α-tubulin in regenerating and control nerves.

Homogenates of 5mm pieces of regenerating and control nerves were transferred to nitrocellulose and the resulting Western blots were probed with mouse monoclonal antibody 6-11-B1 (specific for acetylated tubulin) and visualized with horseradish peroxidase and ECL.

A. Sequential 5mm pieces of regenerating nerve from the dorsal root ganglia to the nerve crush.

B. Sequential 5mm pieces distal to a nerve crush.

C. Sequential 5mm pieces of control nerve equivalent to pieces of regenerating nerve in A.

D. Sequential 5mm pieces of control nerve equivalent to pieces of regenerating nerve in B.

Figure is representative of data from four animals.

1.2 Acetylation of α-tubulin

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Figure 32

Fluorographs of acetylated α-tubulin in regenerating and control nerves.

Homogenates of 5mm pieces of regenerating and control nerves were transferred to nitrocellulose and the resulting Western blots were probed with mouse monoclonal antibody 6-11-B1 (specific for acetylated tubulin) and visualized with horseradish peroxidase and ECL.

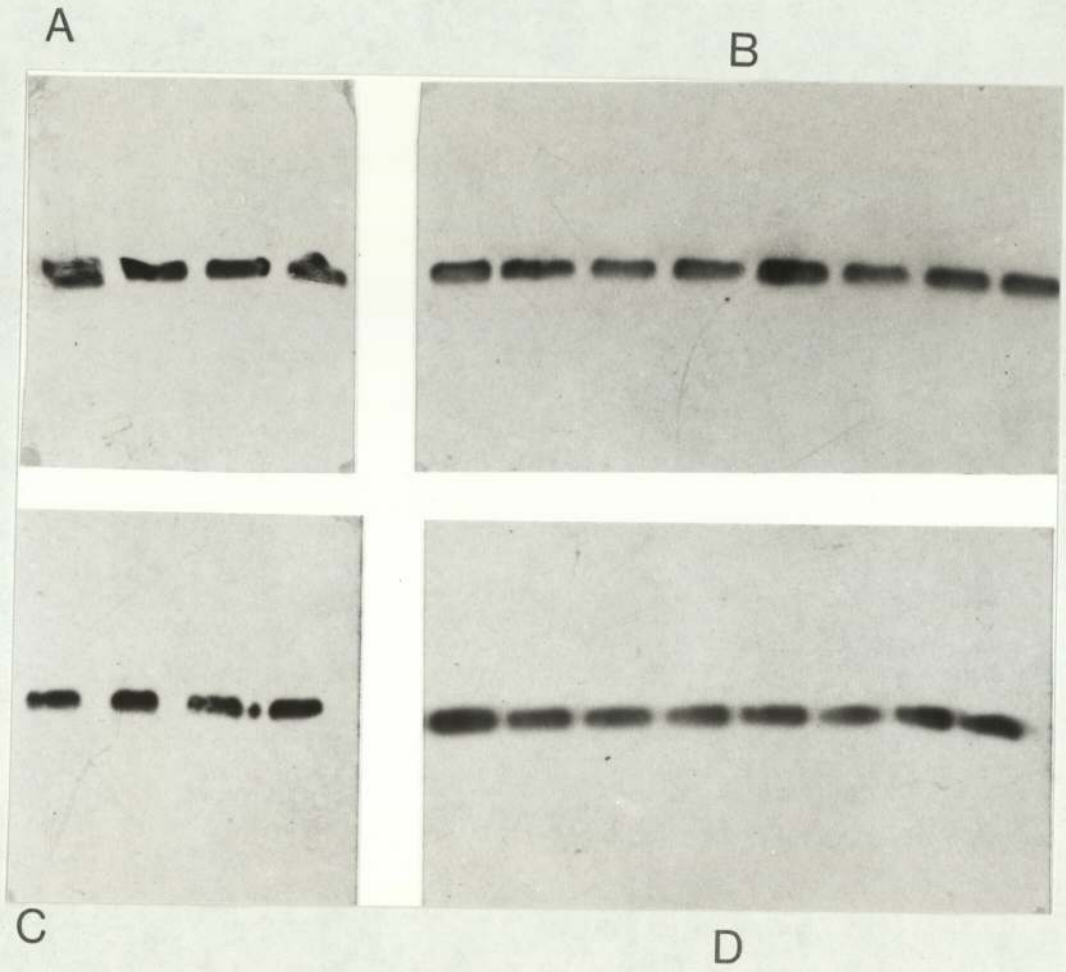
A. Sequential 5mm pieces of regenerating nerve from the dorsal root ganglia to the nerve crush.

B. Sequential 5mm pieces distal to a nerve crush.

C. Sequential 5mm pieces of control nerve equivalent to pieces of regenerating nerve in A.

D. Sequential 5mm pieces of control nerve equivalent to pieces of regenerating nerve in B.

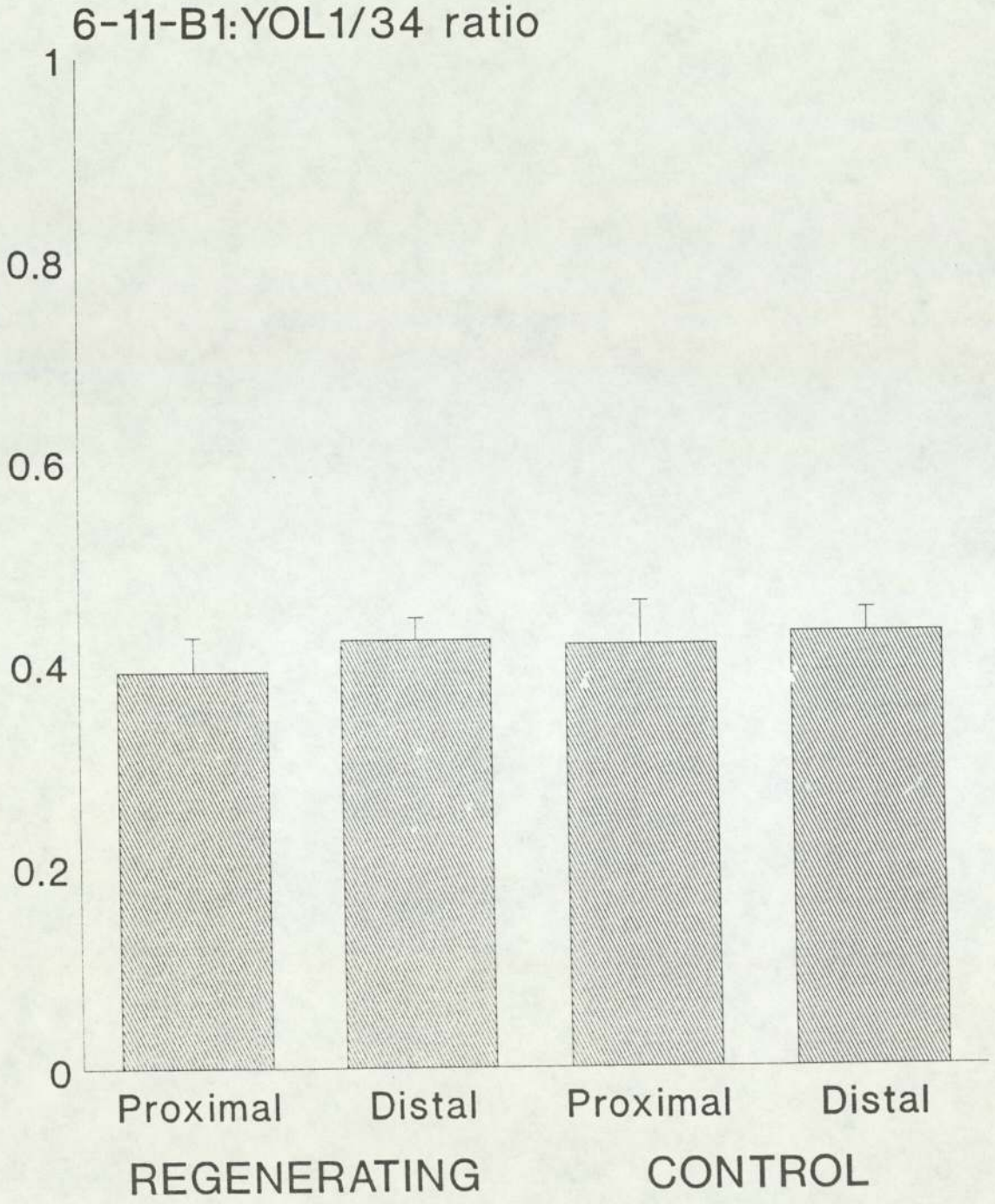
Figure is representative of data from four animals.


A
B
C
D

To determine quantitatively whether the proportion of total α-tubulin that was acetylated changed between pieces of nerve proximal and distal to a crush, or between dorsal root ganglia, the immunoreactivity observed for acetylated tubulin and total α-tubulin in the same nerve pieces was measured by laser densitometry. The ratio of acetylated α-tubulin relative to total α-tubulin was determined in each nerve piece. The mean ratio obtained from proximal nerve pieces and distal nerve pieces, from regenerating and control nerves, were compared with each other and are shown in figure 33. There was no apparent difference ($P > 0.05$) in the proportion of total α-tubulin that was acetylated between nerve pieces proximal or distal to a crush, nor were there any differences observed between regenerating and control nerves (figure 33). There was no difference ($P > 0.05$) in the acetylation of α-tubulin between dorsal root ganglia from regenerating and control nerve.

Figure 33

The ratio of acetylated tubulin to total α-tubulin in pieces of regenerating and control nerve. The immunobinding of 6-11-B1 (specific for acetylated α-tubulin) relative to that of YOL 1/34 (specific for total α-tubulin) was determined in homogenates of pieces of rat sciatic nerve proximal or distal to a nerve crush (regenerating) or from equivalent pieces of uncrushed control nerve. The immunobinding was quantified by laser densitometry and expressed as the mean ratio of immunobinding of 6-11-B1 relative to YOL 1/34 ($\pm$ S.D.). n=4 in each case.


6-11-B1:YOL1/34 ratio
1
0.8
0.6
0.4
0.2
0
Proximal
Distal
Proximal
Distal
REGENERATING
CONTROL

1.2.2 *Dot-immunobinding analysis*

To investigate the possibility that small changes in the acetylation state of tubulin may not be detected by Western blotting experiments, due to the variance associated with protein content of the nerve pieces, the more sensitive dot-immunobinding assay (see METHODS section 5.3.1.) was used. The proportion of α-tubulin that was acetylated in pieces of crushed (regenerating) and uncrushed (control) nerve is shown in figure 34. There was no increase in the proportion of α-tubulin that was acetylated on comparison with pieces of nerve proximal and distal to a crush or on comparison with pieces of regenerating nerve with equivalent pieces of control nerve ($P > 0.05$).

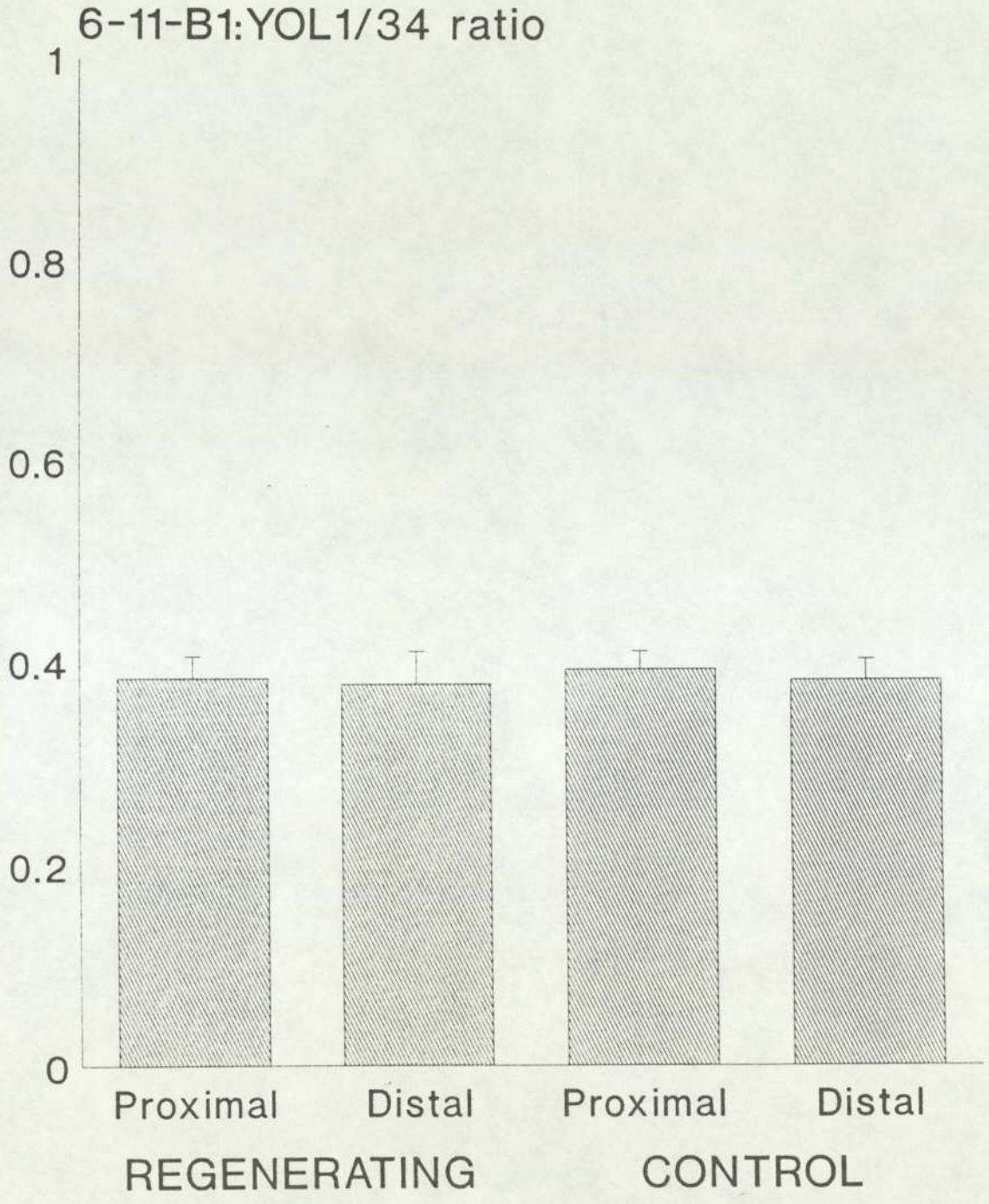
1.2.3 *Acetylase activity*

We wished to ascertain if tubulin acetylase activity could be detected in rat sciatic nerve, and thus determine if a change in its activity occurred as a consequence of neuronal injury.

Fresh subfractionated homogenates of brain and sciatic nerve were incubated with radiolabelled [^{14}C]-sodium acetate in the presence or absence of an exogenous tubulin substrate, over various time periods, and the incorporation of radiolabel measured by scintillation counting (see METHODS section 8). Using either brain or sciatic nerve as a source of acetylase activity, no incorporation of radiolabel into tubulin could be detected either in the presence or absence of phosphocellulose-purified tubulin. The possibility that tubulin acetylase preferred an assembled tubulin substrate was then investigated. Incubations were performed in the presence of taxol

Figure 34

Dot immunobinding of antibody 6-11-B1 (specific for acetylated α-tubulin) relative to that of YOL 1/34 (specific for total α-tubulin) from homogenates of pieces of rat sciatic nerve proximal or distal to a nerve crush (regenerating) or from equivalent pieces of uncrushed (control) nerve. The immunobinding was quantified by laser densitometry and expressed as the mean ratio of 6-11-B1 relative to YOL 1/34 ($\pm$ S.D.) n = 8 in each case.


6-11-B1:YOL1/34 ratio
1
0.8
0.6
0.4
0.2
0
Proximal
Distal
Proximal
Distal
REGENERATING
CONTROL

and GTP to promote the polymerization of purified tubulin. Similarly, in this set of experiments no incorporation of radiolabel could be detected.

The above experiments were repeated using radiolabelled [^{14}C]-acetyl coenzyme A as the source of radiolabel, to overcome the possibility that tubulin acetylase had a preference for the form in which radiolabelled acetate was presented. However, these experiments were unable to detect the presence of any acetylase activity directed against tubulin in either brain or sciatic nerve extracts. No results of these negative experiments are shown.

2 PHOSPHORYLATION OF MICROTUBULE PROTEINS

The experiments outlined in the following chapter were designed to measure the activity of endogenous protein kinases present in pieces of nerve proximal and distal to a crush, and in equivalent pieces of uncrushed control nerve. In particular, the kinase activities present in subfractionated homogenates of nerve pieces were measured under different assay conditions using distinct microtubule protein substrates.

The nerve samples were subfractionated as described in METHODS section 9.1 in order to separate crudely the cytosolic protein kinases from the membrane associated kinases. The microtubule proteins used as exogenous substrates in the phosphorylation assays were prepared as outlined in METHODS section 2; the purity of the samples are shown in figure 35. Purified tubulin is shown with purified MAP-2. The purified proteins did not possess any associated protein kinase activity.

2.1 Phosphorylation of tubulin

Tubulin can be post-translationally phosphorylated by a number of different protein kinases. Examples of some of these include casein kinase II and Ca^{2+}/calmodulin-dependent kinase and cAMP- dependent protein kinase (II). The actions of these protein kinases are thought to modify the biochemical properties of tubulin. Indeed, phosphorylation by either Ca^{2+}/calmodulin kinase or cAMP-dependent protein kinase results in a decrease in the ability of tubulin to polymerize, while phosphorylation of tubulin by casein kinase (II) is linked with increased assembly/polymerization of tubulin. The ability of post-translational phosphorylation

Figure 35

1-dimensional gel electrophoresis of purified tubulin and purified heat stable MAP-2 and Tau.

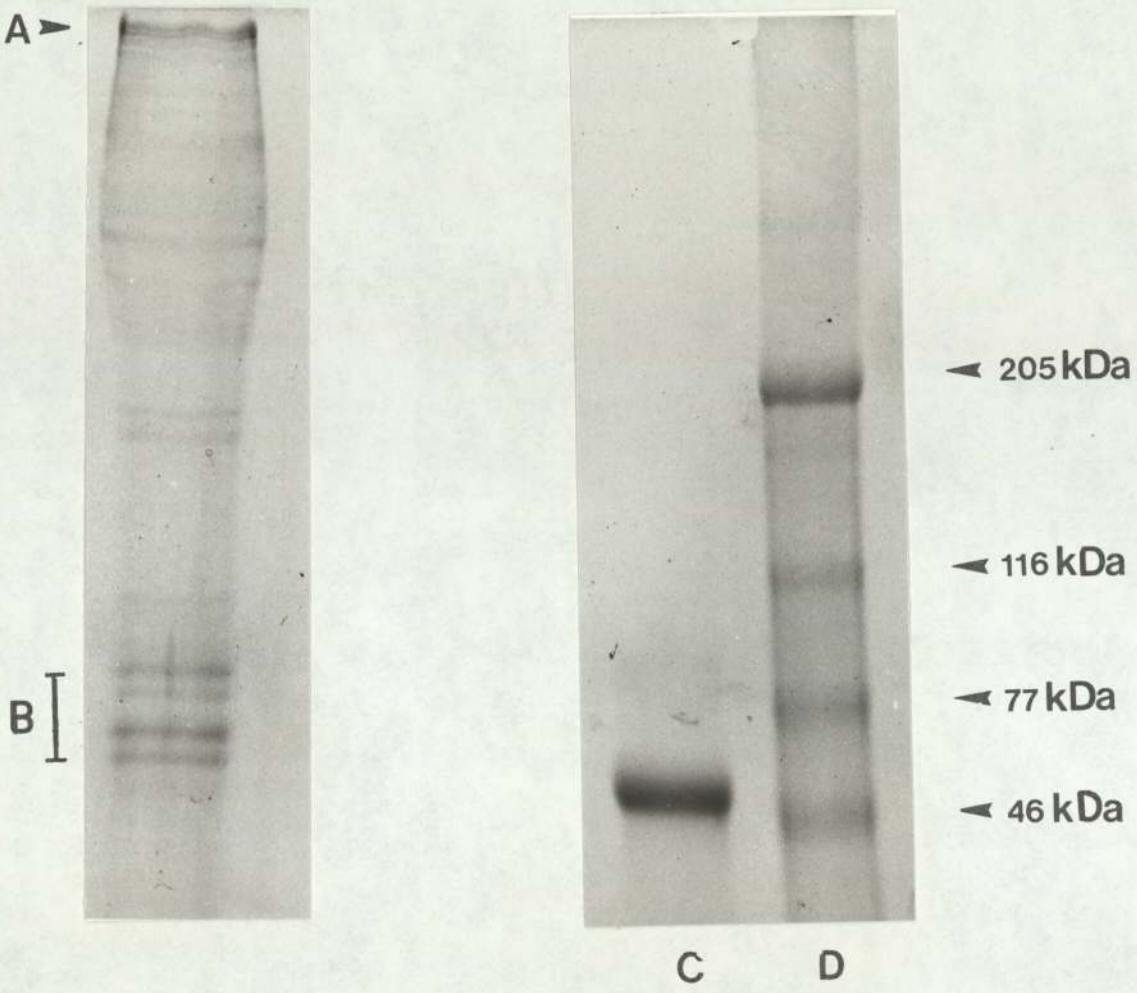
Extracts from purified tubulin and purified heat stable MAPs: MAP-2 and Tau were resolved by 1-dimensional SDS-PAGE on 7.5% gels and visualized by Coomasie Blue staining.

A. MAP-2

B. Tau proteins

C. Purified tubulin

D. Molecular weight markers


A
B
205 kDa
116 kDa
77 kDa
46 kDa
C
D

of tubulin to modify tubulin stability, could therefore be important in the structural reorganization of the cytoskeleton both prior to and during neuronal outgrowth.

The following experiments were designed to measure Mg^{2+}-dependent/Ca^{2+}-independent protein kinase and Ca^{2+}/calmodulin-dependent protein kinase activity present in pieces of rat sciatic nerve, proximal and distal to a crush from regenerating nerve, and their equivalent from uncrushed (control) nerve. The kinase activity that was directed against exogenous purified tubulin was of particular interest. Where no activity was seen, even where autoradiographs were left to expose for up to 21 days i.e. blank autoradiographs, the autoradiographs are not shown, and are indicated in the text as "data not shown".

2.1.1 *Mg^{2+}-dependent and Ca^{2+}-independent protein kinase activity*

In this assay, the conditions were chosen to examine Mg^{2+}-dependent and Ca^{2+}-independent protein kinase activities directed against tubulin.

Homogenates of pieces of rat sciatic nerve, from regenerating and control (uncrushed) nerve, and their respective L4 and L5 dorsal root ganglia, were prepared as outlined in METHODS section 9.1. The homogenates were sub-fractionated by centrifugation into supernatant and particulate fractions to separate cytosolic and membrane-associated protein kinase activities. The supernatant (cytosolic) fractions and particulate (membrane-associated) fractions were processed further, as outlined in METHODS section 9.1, prior to their use. Each fraction was assayed independently for their ability to phosphorylate exogenous purified tubulin under conditions which favour Mg^{2+}-dependent and Ca^{2+}-independent protein kinase

activity. The incorporation of radiolabelled [^{32}P]-phosphate into exogenous purified tubulin and endogenous proteins was examined by SDS-PAGE and autoradiography as outlined in METHODS section 9.3.

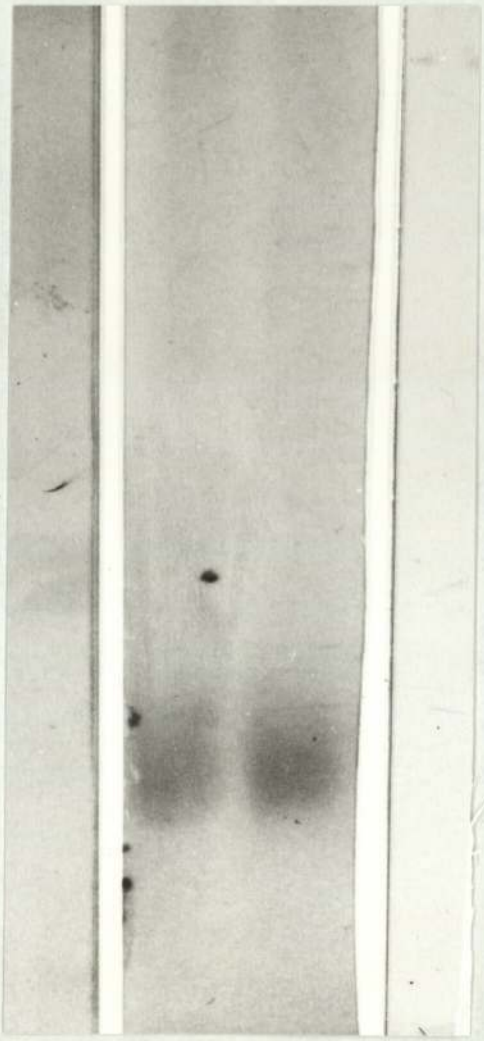
No protein kinase activity directed against exogenous tubulin could be detected in the supernatant subfractions obtained from dorsal root ganglia, or pieces of rat sciatic nerve either proximal or distal to a nerve crush, or their equivalents from uncrushed control nerve (data not shown). Furthermore, no protein kinase activity directed against exogenous purified tubulin could be detected in particulate subfractions obtained from either pieces of rat sciatic nerve or dorsal root ganglia obtained from regenerating and uncrushed control nerve (data not shown).

The possibility that the exogenous tubulin was a poor substrate because it was in a non-assembled form was investigated. In some experiments, the exogenous tubulin substrate was incubated with 10μM taxol and 1mM GTP, to assemble tubulin, prior to use in the protein kinase assay. Using these conditions, we hoped to increase the specificity of the tubulin substrate for casein kinase II, a cytosolic kinase which can phosphorylate tubulin but prefers assembled tubulin as a substrate. In this system, protein kinase activity directed against exogenous purified tubulin was detected in the supernatant subfraction from homogenates of dorsal root ganglia, obtained from regenerating and control nerve, as shown in figure 36. The phosphorylation pattern was analyzed by laser densitometry, as shown in figure 37. There was no significant difference ($p > 0.05$) between the protein kinase activities of dorsal root ganglia supernatant subfractions, between regenerating and control nerve.

Figure 36

Autoradiographs showing ^{32}P-radiolabelled tubulin that has been phosphorylated by Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity in supernatant subfractions of homogenates from dorsal root ganglia of regenerating and control. Incubations were performed in the presence (B, C) and absence (A, D) of purified tubulin with subfractionated homogenates of regenerating (A, B) dorsal root ganglia and control (C,D) dorsal root ganglia together with ^{32}P-ATP, GTP and Taxol, and separated by 1-dimensional SDS-PAGE and visualized by autoradiography.

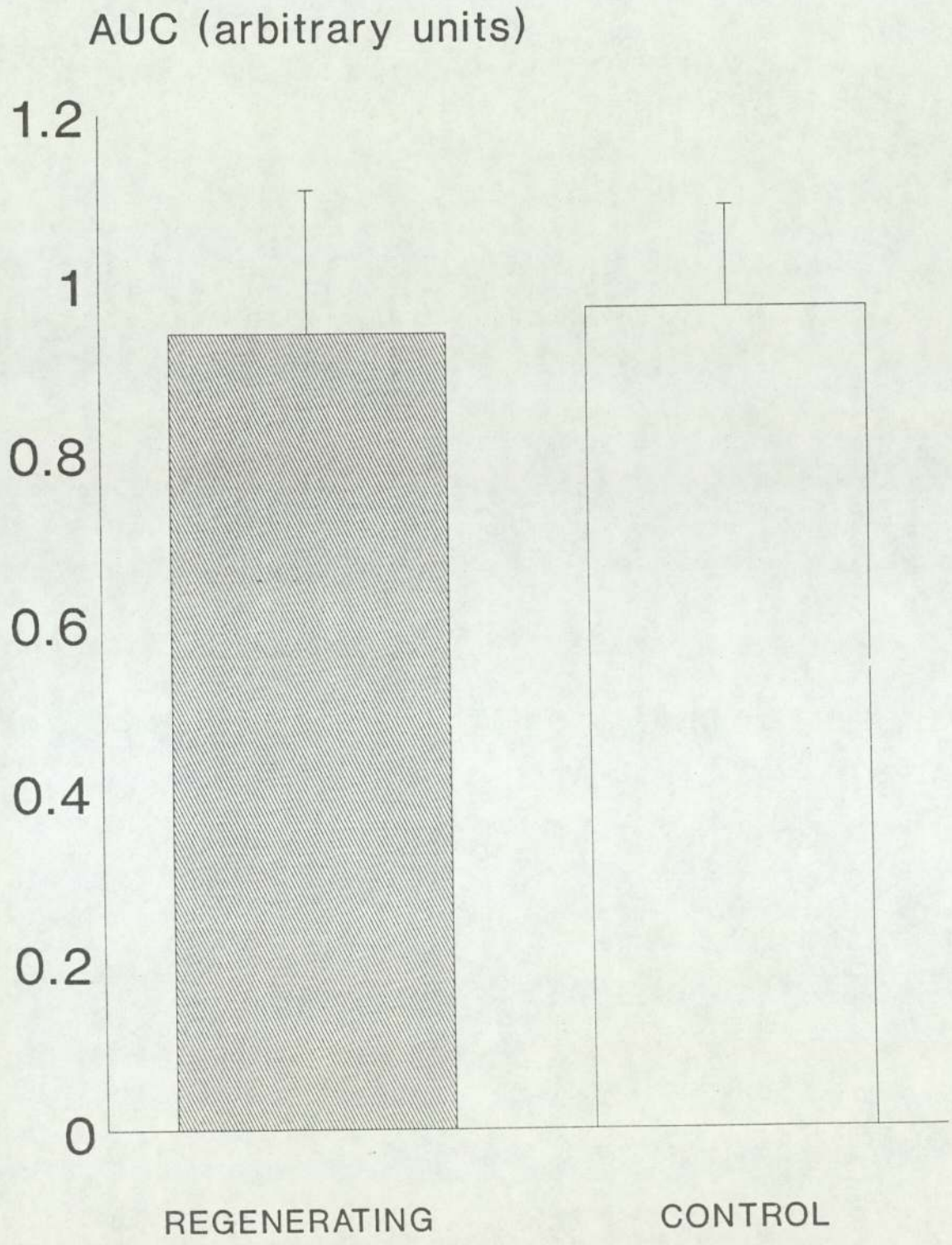
Autoradiographs are typical of four separate experiments.



A B C D

Figure 37

Densitometric analysis of autoradiographs showing ^{32}P-radiolabelled tubulin that has been phosphorylated by Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity in homogenates of regenerating and control dorsal root ganglia. The phosphorylation was quantified by laser densitometry data expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 4 in each case.


AUC (arbitrary units)
1.2
1
0.8
0.6
0.4
0.2
0
REGENERATING
CONTROL

However, I was unable to detect any protein kinase activity directed against the polymerized exogenous tubulin (data not shown) in supernatant subfractions of samples from pieces of rat sciatic nerve, obtained from either regenerating or control nerve. Nor was I able to detect protein kinase activity in particulate subfractions of samples obtained from dorsal root ganglia or pieces of sciatic nerve proximal and distal to a crush, either from regenerating or control nerve.

2.1.2 *cAMP-dependent protein kinase activity*

In this assay, the conditions were chosen to examine cAMP-dependent protein kinase activities directed against tubulin.

Homogenates of pieces of rat sciatic nerve, from regenerating and control (uncrushed) nerve, and their respective L4 and L5 dorsal root ganglia, were prepared as outlined in METHODS section 9.1 in the presence of EGTA. The homogenates were subfractionated by centrifugation into supernatant and particulate fractions to separate cytosolic and membrane-associated protein kinase activities. The supernatant (cytosolic) fractions and particulate (membrane-associated) fractions were processed further, as outlined in METHODS section 9.1, prior to their use. Each fraction was assayed independently for their ability to phosphorylate exogenous purified tubulin under conditions which favour cAMP-dependent protein kinase activity. The incorporation of radiolabelled [^{32}P]-phosphate into exogenous purified tubulin and endogenous proteins was examined by SDS-PAGE and autoradiography as outlined in METHODS section 9.3.

No protein kinase activity directed against exogenous tubulin could be detected in the supernatant subfractions obtained from dorsal root ganglia, or pieces of rat sciatic nerve either proximal or distal to a nerve crush, or their equivalents from uncrushed control nerve (data not shown). Furthermore, no protein kinase activity directed against exogenous purified tubulin could be detected in particulate subfractions obtained from either pieces of rat sciatic nerve or dorsal root ganglia obtained from regenerating or uncrushed control nerve (data not shown).

2.1.3. *Ca^{2+}/calmodulin-dependent protein kinase activity*

In this assay, the conditions were chosen to examine the Ca^{2+}/calmodulin-dependent kinase activities present in samples of rat sciatic nerve which were directed against tubulin.

As described in 2.1.1 and METHODS section 9.1, samples of rat sciatic nerve and dorsal root ganglia, prepared in the absence of EGTA, were assayed independently for their ability to phosphorylate exogenous purified tubulin, under conditions which favour Ca^{2+}/calmodulin- dependent protein kinase activity. The incorporation of radiolabelled [^{32}P]-phosphate into exogenous purified tubulin and endogenous protein was examined by SDS-PAGE and autoradiography as outlined in METHODS section 9.3.

No protein kinase activity directed against exogenous tubulin could be detected in supernatant subfractions from pieces of rat sciatic nerve or dorsal ganglia, obtained from regenerating or uncrushed control nerve (data not shown). Using particulate subfractions in the assay, protein kinase activity was detected, in

all pieces of rat sciatic nerve and dorsal root ganglia that were examined, as shown in figure 38. However, there was no apparent increase in the phosphorylation profile in the presence of exogenous purified tubulin over that in its absence; indeed, the activity detected appeared diffuse and was directed against high molecular weight proteins. The diffuse phosphorylation pattern made analysis by laser densitometry difficult. However the major phosphorylated proteins (indicated by an asterisk on figure 38) did not reveal any differences by eye, between the intensity of the phosphorylation in regenerating and control nerve.

2.2 Phosphorylation of MAPs

Microtubule stability has been shown to be regulated by the binding of microtubule associated proteins (MAPs). In particular, MAP-1, MAP-2 and tau promote microtubule assembly. Post-translational phosphorylation of MAPs reduces their ability to promote assembly, resulting in a net depolymerization of microtubules, thus affecting microtubule stability.

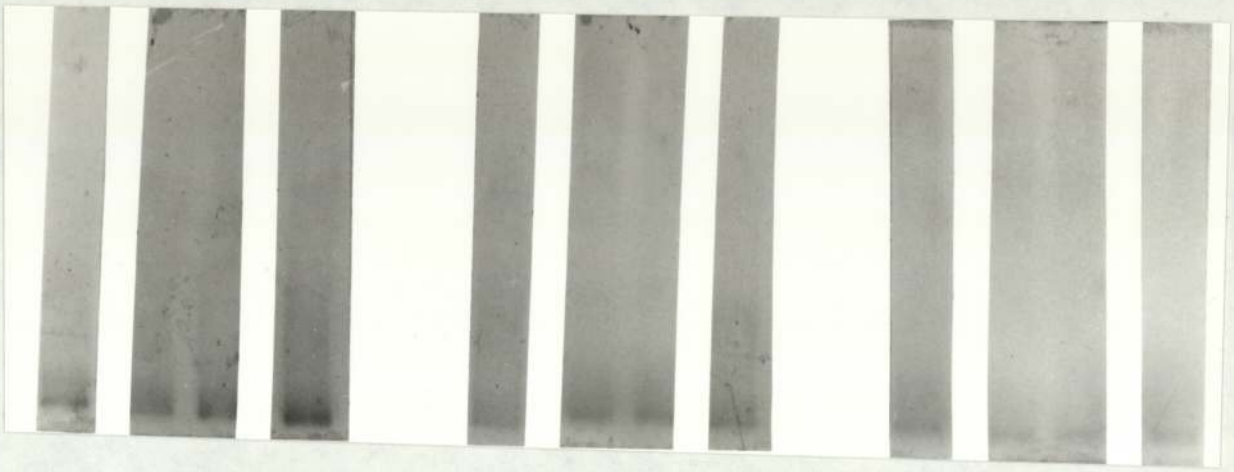
MAP-2 and tau are good substrates for both the Ca^{2+}/calmodulin-dependent protein kinase and cAMP-dependent protein kinase. Cell culture studies have shown that changes in the phosphorylation of MAPs are associated with the growth of neurites. The principal aim of the following experiments was to measure protein kinase activities directed against the heat stable MAPs: MAP-2 and tau, that are present in supernatant and particulate subfractions from regenerating and uncrushed control nerve.

Figure 38

Autoradiographs displaying Ca^{2+}/calmodulin-dependent protein kinase activity in supernatant subfractions of homogenates from dorsal root ganglia, pieces of nerve proximal to a crush and distal to a crush from regenerating and control nerve. Incubations were performed in the presence (+) and absence (-) of purified porcine tubulin with subfractionated homogenates of dorsal root ganglia, pieces of nerve proximal and distal to a crush from regenerating nerve and their equivalents from control nerve, together with ^{32}P-ATP, and separated by 1-Dimensional SDS-Page and visualized by autoradiography.

No.	Nerve piece	Reg/Con	+/- Exog. tubulin
1	Dorsal root ganglia	R	-
2	" " "	R	+
3	" " "	C	+
4	" " "	C	-
5	Proximal to a crush	R	-
6	" " "	R	+
7	" " "	C	+
8	" " "	C	-
9	Distal to a crush	R	-
10	" " "	R	+
11	" " "	C	+
12	" " "	C	-

* indicates activity



1 2 3 4 5 6 7 8 9 10
1 2 3 4 5 6 7 8 9 10 11 12

2.2.1 *Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity*

In this assay, the conditions were chosen to examine Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activities directed against exogenous heat-stable MAPs: MAP-2 and tau.

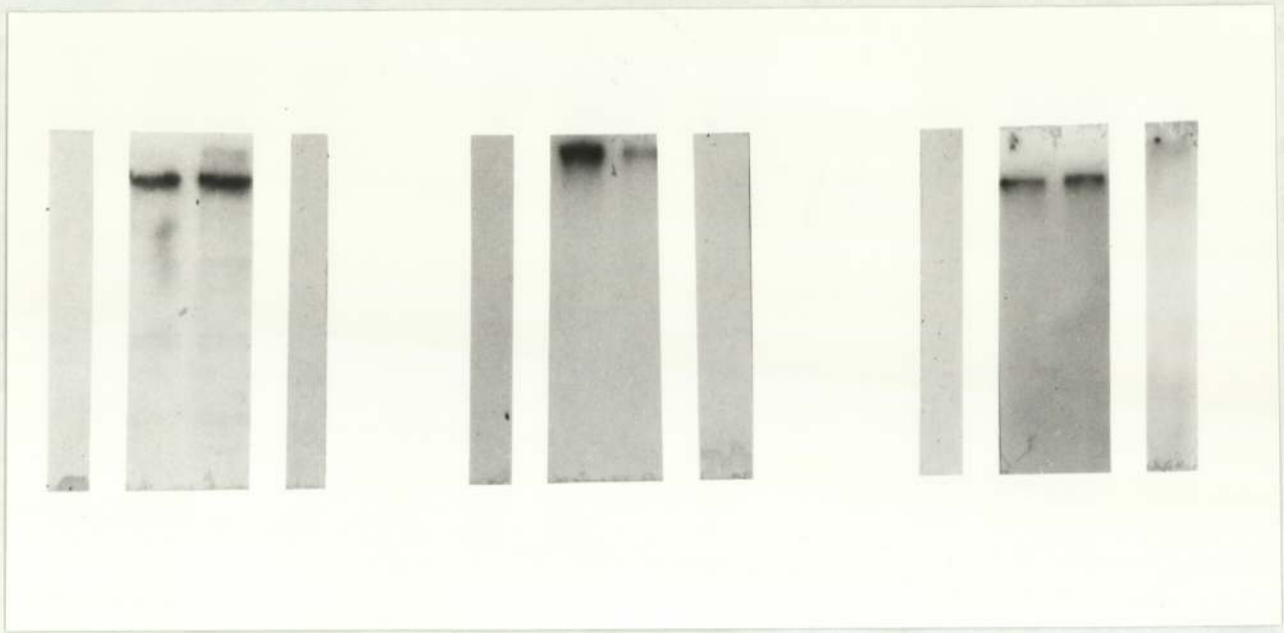
Homogenates of pieces of rat sciatic nerve from regenerating and control (uncrushed) nerve, and their respective L4 and L5 dorsal root ganglia were prepared as outlined in METHODS section 9.1. The homogenates were sub-fractionated by centrifugation into supernatant and particulate fractions to separate cytosolic (supernatant) and membrane-associated (particulate) protein kinase activities, as outlined in METHODS section 9.1. The incorporation of radiolabelled [^{32}P]-phosphate into exogenous purified MAPs and endogenous proteins was examined by SDS-PAGE and autoradiography as outlined in METHODS section 9.3.

Protein kinase activity was detected in the supernatant sub-fraction from homogenates of dorsal root ganglia and pieces of rat sciatic nerve proximal and distal to a crush from regenerating nerve, and also their equivalents obtained from uncrushed control nerve, as shown in figure 39. The incorporation of [^{32}P]-radiolabelled phosphate into a high molecular weight protein occurred only in the presence of exogenous heat-stable MAPs. This was confirmed to be MAP-2 using the techniques outlined in METHODS section 9.4. As shown in figure 40, the MAP-2 immunoreactive pattern matched the phosphorylation pattern obtained from the protein kinase assay. There appeared to be no activity directed towards tau.

Figure 39

Autoradiographs displaying phosphorylation of MAP-2 by Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity in supernatant subfractions of homogenates from dorsal root ganglia and pieces of nerve proximal and distal to a crush from regenerating and control nerve. Incubations were performed in the presence (+) and absence (-) of heat-stable MAPs: MAP-2 and tau with subfractionated homogenates of dorsal root ganglia, pieces of nerve proximal and distal to a crush from regenerating nerve and their equivalents from control nerve, together with ^{32}P-ATP, and separated by 1-dimensional SDS-PAGE and visualized by autoradiography.

No.	Nerve piece	Reg/Con	+/- Exog. MAPs
1	Dorsal root ganglia	R	-
2	" " "	R	+
3	" " "	C	+
4	" " "	C	-
5	Proximal to a crush	R	-
6	" " "	R	+
7	" " "	C	+
8	" " "	C	-
9	Distal to a crush	R	-
10	" " "	R	+
11	" " "	C	+
12	" " "	C	-



1 2 3 4 5 6 7 8 9 10 11 12

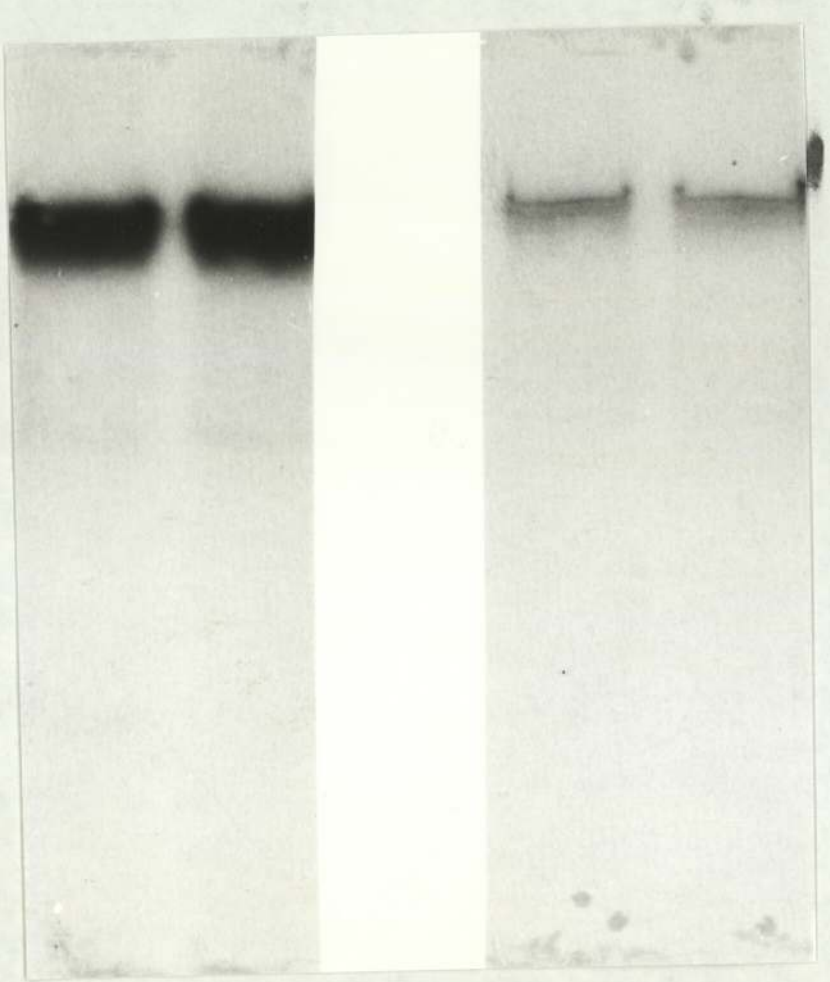
Figure 40

Fluorograph of MAP-2 immunoreactivity compared with the autoradiograph of MAP-2 that has been phosphorylated by Mg^{2+}-dependent and Ca^{2+}-independent protein kinase activity.

Radiolabelled proteins were resolved by 1-dimensional-SDS PAGE and transferred to nitrocellulose. The resulting Western blots were exposed to film for autoradiography. Following autoradiography, the blots were stored until remaining radioactivity had subsided, and then probed with HM-2 antibody against MAP-2, visualized by HRP-conjugated rabbit antimouse IgG with ECL and fluorography.

A. Autoradiograph of MAP-2 phosphorylation (duplicate samples).

B. Fluorograph of MAP-2 immunoreactivity (duplicate samples).


A
B

The protein kinase activity directed towards MAP-2 from supernatant subfractions of pieces of rat sciatic nerve proximal to a nerve crush, appeared to be greater than that in equivalent pieces of uncrushed nerve, whereas the same protein kinase activities in pieces of rat sciatic nerve distal to a nerve crush, and dorsal root ganglia from regenerating nerve, appeared to be similar to those observed in equivalent regions of uncrushed control nerve. These phosphorylation patterns were then analyzed by laser densitometry in order to quantify any differences observed. There was no significant difference ($P > 0.05$) between the protein kinase activities directed towards MAP-2 in dorsal root ganglia of regenerating and uncrushed control nerve, as shown in figure 41. Analysis of the phosphorylation pattern produced by protein kinase activities in supernatant subfractions of pieces of rat sciatic nerve, proximal to a nerve crush, revealed an increased protein kinase activity directed against MAP-2 in regenerating nerve compared to uncrushed control nerve ($P < 0.05$), as shown in figure 42. There was no significant difference ($P > 0.05$) between the protein kinase activities observed in supernatant subfractions of pieces of rat sciatic nerve distal to a crush and their equivalent from uncrushed control nerve, as shown in figure 43.

Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity, directed against the heat-stable MAPs MAP 2 and tau, were measured in particulate subfractions of regenerating and control rat sciatic nerve and dorsal root ganglia. No protein kinase activity could be detected in particulate subfractions of regenerating or control nerve under the assay conditions employed (data not shown).

Figure 41

Densitometric analysis of autoradiographs of MAP-2 phosphorylated by Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity in homogenates of regenerating and control dorsal root ganglia. Phosphorylation was quantified by laser densitometry data expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 8 in each case.

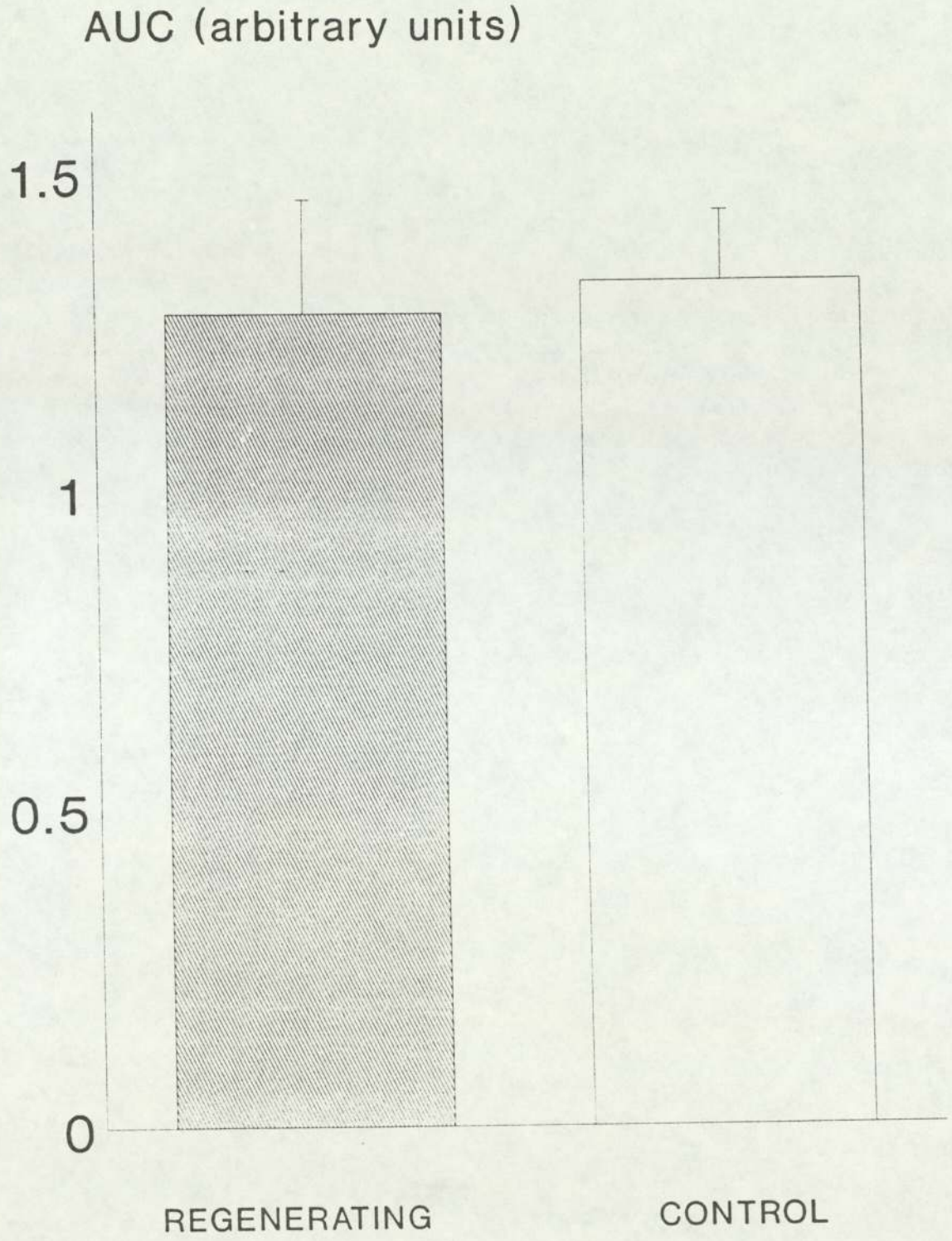

AUC (arbitrary units)
1.5
1
0.5
0
REGENERATING
CONTROL

Figure 42

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated by Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity in homogenates of nerve proximal to a nerve crush (regenerating) and their equivalents from uncrushed (control). Phosphorylation was quantified by laser densitometry data expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 8 in each case. * $P < 0.05$.

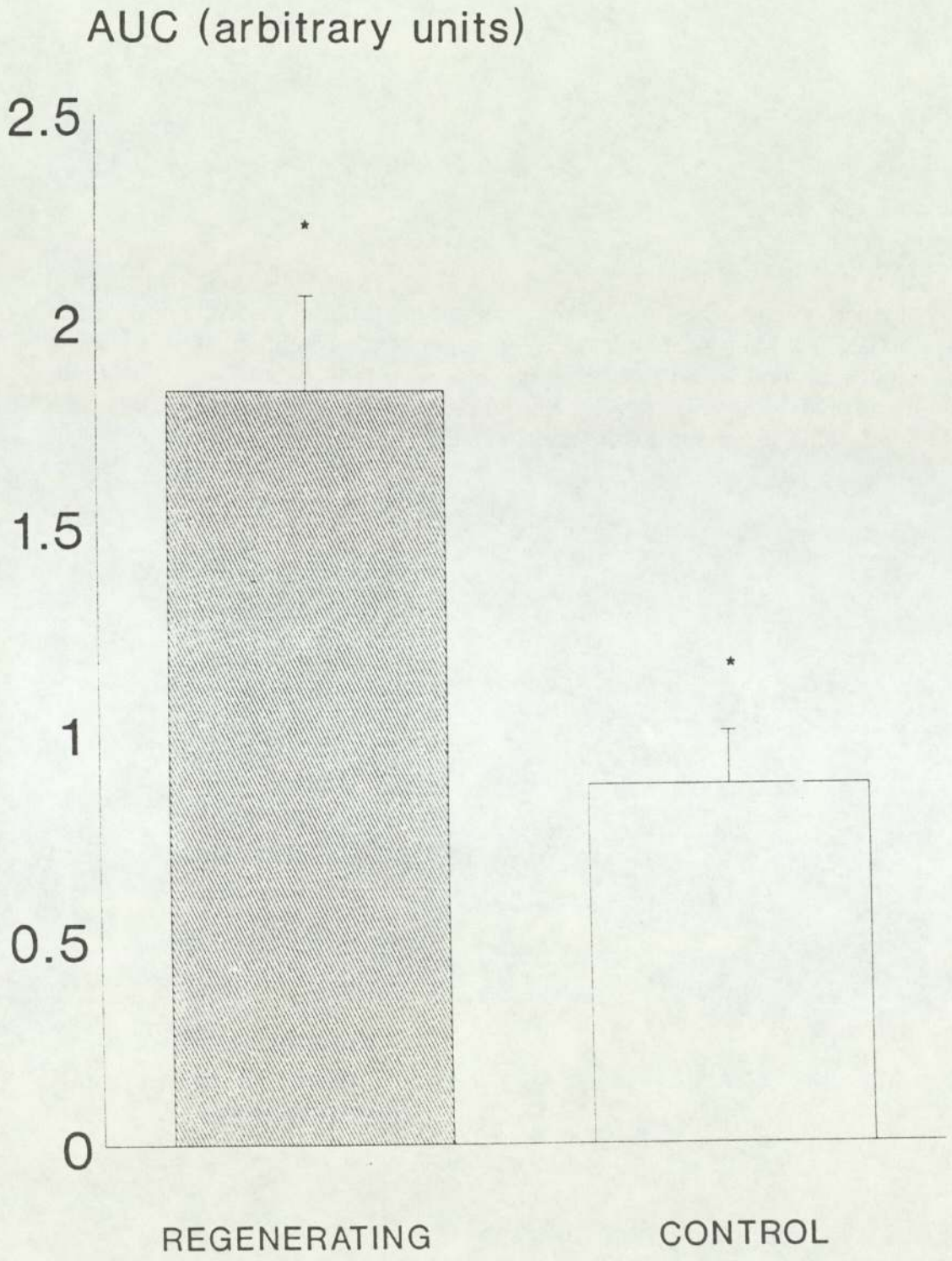
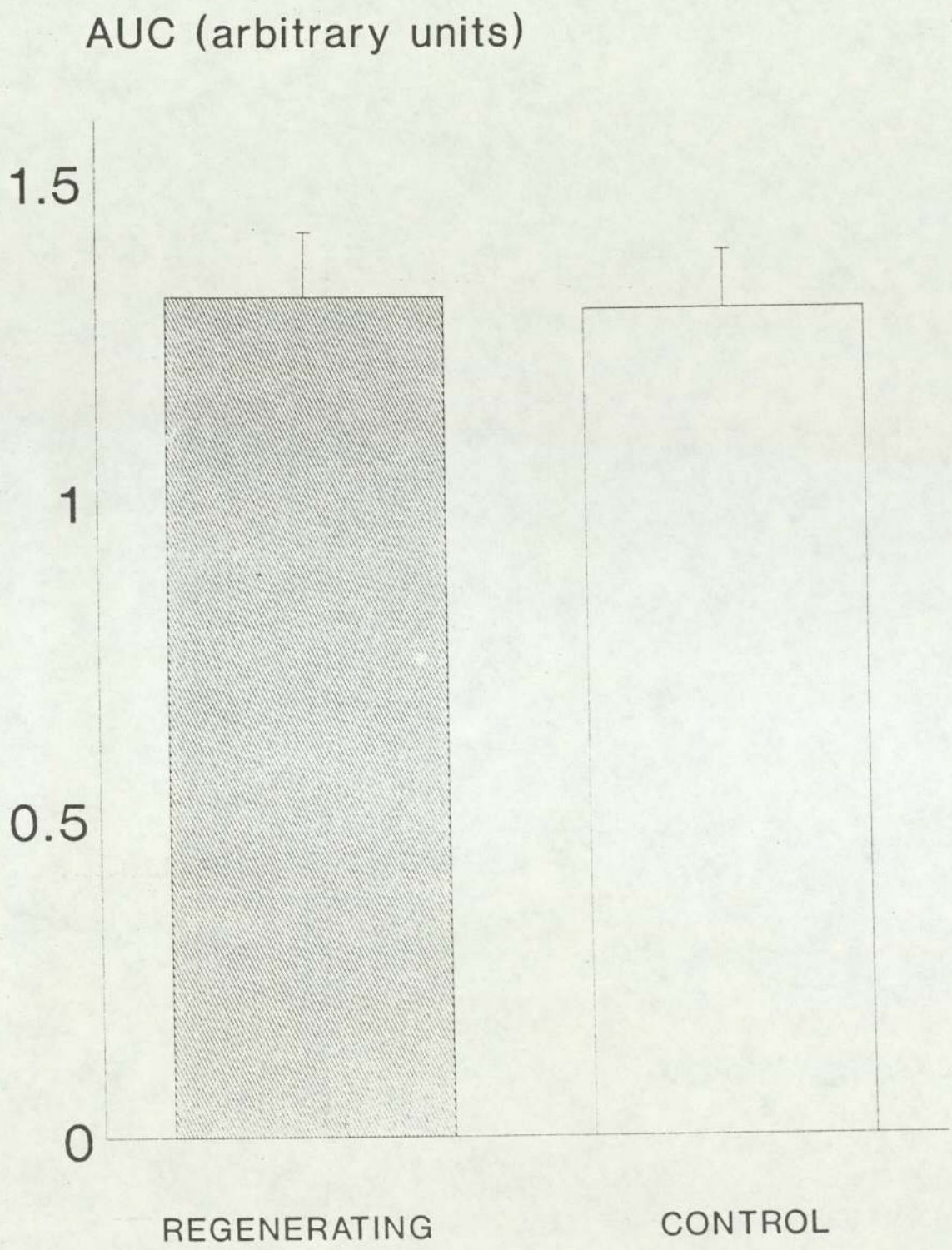

AUC (arbitrary units)
2.5
2
1.5
1
0.5
0
*
*
REGENERATING
CONTROL

Figure 43

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated by Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity in homogenates of nerve distal to a nerve crush (regenerating) and their equivalents from uncrushed (control) nerve. Phosphorylation was quantified by laser densitometry data expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 8 in each case.


AUC (arbitrary units)
1.5
1
0.5
0
REGENERATING
CONTROL

2.2.2 *cAMP-dependent protein kinase activity*

In this assay the conditions were chosen to examine and favour cAMP-dependent protein kinase activity directed against the heat-stable MAPs MAP-2 and tau.

Homogenates of pieces of rat sciatic nerve, from regenerating and control nerve, and their respective L4 and L5 dorsal root ganglia, were prepared as outlined in METHODS section 9.1. The homogenates were subfractionated by centrifugation into supernatant and particulate fractions, to separate cytosolic (supernatant) and membrane-associated (particulate) protein kinase activities, as outlined in METHODS section 9.1. The incorporation of radiolabelled [^{32}P]-phosphate into exogenous heat-stable MAPs and endogenous proteins was examined by SDS-PAGE and autoradiography as outlined in METHODS section 9.3.

Protein kinase activity was detected in all supernatant subfractions of dorsal root ganglia and pieces of rat sciatic nerve, proximal and distal to a nerve crush, from regenerating and uncrushed control nerve, as shown in figure 44. The protein kinase activity was directed against MAP-2 as indicated in figure 45. There appeared to be no activity towards tau. Analysis of the phosphorylation pattern by laser densitometry revealed that the protein kinase activity in the dorsal root ganglia was not significantly different between regenerating and control nerve, as shown in figure 46. Proximal to a crush, there was significantly greater ($P < 0.05$) protein kinase

Figure 44

Autoradiographs displaying phosphorylation of MAP-2 by cAMP-dependent protein kinase activity in supernatant subfractions of homogenates from dorsal root ganglia and pieces of nerve proximal and distal to a crush from regenerating and control nerve. Incubations were performed in the presence (+) and absence (-) of heat stable MAPs: MAP-2 and tau with subfractionated homogenates of dorsal root ganglia, pieces of nerve proximal and distal to a crush from regenerating nerve and their equivalents from control nerve, together with ^{32}P-ATP, and separated by 1-dimensional SDS-PAGE and visualized by autoradiography.

No.	Nerve piece	Reg/Con	+/- Exog. MAPs
1	Dorsal root ganglia	R	-
2	" " "	R	+
3	" " "	C	+
4	" " "	C	-
5	Proximal to a crush	R	-
6	" " "	R	+
7	" " "	C	+
8	" " "	C	-
9	Distal to a crush	R	-
10	" " "	R	+
11	" " "	C	+
12	" " "	C	-

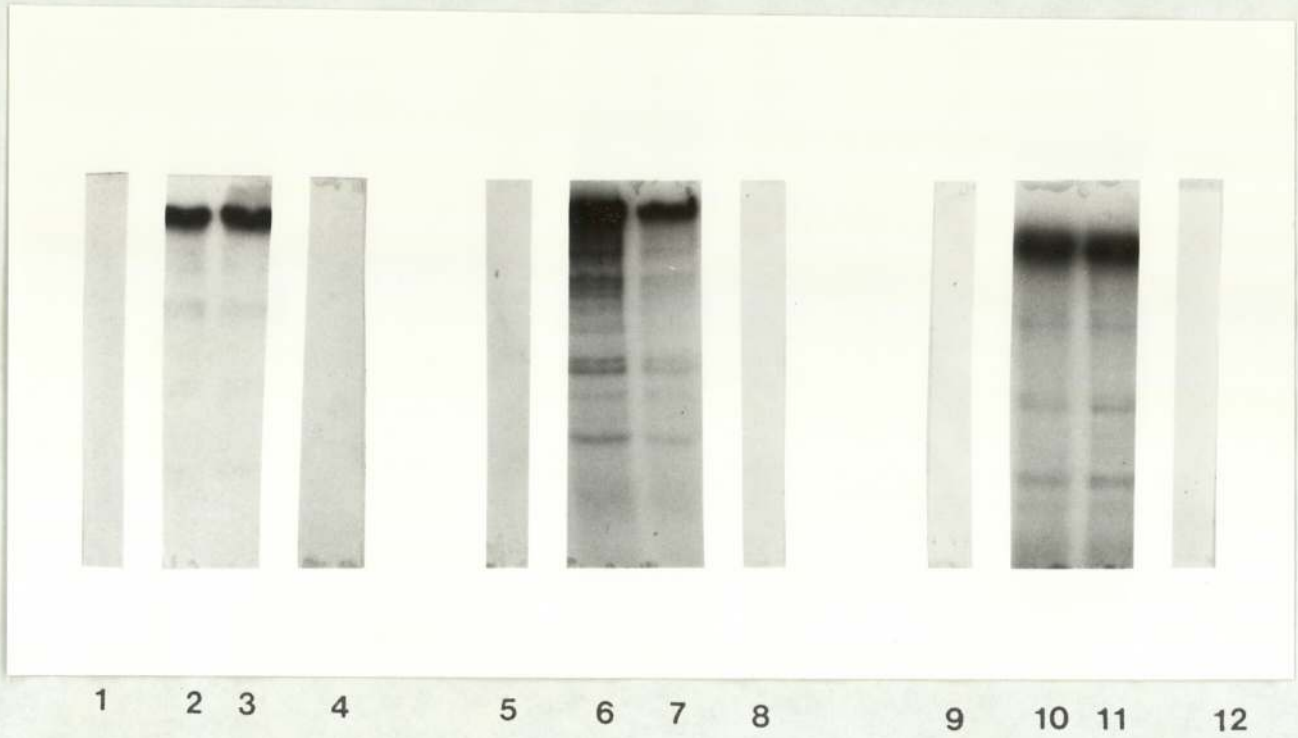

1
2
3
4
5
6
7
8
9
10
11
12

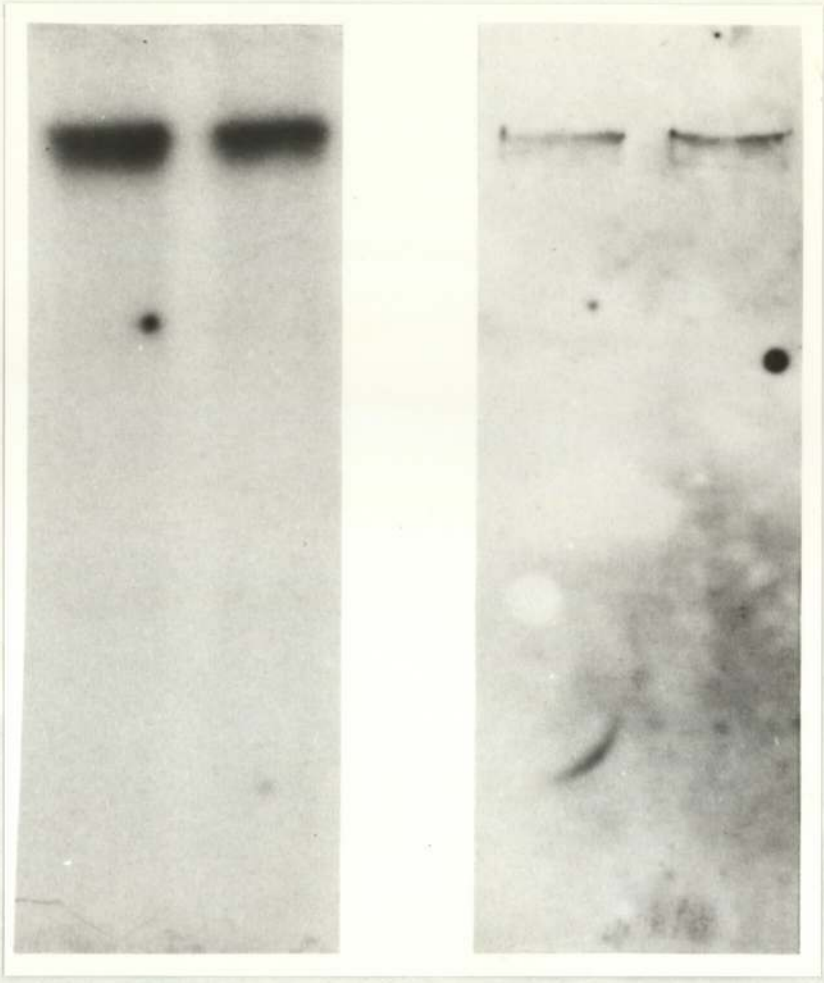
Figure 45

Fluorograph of MAP-2 immunoreactivity compared with the autoradiograph of MAP-2 that has been phosphorylated by cAMP-dependent protein kinase activity.

Radiolabelled proteins were resolved by 1-dimensional-SDS PAGE and transferred to nitrocellulose. The resulting Western blots were exposed to film for autoradiography. Following autoradiography, the blots were stored until remaining radioactivity had subsided, and then probed with HM-2 antibody against MAP-2, visualized by HRP-conjugated rabbit antimouse IgG with ECL and fluorography.

A. Autoradiograph of MAP-2 phosphorylation (duplicate samples).

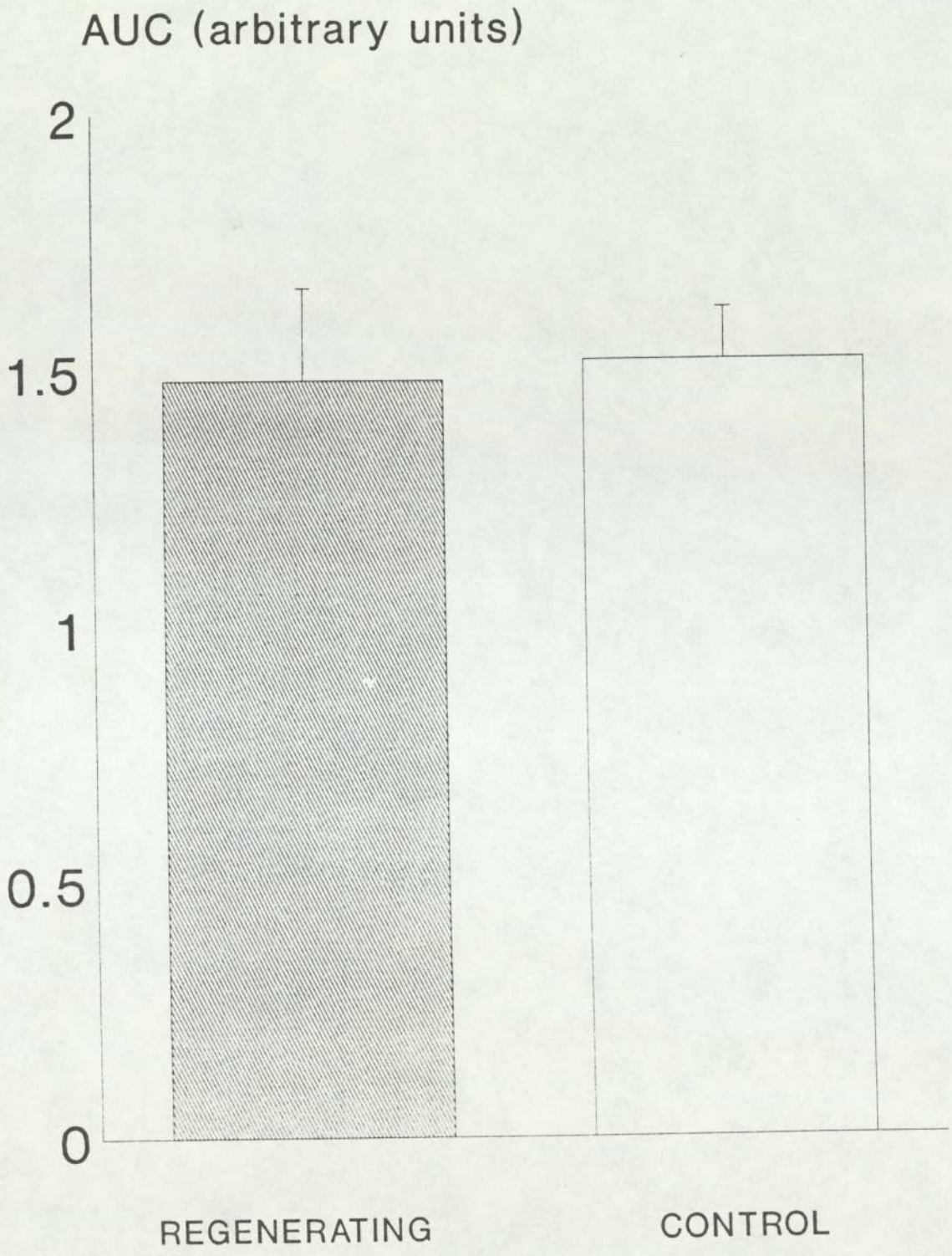
B. Fluorograph of MAP-2 immunoreactivity (duplicate samples).



A B

Figure 46

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated cAMP-dependent protein kinase activity in homogenates of regenerating and control dorsal root ganglia. Phosphorylation was quantified by laser densitometry. Data are expressed as the absorbance (arbitrary units) $\pm$ S.D. $n = 8$ in each case.


AUC (arbitrary units)
2
1.5
1
0.5
0
REGENERATING
CONTROL

activity directed towards MAP-2 in regenerating than that in control nerve (figure 47). In pieces of nerve distal to a crush, there was no significant difference between the protein kinase activity directed against MAP-2 by regenerating or control nerve extracts (figure 48).

The assay conditions used to assay Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity were similar to those used to assay cAMP-dependent protein kinase activity, with the presence of cAMP being the only different factor. Therefore we investigated the possibility that the protein kinase activity observed in section 2.2.1 was the same as that observed here. In addition, we wished to determine whether the presence of cAMP had any effect on the protein kinase activity observed. To examine this, we performed protein kinase assays in the presence and absence of cAMP on supernatant subfractions of pieces of nerve proximal to a nerve crush and their equivalents from uncrushed nerve. As can be seen in figure 49, the presence of cAMP increased the protein kinase activity of both regenerating and control nerve. Analysis by laser densitometry confirmed the difference between the protein kinase activities of regenerating and control nerve in the presence and the absence of cAMP, as shown in figure 50.

Figure 47

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated by cAMP-dependent protein kinase activity in homogenates of nerve proximal to a nerve crush (regenerating) and their equivalents from uncrushed (control). Phosphorylation was quantified by laser densitometry. Data expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 8 in each case. * $P < 0.05$.

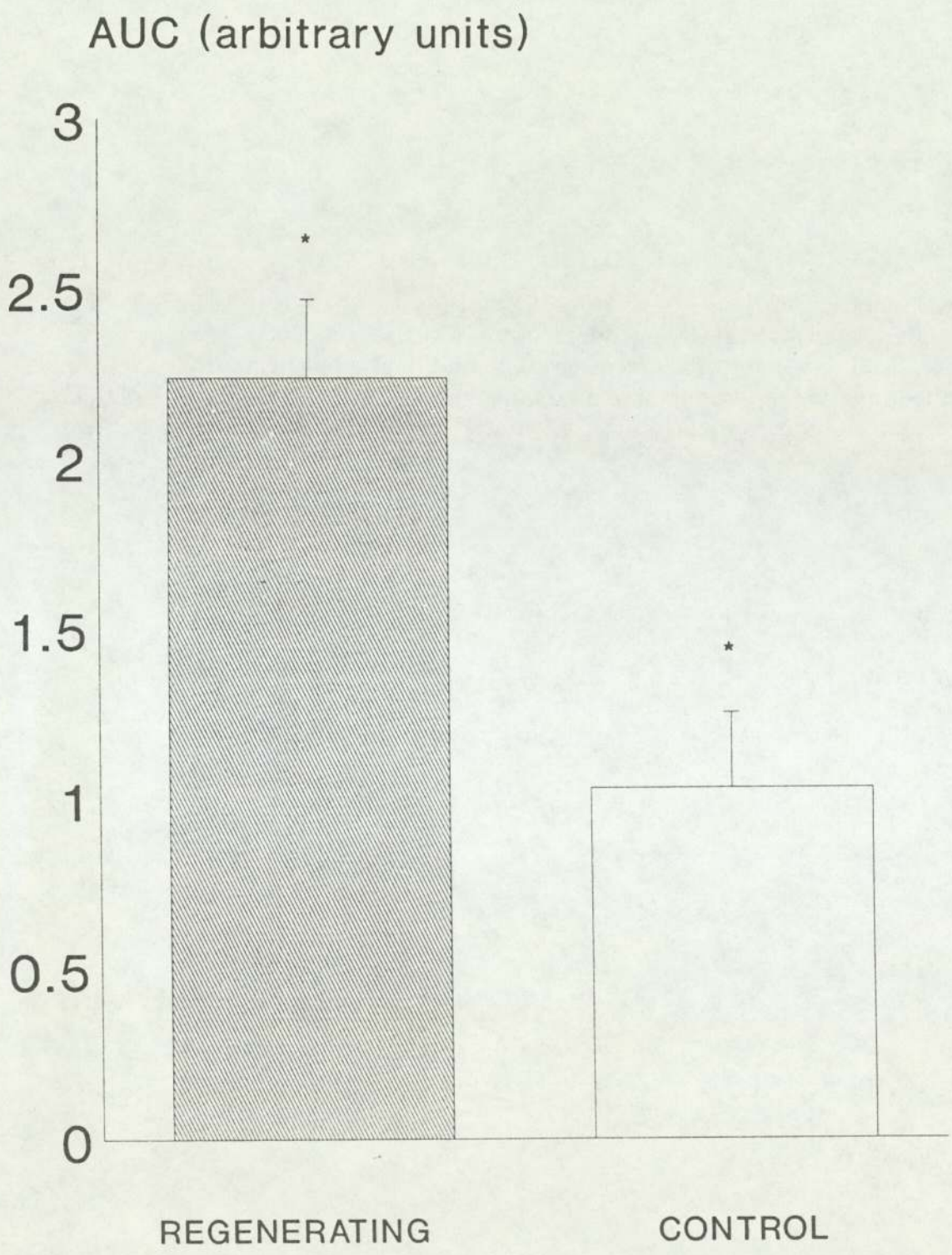

AUC (arbitrary units)
3
2.5
2
1.5
1
0.5
0
*
*
REGENERATING
CONTROL

Figure 48

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated by cAMP-dependent protein kinase activity in homogenates of nerve distal to a nerve crush (regenerating) and their equivalents from uncrushed (control) nerve. Phosphorylation was quantified by laser densitometry. Data expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 8 in each case.

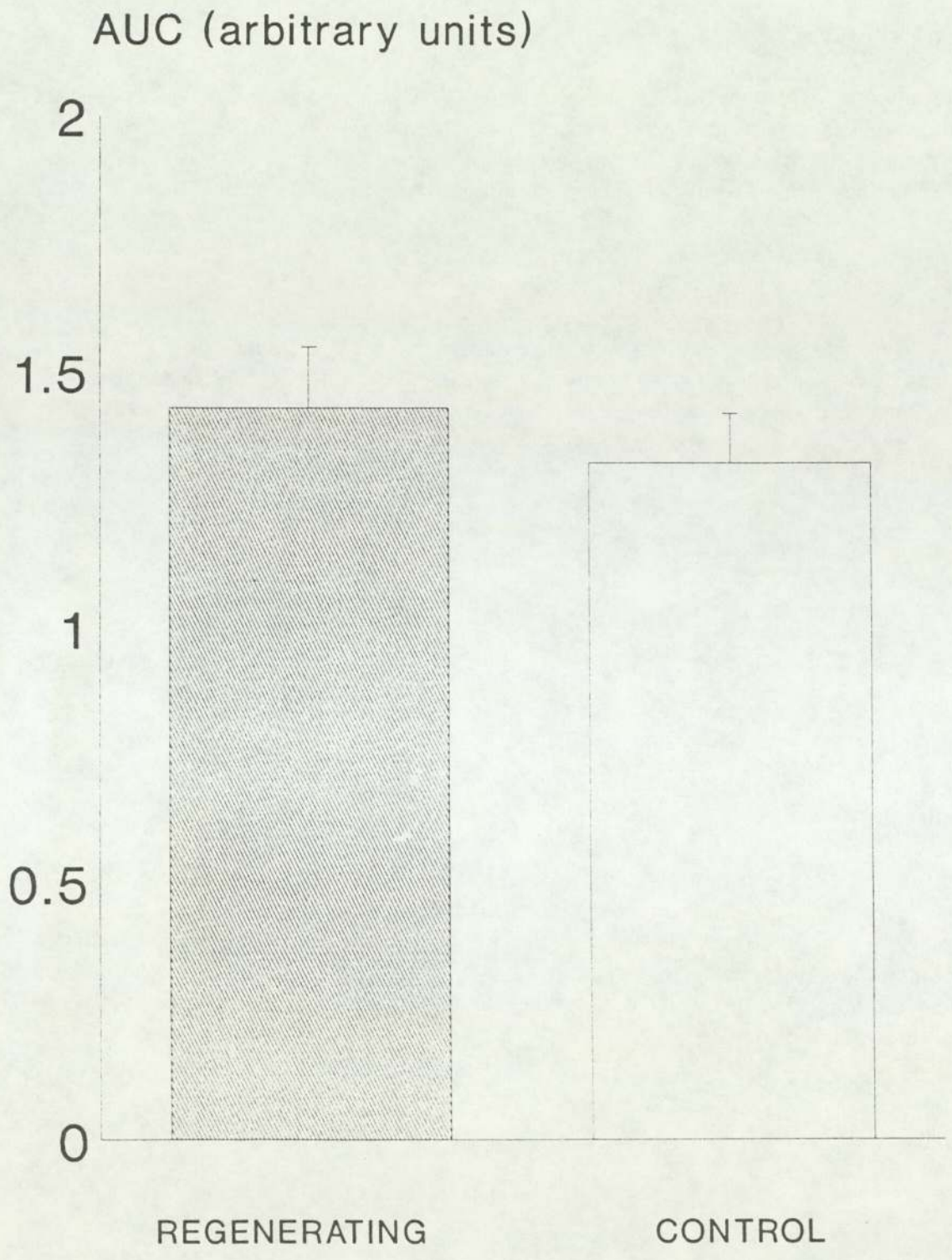

AUC (arbitrary units)
2
1.5
1
0.5
0
REGENERATING
CONTROL

Figure 49

Autoradiograph showing MAP-2 phosphorylated by Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity in the presence (+) or absence (-) of cAMP, in supernatant subfractions of homogenates from pieces of nerve proximal to a nerve crush (regenerating: R) and equivalent pieces from uncrushed nerve (control: C).

\+ in the presence of 10mM cAMP
\- in the absence of 10mM cAMP

Incubations were performed in the presence of exogenous MAP-2, and sub-fractionated homogenates of regenerating and control nerve together with ^{32} P-ATP, Radiolabelled proteins were separated by 1-dimensional SDS-PAGE and visualized by autoradiography.

Figure 50

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated by Mg^{2+}-dependent/Ca^{2+}-independent protein kinase activity, in the presence (+) or absence (-) of 10mM cAMP from subfractionated homogenates of pieces of nerve proximal to a crush (regenerating) and equivalent pieces of uncrushed nerve (control). Phosphorylation was quantified by laser densitometry. Data expressed as the absorbance (arbitrary units)

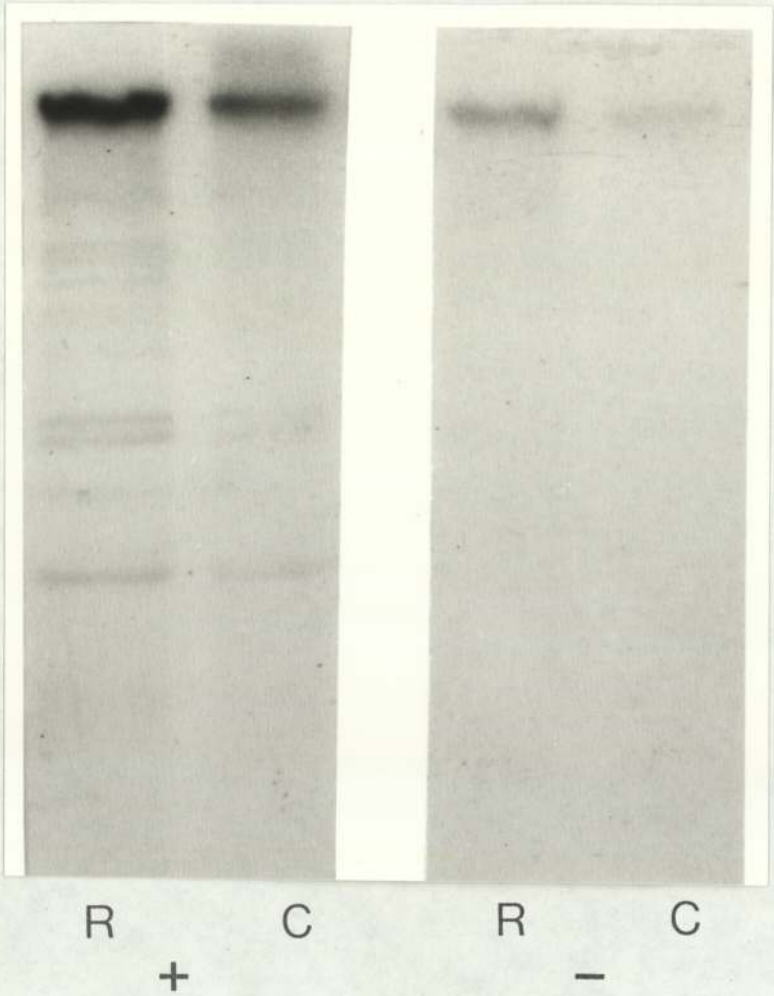
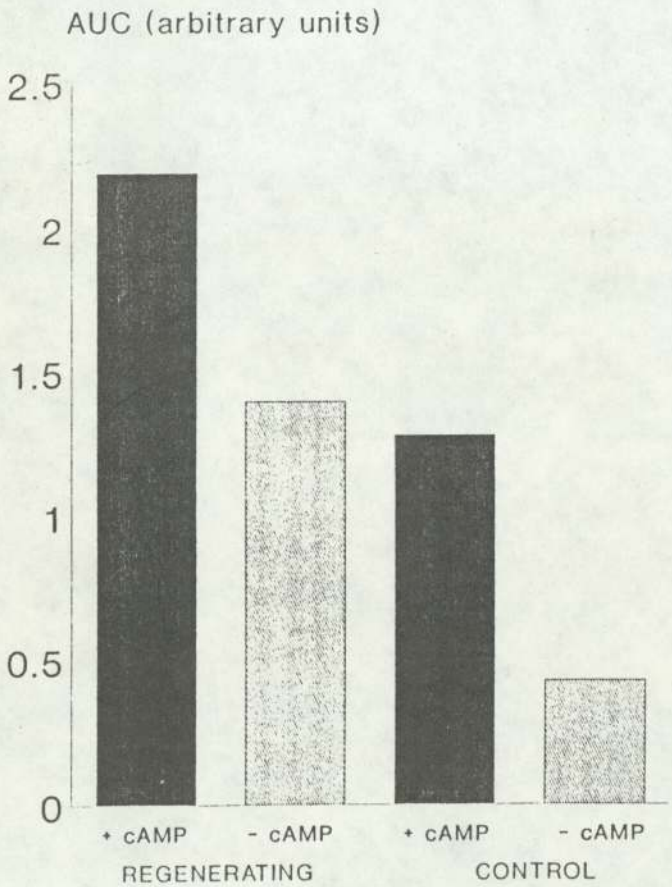

R
C
R
C
+
−

Figure 50


AUC (arbitrary units)
2.5
2
1.5
1
0.5
0
+ cAMP
- cAMP
+ cAMP
- cAMP
REGENERATING
CONTROL

2.2.3 *Ca^{2+}/calmodulin-dependent protein kinase activity*

In this assay, the conditions were chosen to examine the Ca^{2+}/calmodulin-dependent protein kinase activities present in samples of rat sciatic nerve which were directed against exogenous heat-stable MAPs (MAP-2 and tau).

As described in 2.1.1 and METHODS section 9.1 samples of rat sciatic nerve and dorsal root ganglia, prepared in the absence of EGTA, were assayed independently for protein kinase activity which would phosphorylate exogenous purified tubulin, under conditions which favour Ca^{2+}/calmodulin-dependent protein kinase activity. The incorporation of radiolabelled [^{32}P]-phosphate into exogenous purified tubulin and endogenous protein was examined by SDS-PAGE and autoradiography as outlined in METHODS section 9.3.

Protein kinase activity was not detected in supernatant sub-fractions of dorsal root ganglia, or pieces of rat sciatic nerve, proximal and distal to a crush, from either regenerating or control nerve (data not shown). However, protein kinase activity directed against MAP-2 was detected in particulate subfractions of dorsal root ganglia and pieces of rat sciatic nerve proximal and distal to a crush from regenerating nerve and also in equivalent regions of contralateral control nerve, as shown in figure 51. As shown in figure 52, the MAP-2 immunoreactive pattern matched the phosphorylation pattern obtained from the protein kinase assay. However, analysis of the phosphorylation patterns by laser densitometry revealed no significant difference ($P > 0.05$) in the protein kinase activity between any of the regenerating and control nerve pieces (figures 53, 54 and 55 respectively).

Figure 51

Autoradiographs displaying phosphorylation of MAP-2 by Ca^{2+}/calmodulin-dependent protein kinase activity in supernatant subfractions of homogenates from dorsal root ganglia, pieces of nerve proximal to a crush and distal to a crush from regenerating and control nerve. Incubations were performed in the presence (+) and absence (-) of heat-stable MAPs: MAP-2 and tau with subfractionated homogenates of dorsal root ganglia, pieces of nerve proximal and distal to a crush from regenerating nerve and their equivalents from control nerve, together with ^{32}P-ATP, and separated by 1-dimensional SDS-PAGE and visualized by autoradiography.

No.	Nerve piece	Reg/Con	+/- Exog. MAPs
1	Dorsal root ganglia	R	-
2	" " "	R	+
3	" " "	C	+
4	" " "	C	-
5	Proximal to a crush	R	-
6	" " "	R	+
7	" " "	C	+
8	" " "	C	-
9	Distal to a crush	R	-
10	" " "	R	+
11	" " "	C	+
12	" " "	C	-

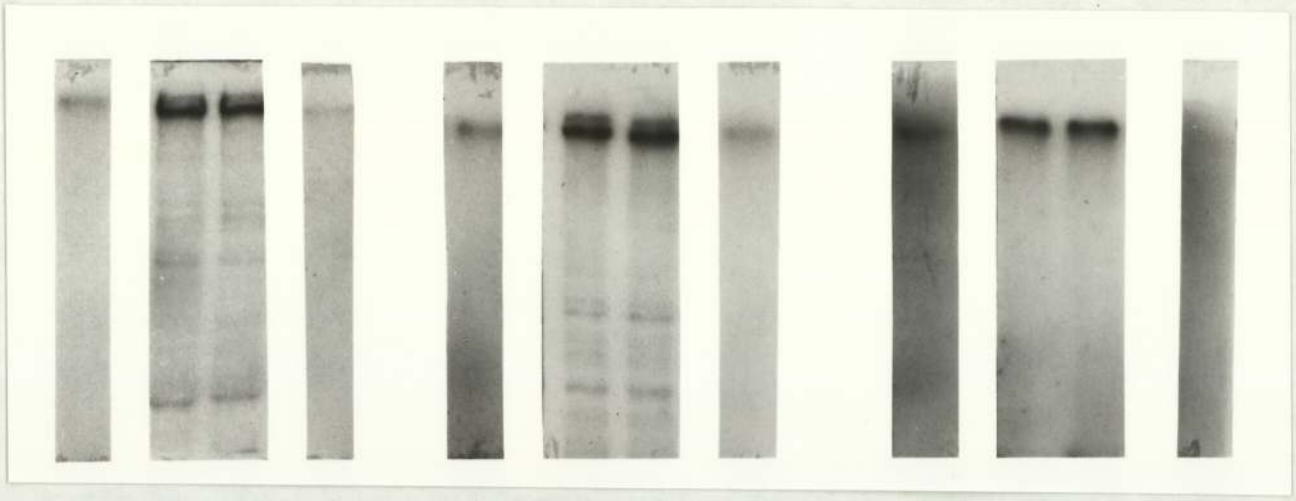
Figure 52

Fluorograph of MAP-2 immunoreactivity compared with the autoradiograph of MAP-2 that has been phosphorylated by Ca^{2+}/calmodulin-dependent protein kinase activity.

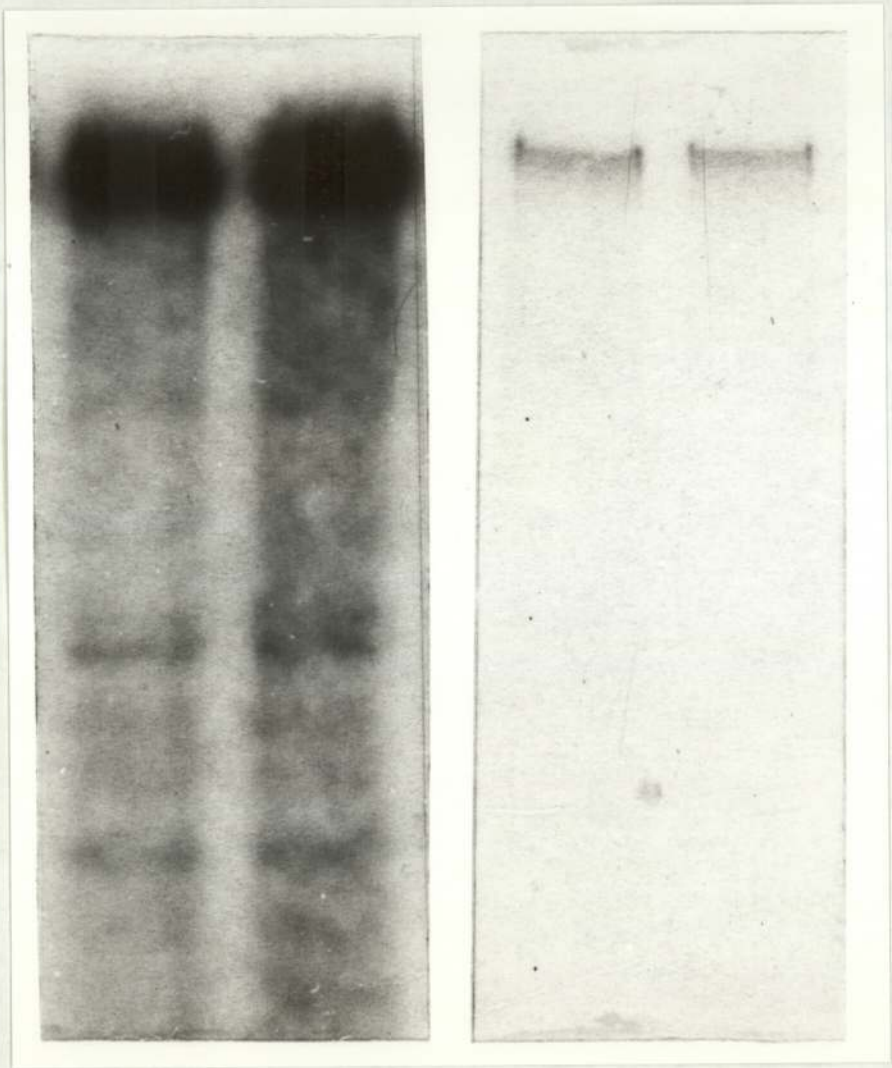
Radiolabelled proteins were resolved by 1-dimensional-SDS PAGE and transferred to nitrocellulose. The resulting Western blots were exposed to film for autoradiography, following autoradiography, the blots were stored until remaining radioactivity had subsided, and then probed with HM-2 antibody against MAP-2, visualized by HRP-conjugated rabbit antimouse IgG with ECL and fluorography.

A. Autoradiograph of MAP-2 phosphorylation (duplicate samples).

B. Fluorograph of MAP-2 immunoreactivity (duplicate samples).



1 2 3 4 5 6 7 8 9 10 11 12



A B

Figure 53

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated by Ca^{2+}/calmodulin-dependent protein kinase in homogenates of regenerating and control dorsal root ganglia. Phosphorylation was quantified by laser densitometry. Data are expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 8 in each case.

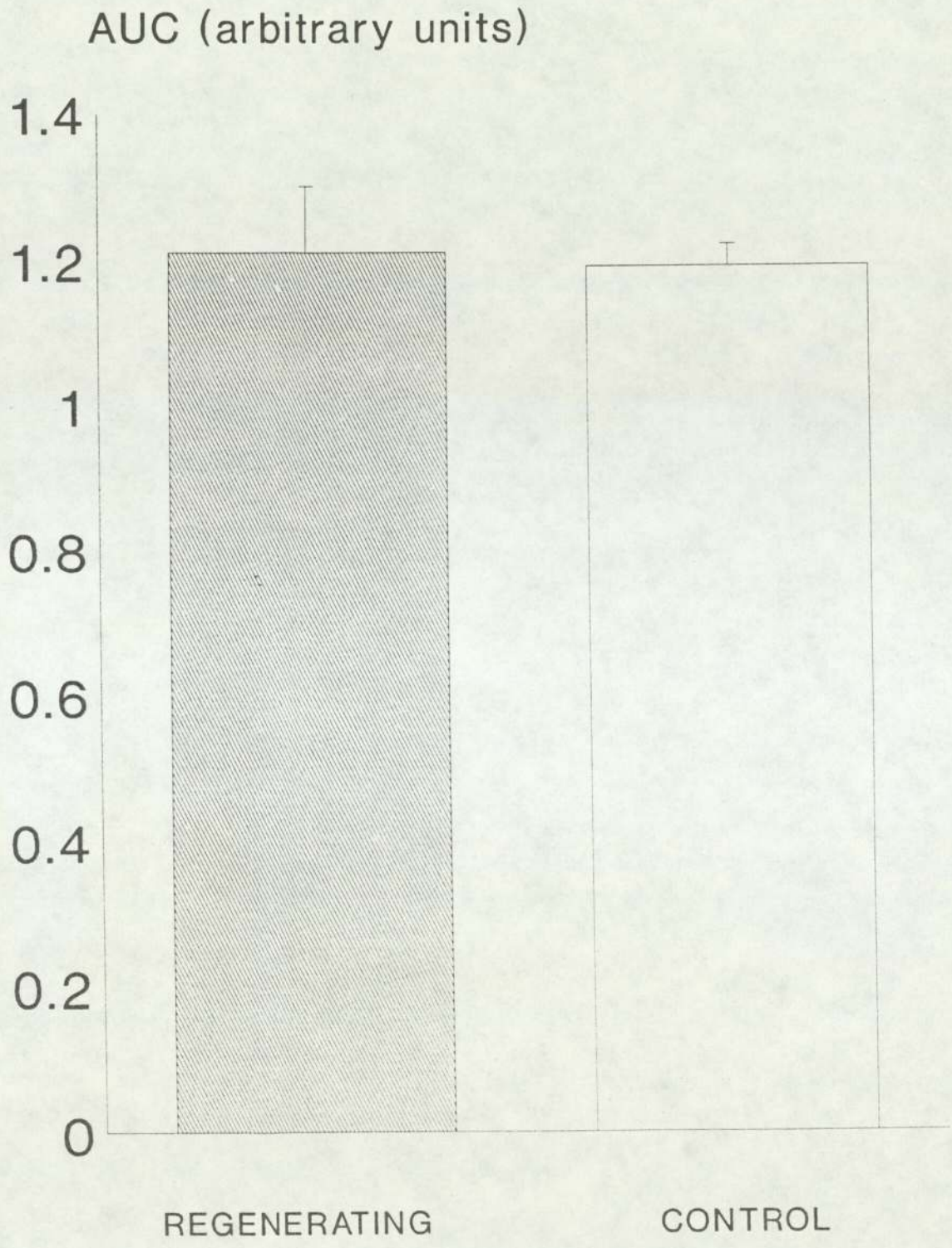

AUC (arbitrary units)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
REGENERATING
CONTROL

Figure 54

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated by Ca^{2+}/calmodulin-dependent protein kinase in homogenates of nerve proximal to a nerve crush (regenerating) and their equivalents from uncrushed (control). Phosphorylation was quantified by laser densitometry. Data are expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 8 in each case.

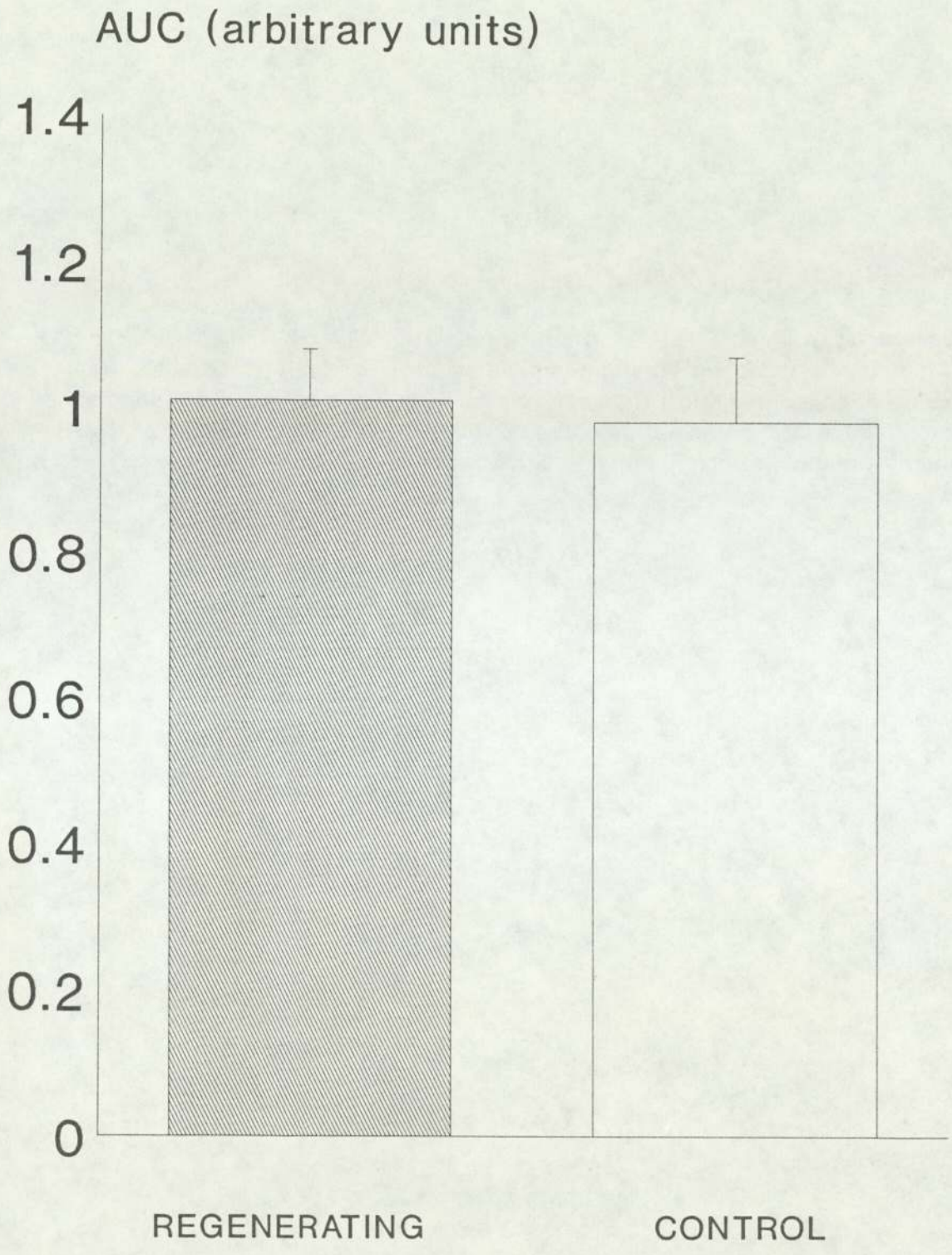
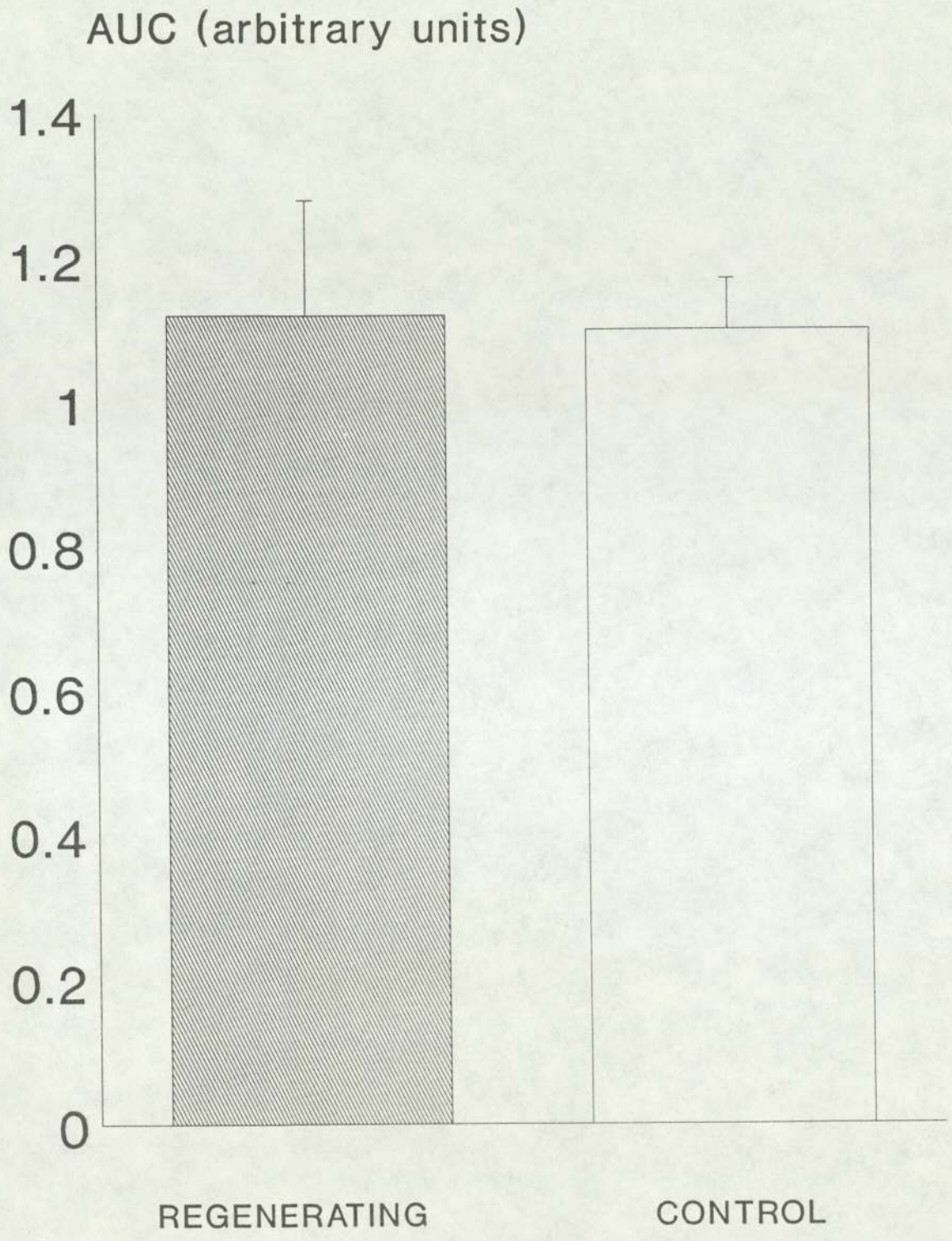

AUC (arbitrary units)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
REGENERATING
CONTROL

Figure 55

Densitometric analysis of autoradiographs of MAP-2 which has been phosphorylated by Ca^{2+}/calmodulin-dependent protein kinase in homogenates of nerve distal to a nerve crush (regenerating) and their equivalents from uncrushed (control). Phosphorylation was quantified by laser densitometry. Data are expressed as the absorbance (arbitrary units) $\pm$ S.D. n = 8 in each case.


AUC (arbitrary units)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
REGENERATING
CONTROL

2.3 Distribution of MAP-2a,2b,2c and MAP-2a,2b between assembled and non-assembled microtubule fractions

The possibility that an alteration in the cAMP-dependent protein kinase activity may result from a shift in the proportion of MAP-2 associated with the assembled and non-assembled microtubule fractions from pieces of regenerating and control nerve was investigated. MAP-2 has a high affinity for the regulatory subunits as well as the catalytic subunits of cAMP-dependent protein kinase. A shift in the association of MAP-2 with microtubules may account for a change in the activity of the kinase.

Pieces of regenerating nerve, proximal and distal to a crush and their equivalents from uncrushed control nerve were solubilized in NP-40 (see METHODS section 3) and separated into assembled (pellet) and non-assembled (supernatant) fractions following centrifugation. The distributions of MAP 2a,2b,2c and MAP 2a,2b between assembled and non-assembled fractions, from regenerating and control nerve pieces were then determined by dot-immunobinding assays using monoclonal antibodies HM-2 and AP-20 directed against MAP 2a,2b,2c and MAP 2a,2b, respectively, as outlined in METHODS 5.3.

2.3.1 *Specificity of the antibodies*

The specificity of the mouse monoclonal antibody directed against MAP-2 was determined using the same method outlined for YOL 1/34 in section 1.1 of this chapter. Purified MAP-2 and sciatic nerve extracts were resolved by 1-dimensional SDS-PAGE and transferred to nitrocellulose. The resultant blots were incubated with HM-2 and AP-20 mouse monoclonal antibodies, then visualized by fluorography. As

shown in figure 56, both antibodies show specificity for MAP-2.

2.3.2 *Immunobinding analysis*

The percentage of MAP-2a,2b,2c that was associated with non-assembled microtubule fractions from pieces of regenerating and control nerve is shown in figure 57. In pieces of regenerating nerve proximal to a crush 95.8 $\pm$ 0.3% of MAP-2a,2b,2c was associated with the non-assembled microtubule fraction, whereas in pieces distal to a crush, 93.9 $\pm$ 0.7% of MAP-2a,2b,2c was associated with the non-assembled microtubule fraction. In control nerve pieces, 94.6 $\pm$ 0.4% of MAP-2a,2b,2c was associated with the non-assembled microtubule fraction equivalent pieces proximal to a crush, while 94.4 $\pm$ 0.5% of MAP-2a,2b,2c was associated with the non-assembled microtubule fraction in pieces of control nerve equivalent to distal to a crush. However, the total MAP-2 (MAP-2a,2b,2c) immunoreactivity associated with the non-assembled fraction was significantly higher ($P<0.05$) in pieces of nerve proximal to a crush on comparison to equivalent pieces of contralateral control nerve and pieces of nerve distal to a crush.

In consideration that MAP-2a and 2b are thought to be localized to the cell body and dendritic processes, it was thought the shift in the distribution of MAP-2 was attributable to MAP-2c differences. However, no access to a specific antibody for MAP-2c was available to us, and therefore we measured MAP-2a and 2b immunoreactivity in the pieces of nerve previously examined for MAP-2a,2b,2c immunoreactivity. If no shift in the distribution could be measured we could attribute the above changes to a shift in the MAP-2c distribution.

Figure 56

Fluorographs of immunoreactivity of HM2 and AP-20, mouse monoclonal antibodies against MAP-2a,2b,2c and MAP2a,2b respectively with whole nerve extract and purified MAP-2

A. Immunoblots probed with HM-2 with sample of purified MAP-2.

B. Immunoblots probed with HM-2 with sample of nerve extract.

C. Immunoblots probed with AP-20 with sample of purified MAP-2.

D. Immunoblots probed with AP-20 with sample of nerve extract.

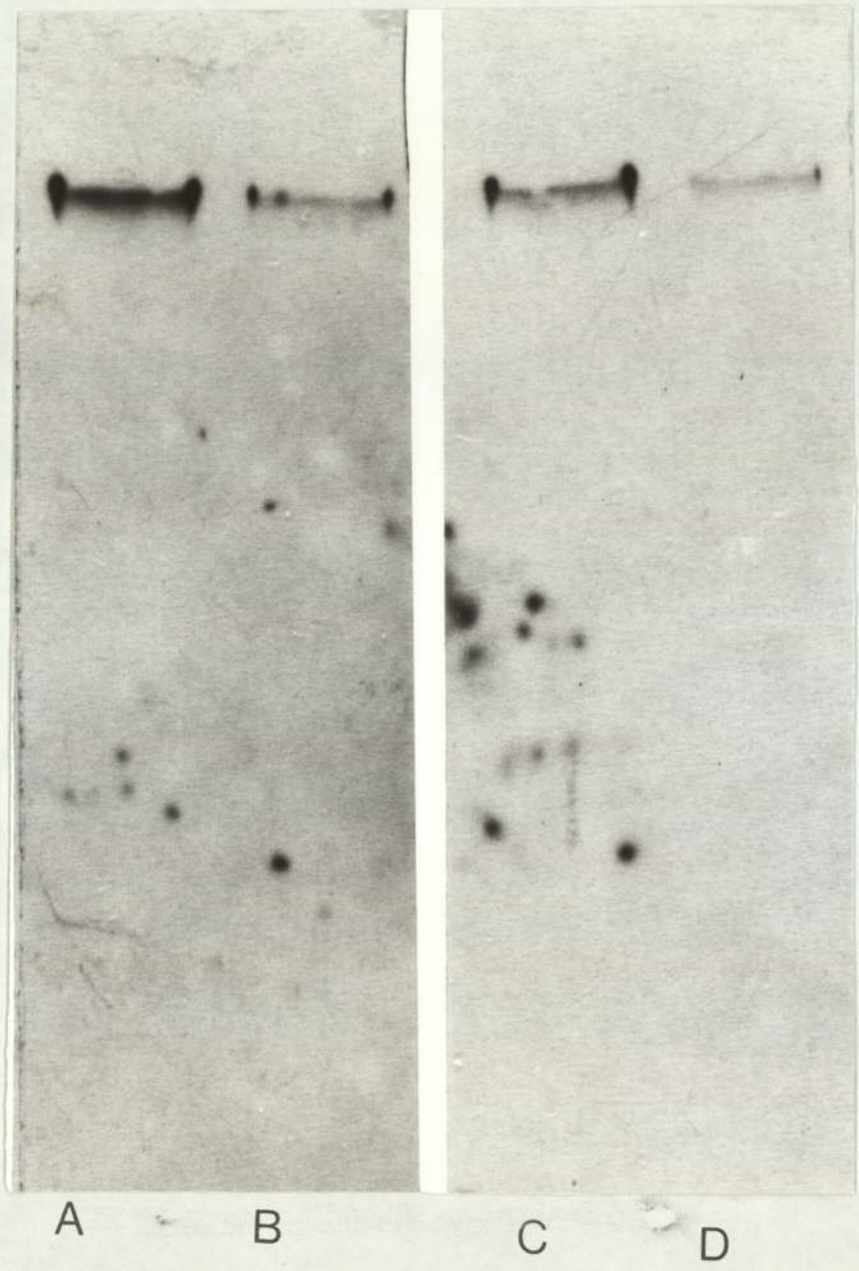
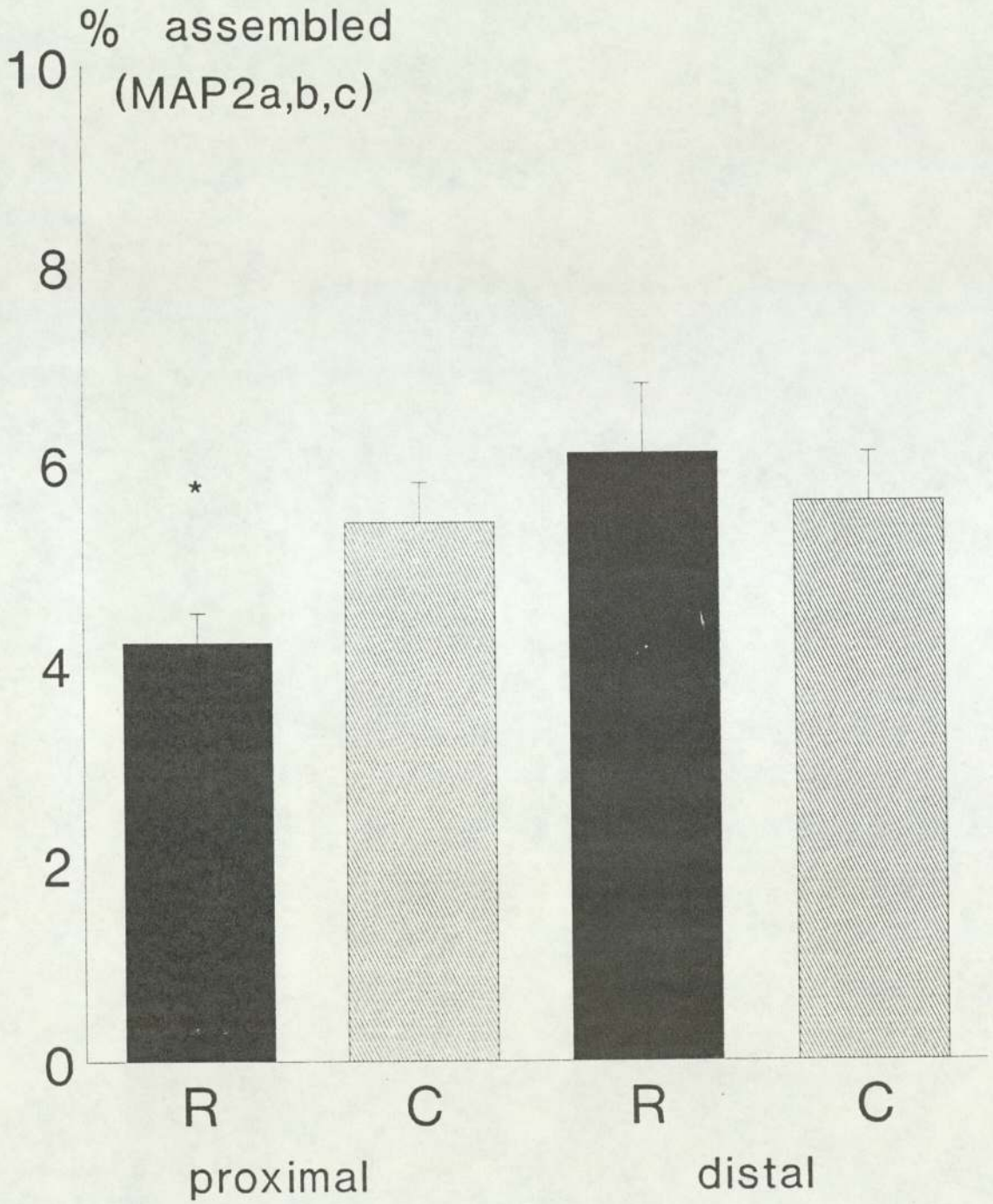

A
B
C
D

Figure 57

Percentage of MAP-2a,2b,2c that was associated with assembled microtubule fractions from pieces of regenerating (R) nerve (1cm) proximal and distal to a nerve crush and their equivalents from uncrushed (control) (C) nerve. The data are expressed as the % $\pm$ S.D. n = 4 in each case. * $P < 0.05$ compared with proximal pieces of contralateral control nerve.

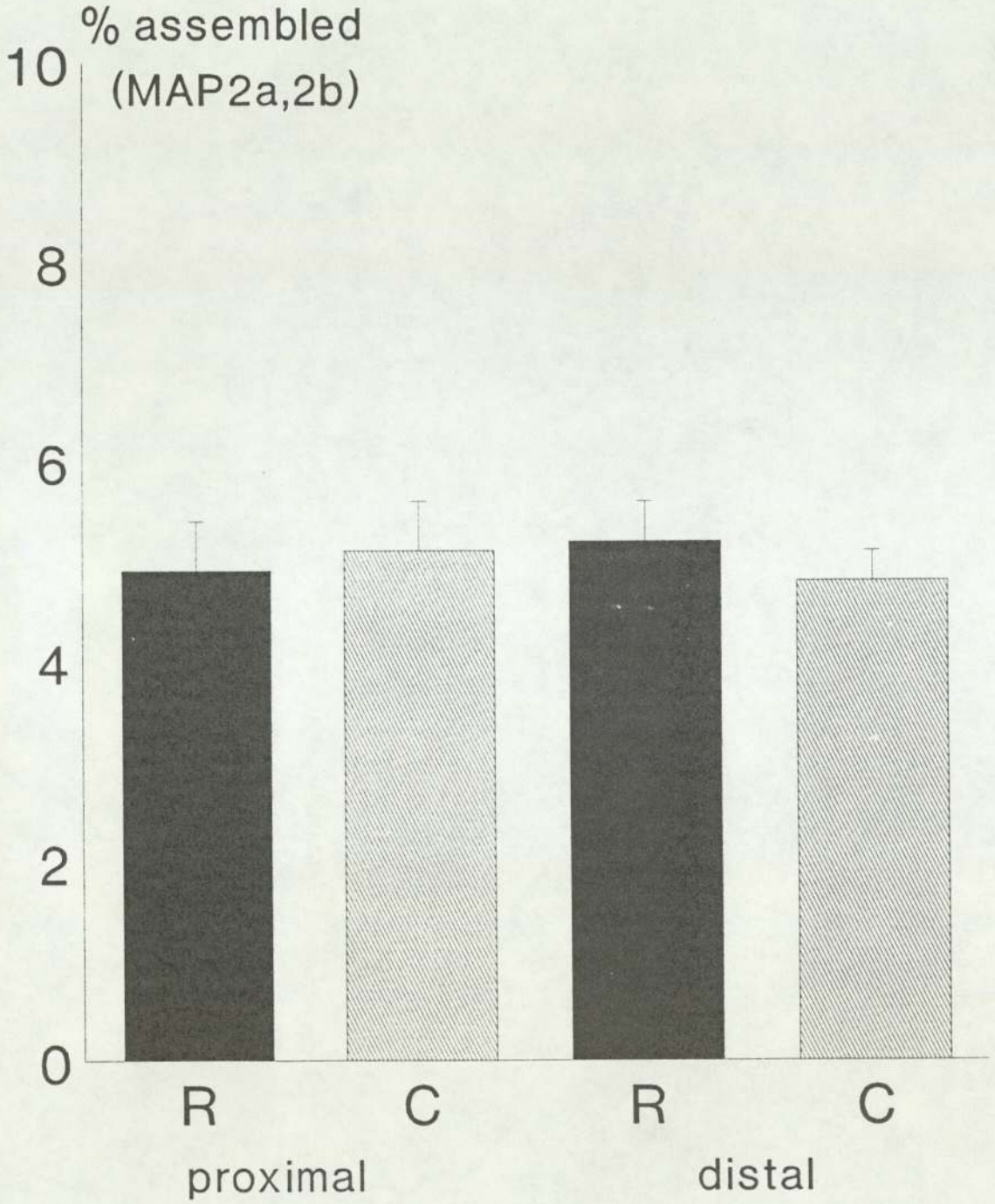

% assembled
(MAP2a,b,c)
10
8
6
4
2
0
*
R
C
R
C
proximal
distal

Dot-blot analysis of the distribution of MAP-2a and 2b between assembled and labile microtubule fractions from pieces of regenerating nerve, proximal and distal to a crush, and equivalent pieces of control nerve was performed using the monoclonal antibody AP-20. The immunoreactivity that could be measured, was also predominantly associated with the labile microtubule fractions; 95.1 $\pm$ 0.5% in nerve pieces proximal to a nerve crush, 94.9 $\pm$ 0.5% in an equivalent piece of control nerve, 94.8 $\pm$ 0.4% in nerve pieces distal to a nerve crush with 95.2 $\pm$ 0.3% in an equivalent piece of control nerve as shown in figure 58. While the immunoreactivity in general, was lower for AP-20 than that of HM-2 there was no significant difference in the distribution between labile and assembled microtubule fractions of MAP-2a and 2b in nerve pieces of regenerating and control nerve examined. While these observations did not measure the actual amounts of MAP-2 present in regenerating and control nerve, they highlighted the differences in the distribution of MAP-2 forms present in the nerve, irrespective of the amount of MAP-2 present.

Thus the shift in the distribution of MAP-2, probably MAP-2c, proximal to the crush in regenerating nerve correlated with a shift in the cAMP-dependent protein kinase activity proximal to a crush in regenerating nerve.

Figure 58

Percentage of MAP-2a,2b that was associated with assembled microtubule fractions from pieces of regenerating (R) nerve (1cm) proximal and distal to a nerve crush and their equivalents from uncrushed (control) (C) nerve. The data are expressed as the % $\pm$ S.D. n = 4 in each case.


% assembled
(MAP2a,2b)
10
8
6
4
2
0
R
C
R
C
proximal
distal

DISCUSSION

The aim of the work presented throughout this thesis, has been to examine and investigate the effect of nerve injury, and subsequent nerve regeneration in the peripheral nervous system, on the post-translational processing of tubulin and microtubule-associated proteins (MAPs).

1 SUITABILITY OF THE NERVE REGENERATION MODEL

The mammalian peripheral nerve regeneration model used in my own study, viz. the rat sciatic nerve has been used previously in our laboratory to demonstrate molecular changes that occur as a consequence of nerve regeneration.

Such alterations include:

1) The induction of ODC during the early stages of nerve regeneration and its impairment in toxic neuropathies (McLean *et al.*, 1987)

2) Changes in the axonal transport of newly synthesized proteins (McLean, 1985).

The rate of regeneration in the sciatic nerve model has been characterized (Gutmann *et al.*, 1942, Bisby, 1978) using two separate methods of study: the 'pinch reflex' test and analysis of the axonal transport of radiolabelled proteins. Both of these methods found the rate of regeneration in sciatic nerve to be approximately 3-4mm/day.

In my own studies, a five day nerve regeneration model was chosen. In this model, we expected regenerating fibres to extend up to 12-15 mm from the site of nerve injury, based on a regeneration rate of 3-4 mm/ day and accounting for a delay in the onset of nerve regeneration of approximately 24-36 h (Bisby, 1978). Furthermore, the slow axonal transport of newly synthesized proteins, in particular tubulin, would only

have travelled approximately 15-20 mm from the cell body, based on the rate of slow axonal transport (Bisby, 1978) during nerve regeneration. The newly synthesized proteins would be about 20-25 mm from the site of nerve injury, which was particularly important when assessing post-translational processing of microtubule proteins 1 cm proximal and 1 cm distal to the site of the nerve injury.

2 TYROSINATION AND ACETYLATION OF α-TUBULIN

2.1 Tyrosination

Previous studies have shown that the tyrosination state of tubulin is altered through the course of development (Rodriguez and Borisy, 1978; Cumming *et al.*, 1984; Arregui and Barra, 1989; Arregui *et al.*, 1991). These observations are also correlated with alterations in the activity of tubulin:tyrosine ligase (TTL), the enzyme responsible for the post-translational tyrosination of α-tubulin during development (Deanin *et al.*, 1977). Furthermore, cell culture studies have shown that the tyrosination state and TTL activity are increased upon cell differentiation and subsequent neurite outgrowth (Deanin *et al.*, 1982; Cambray-Deakin and Burgoyne, 1987).

Distinct subpopulations of microtubules have been distinguished biochemically, by their relative sensitivity to temperature and depolymerizing drugs (Kreis, 1987; Gundersen *et al.*, 1987; Watson *et al.*, 1990). Indeed, observations from these studies suggested tyrosinated microtubules were associated with more dynamic subpopulations of microtubules whereas detyrosinated microtubules were associated with more stable subpopulations of microtubules.

Many of the molecular events that occur during neurite extension in neuronal cell culture are recapitulated during mammalian peripheral nerve regeneration (Goldstein *et al.*, 1988). The purpose of this study was to examine whether any alterations in the post-translational tyrosination of tubulin occurred as a consequence of nerve regeneration and to correlate these changes with observations from cell culture and developmental studies. It was of further interest to try to correlate any changes with the stability of the cytoskeletons of regenerating and control nerve.

Five days after a crush injury, tyrosinated α-tubulin immunoreactivity was measured in pieces of rat sciatic nerve immediately proximal (central side) and distal (peripheral side) to a nerve crush, and in equivalent pieces of contralateral uncrushed nerve. Dot-immunobinding analysis revealed an increase in tyrosinated α-tubulin immunoreactivity in regenerating nerve, localized to the region where regenerating fibres are present. Analysis of sections of similar pieces of regenerating and control nerve by cytofluorimetric scanning and confocal laser microscopy confirmed the biochemical observations (see RESULTS section 1.1.2).

The question arose as to whether the apparent change in the tyrosination of α-tubulin was localized to axons, Schwann cells, or both, and also, whether the increased tyrosination was a consequence of nerve regeneration or nerve degeneration. To address this question I measured the tyrosination state of α-tubulin in pieces of rat sciatic nerve, proximal and distal to a nerve crush that had been ligatured. Both regenerating and ligatured nerve models exhibit Wallerian degeneration resulting in the degeneration of

existing axons and the breakdown of myelin in the distal stump together with a characteristic, transient proliferation of the Schwann cell (Perry *et al.*, 1987). If we were to consider that the increased tyrosination of α-tubulin was a consequence of Schwann cell proliferation, then it might be expected that an increased tyrosination state of α-tubulin would be observed distal to a crush in the ligatured nerve model. In fact, no such observation was made; indeed, there was no difference in the observed immunoreactivity between ligatured and control nerves, for both tyrosinated and total α-tubulin.

During Wallerian degeneration, microtubules and tubulin are believed to be broken down. These findings would therefore suggest that the immunoreactivities i.e. epitopes are conserved following disassembly, and the observed immunoreactivity distal to a ligatured crush is that attributable to pre-existing axons. i.e. reflects what was there previously. The data implies therefore that the change in tyrosination of α-tubulin is a consequence of nerve regeneration, and that the change is axonal, and localized to regenerating fibres. This view is supported by the cytofluorimetry and confocal laser microscopy data. In these studies, the increased immunoreactivity associated with tyrosinated α-tubulin was located in regenerating axons, since the immunofluorescence extended over approximately 8-10 mm distal to the crush, which corresponds to the approximate distance over which regenerating axons would have grown, and where axonal regenerating sprouts were observed in the fluorescence microscope (see RESULTS section 1.1.3).

In some sections of regenerating nerve distal to the crush, the immunofluorescence associated with tyrosinated α-tubulin was observed to increase

initially and then decline (see RESULTS section 1.1.3). This observation correlated with the immunofluorescence profile associated with GAP-43, a fast axonally transported growth associated polypeptide whose expression is increased during nerve regeneration and found in elevated amounts in regenerating fibres. The immunofluorescence associated with tyrosinated α-tubulin was observed to decrease just proximal to the decrease in GAP-43 immunofluorescence (see RESULTS section 1.1.3), further indicative that the change in the tyrosination state of α-tubulin was localized to the region where regenerating fibres are present.

In the regeneration model used to study nerve regeneration in the work presented, there was insufficient time for newly synthesized tubulin to have been transported from the cell bodies to beyond the crush site. Therefore, if the elevation in the tyrosination of α-tubulin was entirely within axons, then it was clearly not the consequence of an increased expression/synthesis of tyrosinated α-tubulin, subsequently transported to the regenerating fibres. Besides, the tyrosination state of α-tubulin proximal to the crush site was no different to that observed in uncrushed control nerves.

Thus, the increase in tubulin tyrosination appears to be generated by post-translational modification of the pre-existing tubulin population and is probably in response to the requirement of a functional alteration of microtubules to support growth cone mobility. The data conforms with the observations from the study on isolated growth cones (Gordon-Weeks and Lang, 1988) which shows a requirement for an increased pool of dynamic (tyrosinated) microtubules for growth cone advance.

The amount of tubulin that exists in an assembled and a non-assembled state did

not differ significantly between regenerating and control nerve i.e. no shift in the assembly state of tubulin could be detected (see RESULTS section 1.1.4). This is consistent with *in vitro* and *in vivo* data from polymerization studies which show that whatever the tyrosination state of α-tubulin, microtubules are able to assemble with the same efficiency (Kumar and Flavin, 1982; Gundersen *et al.*, 1987). The proportion of tubulin in the assembled state was approximately 94% in all pieces of nerve that were examined. This proportion was similar to that reported by Arregui and Barra (1989), for mature chick retina. However, an increase in the percentage of tyrosinated tubulin in the non-assembled state was observed in pieces of regenerating nerve distal to a crush.

Post-translational tyrosination of α-tubulin was originally demonstrated in rat brain (Barra *et al.*, 1973). These workers found that TTL was the enzyme responsible for this modification. The activity of TTL has been found to be developmentally regulated (Deanin *et al.*, 1977; Rodriguez and Borisy, 1978; Barra *et al.*, 1980) and, in cultured cells, increases during the formation of neuronal processes (Deanin *et al.*, 1982). In consideration of these observations, I wished to examine whether the increased tyrosination state of tubulin I had previously observed in regenerating nerve (see RESULTS section 1.1.5) was the consequence of an elevated activity of TTL.

I was unable to measure TTL activity directed against an exogenous tubulin substrate that had been obtained and purified from either porcine brain or rat brain, and I am unable to present evidence for an alteration in the activity of TTL itself in regenerating nerve. It has been suggested that TTL activity measured from sources other than brain may require that the substrate tubulin be from the same source as TTL

itself (Chang and Flavin, 1988). Preliminary experiments to some extent substantiate this observation. The incorporation of [^{3}H]-tyrosine into an exogenous tubulin substrate from porcine or rat brain could be detected using fractionated rat brain extracts as a source of TTL. However, no incorporation of [^{3}H]-tyrosine into exogenous tubulin substrate from porcine or rat brain could be detected using fractionated rat sciatic nerve extracts, that was in excess of that which could be accounted for as incorporation into endogenous peripheral nerve tubulin (see RESULTS section 1.1.5). The obvious limitations of purifying tubulin from sciatic nerve in sufficient amounts to assay TTL activity in rat sciatic nerve, prevented me from performing a TTL assay that could measure the absolute TTL activity.

Nevertheless it was possible to demonstrate, for the first time, that TTL activity is present in the rat sciatic nerve. The TTL activity directed against endogenous peripheral nerve tubulin, already present in subfractionated pieces of rat sciatic nerve for assay, (see RESULTS section 1.1.5) was observed to decrease in pieces of regenerating nerve distal to the crush. The decrease did not reflect a difference in the protein content between pieces of nerve examined, but may reflect a reduction in the substrate availability. An increased tyrosination state of α-tubulin, particularly in the non-assembled form, the preferred substrate of TTL, would essentially reduce substrate availability. In support of this possibility is the observation that there is indeed an increase in the tyrosination state of tubulin in the non-assembled form (see RESULTS section 1.1.4).

If we were to consider that the increase in tyrosinated α-tubulin in regenerating axons was the result of a transient increased expression of TTL, we must assume that

TTL is fast axonally transported to the regenerating axons/growth cone where it must accumulate. In consideration of this argument we might expect to observe an accumulation of TTL and subsequently an increased amount of tyrosinated α-tubulin relative to total α-tubulin proximal to a crush in ligatured nerves. However the data I obtained from my own experiments are inconsistent with this view.

The possibility that the increase in tyrosinated tubulin distal to a crush in regenerating axons is in response to an activation of TTL still remains. The activation would have to occur locally and may be in response to retrogradely transported trophic factors. Nath *et al.*, (1981) presented data which showed that a synthetic peptide chemoattractant, N-formyl-methionyl-leucyl-phenylalanine (FMLP), caused a specific stimulation of tubulin tyrosination mediated via the activation of TTL. Indeed taxol, which polymerizes microtubules and therefore reduces the substrate available for TTL, inhibited the response of FMLP. Furthermore, TTL activity has been found to increase following differentiation with nerve growth factor (NGF: Deanin *et al.*, 1982; Arregui *et al.*, 1991). NGF is a retrogradely transported signal which initiates the regeneration process and is important for the development, survival and extension of neurones (Seckel, 1990). Additionally, NGF has been implicated in the stimulation of other post-translational modifications of other protein e.g. GAP-43 phosphorylation (Meiri and Burdick, 1991), and may therefore be implicated in stimulating TTL activity *in vivo* during mammalian peripheral nerve regeneration.

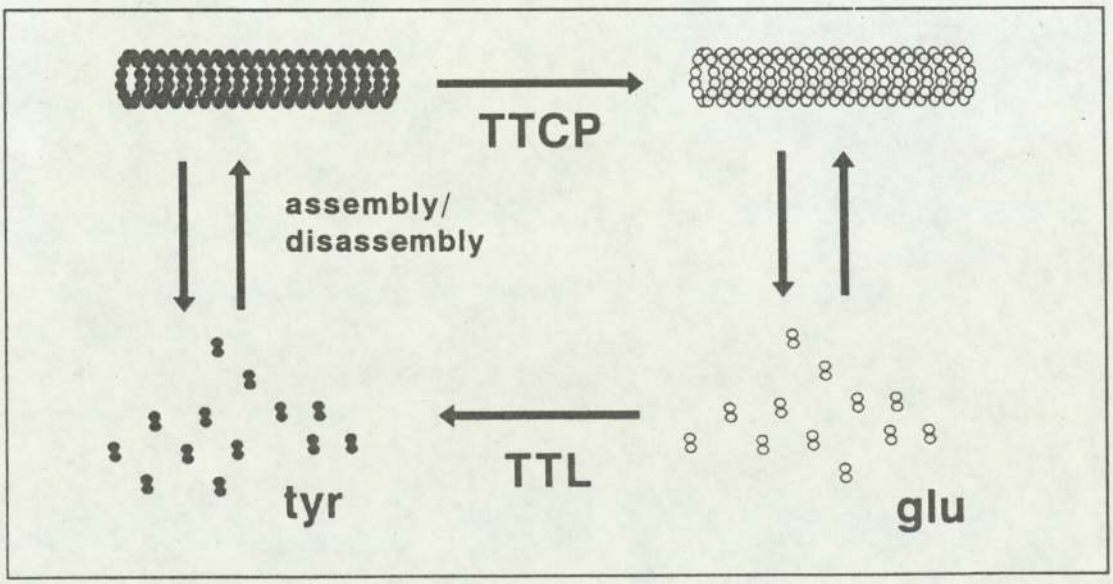
An alternative cause of the increased tyrosination state of α-tubulin in regenerating nerve could be an alteration in the endogenous carboxypeptidase activity

termed TTCP. A decrease in TTCP enzyme activity would result in a decreased rate of detyrosination, effectively perturbing the equilibrium between TTCP and TTL thus leading to an increased tyrosination state of tubulin. It has been shown that in certain instances, TTCP may be inhibited by polyamines (Barra and Argarana, 1982; Barra *et al.*, 1988). One of the earliest changes in metabolic activity that follows peripheral nerve injury is an increase in the activity of the enzyme ornithine decarboxylase (ODC; Gilad and Gilad, 1983; Tetzlaff and Kreutzberg, 1985). ODC is important, and indeed the rate limiting enzyme involved in the decarboxylation of the non-essential amino acid ornithine into putrescine, which is further converted to polyamines, spermidine and spermine, which are essential for at least the early stages of growth and differentiation (Maudsley, 1979). It is conceivable therefore that an increased ODC activity produces an increase in polyamines, which are fast axonally transported to regenerating axons where they may accumulate and modify TTCP activity. In light of these observations, I aimed to determine whether TTCP activity was modified as a consequence of nerve injury and subsequent regeneration. However, TTCP activity could not be detected in sciatic nerve under the conditions employed.

The increased tyrosination state and altered proportion of tubulin in the non-assembled fraction distal to a crush may be a reflection of an increased rate of assembly/disassembly of microtubules in the regenerating axons i.e. increased microtubule turnover. Detyrosination of α-tubulin by endogenous tubulin: tyrosine carboxypeptidase is polymer-specific i.e. assembled tubulin is the preferred substrate. This modification has been shown to accumulate with time (Gundersen *et al.*, 1987; Black *et al.*, 1989); these investigators looked at microtubule turnover in cell culture

and found that when microtubules were allowed to reassemble after complete depolymerization, initially only tyrosinated forms of microtubules could be detected, (Gundersen *et al.*, 1984) and that detyrosinated microtubules reappeared following a delay of ~25 min, similar to that found in cultured muscle cells (37 min; Thompson *et al.*, 1979). This suggested that detyrosination was a post-polymerization post-translational modification whereas tyrosination/retyrosination occurred when tubulin existed in a non-polymeric form. Gundersen *et al.*, 1987 proposed the model for the detyrosination-tyrosination cycle of α-tubulin, shown below. In this model, detyrosination occurs on assembled microtubules whereas tyrosination occurs on non-assembled microtubules.

Adapted from Gundersen *et al*, 1987 J. Cell Biol. **105** 251-264.



This model may be simplified to explain how the rate of microtubule turnover affects the tyrosination state of tubulin, and is shown below. In normal nerve, **MODEL A** represents the cyclic events that may occur with respect to tyrosination /detyrosination of microtubules and the rate of microtubule turnover, whereas **MODEL B** represents the events that may occur in regenerating axons. In using these models it must be assumed the time required by TTL to tyrosinate a population of tubulin subunits in a non-assembled form is *less* than that required by TTCP to detyrosinate an equivalent population of tubulin subunits in an assembled form. In these models we examine the fate of subunits that have exchanged between assembled and non-assembled states. Furthermore, a simplified model of microtubule assembly and disassembly is used to demonstrate the main principle of the models.

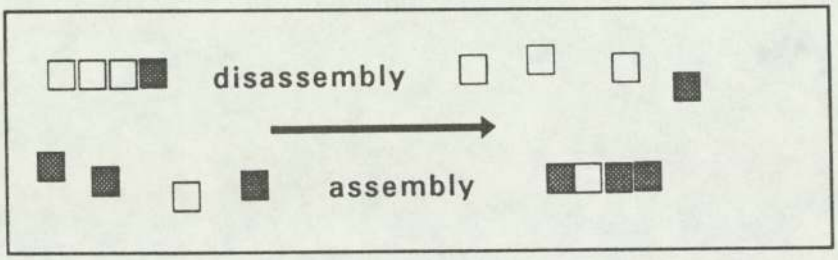
In **MODEL A**, microtubules turn over and tubulin subunits are exchanged between assembled and non-assembled states, as shown in (i) and (iii), and the time between subsequent turnover events is shown in (ii); within this time the assembled and non-assembled tubulin are subjected to TTCP and TTL, respectively. In this model the total polymer mass remains constant, and the turnover rate is slow enough to allow the actions of TTL and TTCP to maintain the levels of tyrosinated and detyrosinated tubulin.

In **MODEL B**, the time between microtubule turnover events is reduced as the rate of microtubule turnover increases, the effect of which is shown in (ii) and (iv), there is insufficient time for TTCP to detyrosinate the same amount of tyrosinated microtubules as shown in (ii) of **MODEL A**. However, there is sufficient time for TTL to tyrosinate the non-assembled detyrosinated tubulin, and in this instance the levels of

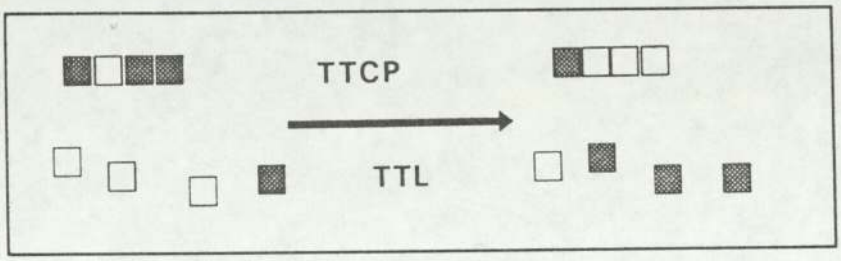
tyrosinated and detyrosinated tubulin are altered, resulting in an increased level of tyrosination.

MODEL A

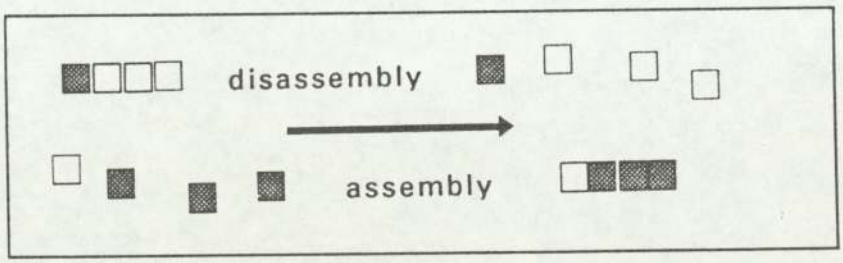
(i)
MICROTUBULE TURNOVER



(ii)
ENZYMATIC TYROSINATION AND DE-TYROSINATION



(iii)
MICROTUBULE TURNOVER

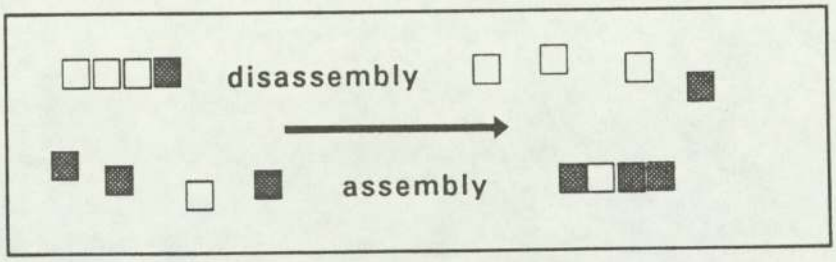


- tyrosinated tubulin subunit

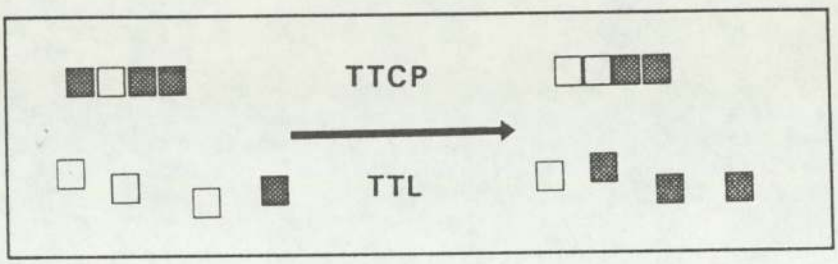
- detyrosinated tubulin subunit

MODEL B

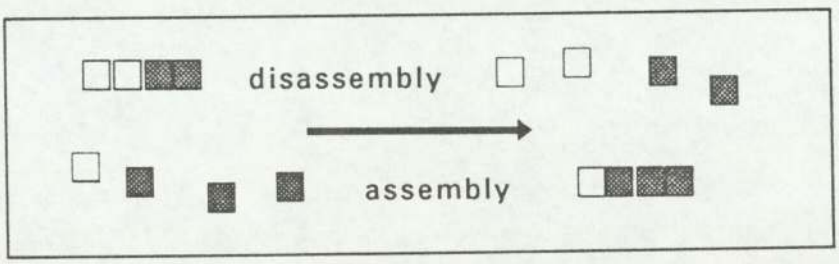
(i)
MICROTUBULE TURNOVER



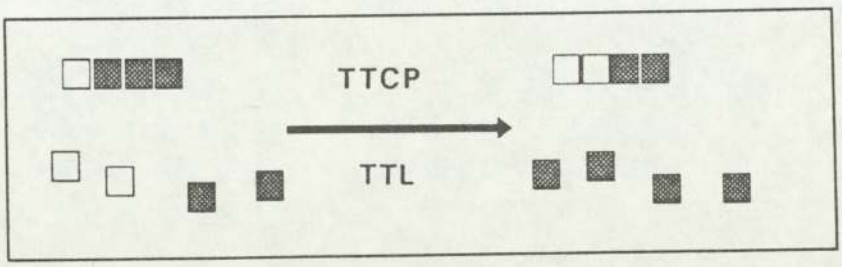
(ii)
ENZYMATIC TYROSINATION AND DE-TYROSINATION



(iii)
MICROTUBULE TURNOVER



(iv)
ENZYMATIC TYROSINATION AND DE-TYROSINATION



▩ - tyrosinated tubulin subunit

□ - detyrosinated tubulin subunit

These models may explain the increase in the tyrosination state of tubulin in regenerating axons, an area where one may expect an increased microtubule turnover in order to generate the increased plasticity of regenerating axons during neuronal pathfinding until the neuronal target is reached.

The increase in the tyrosination state of tubulin in regenerating areas is coupled with an increase in the proportion of tyrosinated tubulin that exists in the non-assembled tubulin fraction, indicating that the increase in the degree of tyrosination of tubulin in regenerating axons is largely in the non-assembled form. This altered tyrosination state may lead to altered stability of the cytoskeleton, in particular microtubules, or it may be the consequence of altered stability of microtubules.

If the tyrosination state of α-tubulin is important in modifying the stability of microtubules, it may be through affecting the interaction of tubulin with MAPs that are known to modify microtubule stability and influence the polarity of neurite outgrowth (Bernhardt *et al.*, 1985; Olmsted, 1986; Ferreira and Caceres, 1989). Indeed, it was proposed by Postingl *et al.*, (1979) that the tyrosination of α-tubulin may modify the secondary structure of tubulin; this could in turn, affect the interaction of tubulin with MAPs. Alternatively, tubulin tyrosination could act as a priming step for another post-translational modification, phosphorylation, in particular tubulin tyrosine phosphorylation. A coupled secondary post-translational modification may equally influence microtubule stability. *In vitro* studies have indicated that phosphorylation of the C-terminal tyrosine of α-tubulin prevents its incorporation into microtubules (Wandosell *et al.*, 1987), and therefore modulates microtubule stability. Moreover a study on isolated growth cones has shown that tyrosine kinases are stimulated during

neurite outgrowth (Matten *et al.*, 1990), resulting in an enrichment of tubulin tyrosine phosphorylation. Although this study did not attempt to reveal a tyrosine kinase specific for α-tubulin C-terminal tyrosine, the possibility that a coupled secondary post-translational phosphorylation of C-terminal tyrosine on α-tubulin occurs and may modify tubulin stability is intriguing. A phenomenon like this might result in an increased pool of tyrosinated tubulin in a non-assembled form.

Evidence suggesting that the tyrosination state of tubulin has no influence on stability comes largely from the study of Paturle *et al.*, (1989). In this study, the workers claimed to separate tyrosinable species of tubulin according to their ability to participate in the tyrosination/detyrosination (post-translational modification) cycle. Using tubulin species they defined as participating in the tyrosination/ detyrosination cycle, these workers examined the differences between detyrosinated and tyrosinated forms, of this species, with respect to their ability to associate with MAPs and their assembly kinetics. Their analysis found no difference between the tyrosinated and detyrosinated forms of this tyrosinable/ detyrosinable species, which led them to conclude that the tyrosination state of tubulin did not affect microtubule stability. Paturle *et al* (1989) separated tyrosinable and non-tyrosinable forms from this species by an immunoaffinity column coupled with YL1/2 (specific for tyrosinated tubulin). The tubulin species used for separation was tyrosinated *in vitro*, using a TTL assay, so that the tubulin species was maximally tyrosinated, prior to separation. However, the pre-existing tyrosinated tubulin was not removed prior to the assay, such that after separation of the apparent tyrosinable and non-tyrosinable species, by the immunoaffinity column, the tyrosinable species contained tubulin tyrosinated by TTL,

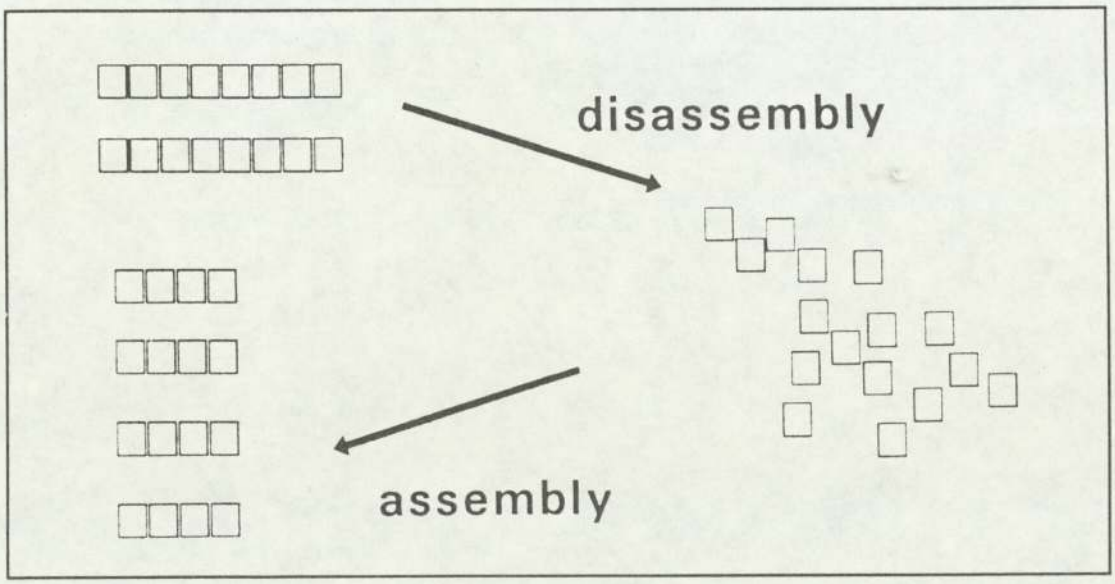
together with any pre-existing tyrosinated tubulin, which may have been generated by translation of tubulin mRNA or post-translationally by TTL. This is an important observation since other workers have found the majority of non-tyrosinable tubulin i.e. species that does not participate in the post-translational tyrosination/detyrosination cycle, is already tyrosinated (Arregui and Barra, 1990).

Thus part of the tyrosinable tubulin species separated in the study of Paturle *et al.*, 1989 may indeed be 'contaminated' with non-tyrosinable tubulin. The significance of this contaminating species in the subsequent analyses of Paturle *et al.*, 1989 is unknown, but in sufficient quantity it may mask differences between post-translationally generated tyrosinated and detyrosinated species on assembly kinetics and their association with other cytoskeletal components. Clearly this is an area worthy of further examination and evaluation, and the possibility that the tyrosination state of tubulin may modify stability still remains.

A study by Tashiro and Komiya, (1991) reported a shift in the stability of microtubules in regenerating nerve. These authors measured microtubule stability in a different way; rather than measure the actual amount of tubulin that was assembled at 37°C, they measured the sensitivity of microtubules to cold and detergent extraction. It is possible that an increased number of shorter microtubules, as opposed to fewer and longer microtubules, are more sensitive to the cold and have a different susceptibility to detergent than longer microtubules; this may explain the increase in the sensitivity of microtubules to the cold observed by Tashiro and Komiya (1991) in regenerating axons i.e. decreased stability. However, according to this hypothesis no change in the total polymer mass would be required to observe this stability change.

According to the dynamic instability model for microtubule assembly/ disassembly, the polymer mass remains constant at steady state but the number of microtubules and length may differ i.e. a decrease in length of microtubules would result in an increased number of microtubules to maintain a constant polymer level. The findings from my own work suggest that there is no overall shift in the total population of tubulin that exists in a polymerized form between regenerating and control nerve. However, of course, there may be a shift in the steady state of tubulin, which results in an increased population of tyrosinated microtubules. The shift in steady state may be associated with a change in microtubule number and length, which account for the apparent stability change observed by Tashiro and Komiya, (1991), as discussed above.

The simplified model shown below outlines the change of longer microtubules to shorter microtubules. In this model two long microtubules disassemble (depolymerize) and then reassemble (polymerize) into four shorter microtubules. The subunits that exist in a polymerized form is maintained following reassembly.



Alternatively, there is the possibility that the tyrosination state of α-tubulin

may influence organelle transport. Miller *et al.*, (1987) defined two distinct microtubule populations: transport microtubules and architectural microtubules. An earlier study by Miller and Lasek, (1985) found the transport microtubule populations to be sensitive to both the cold and treatment with colchicine, a criterion that has been used to define dynamic microtubules, which are predominantly tyrosinated (Falconer *et al.*, 1989). Indeed, one of the mechanochemical ATPases, cytoplasmic dynein, responsible for the transport of organelles interacts with microtubules at the C-terminal of α-tubulin as well as β-tubulin (Paschal *et al.*, 1989). It is possible that the association of this mechanochemical ATPase to microtubules is influenced by the tyrosination state of α-tubulin. An increased tyrosination state of α-tubulin in regenerating axons may reflect a requirement for transport microtubules, rather than architectural microtubules, at the advancing growth cones and in the growing axons.

In summary, my own data are consistent with a recent cell culture study (Ahmad *et al.*, 1993) suggesting that within axons there are regional differences in tubulin tyrosination; in particular, there is an increased level of α-tubulin tyrosination in the region nearest to the advancing growth cone. My work, examining the level of tubulin tyrosination in regenerating sciatic nerve, has shown an increase in the proportion of α-tubulin which is tyrosinated distal to the crush as compared to proximal and to uncrushed nerve. In addition, I have demonstrated that the change is present in regenerating nerve fibres and that the increase in the proportion of tyrosinated tubulin is essentially in the non-assembled tubulin fraction. Furthermore, I have demonstrated that TTL activity can be measured from rat sciatic nerve, although the measurements made reflect the activity against endogenous tubulin substrate. In conclusion, it has been

demonstrated that tubulin tyrosination can be included among the number of growth associated changes that occur in neuronal development *in vivo* and in differentiating cells *in vitro* and which are reiterated during mammalian nerve regeneration.

2.2 Acetylation

The post-translational acetylation of α-tubulin was first described in *Chlamydomonas*, and this process was found to occur on the E amino group of the lysine at residue 40 of the tubulin sequence, (L'Hernault and Rosenbaum, 1983;1985). In addition the acetylation of α-tubulin has been studied in cell culture (Black and Keyser, 1987; Cambray-Deakin and Burgoyne, 1987; Ferreira and Caceres, 1989); in general, acetylated α-tubulin was found to be co-distributed with detyrosinated α-tubulin and predominantly localized in stable microtubule subpopulations (Webster and Borisy, 1989; Falconer *et al.*, 1989; Ferreira and Caceres, 1989). Observations made on PC12 cells differentiated by NGF, showed that acetylated α-tubulin appeared following neuronal maturation.

It has only recently been demonstrated that an acetylase activity is present in brain tissue, with the ability to acetylate tubulin at Lys_{40} (Edde, *et al.*, 1991). The purpose of my own studies was to establish whether any change in the acetylation state of α-tubulin occurred as a consequence of nerve injury in rat sciatic nerve, and also to measure the acetylase activity, if detectable, in pieces of regenerating and control nerve.

Acetylated α-tubulin immunoreactivity was measured five days after a crush injury in pieces of rat sciatic nerve, immediately proximal (control) and distal (peripheral) to a nerve crush and their equivalents from contralateral control nerve. Dot immunobinding analysis revealed no difference in acetylated α-tubulin immunoreactivity between regenerating and control nerve. My efforts to measure acetyl transferase/acetylase activity in rat sciatic nerve were unsuccessful; nor was I able to measure this activity in rat brain. Edde *et al.*, (1991) suggested that acetylation may

only be a minor post-translational modification of tubulin; they proposed that previous attempts to measure acetylated tubulin in neuronal cell cultures incubated with radioactive acetate in the presence of the protein synthesis inhibitor cycloheximide, were inconclusive in that post-translational glutamylation of tubulin rather than acetylation was responsible for the radiolabel uptake into tubulin. It was suggested that radiolabelled acetate was converted into glutamate via acetyl coenzyme A and part of the tricarboxylic acid cycle, and that the radiolabelled glutamate was added to tubulin. However, I was unable to detect either glutamylation or acetylation of tubulin in brain extracts using my assay system. This non-observance may possibly relate to the preparation of extracts for assay. Indeed, the acetylase assays to date have measured activity in whole cell systems i.e. cell culture, that have not been perturbed, with the exception of the assay on mouse brains by Edde *et al.* (1991). The principle difference between the preparations appears to be that Edde *et al.*(1991) purified microtubule proteins from brain by cycles of assembly and disassembly, and examined the acetylase associated with microtubule protein. In our own assay system we examined acetylase activity in crude homogenates or the activity associated with particulate or soluble fractions following subfractionation by centrifugation. It is possible that when Edde *et al.* (1991) partially purified the brain extracts it may well have removed any endogenous deacetylase or acetylase inhibitors present which would affect acetylase activity. In support of this view Greer *et al.*(1985) when characterizing acetylase activity in *Chlamydomonas*, suggested there was a deacetylase or an inhibitor of acetylase present, which regulated the acetylase activity. Our inability to acetylate α-tubulin *in vitro* may therefore relate to the presence of an inhibitor of acetylase and/or a deacetylase enzyme in our sample preparations.

In conclusion, I can present no evidence for a change in the acetylation state of tubulin in rat sciatic nerve, as a consequence of nerve regeneration, nor was I able to detect the presence of endogenous acetylase activity directed against tubulin, though the fact that 6-11B-1 recognizes acetylated tubulin is evidence that an acetylase must exist. As previously mentioned, acetylated tubulin is associated with stable microtubules (Kreis, 1987; Webster and Borisy, 1989; Ferreira and Caceres, 1989) and the post-translational acetylation state of tubulin depends on the stability of microtubules (Baas *et al.*, 1991). I did not detect a shift in the stability of microtubules, with respect to the state of assembly of microtubules between regenerating and control nerve, and therefore the similarity in the acetylation state of tubulin between regenerating and control nerve reflects the observations that the acetylation state of microtubules is dependent on the stability of microtubules.

3 PHOSPHORYLATION OF MICROTUBULE PROTEINS

Several studies have indicated a role for the post-translational phosphorylation of microtubule proteins during cell differentiation and neurite outgrowth in cell culture systems. For example, PC12 cells have been shown to exhibit enhanced MAP-2 phosphorylation following cell differentiation with NGF and during neurite extension (Alletta *et al.*, 1988; Sano *et al.*, 1990; Miyasaka *et al.*, 1990). In a mouse neuroblastoma cell line, an increase in β-tubulin phosphorylation was found to accompany cell differentiation and neurite outgrowth (Gard and Kirschner, 1985; Serrano *et al.*, 1987), a phenomenon also found to occur during neuritogenesis following PC12 cell differentiation in PC12 cells (Alletta *et al.*, 1987). In addition,

some studies on nerve regeneration have indicated that the post-translational phosphorylation of microtubule protein may increase with nerve regeneration. For example, Larivee and Grafstein, (1987) found that tubulin phosphorylation increased during nerve regeneration of the goldfish optic nerve, Watterson *et al.*, (1989) also observed an increase in the phosphorylation of β-tubulin in cervical ganglia during nerve regeneration.

I wished to examine the pattern of phosphorylation of microtubule protein during nerve regeneration in the rat sciatic nerve. Specific antibodies that recognize and differentiate between phosphorylated and non-phosphorylated forms of microtubule protein are not readily available. In the light of these difficulties I decided to screen for specific protein kinase activities that were directed against microtubule protein. A number of protein kinases have been identified with the ability to phosphorylate microtubule proteins, in particular cyclic AMP (cAMP)-dependent protein kinase (II), Ca^{2+}/calmodulin dependent-protein kinase (II) and a casein kinase II-like enzyme. Ca^{2+}/calmodulin-dependent kinase (II) has been shown to phosphorylate both tubulin and MAP-2 *in vitro* (Yamamoto *et al.*, 1985; Goshima *et al.*, 1993) while cAMP-dependent protein kinase (II) has been shown to phosphorylate MAP-2 and to a lesser extent tubulin *in vitro* (Yamamoto *et al.*, 1985). The casein kinase II-like enzyme has the ability to phosphorylate MAP-1 and tubulin (Avila *et al.*, 1988). The aim of my own work was to screen a number of kinase activities directed towards tubulin and MAPs, and to see if kinase activity was related to phenomena observed in cell culture systems (with respect to phosphorylation of microtubule protein).

In my study, I used selective conditions to screen for the activities of casein

kinase (II)-like enzyme, cAMP-dependent protein kinase (II) and Ca^{2+}/calmodulin-dependent protein kinase (II). In consideration that phosphorylation systems are a mixture of protein kinase(s), ATPase(s) and protein phosphatase(s), I attempted to enhance the detection of protein kinase activity, by including inhibitors of ATPases and phosphatases in protein kinase assays.

Protein kinase activity was measured five days after a crush injury, in subfractionated 1cm pieces of rat sciatic nerve immediately proximal or distal to a nerve crush together with their equivalents from contralateral control nerve, or L4,L5 dorsal root ganglia from regenerating and control rat sciatic nerve. Using conditions for screening Mg^{2+}-dependent/Ca^{2+}-independent protein kinase(s) activity, I detected an activity directed against an exogenous tubulin substrate, in the cytosolic fraction of L4,L5 dorsal root ganglia from both regenerating and control nerve. This activity was only seen when the exogenous tubulin was polymerized prior to incubation indicating a preference for assembled tubulin as a substrate (see RESULTS section 2.1.1). The activities were not significantly different between regenerating and control L4,L5 dorsal root ganglia (see RESULTS section 2.1.1). The absence of this from particulate fractions of L4,L5 dorsal root ganglia suggested that the enzyme was associated with the cytosolic fraction.

Under the Mg^{2+}-dependent/Ca^{2+}-independent conditions employed in this work, the protein kinase(s) responsible for the phosphorylation of tubulin could not have been Ca^{2+}/calmodulin dependent protein kinase (II); indeed no protein kinase activity from either cytosolic or particulate fractions could be detected against tubulin when screening for Ca^{2+}/calmodulin-dependent protein kinase (II) activity (see RESULTS section

2.1.3). It also appears unlikely that the phosphorylation of tubulin resulted from tyrosine kinase $pp60^{c\text{-}src}$ activity, which is enriched in neurones and has affinity for tubulin (Matten *et al.*, 1990) because this protein kinase is found in membrane-associated fractions (Matten *et al.*, 1990) and no protein kinase activity directed against tubulin could be detected in the particulate fractions under Mg^{2+}-dependent/Ca^{2+}-independent conditions. While the conditions used in these screening assays were similar to those employed for screening cAMP-dependent protein kinase (II) activity, the addition of cAMP had no effect on the kinase activity directed towards tubulin, over that which was observed in the absence of cAMP. An important observation was that the protein kinase activity towards the exogenous tubulin, was only observed when the exogenous tubulin was in a polymeric form. This indicates that the Mg^{2+}-dependent/Ca^{2+}-independent tubulin phosphorylating activity detected in the L4,L5 dorsal root ganglia is probably the casein kinase (II)-like enzyme described by Serrano *et al.*, (1987). These authors suggest that a casein kinase II-related enzyme is involved in the phosphorylation of tubulin *in vivo* in neuroblastoma cells, and that this protein kinase preferentially phosphorylates polymerized tubulin.

In a previous study (Watterson *et al.*, 1989), protein kinase activity directed against tubulin was detected in superior cervical ganglia. Furthermore, this study found that tubulin phosphorylation was elevated in superior cervical ganglia that regenerated following injury, on comparison to control cervical ganglia. These authors measured the incorporation of radiolabelled phosphate into endogenous proteins, as opposed to the exogenous substrate used in my own study. The method employed by Watterson *et al.*,

does not reflect the actual protein kinase activities; indeed, this method is similar to the method I used to measure TTL activity (previous section 2). Thus, there is the possibility that the substrate may be a limiting factor in their study and that the protein kinase activity itself does not alter, as observed for Mg^{2+}-dependent/Ca^{2+}-dependent protein kinase activity in dorsal root ganglia, from my own work. In support of this, are the observations made by Watterson *et al.*, (1989) at time periods during regeneration. The phosphorylation states (reflected by the incorporation of radiolabel) of individual proteins relative to one another, differ during different times of regeneration. Indeed, the increased tubulin phosphorylation observed by Watterson *et al.*, may reflect differences in the expression of tubulin between regenerating and control superior cervical ganglia. The expression of specific tubulin isotypes have been shown to increase during nerve regeneration (Hoffman and Cleveland, 1988; Miller *et al.*, 1989). This may also account for observations made on neuronal cultures, where tubulin phosphorylation was found to increase upon differentiation (Gard and Kirschner, 1985; Serrano *et al.*, 1987). Thus the elevated tubulin phosphorylation may be likened to a translational rather than a post-translational difference in the processing of tubulin. My own data support this.

As previously mentioned above using conditions to screen for cAMP-dependent protein kinase (II) and Ca^{2+}/calmodulin-dependent protein kinase (II), no activity towards tubulin could be detected in either the cytosolic or particulate fractions. In the study of Watterson *et al.* (1989) cAMP-dependent protein kinase activity directed against tubulin was detected in superior cervical ganglia. There is conflicting evidence to whether cAMP-dependent protein kinase (II) phosphorylates tubulin; Sloboda *et*

al.(1975) suggests cAMP-dependent protein kinase (II) does not phosphorylate tubulin, whereas Yamamoto *et al.* (1985) provide evidence that it does. In my own study, I was unable to detect cAMP-dependent protein kinase (II) activity directed against exogenous tubulin, but I did detect cAMP-dependent protein kinase (II) activity directed against the microtubule associated protein MAP-2 (see RESULTS section 2.2.2). This apparent difference may reflect the affinity of cAMP-dependent protein kinase (II) for MAP-2 and tubulin; kinetic studies on MAP-2 and tubulin phosphorylation by cAMP-dependent protein kinase (II) support this view (Yamamoto *et al.*, 1985). In this study, Yamamoto *et al.* found that MAP-2 was able to incorporate up to 8-fold more phosphate residues than tubulin, together with a 16-fold greater V_{max}. This suggests that it would be easier to detect MAP-2 phosphorylation above that of tubulin phosphorylation, over a particular time period.

In summary, screening for Mg^{2+}-dependent/Ca^{2+}-independent protein kinase, activity in L4,L5 dorsal root ganglia, and pieces of rat sciatic nerve proximal and distal to a crush and equivalent pieces of control nerve, has revealed the presence of a protein kinase activity, in L4,L6 dorsal root ganglia, directed against tubulin that is unaltered as a consequence of nerve regeneration. This protein kinase activity appears to be related to casein kinase (II).

MAP-2 may be phosphorylated by a number of kinases, in particular, type (II) cAMP-dependent protein kinase (II) (Vallee *et al.*, 1980; Yamamoto *et al.*, 1985), Ca^{2+}/calmodulin kinase (II) (Yamamoto *et al.*, 1983;1985; Schulman, 1984), protein kinase C (Tsuyami, 1986., Hoshi *et al.*, 1988a,b) and growth factor sensitive MAP-2 kinase(s) (Landreth *et al.*, 1990; Sano *et al.*, 1990; Miyasaka *et al.*, 1990). Indeed,

MAP-2 can undergo multiple phosphorylation, with distinct phosphorylation sites which are characteristic of the protein kinase that is responsible (Walaas and Nairn, 1989).

Ca^{2+}/calmodulin-dependent protein kinase (II) has been characterized partially in rat brain (Fukunaga *et al.*, 1982) and has been localized in both cytosolic and particulates fractions obtained from brain (DeCamilli *et al.*, 1984). Ca^{2+}/calmodulin dependent protein kinase (II) may phosphorylate tubulin as well as MAP-2. Indeed phosphorylation of tubulin modifies its association with phospholipid vesicles (Hargreaves *et al.*, 1986) which underlies a role for this protein kinase in cytoskeleton-membrane interactions.

Under conditions used for screening Ca^{2+}/calmodulin-dependent protein kinase (II) activity directed towards MAP-2, activity was detected in particulate fractions obtained from all pieces of rat sciatic nerve, from regenerating and control nerve (see RESULTS section 2.2.3). As previously mentioned, no activity was detected against tubulin.

There was no significant difference in the protein kinase activities between regenerating and control nerve (see RESULTS section 2.2.3). The presence of activity directed against MAP-2 (but not tubulin) may possibly relate to differences in the affinity of the Ca^{2+}/calmodulin-dependent protein kinase (II) for tubulin and MAP-2. Indeed, Yamamoto *et al.*, 1985, studied the affinity and kinetics for this protein kinase directed against tubulin and MAP-2. Like cAMP-dependent protein kinase, Ca^{2+}/calmodulin-dependent protein kinase (II) had a higher affinity for MAP-2 than tubulin (Yamamoto *et al.*, 1985). Similarly MAP-2 incorporated 5-fold more phosphate

than tubulin by Ca^{2+}/calmodulin-dependent protein kinase (II). Thus, in the presence of an excess of exogenous substrate, the detection of MAP-2 would be greater than tubulin, over a set time period.

The inability to detect Ca^{2+}/calmodulin-dependent protein kinase (II) activity in the cytosolic fraction, may suggest the presence of Ca^{2+}-activated proteases and/or phosphatases. Ca^{2+}-activated phosphatases could dephosphorylate the substrate phosphorylated by Ca^{2+}/calmodulin-dependent kinase (II) resulting in no net incorporation of radiolabel into the exogenous substrate. For example the calcium and calmodulin binding protein calcineurin (Klees *et al.*, 1979) is a protein phosphatase, particular in the dephosphorylation of MAP-2 as well as tubulin (Goto *et al.*, 1985). Calcineurin dephosphorylates MAP-2 universally in a non-site-specific manner, particularly MAP-2 that has been phosphorylated by Ca^{2+}/calmodulin-dependent protein kinase (II) (Goto *et al.*, 1985). Calcineurin was purified by Tallant *et al.*, (1983) and found to be associated in both the cytosolic and particulate fractions from brain extracts. However, these workers suggested calcineurin activity associated with the particulate fraction is frequently not fully accessible; indeed, measurement of calcineurin activity in the particulate fraction was difficult if not impossible (Tallant *et al.*, 1983). If the inability to measure any incorporation of radiolabel into MAP-2 or tubulin in cytosolic fractions was due to the presence of calcineurin, then the difficulty in extracting calcineurin activity in particulate fractions may explain why radiolabel incorporation was detected in particulate fractions. Other workers have measured Ca^{2+}/calmodulin-dependent protein kinase (II) activity in the cytosolic fraction of brain tissue (Yamamoto *et al.*, 1983; 1988). However, the protein kinase activity in the cytosolic fraction was extracted by detergent, the presence of which has been suggested to inhibit calcineurin

ctivity (Tallant *et al.*, 1983).

A recent study of the over-expression of Ca^{2+}/calmodulin protein kinase (II) ansfectants and their phenotypic effect has been performed on neuroblastoma cell lines Goshima *et al.*, 1993). The principal findings of this study were that transfection of euroblastoma cell lines with Ca^{2+}/calmodulin protein kinase (II) enhanced neurite utgrowth and growth cone mobility. However, these phenotypic effects were only bserved when Ca^{2+}/calmodulin-dependent protein kinase (II) existed in a Ca^{2+}/calmodulin-independent state, which occurs when the former is autophosphorylated n its threonine$_{286}$ residue. The purpose of this is thought to prolong the activities of he kinase after the intracellular Ca^{2+} concentration is reduced to resting levels (Ohsako *et al.*, 1991).

It is possible that the activity of Ca^{2+}/calmodulin-dependent protein kinase (II) in mammalian nerve is modulated by autophosphorylation as occurs in neuroblastoma cell lines. My own work suggested that this protein kinase activity obtained from particulate fractions of regenerating and control nerve was unaltered (see RESULTS section 2.2.3). This finding may reflect the free Ca^{2+} concentration used in the assay,leading the activities of Ca^{2+}/calmodulin protein kinase to be similar in regenerating and control nerve, and both maximally stimulated. *In vivo* however, the Ca^{2+} influx and intracellular calmodulin levels may well be different between regenerating and control nerve, and although the absolute levels of Ca^{2+}/calmodulin-dependent protein kinase (II) could be similar, the state of autophosphorylation, a prerequisite of the cellular function of this kinase (Goshima *et al.*, 1993), may be different between regenerating and control nerve dependent or the Ca^{2+} and calmodulin levels. Le Vine *et al.*, (1986) indicates that at the cellular level of calmodulin, the stimulation of autophosphorylation

is a cooperative function of Ca^{2+} levels; thus, a small change in the intracellular concentration of Ca^{2+} could result in a change in the autophosphorylation state. Furthermore, mobile growth cones in cultured neurones from embryonic rat brain have higher intracellular Ca^{2+} levels than non-mobile growth cones. Thus, different activation states of Ca^{2+}/calmodulin-dependent protein kinase (II) may exist between regenerating and control nerve, dependent on intracellular concentration of Ca^{2+} and calmodulin. Using this principle, the assay system used in my study would be unable to detect a difference in the state of activation of Ca^{2+}/calmodulin-dependent protein kinase (II) between regenerating and control rat sciatic nerve.

In summary, on screening for Ca^{2+}/calmodulin-dependent protein kinase (II) activity there is no apparent difference between the levels of Ca^{2+}/calmodulin-dependent protein kinase (II) in particulate fractions of pieces of regenerating and control nerve. However this may not reflect the *in vivo* activities of Ca^{2+}/calmodulin-dependent protein kinase (II) as discussed earlier. The inability to measure Ca^{2+}/calmodulin-dependent protein kinase (II) activity in cytosolic fractions may reflect the presence of Ca^{2+}-activated phosphatases e.g. calcineurin. This may, in part, reflect the importance of phosphatases in phosphorylation systems and cytoskeletal modification, and is worthy of further investigation.

Using conditions for screening Mg^{2+}-dependent/Ca^{2+}-independent protein kinase(s), I detected activity directed against exogenous MAP-2 in cytosolic fractions from pieces of regenerating and control nerve (see RESULTS section 2.2.1). Furthermore, the activity in pieces of nerve proximal to a crush was higher than that in pieces of contralateral control nerve. The addition of cAMP to the assay system, in order to screen additionally for cAMP-dependent protein kinase activity, increased the

observed kinase activity directed against MAP-2, and maintained the difference in activity between pieces of regenerating nerve proximal to a crush and equivalent pieces of contralateral control nerve (see RESULTS section 2.2.2). The possibility arose that the kinase activities observed under Mg^{2+}-dependent/Ca^{2+}-independent conditions in the presence or absence of cAMP might be mediated by one or more protein kinases.

Nerve growth factor (NGF)-sensitive protein kinase displays Ca^{2+}-independent activity, has been detected in cell culture upon cell differentiation and can phosphorylate MAP-2 (Landreth *et al.*, 1990). Analysis of the subcellular distribution of this kinase revealed the kinase was enriched in crude cytosolic fractions, (Landreth *et al.*, 1990). However, this serine-threonine kinase discovered by Landreth *et al.*, (1990) was found to catalyze the incorporation of less than 1 mol phosphate/mol MAP-2, whereas the serine-residue specific cAMP-dependent protein kinase catalyzed an incorporation in excess of 12 mol phosphate/ mol MAP-2 (Yamamoto *et al.*, 1985). If NGF-sensitive protein kinase was the kinase activity present in the absence of cAMP, then the addition of cAMP would be expected to mask the activity of NGF-sensitive protein kinase upon the stimulation of cAMP-dependent protein kinase (II). Either that, or NGF-sensitive protein kinase and cAMP-dependent protein kinase (II) both exhibited an increased activity in pieces of regenerating nerve proximal to a crush.

There are observations to suggest MAP-2 is not a physiologically relevant substrate for NGF-sensitive kinase, described by Landreth *et al.*, (1990). Greene *et al.* (1983) found PC12 cells contained very little (if any) MAP-2 which when present, was a minor constituent of fully differentiated PC12 cells. Furthermore, there is evidence which indicates that NGF stimulates cAMP-dependent protein kinase (II) activity in cultured cells (Brady *et al.*, 1990), and that this enzyme has a preference and high

affinity for MAP-2 as a substrate (Vallee, 1980; Theurkauf and Vallee, 1982). This would suggest that the Mg^{2+}-dependent/Ca^{2+}-independent activity in the absence of cAMP is not the NGF-sensitive protein kinase described by Landreth *et al.*, (1990) but is likely to be cAMP-dependent protein kinase (II) itself, and that the addition of cAMP merely enhances the observed activity directed against MAP-2.

In support of a role for cAMP (type II)-dependent protein kinase in neuronal function, particularly cytoskeletal function, is the ability of this kinase to phosphorylate MAP-2 *in vivo*, and thereby modify microtubule stability (Jameson *et al.*, 1980; Diaz-Nido *et al.*, 1990) and the interaction of MAP-2 with actin microfilaments resulting in the inhibition of actin gelation (Seldon and Pollard, 1983). Thus MAP-2 phosphorylation mediated by cAMP-dependent protein kinase can influence neuronal plasticity and may possibly be of importance during nerve regeneration, by allowing changes in cytoskeletal architecture in particular, through its effect on the assembly and disassembly of microtubules in response to changes in the local environment, .

The localization of the increased protein kinase activity was odd because it was not an area where regeneration was occurring and no change in protein kinase activity was observed in pieces distal to crush where axons were actively regenerating. It is unlikely to be the result of an alteration in the expression of protein kinases, otherwise one might expect to see an increased kinase activity in dorsal root ganglia of regenerating nerve, which was not evident from my own data. One possible explanation of this phenomenon is that the increased protein kinase activity reflects an accumulation of the protein kinase(s). If the increase in protein kinase activity proximal to a crush was the consequence of protein kinase accumulation at the injury site, one might expect a drop in the kinase activity compared with controls in regenerating axons distal to the

crush. However, no such observation was made from my own study.

An alternative explanation for the apparent increase in the protein kinase activity, is a shift in the subcellular distribution of the protein kinase e.g. from the membrane to the cytosol. The cAMP-dependent protein kinase (II) is associated with both membranes and the cytosol (Rubino *et al.*, 1989), and a shift in the subcellular distribution could explain the observed phenomena. In my study I was unable to extract and measure cAMP-dependent protein kinase activity in the crude, membrane-associated particulate fraction, thus we were unable to measure directly any shift in the protein kinase activity.

The binding of the regulatory subunits on cAMP-dependent protein kinase (II) to specific cellular proteins may be a mechanism for the localization of the protein kinase (Leiser *et al.*, 1986). The regulatory subunits have a high affinity for MAP-2, the localization of cAMP-dependent protein kinase (II) may therefore reflect the localization of MAP-2. Furthermore, MAP-2 associates with microtubules in a manner that is dependent on its phosphorylation state. The dissociation of MAP-2 from microtubules, would release free catalytic subunits on cAMP-dependent protein kinase associated with MAP-2, effectively increasing cAMP-dependent protein kinase activity. Although no antibodies specific for phosphorylated forms of MAP-2 are available, I was able to measure the distribution of MAP-2 between assembled and non-assembled microtubules. In view of this, I wished to measure the distribution of MAP-2 between assembled and non-assembled microtubules in pieces of regenerating nerve proximal and distal to a crush, and equivalent regions of contralateral control nerve. A shift in the distribution of MAP-2 between assembled and non-assembled microtubule could explain the change in cAMP-dependent protein kinase activity.

MAP-2 is a group of related proteins, the main isoforms are MAP-2a and MAP-2b, and a isoform with a lower molecular weight MAP-2c (Couchie and Nunez, 1985; Garner *et al.*, 1988). The intracellular distribution of MAP-2c is fairly ubiquitous (Watson *et al.*, 1991) whereas MAP-2a,2b appears to be preferentially localized in dendritic regions of the cell body, although it has been found in axons (Papasozomenos *et al.*, 1985; Ferreira *et al.*, 1987; Watson *et al.*, 1991). This compartmentalization may reflect differential rates of turnover of MAP-2 (Okabe and Hirokawa, 1989).

Using an immunological method to detect the distribution of MAP-2 between assembled and non-assembled states of microtubules, I detected MAP-2 immunoreactivity in all pieces of nerve examined (see RESULTS section 2.3). MAP-2 was predominantly associated with the non-assembled fractions, which would suggest the catalytic subunits of cAMP-dependent protein kinase (II) were predominantly free. There was no shift in the distribution of MAP-2a and MAP-2b associated with assembled and non-assembled microtubules between regenerating and control nerve (see RESULTS section 2.3). However, the distribution of MAP-2c was found to alter between assembled and non-assembled microtuble protein in pieces of nerve proximal to a crush compared to equivalent pieces of control nerve (see RESULTS section 2.3). There was a statistically significant ($P < 0.05$) increase in the amount of MAP-2c associated with the non-assembled microtubule fraction in pieces of regenerating nerve, proximal to the crush, compared to equivalent pieces of control nerve.

The physiological significance of this observation is obscure since only a 1.2% shift from the assembled to the non-assembled fraction was observed. Furthermore, it was puzzling how greater than 90% of the MAP-2 immunoreactivity was associated with the disassembled fraction, since MAP-2 has been shown to stabilise microtubules.

It is possible that MAP-2 in the axons is highly phosphorylated which may inhibit microtubule assembly, as suggested by Burns *et al.*, (1984). This may explain why the observed MAP-2 immunoreactivity was predominantly associated with the disassembled fraction. An alternative explanation, may be that either (i) detergent extraction causes displacement of MAP-2 from peripheral nerve microtubules or (ii) the sciatic nerve MAP-2- like protein is less likely to interact with microtubules. Precisely the same distribution of MAPs is seen by Tashiro and Komiya in their Triton-extracted microtubule preparations (Komiya, Y., personal communication). However, the observation is consistent with the idea that an increase in MAP-2 associated with the non-assembled fraction would affect the availability of the catalytic subunits of cAMP-dependent protein kinase (II).

In summary, the increase in cAMP-dependent protein kinase (II) may be the consequence of a number of factors, such as accumulation, a shift in the subcellular distribution of the protein kinase, or modification of the activity dependent on its association to cellular proteins e.g. MAP-2, together with other phenomenon which as yet are unknown. The physiological importance of the observed changes is not known, but it is likely to be related in some way to the control of microtubule integrity or another cellular process that is dependent on cAMP-dependent protein kinase (II).

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